Uncertainty Quantification of Sparse Travel Demand Prediction with Spatial-Temporal Graph Neural Networks

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ABSTRACT

Origin-Destination (O-D) travel demand prediction is a fundamental challenge in transportation. Recently, spatial-temporal deep learning models demonstrate the tremendous potential to enhance prediction accuracy. However, few studies tackled the uncertainty and sparsity issues in fine-grained O-D matrices. This presents a serious problem, because a vast number of zeros deviate from the Gaussian assumption underlying the deterministic deep learning models. To address this issue, we design a Spatial-Temporal Zero-Inflated Negative Binomial Graph Neural Network (STZINB-GNN) to quantify the uncertainty of the sparse travel demand. It analyzes spatial and temporal correlations using diffusion and temporal convolution networks, which are then fused to parameterize the probabilistic distributions of travel demand. The STZINB-GNN is examined using two real-world datasets with various spatial and temporal resolutions. The results demonstrate the superiority of STZINB-GNN over benchmark models, especially under high spatial-temporal resolutions, because of its high accuracy, tight confidence intervals, and interpretable parameters. The sparsity parameter of the STZINB-GNN has physical interpretation for various transportation applications.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**.

KEYWORDS

Spatial-temporal Sparse Data, Uncertainty Quantification, Graph Neural Networks, Travel Demand

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1 INTRODUCTION

It has attracted a lot of attention to predict the Origin-Destination (O-D) matrices of travel demand [10, 12, 40]. This task is challenging because the characteristics of the demand data vary with mobility services. For example, the O-D matrices for the subway stations are generally dense and probably satisfy the continuous Gaussian distribution, which implicitly underlies deterministic deep learning models. But for ride-hailing or bike-sharing services, the O-D zones could be much more granular, which lead to sparse and discrete entries in the O-D matrices. This sparsity issue becomes even more severe in the O-D matrices with high spatial-temporal resolution, because the initially dense O-D matrices could become much more diluted. As the mobility companies are increasingly adopting real-time interventions, the accurate prediction of the sparse and discrete O-D matrices at a high spatial-temporal resolution could significantly improve the quality of mobility services.

Although the prediction of dense and low-resolution O-D matrices has been extensively studied using deep learning (DL) models, few studies addressed the sparsity issue in the high-resolution travel demand data. The continuous data entries in the dense O-D matrices could follow Gaussian distributions [12, 13, 21, 30, 45, 46]. However, a large number of zero entries in a sparse O-D matrix evidently deviate from the Gaussian assumption. When the O-D matrix is sparse and dispersed with integer values, discrete distributions like the negative binomial distribution, would be more appropriate. Moreover, an enormous number of zeros could naturally emerge when the data set has a high spatial-temporal resolution. These zeros are important for transportation management, because they indicate areas with particularly low demand. Therefore, a successful prediction model should capture explicitly the zeros in the sparse

matrix and quantify their uncertainty, thus guiding the service allocation and management decisions.

To address sparsity, we propose a Spatial-Temporal Zero-Inflated Negative Binomial Graph Neural Network (STZINB-GNN) to quantify uncertainty and enhance prediction performance. We utilize zero-inflated negative binomial (ZINB) distributions to capture the enormous number of zeros in sparse O-D matrices, and the negative binomial (NB) distribution for each non-zero entry. Different from the variational autoencoder models, we design the spatial-temporal embedding with an additional parameter π to learn the likelihood of the inputs being zero. Our model utilizes the representation power of spatial-temporal graph neural networks to fit the parameters of probabilistic distributions. We compare a variety of probabilistic layers to assess the effectiveness of π in capturing sparsity. Empirically, we demonstrate the superiority of our model in the data set with fine-grained spatial-temporal resolution. Our main contributions include:

(1) We propose the STZINB-GNN to quantify the spatial-temporal uncertainty of O-D travel demand using a parameter π to learn data sparsity
(2) The parameters of the probabilistic GNNs successfully quantify the sparse and discrete uncertainty particularly in high-resolution data sets
(3) We demonstrate that the STZINB-GNN outperforms other models by using two real-world travel demand datasets with various spatial-temporal resolutions

The paper is organized as follows. Section 2 summarizes recent studies related to DL models for travel demand prediction, sparse data modeling, and uncertainty quantification. Section 3 defines the research question and develops the model. Section 4 introduces the dataset used for the case study, the evaluation metrics, and the experimental results. Section 5 concludes the paper and discuss future research.

2 LITERATURE REVIEW

2.1 Travel demand prediction with spatial-temporal deep learning

Travel demand prediction is a fundamental task in transportation applications. Based on the spatial division of the area of interest into zones, travel demand prediction is usually associated with the prediction of flow from the origin zone to the destination zone pairs, in the form of O-D matrices [12]. The main challenges for O-D matrix prediction relate to time series prediction and spatial correlation detection. Recently these challenges have been addressed using spatial-temporal deep learning techniques. Convolution Neural Network (CNN) based techniques are applied to extract the spatial patterns because O-D matrices are usually modelled on urban grids [22, 41, 43]. Noticeably, the CNN and its variants discover spatial patterns in Euclidean space [39]. The O-D matrices naturally possess the graph structure where origin or destination regions are usually regarded as the nodes and the O-D pairs are the edges. Recent work also applied graph neural networks (GNNs) on travel demand estimation to capture non-Euclidean correlations [4, 40]. On the other hand, Ke et al. [12] regarded O-D pairs, instead of origin and destination regions, as nodes and applied multi-graph convolutional network to predict ride-sourcing demand. Other studies used deep learning to predict the individual travel behavior by focusing on interpretation and architectural design [32, 34]. The power of deep learning arises from its representation learning capacity and its new statistical foundation [33]. However, both CNN and GNN-based approaches treat the O-D matrix entries as continuous and only focus on coarse temporal resolutions, like 60 minutes. Very few studies discussed the challenge to analyze the sparse O-D matrices with high spatial-temporal resolution.

Sparse travel demand data are different from the missing data. Zeros in sparse data mean no trips, but the sensors work properly. Missing data, on the other hand, are unobservable, potentially due to sensor malfunction. Wang et al. [35] applied the spatial-temporal Hankelization and tensor factorization to estimate traffic states using only 17% of the matrix entries. However, tensor factorization is a transductive method, which needs recalculation when new data come in. It is not suitable for short-term prediction. To differentiate our contributions from previous work, we will emphasize our model's spatial-temporal interpretability and uncertainty quantification in travel demand prediction.

2.2 Uncertainty of sparse travel demand prediction

Besides the average value, it is widely acknowledged that models should also predict uncertainty [26, 44]. The deterministic models that dominate the majority of the research implicitly assume homoskedasticity (i.e. common variance), which significantly simplifies the variance structure [2, 6]. However, it is also critical to capture uncertainty of sparse travel demand using deep learning. Rodrigues and Pereira [27] explored uncertainty quantification by training a CNN-LSTM model to fit both the mean and the quantiles. They showed the power of combining the spatial and temporal embedding to predict various distributions, but they did not investigate the sparse scenario. In non-DL models, Jang [8] showed that travel demand follows a zero-inflated negative binomial distribution. Rojas et al. [28] also noted that using a zero-inflation model is promising for modeling intermittent travel demand. However, recent demand prediction papers sidestepped this challenge by choosing a low resolution, such as 60 minutes [11, 12]. It is challenging to consider the higher resolution, such as 5min, because a deterministic DL model is no longer appropriate. Hence, it could be a viable alternative to combine the zero-inflated distribution and the deterministic DL to analyze the sparse O-D matrices.

It is also critical to design metrics to evaluate the quality of uncertainty quantification. Khosravi et al. [14] proposed mean prediction interval width and prediction interval coverage probability to quantify data uncertainty. Sankararaman and Mahadevan [29] introduced a likelihood-based metric, which is also an effective indicator to measure the alignment of the predicted and ground truth data distributions. Based on the existing research gap, our paper seek to design a model that formulates the sparse travel demand prediction problem with proper spatial-temporal pattern recognition and tight prediction interval bounds.

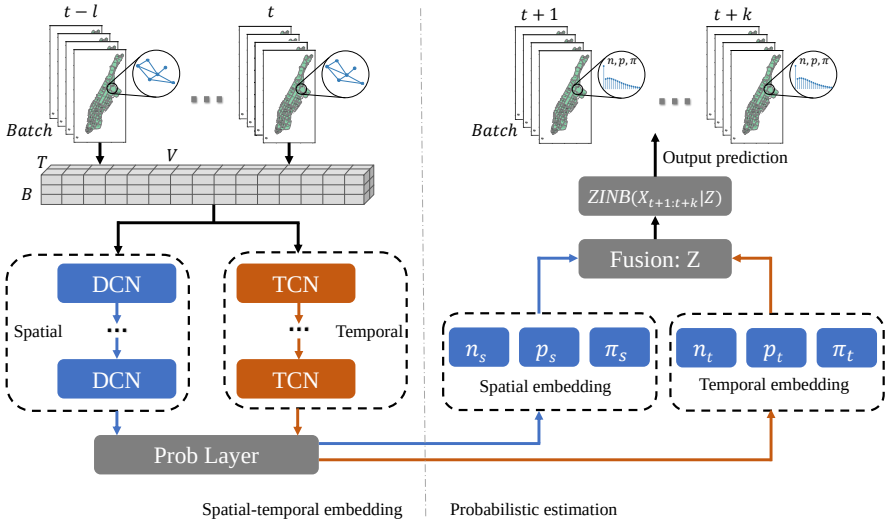


Figure 1: Framework of STZINB-GNN model.

3 METHODOLOGY

3.1 Problem Description

Our model predicts the future expected travel demand and confidence interval of each O-D pair with k time windows ahead, using m origins and u destinations along with the travel demand in time periods (windows) of length T minutes. It is a sequence-to-sequence prediction task. Different from the previous work that treated the locations of origins or destinations as vertices, we build the O-D graph $\mathcal{G} = (V, E, A)$ where V represents the O-D pair set, E denotes the edge set, and $A \in \mathbb{R}^{|V|\times|V|}$ is the adjacency matrix describing the relationship between O-D pairs [12]. It is clear that $|V| = m \times u$, and the O-D graph is fully connected. Let x_{it} denote the trips of the i^{th} O-D pair in the t^{th} time window, where $i \in V$, $x_{it} \in \mathbb{N}$. Then $X_t \in \mathbb{N}^{|V|\times T}$ denote the demand for all O-D pairs in the t^{th} time window, with x_{it} as its entry. Our goal is to leverage historical records $X_{1:t}$ as the data inputs to predict the distribution of $X_{t:t+k}$ (i.e. the demand for the next k time windows), thus analyzing the expectation and confidence intervals of the future demand.

3.2 Zero-Inflated Negative Binomial (ZINB) Distribution

We assume that the inputs follow the ZINB distribution [9, 28]. A random variable that follows NB distribution has a probability mass function f_{NB} as:

$$f_{NB}(x_k; n, p) \equiv Pr(X = x_k) = \binom{x_k + n - 1}{n - 1} (1-p)^{x_k} p^n. \quad (1)$$

where n and p are the shape parameters that determine the number of successes and the probability of a single failure respectively. However, the real-world data often have many observations with zeros and overdispersion [18]. The exploded number of zeros exacerbates the parameter learning of the NB distribution. A new parameter π is therefore introduced to learn the inflation of zeros, leading to the ZINB distribution. Formally, its probability mass function can be described as:

$$f_{ZINB}(x_k; \pi, n, p) = \begin{cases} \pi + (1-\pi) f_{NB}(0; n, p) & \text{if } x_k = 0 \\ (1-\pi) f_{NB}(x_k; n, p) & \text{if } x_k > 0 \end{cases}. \quad (2)$$

Two steps are needed to generate the distributions of the data: they are either zeros with probability π or non zeros with probability $1 - \pi$ following the NB distribution. The parameters π, n, p in the probability distributions are parameterized by spatial-temporal GNNs.

3.3 STZINB-GNN

We introduce STZINB-GNN, a generalizable deep learning architecture, to capture spatial-temporal correlations $Z = \mathcal{M}(X_{1:t}, A)$ under the ZINB assumption for each O-D matrix entry and predict the future k time windows ahead with $f_{ZINB}(X_{t+1:t+k}|Z)$. $\mathcal{M}$ is the proposed model framework that takes the historical demand $X_{1:t} \in \mathbb{Z}^{|V|\times tT}$ and adjacency matrix $A \in \mathbb{R}^{|V|\times|V|}$ as inputs, and learns the parameter embedding $Z \in \mathbb{R}^{|V|\times k\times 3}$ of the future demand $X_{t+1:t+k} \in \mathbb{R}^{|V|\times k}$ with:

$$\begin{aligned} &f_{ZINB}(X_{t+1:t+k}|n_{t+1:t+k}, p_{t+1:t+k}, \pi_{t+1:t+k}) \\ &= f_{ZINB}(X_{t+1:t+k}|\mathcal{M}(X_{1:t}, A)) = f_{ZINB}(X_{t+1:t+k}|Z). \end{aligned} \quad (3)$$

The overall architecture of the STZINB-GNN is shown in Figure 1. We convert a batch of size B of input O-D-T tensor at j^{th} time window $X_{j:j+B} \in \mathbb{Z}^{|V|\times BT}$ into a tensor $\mathcal{X}_j \in \mathbb{Z}^{|V|\times B\times T}$ as the model input. We use Diffusion Graph Convolution Networks (DGCNs) to capture the spatial adjacency of O-D pairs, and use Temporal Convolutional Networks (TCNs) for temporal correlation. The outputs of the DGCNs and TCNs, including the spatial embedding n_s, p_s, π_s and temporal embedding n_t, p_t, π_t, are used to parameterize the ZINB distribution. These spatial and temporal embeddings contain the independent estimation of the ZINB parameters of their spatial and temporal locality. We then fuse the n_s, p_s, π_s and n_t, p_t, π_t into Z using the Hadamard product, which can be replaced by other non-linear operations like a fully-connected layer. By fusing the

spatial and temporal embeddings of the distribution parameters, we obtain the final ZINB parameter set Z that fulfills

$$f_{ZINB}(X_{t+1:t+k}|n_{t+1:t+k}, p_{t+1:t+k}, \pi_{t+1:t+k}) = f_{ZINB}(X_{t+1:t+k}|Z). \tag{4}$$

Notice that the fused Z in Figure 1 can be interpreted as the parameter set of the future demand distribution. This procedure is similar to the variational autoencoders (VAEs) concept, where we learn the latent shape variables that formulate the distributions [7, 15, 16]. The theory of VAEs is used to introduce deep learning techniques into the statistical domain of variational inference, which provides more powerful representation of latent variables. Our encoding component uses the spatial-temporal embedding architecture with an additional sparsity parameter and the decoding part is related to the probabilistic estimation of future demand.

To address the problem that zeros will give infinite values for the KL-divergence based variational lower bound, we directly use the negative likelihood as our loss function to better fit the distribution into the data. Let y be the ground-truth values corresponding to one of the predicted matrix entries with parameters n, p, π from Z. The log likelihood of ZINB is composed of the $y = 0$ and $y > 0$ parts, and can be approximated as [23, 24]:

$$LL_y = \begin{cases} \log \pi + \log(1-\pi)p^n & \text{when} \quad y = 0 \\ \log 1 - \pi + \log\Gamma(n+y) - \log\Gamma(y+1) & \\ \quad - \log\Gamma(n) + n\log p + y\log(1-\pi) & \text{when} \quad y > 0 \end{cases}, \tag{5}$$

where π, n, p are also selected and calculated according to the index of $y = 0$ or $y > 0$, and Γ is the Gamma function. The final negative log likelihood loss function is given by:

$$NLL_{STZINB} = -LL_{y=0} - LL_{y>0}. \tag{6}$$

Note that our model can also be generalized to other distributions by modifying the probability layer into other distributions and using Z to represent the related shape parameter sets. For example, if we use Gaussian distribution as our model, we can parameterize the probability layer using the spatial and temporal embedding of the mean and variance, thus quantifying the data uncertainty that follows the Gaussian distribution. Using the flexibility of the probability layer, we design a variety of benchmark models to compare to the ZINB model. Our scripts can be found in Github[1].

3.4 Adjacency Matrix for O-D Pairs

This study uses O-D pairs as vertices, different from the previous GNN approaches that use regions as vertices [4, 5, 30, 41]. Therefore, we need to model spatial correlations of O-D pairs and construct the adjacency matrix in a different way. Intuitively, the O-D pairs with similar origins or destinations are also close in a graph representation, because passengers are likely to transfer between adjoining O-D pairs rather than remote ones. Inspired by the work of Ke et al. [12], we formulate our adjacency matrix as:

$$\begin{aligned} A^O_{i,j} &= haversine(lng^O_i, lat^O_i, lng^O_j, lat^O_j)^{-1}, \forall i, j \in V \\ A^D_{i,j} &= haversine(lng^D_i, lat^D_i, lng^D_j, lat^D_j)^{-1}, \forall i, j \in V \\ A_{i,j} &= \sqrt{\frac{1}{2}((A^O_{i,j})^2 + (A^D_{i,j})^2)}, \end{aligned} \tag{7}$$

where $lng^O_i, lat^O_i, lng^O_j, lat^O_j$ are the longitudes and latitudes of the origins of O-D pair i, j, and $lng^D_i, lat^D_i, lng^D_j, lat^D_j$ are for the destinations similarly. The function $haversine(\cdot)$ takes the longitudes and latitudes of two geographical points and calculate their distance on Earth. The basic idea is to leverage the O-D pairs' geographical adjacency by averaging the origin and destination similarity. It is clear that the order of i, j does not affect the output of the *haversine* function, which means A is symmetric. The final adjacency matrix A is the quadratic mean of A^O and A^D, where the distances between origins or destinations have the same influence in the adjacency matrix. Future studies can assign different weights to the origins or destinations in constructing the adjacency matrix, or even combine with demographic graphs to enrich the information of A. Since this paper focuses on uncertainty quantification and interpretability of the model, we use a simple construction of A to prevent any distraction.

3.5 Diffusion Graph Convolution Network

In order to capture the stochastic nature of flow dynamics among O-D pairs, we model the spatial correlations as a diffusion process [1, 20]. Introducing the diffusion process facilitates the learning of the spatial dependency from one O-D pair to another. The process is characterized by a random walk on the given graph with a probability $\alpha \in [0, 1]$ and a forward transition matrix $\tilde{W}_f = A/rowsum(A)$ [37]. After sufficiently large time steps, the Markovian property of the diffusion process guarantees it to converge to a stationary distribution $\mathcal{P} \in \mathbb{R}^{|V|\times|V|}$. Each row of $\mathcal{P}$ stands for the probability of diffusion from that node. The stationary distribution can be calculated in closed form [31]:

$$\mathcal{P} = \sum_{k=0}^{\infty} \alpha(1-\alpha)^k (D_O^{-1}A)^k, \tag{8}$$

where k is the diffusion step, which is usually set to finite number K. Variable k is only reused for demonstration of the diffusion process. We can similarly define the backward diffusion process with backward transition matrix $\tilde{W}_b = A^T/rowsum(A^T)$. Since our adjacency matrix A is symmetric, $\tilde{W}_f = \tilde{W}_b$. The forward and backward diffusion processes model the dynamics of passengers shifting from one O-D pair to another, like the shifting from school-home trip to home-market trip. The building block of DGCN layer can be written as [37]:

$$H_{l+1} = \sigma(\sum_{k=1}^{K} T_k(\tilde{W}_f)H_l\Theta^k_{f,l} + T_k(\tilde{W}_b)H_l\Theta^k_{b,l}), \tag{9}$$

where H_l represents the l^{th} hidden layer; the Chebyshev polynomial $T_k(X) = 2XT_{k-1}(X) - T_{k-2}(X)$ is used to approximate the convolution operation in DGCN, with boundary conditions $T_0(X) = I$ and $T_1(X) = X$ (note that the Chebyshev polynomial is used to approximate the diffusion $\mathcal{P}$ instead of using the closed form); learned parameters of the l^{th} layer $\Theta^k_{f,l}$ and $\Theta^k_{b,l}$ are added to control how each node transforms the received information; σ is the activation function (e.g. ReLU, Linear). In our model we stack 3 DGCN layers to better capture the O-D dynamics.

[1] https://github.com/ZhuangDingyi/STZINB

3.6 Temporal Convolutional Network

As Wu et al. [38] point out, the advantages of TCNs compared with recurrent neural networks (RNNs) include: 1) TCNs can use the sequences with varying length as inputs, which is more adaptive to different time resolutions and scales; 2) TCNs have a lightweight architecture and fast training [3]. The general idea of TCNs is to apply a shared gated 1D convolution with width w_l in the l^{th} layer in order to pass the information from w_l neighbors at the current time point. Each TCN layer H_l receives the signals from the previous layer H_{l-1} and is updated using [19]:

$$H_l = f(\Gamma_l * H_{l-1} + b), \tag{10}$$

where Γ_l is the convolution filter for the corresponding layer, $*$ is the shared convolution operation, and b stands for the bias. If previous hidden layer follows $H_{l-1} \in \mathbb{R}^{B\times|V|\times w_{l-1}}$, then the convolution filter is $\Gamma_l \in \mathbb{R}^{w_l\times w_{l-1}}$ so that $H_l \in \mathbb{R}^{B\times|V|\times w_l}$. If there is no padding in each TCN layer, it is clear that $w_l < w_{l-1}$. TCN is also a type of sequence-to-sequence model, which can directly output prediction for future target windows in a row. Furthermore, the TCN receptive field is flexible, which can be controlled by the number and kernels of Γ_l. It is useful in our following discussion about scaling our model with different temporal resolutions. We also stack 3 TCN layers.

4 NUMERICAL EXPERIMENTS

4.1 Data

In this session, we assess the model performance using two real-world datasets from Chicago Data Portal (CDP) [2] and Smart Location Database (SLD) [3]. The **CDP dataset** contains the trip records of Transportation Network Providers (ride-sharing companies) in the Chicago area. The city of Chicago is divided into 77 zones and the trip requests with pick-up and drop-off zone are recorded every 15min. We use 4-month observations from September 1st, 2019 to December 30th, 2019. The dataset is divided into various spatial resolutions to facilitate the discussion of model performance. We randomly select 10×10 O-D pairs with the same time period as spatially sparse data sample; The **SLD dataset** is also used in the work of Ke et al. [12]. Specifically, we select the For-Hire Vehicle (FHV) trip records in the Manhattan area (divided into 67 zones by administrative zip codes) from January 2018 to April 2018. This SLD data have similar features as the CDP dataset, including timestamps, and pick-up and drop-off zone IDs. As the dataset includes the timestamps of the individual trip information, we vary the temporal resolution (5min, 15min, and 60min intervals) to test our model performance. In addition to the full O-D matrix (67×67), we also sample 10×10 small O-D pair samples to align with the CDP dataset. We use 60% of the data for training, 10% for validation, and the last 30% for testing. Table 1 summarizes the data scenarios used in the analysis.

The zero rate increases from 50% to 88% as the temporal resolution for SLD dataset increases from 60 minutes to 5 minutes. As shown in Figure 2, the distribution of O-D flows is quite skewed because most of the O-D flows are concentrated around small values. The O-D flows at a 60min resolution have several observations larger than 8, but at 5min resolution, the lowest value is 3 trips within the 5-minute interval. Such sparse (and small value) O-D matrix is a challenge for most of the deterministic deep learning models.

Name	Resolution	# of O-D pairs	Data size	Zero rate
CDP_SAMP10	15min	10×10	(100, 11521)	81%
SLD_SAMP10	15min	10×10	(100, 11520)	54%
SLD_5min	5min	67×67	(4489, 34560)	88%
SLD_15min	15min	67×67	(4489, 11520)	70%
SLD_60min	60min	67×67	(4489, 2880)	50%

Table 1: Data division summary.

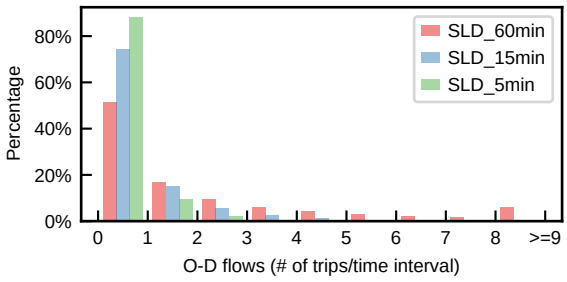


Figure 2: Distribution of travel demand in the SLD dataset per 5, 15, and 60min intervals.

4.2 Evaluation Metrics

We use the metrics for both the point estimate statistics and distributional characteristics to evaluate and compare the performance of the various models. The prediction accuracy of the expected median value (i.e. point estimate accuracy) is evaluated using the Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{k|V|}\sum_{i=1}^{k|V|}|x_i - \hat{x}_i|, \tag{11}$$

where $\hat{x}_i$ and x_i are the predicted and ground-truth values of i^{th} data point respectively. To evaluate the estimated uncertainty, we use Mean Prediction Interval Width (MPIW) on the 10%-90% confidence interval [14]:

$$\text{MPIW} = \frac{1}{k|V|}\sum_{i=1}^{k|V|}(U_i - L_i). \tag{12}$$

where L_i and U_i correspond to the lower and upper bound of the confidence interval for observation i. The definition of MPIW can be extended to other quantiles of the data. Tighter (smaller) prediction intervals are more desirable. Apart from MPIW, we also use the Kullback-Leibler Divergence (KL-Divergence) to assess how close the model output distribution is to the test set distribution [17]. We define the KL-Divergence as:

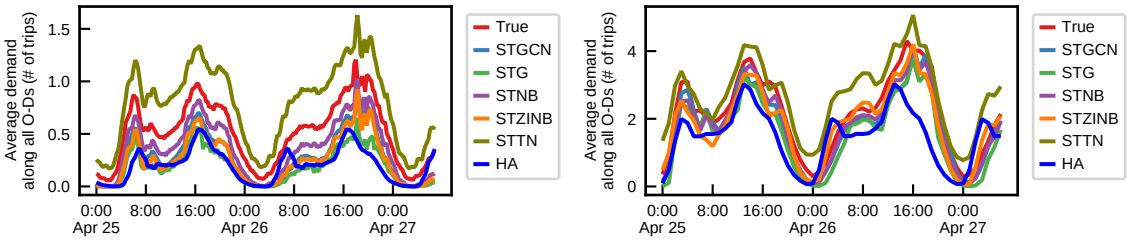
$$\text{KL-Divergence} = \frac{1}{k|V|}\sum_{i=1}^{k|V|}(\hat{x}_i \log\frac{\hat{x}_i + \epsilon}{x_i + \epsilon}), \tag{13}$$

[2] https://data.cityofchicago.org/Transportation/Transportation-Network-Providers-Trips/m6dm-c72p

[3] https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Table 2: Model comparison under different metrics. X/Y values correspond to the mean/median values of the distribution.

Data scenario	Metrics	STZINB-GNN	STNB-GNN	STG-GNN	STTN-GNN	HA	STGCN
CDP_SAMP10	MAE	0.368/**0.366**	0.382/0.379	0.409/0.409	0.432/0.606	0.522	0.395
	MPIW	**1.018**	1.020	2.407	2.089	/	/
	KL-Divergence	**0.291**/0.424	0.342/0.478	0.435/0.435	1.058/0.928	1.377	0.897
	True-zero rate	0.796/0.788	0.796/0.788	0.790/0.790	0.758/0.764	0.759	**0.800**
	F1-Score	**0.848**/0.846	**0.848**/0.841	0.818/0.818	0.842/0.846	0.809	0.840
SLD_SAMP10	MAE	0.663/0.666	0.627/**0.616**	0.630/0.630	0.695/0.665	0.697	0.630
	MPIW	**1.310**	3.628	2.604	1.931	/	/
	KL-Divergence	0.518/**0.507**	0.980/1.662	1.022/1.022	3.578/3.052	0.978	0.768
	True-zero rate	0.499/**0.502**	0.465/0.418	0.461/0.461	0.308/0.336	0.364	0.478
	F1-Score	**0.567**/0.566	0.556/0.552	0.555/0.555	0.477/0.500	0.456	0.563
SLD_5min	MAE	0.149/0.150	0.147/**0.144**	0.155/0.155	0.155/0.155	0.149	0.159
	MPIW	**0.094**	1.249	0.922	0.741	/	/
	KL-Divergence	0.015/0.014	0.042/0.145	**0.001/0.001**	0.001/0.001	0.060	0.056
	True-zero rate	**0.879/0.879**	0.875/0.866	0.877/0.877	0.877/0.877	0.874	0.874
	F1-Score	**0.882/0.882**	0.880/0.878	0.879/0.879	0.879/0.879	0.876	0.879
SLD_15min	MAE	0.370/0.372	0.351/**0.342**	0.356/0.356	0.365/0.356	0.418	0.373
	MPIW	**0.603**	2.283	1.353	1.215	/	/
	KL-Divergence	0.167/**0.156**	0.357/0.704	0.353/0.353	1.445/1.211	0.445	0.395
	True-zero rate	0.725/**0.727**	0.710/0.684	0.709/0.709	0.632/0.648	0.703	0.708
	F1-Score	**0.751**/0.750	0.746/0.745	0.750/0.750	0.716/0.726	0.744	0.750
SLD_60min	MAE	1.040/1.067	0.958/**0.947**	1.199/1.199	1.275/1.254	1.014	0.997
	MPIW	3.277	5.753	2.282	**1.592**	/	/
	KL-Divergence	0.982/1.270	**0.926**/0.963	2.176/2.176	4.120/3.734	2.421	1.114
	True-zero rate	0.458/**0.476**	0.443/0.425	0.390/0.390	0.288/0.308	0.447	0.438
	F1-Score	0.536/0.537	**0.538**/0.534	0.479/0.479	0.407/0.423	0.490	**0.538**



(a) SLD_15min (b) SLD_60min

Figure 3: Prediction results in the SLD_15min and SLD_60min scenarios. Results are averaged over the spatial dimension (i.e. all O-D pairs)

Since many x_i and $\hat{x}_i$ values are likely to be zeros, a small perturbation $\epsilon = 10^{-5}$ is used to avoid numerical issue because of division by 0. Since the KL-Divergence measures the difference between two distributions, smaller values are desirable.

To compare the model performance on the discrete O-D matrix entries, we use the true-zero rate and F1-score measurements. The true-zero rate quantifies how well the model replicates the sparsity in the ground-truth data. The F1-score, on the other hand, measures the accuracy of discrete predictions. Even though the F1-score is designed for classification models, we can still consider the discrete values as multiple labels and define the precision and recall accordingly [25]. Larger true-zero rate and F1-score values indicate better model performance.

4.3 Model Comparison

In order to explore the advantages of the STZINB-GNN, we compare the STZINB-GNN results against three other models: (1) **Historical Average (HA)** serves as the statistic baseline. It is calculated by averaging the demand in the same daily time intervals (e.g. 8:00AM-8:15AM) from the historical data to predict the one-step ahead future demand. (2) **Spatial-Temporal Graph Convolutional Networks (STGCN)**[4] is the state-of-the-art deep learning model for traffic prediction [42]. STGCN also uses graph convolution for spatial embedding and temporal convolution for temporal embedding. However, it fails to quantify the demand uncertainty and only produces point estimates; (3) **Models with probabilistic assumptions** in the spatial-temporal embedded probabilistic layer: negative binomial (STNB-GNN), Gaussian (STG-GNN), and truncated normal (STTN-GNN), as shown in Figure 1. These three

[4]https://github.com/FelixOpolka/STGCN-PyTorch

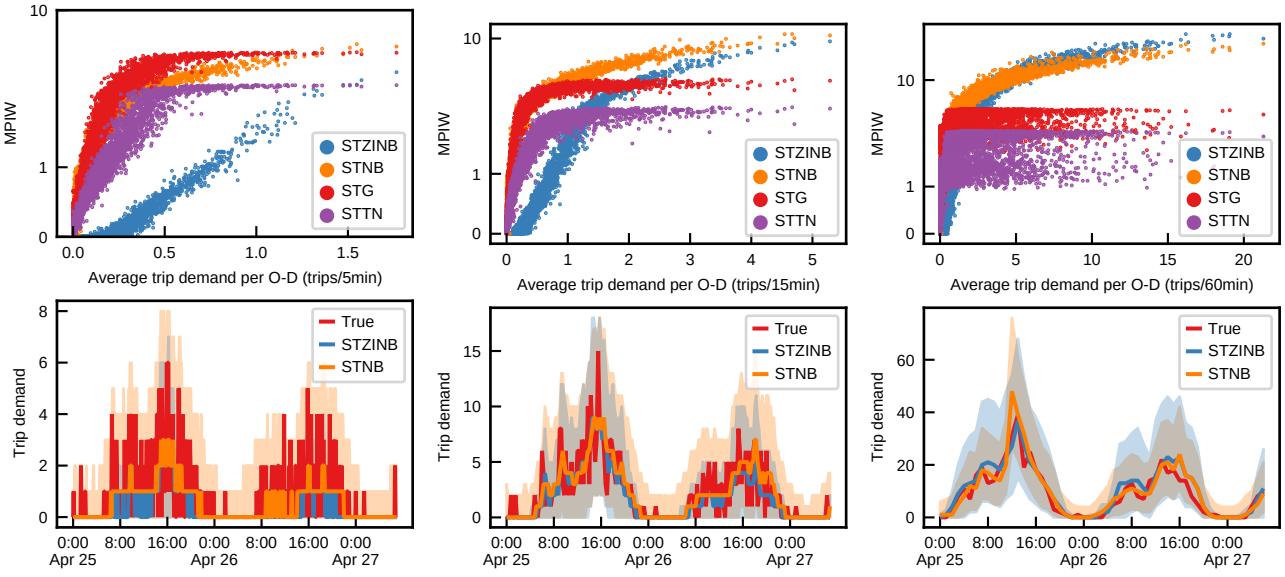
distributions have two parameters, less than the three parameters of STZINB-GNN. The other components and parameters are the same across these models.

STZINB-GNN outperforms other models when the O-D matrix resolution is high but performs worse when the resolution becomes coarser. Table 2 highlights the fields in bold to indicate the best performance for each data set. For fair comparison, the outputs from the continuous-output models, including STG-GNN, STTN-GNN, HA, and STGCN, were rounded to the closest integer to compare to the STNB-GNN and STZINB-GNN. As shown in Table 2, when the spatial and temporal resolutions are low, like the CDP_SAMP10 and SLD_SAMP10 cases, STZINB-GNN performs similarly to STNB-GNN. In the sparsest scenario, SLD_5min (with 88% zeros), STG-GNN, STTN-GNN, and STGCN fail to effectively capture the skewed data distribution, leading to low true-zero rates and F1-scores. The STZINB-GNN, on the other hand, successfully learns the sparsity of the data. Its prediction has a very small MPIW, nearly 10 times smaller than the other models. When the temporal resolution decreases, the STNB-GNN, STTN-GNN, and STGCN start to outperform STZINB-GNN. In the SLD_60min case, STNB-GNN fits the data better than the STGCN. The STZINB-GNN, on the other hand, only predicts the true-zero entries better. No single model dominates others under all resolution levels [36].

The negative binomial distribution in the probability layer can effectively capture the discrete values of the travel demand. Figure 3 illustrates the predicted and ground-truth average travel demand in the SLD (New York) data for two consecutive days, with Figure 3(a) for the SLD_15min and Figure 3(b) for the SLD_60min. With finer resolutions, the STNB-GNN and STZINB-GNN models are more likely to accurately predict the average travel demand. When the resolution decreases, like in the 60min case, all the deep learning models deliver similar performance. This 60-min resolution is commonly used in the majority of the deep learning studies, but our results demonstrate the importance of discrete probabilistic assumptions when the temporal unit is shorter than 60 minutes.

4.4 Uncertainty Quantification

The STZINB-GNN provides superior performance in quantifying uncertainty, because it uses much narrower MPIW for a fixed probability coverage, particularly when the temporal resolution is high. Figure 4 visualizes the scatter plots for the MPIW and the groundtruth travel demand of the 4489 O-D pairs at three temporal resolutions. Figure 4(a) demonstrates that the STZINB-GNN leads to significantly smaller MPIW than the other models in the granular 5min resolution case. This is because the average travel demand for each O-D pair is less than two in the SLD_5min case and most of them are zeros. The introduction of the sparsity parameter π in STZINB-GNN effectively captures the zeros. The STG-GNN and STTN-GNN are not able to capture the skewness of the data distribution, leading to large MPIWs. However, when the temporal resolution decreases, the zero-inflation with the sparsity parameter π becomes less important. Figures 4(b) and 4(c) show that the SLD_15min and SLD_60min cases have larger average travel demand per 15 and 60 minutes time interval. In these two cases, the



(a) SLD_5min case (b) SLD_15min case (c) SLD_60min case

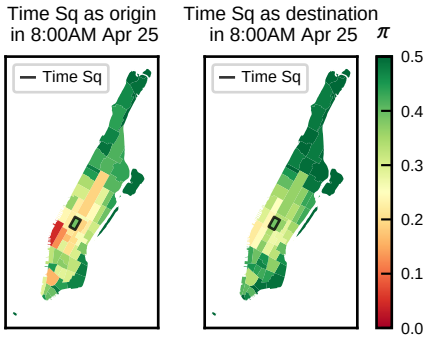
Figure 4: Model prediction uncertainty at various resolutions (SLD_5min to SLD_60min). The top row shows the MPIW needed to predict the demand of each O-D pair at different resolutions. The bottom row represents the prediction with uncertainty for a specific O-D pair, from Midtown Center to Union Square, in April 25 and 26, 2018. This O-D pair is selected because it has the largest trip flow.

STZINB-GNN has smaller MPIW only when the average travel demand is small. The STG-GNN and STTN-GNN outperform STZINB-GNN when the average demand is large. However, their output prediction distributions are not stable, because the same travel demand value corresponds to different MPIW with a large variance.

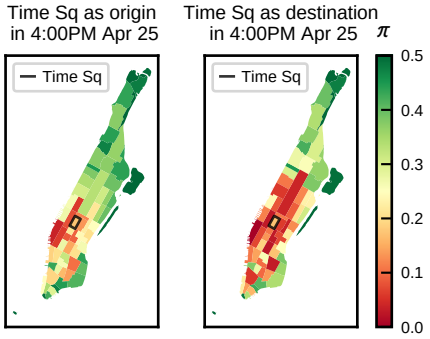
The bottom row of Figure 4 compares the expectation and MPIW between STZINB-GNN and STNB-GNN for a specific O-D pair that has the largest demand flow in the selected time range. In the SLD_5min and SLD_15min cases, the STZINB-GNN provides more compact confidence intervals than the STNB-GNN, even when its point prediction is less accurate. The results are consistent with the results in Table 2 and the MPIW comparison in the top row of Figure 4.

4.5 Interpretation of the Sparsity Parameter π

The sparsity parameter π in the ZINB distribution measures how likely a zone has zero demand. Note that each of our predicted O-D pair has the parameter π, which can capture the inflow and outflow sparsity level of a zone. Since O-D relationship is hard to illustrate in the map, we focus on a specific zone and project the O-D activities when the zone is selected as the origin or destination. Figure 5 shows the O-D activity (i.e. the parameter π) heatmap of Time Square, during the morning and evening peaks of April 25, 2018. It can be found that spatial locality exists, where communities are more likely to commute to their neighbors. Moreover, the temporal patterns also vary. As shown in Figure 5, many visitors explore the neighboring regions in the evening but are very inactive during the morning peak. Therefore, the sparsity parameter π renders the STZINB-GNN highly interpretable. It is very important in the transportation decision-making or the operation manager to use the sparsity parameter π to assign mobility service. Our framework has the potential extended to other prediction tasks that have many zeros and are sensitive to events, like incident precaution or paratransit service for the disabled people.



(a) Morning peak



(b) Evening peak

Figure 5: Morning and evening peaks outflow from and inflow to Time Square. We project the O-D pairs into the map according to Time Square as the origin or the destination zone. Red zones stand for small π values, meaning high possibility to have trips generated there and green regions otherwise.

5 CONCLUSION

In this paper, we propose a generalizable spatial-temporal GNN framework to predict the probabilistic distribution of sparse travel demand and quantify its uncertainty. We introduce the zero-inflated negative binomial distribution with a sparsity parameter π. We use spatial diffusion graph neural networks to capture spatial correlation and temporal convolutional networks to capture temporal dependency. The STZINB-GNN framework embeds the spatial and temporal representation of the distribution parameters respectively and fuses them to obtain the distribution for each spatial-temporal data point. The STZINB-GNN is evaluated using two real-world datasets with different spatial and temporal resolutions. We find that the STZINB-GNN outperforms the baseline models when the data are represented in high resolutions but performs worse when the resolution becomes coarser. This is also reflected in the prediction interval, where the STZINB-GNN has tight confidence intervals. Since the parameter π has physical interpretation, our model could help transportation decision makers to efficiently assign mobility services to zero or non-zero demand areas. This framework has the potential to be extended to other prediction tasks that use highly sparse data points, such as anomaly detection and accident prediction.

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Uncertainty Quantification via Spatial-Temporal Tweedie Model for Zero-inflated and Long-tail Travel Demand Prediction

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ABSTRACT

Understanding Origin-Destination (O-D) travel demand is crucial for transportation management. However, traditional spatial-temporal deep learning models grapple with addressing the sparse and long-tail characteristics in high-resolution O-D matrices and quantifying prediction uncertainty. This dilemma arises from the numerous zeros and over-dispersed demand patterns within these matrices, which challenge the Gaussian assumption inherent to deterministic deep learning models. To address these challenges, we propose a novel approach: the Spatial-Temporal Tweedie Graph Neural Network (STTD). The STTD introduces the Tweedie distribution as a compelling alternative to the traditional 'zero-inflated' model and leverages spatial and temporal embeddings to parameterize travel demand distributions. Our evaluations using real-world datasets highlight STTD's superiority in providing accurate predictions and precise confidence intervals, particularly in high-resolution scenarios. GitHub code is available online[1].

CCS CONCEPTS

• **Computing methodologies** → **Planning under uncertainty**.

KEYWORDS

Spatial-temporal Sparse Data, Uncertainty Quantification, Graph Neural Networks, Travel Demand Prediction

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1 INTRODUCTION

Efficient urban transportation hinges upon a balanced travel supply and demand, a balance greatly aided by accurate O-D travel demand forecasting [6, 12, 29]. This precision in prediction allows for dynamic resource allocation, reducing wait times, and boosting service provider profitability [7, 9, 16, 33]. Thus, improving the model accuracy is the main focus of the travel demand prediction domain. However, the task is not without its challenges, owing to the intricate spatial-temporal interdependencies [4, 10] and fluctuating nature of travel demand. While regions with dense demand, like airports and hospitals, generally present data that adheres to a Gaussian distribution—a core assumption of numerous prediction models—[11, 15], the opposite is true for areas with sparse and discrete O-D demand, such as educational institutions and government premises [1, 18]. Such deviations from the Gaussian assumption further complicate forecasting. Moreover, the problem of data sparsity is exacerbated when accounting for the disparity in urban demand across different regions at high spatial-temporal resolutions, like at 5min intervals [35]. A plethora of zero values, signifying the absence of trips, along with a long-tail distribution at higher demand levels, result in the skewness, discrepancy, and large variance in the data distribution [24, 25, 34]. Hence, accurately interpreting zeros, capturing long-tail distributions, and understanding non-negative discrete values become paramount for robust demand forecasting. Classic deep learning methods, such as Convolutional Neural Networks (CNNs) and Long Short-Term Memory networks (LSTMs), have tackled O-D matrix prediction by exploiting spatial and temporal dependencies [3, 5, 14, 17, 28, 30, 32]. Recent works have introduced Graph Neural Networks (GNNs), which leverage the graph-like structure of O-D matrices to uncover non-Euclidean correlations [8, 29]. Despite the respective merits, all these models mainly treat O-D matrix entries as continuous variables, with a primary focus on coarse temporal resolutions. They usually simplify variance structures by assuming homoskedasticity (constant variance) and predominantly output expected average travel demand values. These approaches could overlook critical features and may fail to sufficiently account for potential deviations and real-world uncertainties [19, 35]. Recent research [23, 35] integrate

zero-inflation statistic models with deep learning frameworks might hold promise for modelling sporadic travel demand, while they still fall short in long-tail distribution in sparse O-D demand as they mainly focus on 'happens or not" instead of such actual values.

In this paper, we propose the Spatial-Temporal Tweedie Graph Neural Network (STTD)—a comprehensive solution designed for joint numeric prediction and uncertainty quantification. Our main contributions can be summarized as follows:

- We integrate the Tweedie distribution to model demand, replacing the traditional two-part zero-inflated model, thereby effectively capturing the zero-inflation and long-tail non-zero characteristics of O-D travel data.
- The proposed combination is adept at quantifying the spatial-temporal uncertainty inherent in sparse travel demand data.
- We validate the superiority of the STTD through experiments on two real-world travel demand datasets, tested across various spatial-temporal resolutions and performance metrics.

The paper is organized as follows. Section 2 defines the research question and develops the model. Section 3 introduces the dataset used for the case study, the evaluation metrics, and the experimental results. Section 4 concludes the paper and discusses future research.

2 METHODOLOGY

2.1 Problem Description

The primary objective of our model is to fit parameters that capture the future travel demand distributions for each O-D pair over a span of k future time windows. The model accomplishes this by leveraging data from m origins, u destinations, and the corresponding travel demand within periods of length T minutes, rendering the task essentially a sequence-to-sequence prediction. Unlike previous work that defined the locations of origins or destinations as vertices, we adopt a more effective approach and directly construct the O-D flow graph $\mathcal{G} = (V, E, A)$. Within this graph, $|V| = m \times u$ signifies the set of O-D pairs, E designates the set of edges, and adjacency matrix $A \in \mathbb{R}^{|V|\times|V|}$ calculated by geodistance between O-D pairs, same as [12, 35]. Our approach provides a more nuanced understanding of the spatial-temporal intricate interrelationships present in urban travel demand.

We let x_{it} represent the trips that occur at the i^{th} O-D pair in the t^{th} time window, where $i \in V$ and $x_{it} \in \mathbb{N}$. Our approach primarily considers the individual instances of travel demand at different intervals, all of which collectively model the term x_{it}. Subsequently, $X_t \in \mathbb{N}^{|V|\times T}$ designates the demand for all O-D pairs in the t^{th} time window, with x_{it} as its entry. The objective is to utilize historical records $X_{1:t}$ as the inputs for training data, aided by the graph structure A, to predict the probabilistic density function $f(X_{t+1:t+k})$ of the distribution of $X_{t+1:t+k}$—that is, the travel demand distribution for the next k time windows. This prediction allows us to analyze the expected values and confidence intervals of travel demands.

2.2 Tweedie (TD) Distribution

Denote f_{TD} as the probability mass function of future O-D travel demand x_{it} out model outputs. It follows Tweedie distribution, which is in the form of: $f_{\mathrm{TD}}(x_{it}|\theta,\phi) \equiv a(x_{it},\phi)\exp\left[\frac{x_{it}\theta-\kappa(\theta)}{\phi}\right]$. Here, $\theta \in \mathbb{R}$ represents the natural parameter, while $\phi \in \mathbb{R}^+$ is the dispersion parameter. The normalizing functions $a(\cdot)$ and $\kappa(\cdot)$ correspond to parameters ϕ and θ, respectively [2, 22]. Functions' details will be provided later. In the context of the Tweedie distribution, the mean and variance of a random variable x are given by the following expressions: $E(x) = \mu = \kappa'(\theta), Var(x) = \phi\kappa''(\theta)$, where $\kappa'(\theta)$ and $\kappa''(\theta)$ denote the first and second derivatives of $\kappa(\theta)$, respectively. Here, $\mu \geq 0$ is the mean parameter. The Tweedie family incorporates many significant distributions based on different index parameter ρ. This includes the Normal ($\rho = 0$), Poisson ($\rho = 1$), Gamma ($\rho = 2$), Inverse Gaussian ($\rho = 3$), and Compound Poisson-Gamma distribution ($1 < \rho < 2$) [20–22]. The Compound Poisson-Gamma distribution is particularly useful due to its ability to parameterize zero-inflated and long-tail data. When $1 < \rho < 2$, the demand x_{it} can be expressed as shown in Eq.1:

$$x_{it} = \begin{cases} 0 & \text{if no trips,} \\ \sum_{j=1}^{L_{it}} l_{it}^{(j)} = l_{it}^{(1)} + l_{it}^{(2)} + \cdots + l_{it}^{(L_{it})} & \text{else.} \end{cases} . \quad (1)$$

where L_{it}, the number of time slices within the time window, follows a Poisson distribution $Pois(\lambda)$ with mean λ. The number of trips, $l_{it}^{(j)}$, are independent gamma random variables denoted by $Gamma(\alpha,\gamma)$ with mean $\alpha\gamma$ and variance $\alpha\gamma^2$. In this way, x_{it} is formed by the aggregation of discreet count, and we introduce L_{it}, $l_{it}^{(j)}$, and $x_{it} = \sum_{j=1}^{L_{it}} l_{it}^{(j)}$ to align with the Tweedie distribution definition[2]. If no trips occur, then $x_{it} = 0$, and the probability mass at zero for travel demand is $P(x_{it} = 0) = \exp(-\lambda)$ [2]. Otherwise, x_{it} is computed as the sum of L_{it} independent Gamma random variables. We re-parameterize the Tweedie distribution where $\theta = \mu^{1-\rho}/(1-\rho)$, and $\kappa(\theta) = \mu^{2-\rho}/(2-\rho)$ as:

$$f_{\mathrm{TD}}(x_{it}|\theta,\phi) \equiv f_{\mathrm{TD}}(x_{it}|\mu,\phi,\rho) = a(x_{it},\phi,\rho)e^{\left[x_{it}\frac{\mu^{1-\rho}}{\phi(1-\rho)} - \frac{\mu^{2-\rho}}{\phi(2-\rho)}\right]},$$

with the normalizing function $a(x_{it},\phi,\rho)$ defined as: $a(x_{it},\phi,\rho) = \begin{cases} 1 & \text{if } x_{it} = 0, \\ \frac{1}{x_{it}}\sum_{j=1}^{\infty} \frac{x_{it}^{-j\alpha}(\rho-1)^{\alpha j}}{\phi^{j(1-\alpha)}(2-\rho)^j j!\Gamma(-j\alpha)} & \text{if } x_{it} > 0. \end{cases}$. In this definition, μ, ϕ, and ρ are the key parameters determining the probability and expected value of travel demands. The parameters in the Gamma and Poisson distributions, namely λ, α, γ, can be computed using μ, ϕ, and ρ: $\lambda = \frac{1}{\phi}\frac{\mu^{2-\rho}}{2-\rho}, \alpha = \frac{2-\rho}{\rho-1}, \gamma = \phi(\rho-1)\mu^{\rho-1}$. The choice of the Tweedie distribution is driven by the characteristics of travel demand data. In practical terms, a specific time window may span different durations, such as 60 minutes, 15 minutes, or even as brief as 5 minutes. However, the distribution of trips during these periods can exhibit significant variations in both spatial and temporal dimensions. By incorporating more granular intervals into the model, it is possible to better capture the variability and heterogeneity of demand within the given time window. The Tweedie distribution effectively models zero-inflated and long-tail data distributions, making it particularly suited to this context.

2.3 Learning Framework and Loss Function

We utilize Diffusion Graph Convolution Network (DGCN) and Temporal Convolutional Network (TCN) as Spatial-Temporal Graph Encoder $\mathcal{ST}$ [27, 35]. Thus, node spatial-temporal embedding $\mathcal{Z}$ can be denoted as: $\mathcal{Z} = \mathcal{ST}_\Theta(X_{1:t}, A)$. where $\mathcal{Z}_i \in \mathbb{R}^{F'}$ is the

[2]For implementation purposes, x_{it} remains the model input. We clarify the specifics of x_{it} to align with the structure necessitated by the Tweedie distribution

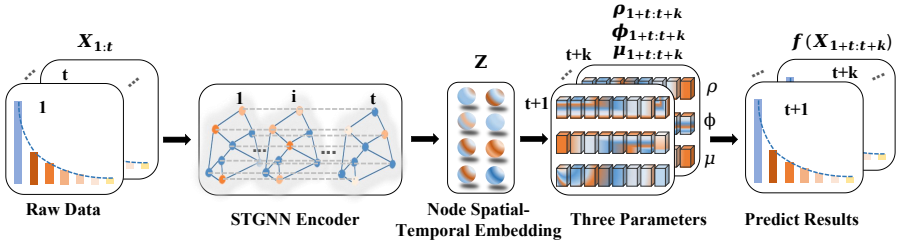

Figure 1: Framework of STTD model.

spatial-temporal embedding of the i^{th} O-D pair. Thus, the three parameters $\mu_{t+1:t+k}, \phi_{t+1:t+k}, \rho_{t+1:t+k}$ defining the Tweedie distribution can be computed as:

$$\begin{aligned}\mu_{t+1:t+k} &= \text{ReLU}(W_\mu \cdot \mathcal{Z} + b_\mu)\\ \phi_{t+1:t+k} &= \text{ReLU}(W_\phi \cdot \mathcal{Z} + b_\phi) + \epsilon\\ \rho_{t+1:t+k} &= \text{Sigmoid}(W_\rho \cdot \mathcal{Z} + b_\rho) + 1 + \epsilon\end{aligned} \tag{2}$$

where $\mu \in [0, +\infty), \phi \in (0, +\infty), \rho \in (1, 2)$. Learnable weight matrices $W_\phi, W_\rho, b_\mu, b_\phi, b_\rho \in \mathbb{R}^{F' \times k}$ and $\lim \epsilon \to 0$ is the minimum value. In order to fully predict travel demands, let x^* be one of the predicted travel demand Tweedie distributions from $f_{TD}(X_{t+1:t+k})$ with parameters μ, ϕ, ρ (notations μ, ϕ, ρ are reused for clearer formula). The learning objective of the whole model can be represented as the maximum log-likelihood function: $\max \log f_{TD}(x^*|\mu, \phi, \rho)$ and directly use the negative likelihood as our loss function to better fit the distribution into the data. The log-likelihood of TD is composed of the $x^* = 0$ and $x^* > 0$:

$$\mathcal{L}_{TD} = -\log f_{TD}(x^* > 0|\mu, \phi, \rho) - \log f_{TD}(x^* = 0|\mu, \phi, \rho) + \lambda\Theta^2.$$

where Θ is model parameters and λ is weight-parameter for L2 Normalization. Moreover, for $x^* > 0$:

$$\log f_{TD}(x^* > 0|\mu, \phi, \rho) = \frac{1}{\phi}\left(x^* \frac{\mu^{1-\rho}}{1-\rho} - \frac{\mu^{2-\rho}}{2-\rho}\right) + \log a(x^* > 0, \phi, \rho)$$

$$\geq \frac{1}{\phi}\left(x^* \frac{\mu^{1-\rho}}{1-\rho} - \frac{\mu^{2-\rho}}{2-\rho}\right) - \log(j_{max}\sqrt{-\alpha x^*}) + j_{max}(\alpha - 1)$$

where $j_{max} = \frac{x^{*2-\rho}}{(2-\rho)\phi}, \alpha = \frac{2-\rho}{1-\rho} < 0$. As for $x^* = 0$, $\log f_{TD}(x^* = 0|\mu, \phi, \rho) = \frac{1}{\phi}\left(-\frac{\mu^{2-\rho}}{2-\rho}\right)$, we select and calculate μ, ϕ, ρ according to the index of $x^* = 0$ or $x^* > 0$. We optimize the lower bound of $\mathcal{L}_{TD}$ during training to avoid calculation of summation formula. The whole framework is illustrated in Figure 1.

3 NUMERICAL EXPERIMENTS

3.1 Experiment Setup

Datasets: Chicago Data Portal (CDP)[3] and Smart Location Database (SLD)[4] datasets. The CDP dataset includes trip records of ride-sharing companies in Chicago's 77 zones every 15 minutes, from 01/09/2019 to 30/12/2019. We randomly select 10×10 grids of O-D pairs from CDP. The SLD dataset encapsulates For-Hire Vehicle trip records in 67 Manhattan administrative zones. We alter temporal resolution (5/15/ 60-minute intervals) and sample 10×10 / 67×67 O-D pairs to gauge our model's performance. Both datasets were used in previous work [12, 35]. **Evaluation Metrics**: (1) Point estimates: Mean Absolute Error (MAE), which measures the accuracy of the mean or median value of the predicted Tweedie distributions. (2) Distributional uncertainty: Mean Prediction Interval Width (MPIW) and Prediction Interval Coverage Probability (PICP) within the 10%-90% confidence interval. MPIW averages the width of the confidence interval, while PICP quantifies the percentage of actual data points within confidence intervals. Additionally, KL-Divergence is applied to evaluate the similarity between the predicted and real data distributions. (3) Discrete demand prediction: true-zero rate and F1-score. The true-zero rate measures the model's fidelity in reproducing data sparsity, and the F1-score gauges the accuracy of discrete predictions. In general, Lower MPIW and KL-Divergence values are favorable while larger true-zero rate, PICP, and F1-score values denote superior model performance. **Baselines**: Historical Average (HA), Spatial-Temporal Graph Convolutional Networks (STGCN)[5][31], Spatial-Temporal Graph Attention Networks (STGAT), the state-of-the-art probabilistic models under zero-inflated negative binomial assumptions (STZINB) and other methods such as negative binomial (STNB), truncated normal (STTN) [6]. We also evaluate our methods with different index parameter ρ including $\rho \in [1, 2]$ (STTD), $\rho = 0$ (STG), $\rho = 1$ (STP), $\rho = 2$ (STGM) and $\rho = 3$ (STIG). STTD and STZINB are three-parameter models while other probabilistic components are two-parameter models. **Reproducibility**: Parameters of baselines are optimized using Adam Optimizer[13] with L_2 regularization and a dropout rate of 0.2. The GNN in STTD and baselines are all two-layered with hidden unit equal 42. We also employ the early-stopping strategy with patience equals 10 to avoid over-fitting. We split the data into 60% for training, 10% for validation, and 30% for testing. All graph adjacency matrices are built following [35].

3.2 Model Comparison

We carry out experiments on five distinct travel demand scenarios, presenting the prediction results in Table 1. Here, the best and second-best scores are highlighted with bold and underlined values, respectively. As a general observation, STTD outperforms all baseline methods in the majority of cases when $\rho \in [1, 2]$. Other models from the Tweedie family tend to surpass probabilistic methods while also exceeding other state-of-the-art deep learning techniques. For instance, STTD's numerical accuracy improvements reach up to 3.47% on SLDSAMP10 in terms of Mean Absolute Error (MAE). Moreover, STTD realizes improvements in Mean Predictive Interval Width (MPIW) of 327.66%, 92.22%, and 67.02% on SLD_15min, CDPSAMP10, and SLD_5min scenarios respectively, when compared to STZINB and STNB. This suggests that STTD is capable of generating more precise predictions while maintaining a narrower confidence interval, a conclusion also supported by PICP

[3]https://data.cityofchicago.org/Transportation/Transportation-Network-Providers-Trips/m6dm-c72p
[4]https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page
[5]https://github.com/FelixOpolka/STGCN-PyTorch
[6]https://github.com/ZhuangDingyi/STZINB

Table 1: Model comparison under different metrics. X/Y values correspond to the mean/median values of the distribution.

Dataset	Metrics	STTD ($1 < \rho < 2$)	STG ($\rho = 0$)	STP ($\rho = 1$)	STGM ($\rho = 2$)	STIG ($\rho = 3$)	STZINB	STNB	STTN	STGCN	STGAT	HA
CDPSAMP10	MAE	**0.358**/0.363	0.409/0.409	0.429/0.453	0.479/0.472	0.376/0.384	0.368/0.366	0.382/0.379	0.432/0.606	0.395	0.397	0.522
	MPIW	**0.059**	2.407	0.758	1.154	1.032	1.018	1.020	2.089	/	/	/
	PICP	**0.976**	0.874	0.948	0.792	0.865	0.958	0.957	0.065	/	/	/
	KL-Divergence	0.184/0.228	0.435/0.435	0.138/0.157	0.120/0.128	0.113/**0.106**	0.291/0.424	0.342/0.478	1.058/0.928	0.897	1.169	1.377
	True-zero rate	**0.803**/0.799	0.790/0.790	0.503/0.552	0.791/0.797	0.797/0.797	0.796/0.788	0.796/0.788	0.758/0.764	0.800	0.787	0.759
	F1-Score	**0.862**/0.844	0.818/0.818	0.619/0.642	0.832/0.843	0.822/0.836	0.848/0.846	0.848/0.841	0.842/0.846	0.840	0.846	0.809
SLDSAMP10	MAE	0.648/0.670	0.627/0.630	0.649/0.654	0.658/0.660	0.638/0.640	0.663/0.666	0.627/**0.616**	0.695/0.665	0.630	0.678	0.697
	MPIW	**0.965**	2.604	1.063	1.523	1.466	1.310	3.628	1.931	/	/	/
	PICP	0.921	0.811	0.907	0.481	0.502	0.942	**0.943**	0.219	/	/	/
	KL-Divergence	**0.102**/0.113	1.022/1.022	0.111/0.122	0.130/0.130	0.123/0.113	0.518/ 0.507	0.980/1.662	3.578/3.052	0.768	0.754	0.978
	True-zero rate	**0.545**/0.529	0.461/0.461	0.504/0.508	0.522/0.531	0.507/0.503	0.499/ 0.502	0.465/0.418	0.308/0.336	0.478	0.508	0.364
	F1-Score	0.611/0.605	0.555/0.555	0.568/0.568	0.659/**0.661**	0.657/0.653	0.567/0.566	0.556/0.552	0.477/0.500	0.563	0.498	0.456
SLD_5min	MAE	**0.139**/0.147	0.155/0.155	0.146/0.149	0.145/**0.139**	0.142/0.143	0.149/0.150	0.147/ 0.144	0.155/0.155	0.159	0.162	0.149
	MPIW	0.031	0.922	**0.016**	0.911	1.224	0.094	1.249	0.741	/	/	/
	PICP	0.973	0.895	**0.981**	0.924	0.888	0.968	0.969	0.037	/	/	/
	KL-Divergence	**0.001**/**0.001**	**0.001**/**0.001**	0.003/0.004	0.004/0.003	0.003/0.002	0.015/0.014	0.042/0.145	**0.001**/**0.001**	0.056	0.053	0.060
	True-zero rate	**0.884**/0.883	0.877/0.877	0.880/0.879	0.878/0.872	0.866/0.871	0.879/0.879	0.875/0.866	0.877/0.877	0.874	0.851	0.874
	F1-Score	**0.905**/0.894	0.879/0.879	0.884/0.839	0.884/0.882	0.884/0.884	0.882/0.882	0.880/0.878	0.879/0.879	0.879	0.868	0.876
SLD_15min	MAE	**0.337**/0.354	0.356/0.356	0.357/0.360	0.362/0.361	0.360/0.360	0.370/0.372	0.351/ 0.342	0.365/0.356	0.373	0.394	0.418
	MPIW	**0.141**	1.353	1.153	0.623	0.648	0.603	2.283	1.215	/	/	/
	PICP	0.938	0.855	**0.962**	0.724	0.715	0.956	0.959	0.120	/	/	/
	KL-Divergence	**0.058**/0.061	0.353/0.353	0.061/0.064	0.063/0.067	0.062/0.071	0.167/ 0.156	0.357/0.704	1.445/1.211	0.395	0.434	0.445
	True-zero rate	0.727/**0.729**	0.709/0.709	0.728/0.721	0.722/0.712	0.724/0.701	0.725/ 0.727	0.710/0.684	0.632/0.648	0.708	0.704	0.703
	F1-Score	**0.796**/0.778	0.750/0.750	0.776/0.722	0.774/0.775	0.772/0.786	0.751/0.750	0.746/0.745	0.716/0.726	0.750	0.753	0.744
SLD_60min	MAE	**0.915**/0.928	1.199/1.199	0.942/0.956	0.937/0.971	0.952/0.933	1.040/1.067	0.958/ 0.947	1.275/1.254	0.997	0.987	1.014
	MPIW	2.241	2.282	2.576	**1.372**	1.431	3.277	5.753	1.592	/	/	/
	PICP	**0.964**	0.557	0.872	0.524	0.533	0.947	0.930	0.133	/	/	/
	KL-Divergence	**0.337**/0.395	2.176/2.176	0.378/0.421	0.363/0.342	0.365/0.352	0.982/1.270	0.926/0.963	4.120/3.734	1.114	2.053	2.421
	True-zero rate	**0.505**/0.488	0.390/0.390	0.474/0.429	0.453/0.449	0.354/0.379	0.458/ 0.476	0.443/0.425	0.288/0.308	0.438	0.416	0.447
	F1-Score	**0.617**/0.608	0.479/0.479	0.610/0.569	0.613/0.609	0.604/0.615	0.536/0.537	0.538/0.534	0.407/0.423	0.538	0.488	0.490

values. Typically, a model necessitating a larger confidence interval is indicative of a higher PICP, aiming to encapsulate all ground truth data points. However, STTD manages to sustain strong performance in both PICP and MPIW, implying the generated confidence intervals adeptly capture the underlying data distribution while remaining relatively narrow. In terms of capturing sparsity, STTD surpasses other methods on metrics such as KL-Divergence, true-zero rate, and F1-Score in most cases, signifying STTD's efficiency in capturing zero values. This attribute is vital for sparse travel demand prediction tasks where accurate and confident prediction results are paramount for effective decision-making. Three-parameter models like STTD and STZINB outshine two-parameter models and 'one-parameter' methods such as STGCN and STGAT. It's noted that no single model dominates others across all resolution levels, as per Wolpert's "No Free Lunch" theorem [26].

3.3 Parameters Visualization

We visualize the learned parameters ϕ, ρ, μ and real values x by 3D surface plots in Figure 2 on CDPSAMP10 and SLDSAMP10 test sets. As the plots provided, it is evident that for long-tailed data, the learned values of ρ are greater (closer to 2), and a reason stands that the main part of loss function $x^* \frac{\mu^{1-\rho}}{1-\rho} (x^* > 0)$ will be punished when predicted x^* lies in long-tail while the predict approaches zero, making the loss function converts sharply. While for zero-valued data, the learned parameter ϕ is huge. Thus it shows that the model can capture zero-inflated and long-tail data effectively from uncertainty aspect. Furthermore, as the model parameter μ serves as the distribution mean, its color distribution matches the value of x, indicating that the model is good at mean point estimation.

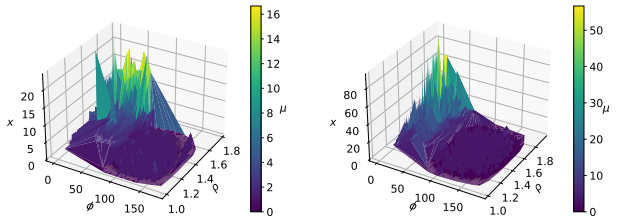

Figure 2: Surface plots for learned STTD parameters (μ, ϕ, ρ) on CDPSAMP10 (left) and SLDSAMP10 (right) test sets.

4 CONCLUSION

In this work, we introduced the Spatial-Temporal Tweedie Distribution (STTD), a novel neural probabilistic graph-based deep learning model aimed at quantifying spatial-temporal uncertainty. STTD leverages the power of the Tweedie distribution, adept at handling zero-inflated and long-tail data. In conjunction with this, we utilize the capabilities of Spatial-Temporal Graph Neural Networks to effectively encode three key uncertainty parameters. This strategy allows STTD to capture the complex, intertwined spatial-temporal dependencies, and intrinsic uncertainties at each data point. We validated the performance through extensive experiments across five representative scenarios, with a keen focus on point estimation and uncertainty measurement. Our results underscore the model's robustness and effectiveness, setting a new benchmark in the field. Our approach brings to light the significance of rigorous uncertainty quantification within spatial-temporal deep learning frameworks, providing a robust platform for future research and development and also contributing to more efficient and reliable transport management systems.

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SAUC: Sparsity-Aware Uncertainty Calibration for Spatiotemporal Prediction with Graph Neural Networks

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Abstract

Quantifying uncertainty is crucial for robust and reliable predictions. However, existing spatiotemporal deep learning mostly focuses on deterministic prediction, overlooking the inherent uncertainty in such prediction. Particularly, highly-granular spatiotemporal datasets are often sparse, posing extra challenges in prediction and uncertainty quantification. To address these issues, this paper introduces a novel post-hoc Sparsity-aware Uncertainty Calibration (SAUC) framework, which calibrates uncertainty in both zero and non-zero values. To develop SAUC, we firstly modify the state-of-the-art deterministic spatiotemporal Graph Neural Networks (ST-GNNs) to probabilistic ones in the pre-calibration phase. Then we calibrate the probabilistic ST-GNNs for zero and non-zero values using quantile approaches. Through extensive experiments, we demonstrate that SAUC can effectively fit the variance of sparse data and generalize across two real-world spatiotemporal datasets at various granularities. Specifically, our empirical experiments show a 20% reduction in calibration errors in zero entries on the sparse traffic accident and urban crime prediction. Overall, this work demonstrates the theoretical and empirical values of the SAUC framework, thus bridging a significant gap between uncertainty quantification and spatiotemporal prediction.

CCS Concepts

• **Information systems** → **Data mining**; • **Computing methodologies** → *Machine learning approaches.*

Keywords

Uncertainty Quantification, Calibration Methods, Spatiotemporal Sparse Data, Graph Neural Networks

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1 Introduction

Spatiotemporal Graph Neural Networks (ST-GNNs) have been instrumental in leveraging spatiotemporal data for various applications, from weather forecasting to urban planning [10, 40, 44]. However, most of the models narrowly focus on deterministic predictions, which overlooks the inherent uncertainties associated with the spatiotemporal phenomena. For various safety-related tasks, such as traffic accidents and urban crime prediction, it is crucial to quantify their uncertainty, thus proactively preventing such phenomena from happening. Despite its importance, uncertainty quantification (UQ) remains relatively understudied in the spatiotemporal context, which can lead to erroneous predictions with severe social consequences [1, 8, 30].

Recent research has addressed UQ in spatiotemporal modeling [14, 36, 47], but often fails to account for the sparsity and asymmetrical distribution present in detailed spatiotemporal data. This sparsity, marked by an abundance of zeros, becomes essential when working with high-resolution data. Additionally, the need for granular forecasting in event precaution often leads to more diluted inputs, making sparsity unavoidable. Thus, effectively managing aleatoric (data) uncertainty is crucial in these forecasting tasks.

Prior ST-GNNs with basic UQ functionalities typically focus on assuming data distributions (e.g., Gaussian) and learning these distributions' parameters for spatiotemporal predictions [15, 36, 47]. A key limitation is their inadequate evaluation of UQ reliability using robust calibration metrics. Research by [10, 15, 37, 47] often emphasizes high accuracy and narrow prediction intervals as indicators of effective UQ. This approach overlooks the crucial relationship between data variance and prediction intervals (PI), which is vital for addressing aleatoric uncertainty. As highlighted by [17], most deep learning models are not fully calibrated for high UQ quality, leaving a significant research gap to be filled.

This paper introduces a Sparsity-Aware Uncertainty Calibration (SAUC) framework, calibrating model outputs to align with true

data distributions, which can be adapted to any ST-GNN model. Inspired by the zero-inflated model [2, 9], we partition zero and non-zero predictions and apply separate Quantile Regression (QR) models to ensure the discrepancy of prediction and true values lie within the PIs. This post-hoc approach, unlike uncertainty-aware spatiotemporal prediction models [29], offers flexibility in applying it to existing models without necessitating architectural changes or full model retraining. Consistent with prior studies on regression task calibration, we formulate a modified calibration metric tailored for PI-based calibration within asymmetric distributions. To investigate our model's generalizability, we adapt three prevalent numeric-output ST-GNN models with negative binomial (NB) distributions to accommodate sparse settings without compromising prediction accuracy. Tested on two real-world spatiotemporal datasets with different levels of sparsity, our SAUC demonstrates promising calibration results, thus assisting in reliable decision-making and risk assessment. The main contributions of this paper are threefold:

- We have developed a novel post-hoc uncertainty calibration framework tailored for predictions on sparse spatiotemporal data. This approach is designed to be compatible with current GNN models and is suitable for any probabilistic outputs.
- We focus on calibrating asymmetric distributions, like the NB distributions, using quantile regression. This approach deviates from conventional mean-variance UQ methods. Additionally, we have developed new calibration metrics based on prediction interval width.
- We conduct experiments on real-world datasets with different temporal resolutions to demonstrate promising calibration results of our SAUC, especially for zero values. We observe a roughly **20% reduction** in calibration errors relative to leading baseline models, underscoring its efficacy for safety-critical decision-making.

2 Related Work

2.1 Spatiotemporal Prediction

Spatiotemporal prediction through deep learning offers a powerful tool for various applications. Among them, GNNs have become prevalent in recent years [40–42, 44]. Most existing models focus on predicting the deterministic values without considering the associated uncertainty, which leaves a noticeable research gap concerning UQ. Such a limitation impacts the reliability of predictions, as the unaccounted uncertainty can lead to deviations from the true data variance.

Bayesian and the frequentist approaches are the two main existing methods for spatiotemporal UQ [36, 46]. While the Bayesian approach offers profound insights through techniques like Laplace approximation [38], Markov Chain Monte Carlo [27], and Variational Inference [16], it suffers from computational complexities and approximation inaccuracies. In contrast, the frequentist approach exploits Mean-Variance Estimation for computational efficiency and model flexibility, accommodating both parametric and non-parametric methods [36]. Recently, many frequentist approaches are proposed by modifying the last layers of the ST-GNN model outputs [10, 15, 29]. However, the application of these methods to sparse data is challenged by data dependencies and distribution constraints [11, 39]. For example, sparse data frequently deviate from models' mean-variance assumptions, favoring other right-skewed distributions like NB over Gaussian, and traditional metrics like mean squared errors can be skewed by sparse datasets' non-zero values.

The use of zero-inflated distributions with spatiotemporal neural networks has been introduced to handle zero instances and non-normal distribution in sparse data [47]. However, this approach has limitations in extreme scenarios and non-time-series contexts, impacting predictive accuracy and system robustness [28]. Therefore, the urgent need to quantify and calibrate uncertainty in the sparse component of spatiotemporal data is evident, yet remains largely unexplored.

2.2 Uncertainty Calibration

Although relatively understudied in the spatiotemporal context, UQ has been examined by many machine learning studies. Among all the UQ approaches, calibration is a post-hoc method that aims to match the predictive probabilities of a model with the true probability of outcomes, developing robust mechanisms for uncertainty validation [26].

Many techniques have been developed to calibrate pre-trained classifiers, mitigating the absence of inherent calibration in many uncertainty estimators [12, 19, 32]. Calibration for regression tasks, however, has received less attention [17, 34]. Post-hoc calibration methods for classifications, such as temperature scaling, Platt scaling, and isotonic regression, have successfully been adapted for regression [18]. Recent research [8] further demonstrated quantile regression-based calibration methods and their extensions. These post-hoc methods offer greater flexibility and wider applicability to various probabilistic outputs compared to in-training methods, without necessitating model retraining [14, 29].

In spatiotemporal prediction contexts, where accurate model outputs are important for applications like traffic accident forecasting and crime prediction [44], the significance of uncertainty calibration is often overlooked. Since spatial and temporal granularity of the data could vary, there could exist different levels of sparsity in the data [47]. It is essential to discern confidence for zero and non-zero segments due to their varied implications. In response, we propose the SAUC framework to manage asymmetric distributions and ensure variances of true targets align with estimated PIs [34].

3 Problem Formulation

3.1 ST-GNNs for Prediction

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ where $\mathcal{V}$ represents the set of nodes (locations), $\mathcal{E}$ the edge set, and $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ the adjacency matrix describing the relationship between nodes. We denote the spatiotemporal dataset $\mathcal{X} \in \mathbb{R}^{|\mathcal{V}|\times t}$ where t is the number of time steps. The objective is to predict the target value $\mathcal{Y}_{1:|\mathcal{V}|,t:t+k}$ of future k time steps given all the past data up to time t, which is $\mathcal{X}_{1:|\mathcal{V}|,1:t}$. The full dataset is partitioned by timesteps into training, calibration (i.e., validation), and testing sets. The three data partitions are denoted as $\mathcal{X}_T$, $\mathcal{X}_S$, and $\mathcal{X}_U$, respectively. For example, the training set $\mathcal{X}_{1:|\mathcal{V}|,1:t}$ is denoted as $\mathcal{X}_T$. The associated target values are

represented as $\mathcal{Y}_T$, $\mathcal{Y}_S$, and $\mathcal{Y}_U$, with subscripts indicating the corresponding sets.

The objective of the ST-GNN models is to design model f_θ, parametrized by θ, which yields:

$$\hat{\mathcal{Y}}_d = f_\theta(\mathcal{X}_d; \mathcal{G}), \forall\, d \in \{T, S, U\}, \tag{1}$$

where $\hat{\mathcal{Y}}_d$ is the predicted target value, and the subscript represents the corresponding data set. Note that θ is fixed once f_θ is trained on the set T.

With emerging interests in quantifying the data and model uncertainty of ST-GNNs, researchers shift towards designing probabilistic GNNs to predict both mean values and PIs [5, 36]. Previous probabilistic ST-GNN studies often assume $\mathcal{X}$ and $\mathcal{Y}$ are independent and identically distributed random variables, i.e., denoted as X_i and Y_i to describe data distributions. The last layer of the neural networks architectures is modified to estimate the parameters of the assumed distributions[15, 47]. For this study, we use the NB distribution to discuss the sparse and discrete data distribution.

The NB distribution of each predicted data point is characterized by shape parameter μ and dispersion parameter α, and $\hat{\mu}_i \in \hat{M}_d$ denotes the mean value of the predicted distribution. Previous work has applied prediction layers with the corresponding likelihood loss after ST-GNN encoding to study the parameter sets of the predicted data points [15, 47]. The set of predicted μ and α corresponding to all elements of $\mathcal{Y}$ are denoted as $\hat{M}$ and $\hat{H}$. Therefore, Equation 1 can be rewritten in the format of NB distribution-based probabilistic ST-GNNs as:

$$(\hat{M}_d, \hat{H}_d) = f_\theta(\mathcal{X}_d; \mathcal{G}), \forall d \in \{T, S, U\}. \tag{2}$$

Subsequently, the objective function is adjusted to incorporate the relevant log-likelihood, as denoted in Equation 12.

It is important to note that the selection of the distribution is not the primary focus of this paper. Our goal is to present a general framework to calibrate probabilistic outputs for sparse spatiotemporal data, as long as the predicted distributions are provided. Implementations for other distributions are included in Section 5.7, but the NB distribution is used for illustration in this study.

3.2 Prediction Interval Calibration

Generally, for the frequentist approach, the spatiotemporal prediction models output the sets of location and shape parameters $\hat{M}_d, \hat{H}_d, \forall d \in \{T, S, U\}$ [17, 22, 33]. We construct the prediction intervals of the i-th data point as $\hat{\mathcal{I}}_i = [\hat{l}_i, \hat{u}_i]$ with $\hat{l}_i$ and $\hat{u}_i$ denoting the 5th- and 95th-percentiles, respectively. Notably, our discussion limits the prediction intervals to the 5% and 95% range. Based on [17], our target is to ensure a prediction model f_θ to be calibrated, i.e.,

$$\frac{\sum_{i=1}^{|U|} \mathbb{I}\{y_i \le \hat{F}_i^{-1}(p)\}}{|U|} \xrightarrow{|U|\to\infty} p,\ \forall p \in \{0.05, 0.95\}, \tag{3}$$

where p stands for the targeting probability values, $\mathbb{I}$ denotes the indicator function, and $\hat{F}_i(\cdot)$ is the CDF of the NB distributions parameterized by $\hat{\mu}_i$ and $\hat{\alpha}_i$. The primary goal is to ensure that, as the dataset size increases, the predicted CDF converges to the true one. Such an alignment is congruent with quantile definitions, suggesting, for instance, that 95% of the genuine data should fall beneath the 95% percentile. Rather than striving for a perfect match across $\forall p \in [0, 1]$, we mainly focus on calibrating the lower bound and upper bound.

Empirically, we leverage the calibration set S to calibrate the uncertainty in data. The empirical CDF for a given value $p \in \hat{F}_i(y_i)$ is computed [17, 23]:

$$\hat{\mathcal{P}}(p) = \frac{|\{y_i | \hat{F}_i(y_i) \le p, i = 1, \ldots, |S|\}|}{|S|}. \tag{4}$$

The calibration process entails mapping a calibration function C, such as isotonic regression or Platt scaling, to the point set $\{(p, \hat{\mathcal{P}}(p))\}_{i=1}^{|S|}$. Consequently, the composition $C \circ \hat{F}$ ensures that $\hat{\mathcal{P}}(p) = p$ within the calibration dataset. This composition can then be applied to the test set to enhance the reliability of predictions.

4 Sparsity-Aware Uncertainty Calibration

4.1 QR for Uncertainty Calibration

Unlike traditional regression techniques that target the conditional mean, QR predicts specific quantiles, which are essential for calibrating PIs. Given $\hat{M}_S$ and $\mathcal{Y}_S$ from the calibration set S, the linear QR model can be expressed as:

$$Q_{\mathcal{Y}_S | \hat{M}_S}(p) = \hat{M}_S^T \beta(p), \tag{5}$$

where $\beta(p)$ signifies the coefficient associated with quantile p. The goal here is to minimize the Pinball loss [8], which is defined as:

$$L_{\text{pinball}}(y, \hat{y}, p) = \begin{cases} (p-1) \cdot (y - \hat{y}), & \text{if } y < \hat{y} \\ p \cdot (y - \hat{y}), & \text{if } y \ge \hat{y} \end{cases}. \tag{6}$$

In our context, we focus on the cases of $p = 5\%$ and 95%, and thus the quantile regressions output calibrated PIs correspond to $[Q(0.05), Q(0.95)]$.

While the NB distribution is frequently used for estimating sparse data uncertainty in previous work [7, 15, 31, 47], it can still be prone to misspecification, particularly when assumptions about data characteristics are less precise. In such instances, non-parametric (not directly modeling the distribution parameters) and distribution-agnostic (not reliant on specific distribution assumptions) methods like QR emerge as robust alternatives, compared to a direct scaling or fitting of the dispersion parameter [6, 21]. QR's resilience to data assumptions helps mitigate potential errors in model specification, a critical advantage especially in handling datasets with heavy tails where significant risks or events are concentrated—features sometimes not fully captured by parametric models [13].

QR is also adept at addressing heteroscedasticity, where variance shifts with the response variable, a situation often oversimplified by a single dispersion parameter in parametric calibration methods [3, 45]. Its ability to handle sparse datasets and accommodate variance fluctuations, particularly in scenarios with numerous zeros, further underscores its utility. Enhanced with basis expansion techniques, QR can discern complex patterns across various quantiles, making it an ideal choice for scenarios where zero outcomes are prevalent and a nuanced understanding of data patterns and risks is crucial.

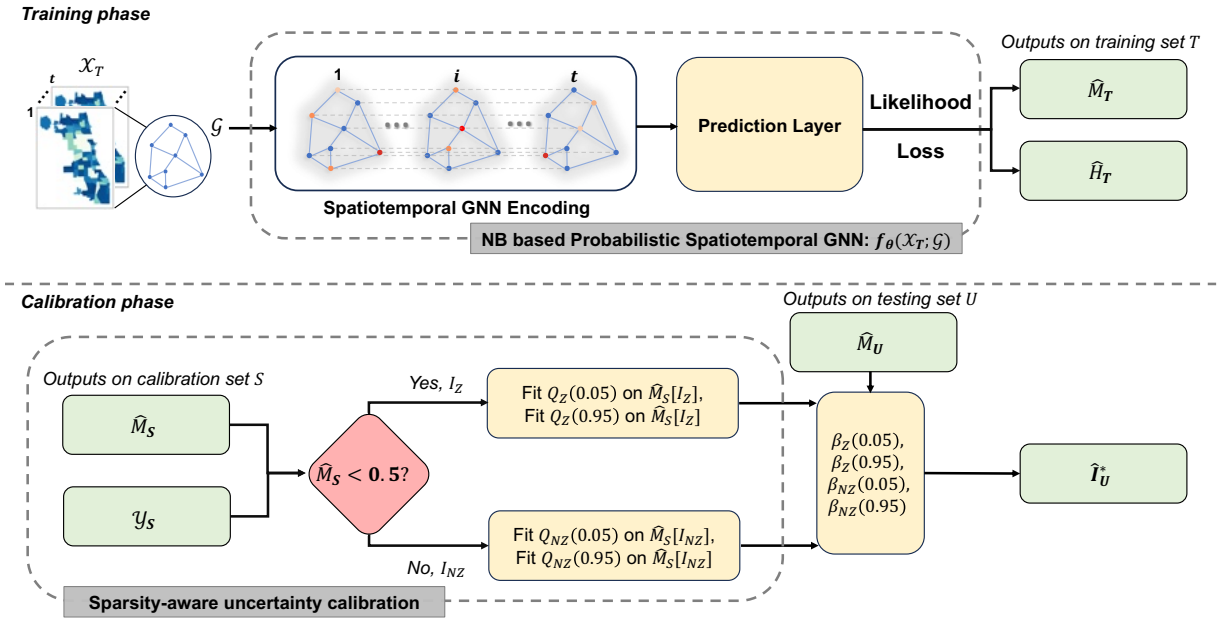


Figure 1: Experiments and modules of the SAUC framework. We modify existing ST-GNN models into probabilistic ones with NB distribution parameters as the outputs. Sparsity-aware uncertainty calibration is then conducted to obtain the quantile regression results and calibrated confidence interval for forecasting. These two steps can be easily and practically generalized to all existing ST-GNN models.

4.2 SAUC Framework

Our SAUC framework is a two-step post-hoc calibration framework tailored for sparse data, as detailed in Figure 1. In the training phase, we trained the modified NB-based probabilistic ST-GNNs to obtain the model parameters θ (modifications detailed in Section 5.2). In the calibration phase, this training yields $\hat{M}_S, \hat{H}_S$ on the calibration set for uncertainty calibration. We partition the calibration set into N bins based on the granularity of data variance we are interested in. We set $N = 15$ by default, leaving its selection discussed in Section 5.6. Inspired by zero-inflated models, predictions in the calibration set S are bifurcated based on indices: $I_Z = \{\hat{M}_S < 0.5\}$ for prediction values nearing zero, and $I_{NZ} = \{\hat{M}_S \geq 0.5\}$ for non-zero predictions. The value of 0.5 is a natural threshold for rounding operations to either 0 or 1. Following this partitioning, two QR models, Q_Z and Q_{NZ}, are trained separately, calibrating quantiles $p = \{.05, .95\}$ to fine-tune the mean and PI predictions. The calibrated model is then applied to predict model outputs on the testing set. The pseudocode is provided in Algorithm 1, and its implementation can be found in the anonymous GitHub [1].

The SAUC framework focuses on calibrating PIs rather than modifying distribution parameters, aligning with the principles of previous post-hoc calibration methods like isotonic regression and Platt scaling [17, 25]. These methods are designed to refine a model's outputs without directly altering internal parameters such as the parameters μ and α of NB distributions. This strategy ensures a distinct separation between modeling and calibration, enhancing flexibility and broad applicability. It avoids complicating the calibration process and potentially impacting the core modeling objectives. Additionally, post-hoc methods have demonstrated practical effectiveness and computational efficiency in improving the probabilistic accuracy of predictions, without necessitating retraining or modifications to the original model.

Song et al. [33] argued that quantile regressions might not retain true moments for the targets with similar predicted moments because of the global averaging effects. Our SAUC framework addresses this limitation because it creates local segments based on

Algorithm 1 SAUC Framework

Require: $\hat{M}_S, \mathcal{Y}_S, N, \hat{M}_U, p \leftarrow \{.05, .95\}$
1: Define the bin thresholds $\mathcal{T}_S$ using percentiles of $\mathcal{Y}_S$.
2: Initialize U^*, L^* in the same size as $\hat{M}_U$.
3: **for** each bin i in $[1, N]$ **do**
4: Extract indices I of instances in bin i based on $\mathcal{T}_S$.
5: Split I to I_{NZ} for $\hat{M}_S \geq 0.5$ and I_Z for $\hat{M}_S < 0.5$.
6: **for** each set i in $\{I_{NZ}, I_Z\}$ **do**
7: **for** each value q in p **do**
8: Fit $Q(q)$ on $\hat{M}_S[i]$ and $\mathcal{Y}_S[i]$ based on Equation 5 and obtain $\beta(q)$.
9: **end for**
10: $L^*[i] \leftarrow \hat{M}_U^T \beta(.05)$, $U^*[i] \leftarrow \hat{M}_U^T \beta(.95)$
11: **end for**
12: **end for**
13: $\mathcal{I}^* \leftarrow [L^*, U^*]$.
14: **return** $\mathcal{I}^*$.

[1] https://github.com/AnonymousSAUC/SAUC

zero and non-zero values. The zero segment, potentially dominant, requires precision in calibration due to its binary implications in real-world applications. For the non-zero subset, the aim is to ensure that quantiles mirror actual event frequencies, resonating with the typical research focus on discerning relative risks or magnitudes of events.

4.3 Calibration Metrics for Asymmetric Distributions

4.3.1 Expected Normalized Calibration Error. While calibration errors are clearly established for classification tasks using Expected Calibration Error (ECE) [12, 19, 26], the definition is much less studied in the regression context, where prediction errors and intervals supersede accuracy and confidence. Based on Equation 3 and 4, we aim to calibrate the length of the prediction interval with the standard error, i.e.,

$$\left[\mathbb{E}[(\hat{\mu}_i - Y_i)^2 | \hat{\mathcal{I}}_i = \mathcal{I}_i]\right]^{1/2} = c|\mathcal{I}_i|, \ \forall i \in \{1, \ldots, |U|\}. \tag{7}$$

Note that $\mathcal{I}_i = [F_i^{-1}(.05), F_i^{-1}(.95)]$ is the realization of the predicted 5%-95% confidence interval and $|\mathcal{I}_i|$ denotes its width. Calibrating the model's standard errors with the 90% confidence interval width reveals a linear relationship with coefficient c, specifically for symmetric distributions like Gaussian. As an approximation, we continue to use the z-score of Gaussian distribution to compute the coefficient c. The width of the 5%-95% confidence interval is roughly $2 \times 1.645 = 3.29$ of the standard error, inversely proportional to c. Empirical results, illustrated in Figure 2, support a linear correlation between the ratio of the predicted interval $\hat{\mathcal{I}}$ and the estimated standard error of the distribution. In our experiments, this slope ranges from 0.3 to 0.4, aligning closely with our theoretical assumption. Consequently, this leads to a value of $c \approx \frac{1}{3.29} \approx 0.303$.

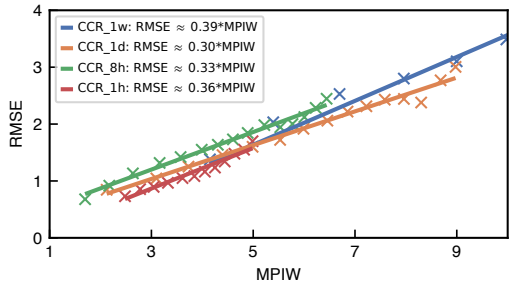


Figure 2: Explore the linear correlation between MPIW and RMSE using different cases of the CCR data.

Therefore, we could diagnose the calibration performance by comparing value differences on both sides of the equation. Similar to Algorithm 1, we partition the indices of test set samples into N evenly sized bins based on sorted predicted PI width, denoted as $\{B_j\}_{j=1}^{N}$. Each bin corresponds to a PI width that falls within the intervals: $[\min_{i \in B_j}\{|\mathcal{I}_i|\}, \max_{i \in B_j}\{|\mathcal{I}_i|\}]$. Note that the intervals are non-overlapping and their boundary values are increasing. Based on the discussion of Levi et al. [23] and Equation 7, for j-th bin, we define the Expected Normalized Calibration Error (ENCE) based on Root Mean Squared Error (RMSE) and Mean Prediction Interval Width (MPIW):

$$ENCE = \frac{1}{N}\sum_{j=1}^{N}\frac{|c \cdot MPIW(j) - RMSE(j)|}{c \cdot MPIW(j)}, \tag{8}$$

where the *RMSE* and *MPIW* are defined as:

$$RMSE(j) = \sqrt{\frac{1}{|B_j|}\sum_{i \in B_j}(\hat{\mu}_i - y_i)^2}, \tag{9}$$

$$MPIW(j) = \frac{1}{|B_j|}\sum_{i \in B_j}|\hat{\mathcal{I}}_i|. \tag{10}$$

The proposed ENCE normalizes calibration errors across bins, sorted and divided by $c \cdot MPIW$, and is particularly beneficial when comparing different temporal resolutions and output magnitudes across datasets. The $c \cdot MPIW$ and observed *RMSE* per bin should be approximately equal in a calibrated forecaster.

4.3.2 Coverage. In uncertainty calibration, coverage directly assesses the alignment between predicted uncertainty intervals and the true variability of the data. The coverage is calculated as how many points fall within the confidence interval, written as:

$$Coverage_{0.05-0.95} = \frac{1}{|U|}\sum_{i \in U} 1_{(y_i \geq \hat{F}_i^{-1}(0.05) \ \& \ y_i \leq \hat{F}_i^{-1}(0.95))}, \tag{11}$$

where $\hat{F}_i^{-1}(0.05)$ and $\hat{F}_i^{-1}(0.95)$ are the predicted lower and upper confidence interval bounds, respectively. The expected coverage should be 0.9 for an ideal calibration. A coverage value closer to the ideal value indicates that the predicted intervals encompass a large proportion of the observed data points, signifying that the uncertainty estimates are accurate and reliable.

Both ENCE and coverage evaluate probabilistic model performance but focus on different aspects: 1) Coverage is calculated at the point distribution level, while ENCE operates at the binning level, allowing detailed analysis across different ranges of predicted uncertainty; 2) Coverage measures the proportion of observed outcomes within PIs, assessing the model's ability to capture data variability directly. Meanwhile, ENCE evaluates the deviation between expected and actual calibration, providing a normalized error metric for accuracy and sharpness. As an extension of the ECE in previous calibration work [12, 17, 19, 32, 34], ENCE could adapt to distributions for sparse data like NB distributions while keeping the theoretical meaning of calibration error.

5 Experiments

5.1 Data Description

Two spatiotemporal datasets are used: (1) Chicago Traffic Crash Data (CTC)[2] sourced from 277 police beats between January 1, 2016 and January 1, 2023. The CTC data records show information about each traffic crash on city streets within the City of Chicago limits and under the jurisdiction of Chicago Police Department; (2) Chicago Crime Records (CCR)[3] derived from 77 census tracts spanning January 1, 2003 to January 1, 2023. This dataset reflects reported incidents of crime (with the exception of murders where

[2]https://data.cityofchicago.org/Transportation/Traffic-Crashes-Crashes/85ca-t3if
[3]https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2

data exists for each victim) that occurred in the City of Chicago. Despite both CTC and CCR data originating from the Chicago area, their disparate reporting sources lead to different spatial units: census tracts for CCR and police beats for CTC. Both datasets use the first 60% timesteps for training, 20% for calibration and validation, and 20% for testing. Full data descriptions can be found in Table 1.

Dataset	Resolution	Size	Sparsity	Mean	Max
CCR	1-hour	(77, 175321)	67%	0.3	59
	8-hour	(77, 21916)	18%	2.7	206
	1-day	(77, 7306)	4%	8	223
	1-week	(77, 1044)	< 0.1%	55.8	539
CTC	1-hour	(277, 61369)	96%	< 0.1%	5
	8-hour	(277, 7672)	76%	0.4	9
	1-day	(277, 2558)	47%	1.1	15
	1-week	(277, 366)	7%	7.3	45

Table 1: Characteristics of various datasets showing variation in sparsity at different temporal resolutions. Dataset sizes are represented as (spatial, temporal) dimension pairs. Sparse cases are marked grey.

The temporal resolutions of the datasets are varied to demonstrate the ubiquitous sparsity issue and its practical significance in spatiotemporal analysis. Four temporal resolutions, 1 hour, 8 hours, 1 day, and 1 week are created for CTC and CCR datasets. We designate the 1-hour and 8-hour cases of Crash and Crime datasets as sparse instances, due to a higher prevalence of zeros.

The histogram in Figure 3 shows traffic crash counts in all police beats across different temporal resolutions. Notably, higher resolutions show noticeable skewness and clustering of values at lower counts, illustrating the sporadic nature of traffic incidents.

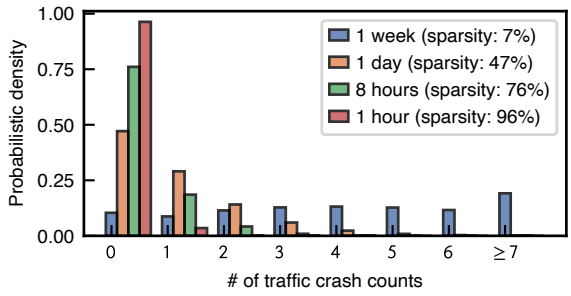


Figure 3: Histogram of number traffic crashes in the CTC data in different temporal resolutions

In the spatiotemporal data analysis, when we focus on more granular and responsive modeling, higher spatial or temporal resolutions are unavoidable. This observation emphasizes the importance of equipping predictive models with capabilities to adeptly manage sparse data, thereby more accurately mirroring real-world temporal patterns.

For adjacency matrices, we calculate geographical distances d_{ij} between the centroids of regions i and j, representing census tracts or police beats. This distance is then transformed into a similarity measure, $\mathbf{A}_{ij} = e^{-d_{ij}/0.1}$, where 0.1 is a scaling parameter, thus forming an adjacency matrix [42].

5.2 Modified Spatiotemporal Models

To demonstrate our model's ability to generalize, we modify three popular ST-GNN forecasting models: Spatio-temporal Graph Convolution Network (STGCN) [42], Graph WaveNet (GWN) [41], and Dynamic Spatial-Temporal Aware Graph Neural Network (DSTAGNN) [20]. These models, originally designed for numerical outputs, are adapted to output the location parameter μ and dispersion parameter α of NB distributions. We add two linear layers to the neural network structures to learn these parameters separately, renaming the models STGCN-NB, GWN-NB, and DSTAGNN-NB, respectively. While NB distributions are used as an example, the output parameters are free to be adjusted based on data assumptions. Lastly, we replace the mean square error loss with the likelihood loss defined in Equation 12:

$$\begin{aligned}\mathcal{L}(\mu,\alpha,y) = &-\left[y\cdot\log\left(\frac{\mu+\epsilon}{\mu+\alpha+2\epsilon}\right)\right]\\ &-\Gamma(y+\alpha+\epsilon)+\Gamma(y+1)+\Gamma(\alpha+\epsilon),\\ &-\alpha\cdot\log\left(\frac{\alpha+\epsilon}{\mu+\alpha+2\epsilon}\right)+\lambda\cdot||\alpha||^2\end{aligned} \tag{12}$$

where y is the target variable, μ and α are model outputs, Γ is the gamma function, λ is the regularization parameter, and ϵ is a small constant added to improve numerical stability. Equation 12 is derived from the NB likelihood controlled by μ and α. Similar likelihood loss could be derived when choosing other distributions.

The loss function ensures the predicted NB distribution aligns well with the true target values. It penalizes predictions where the mean of the distribution (μ) is far from the actual target values. Furthermore, it incorporates the shape of the distribution via the dispersion parameter (α) to capture the variability in the data. The loss function also includes a regularization term to prevent overfitting by discouraging overly large values of α.

All our experiments are implemented on a machine with Ubuntu 22.04, with Intel(R) Core(TM) i9-10980XE CPU @ 3.00GHz CPU, 128GB RAM, and NVIDIA GeForce RTX 4080 GPU.

Dataset	STGCN	GWN	DSTAGNN
CCR_1h	0.637 (0.703)	0.962 (0.766)	1.226 (1.117)
CCR_8h	2.386 (2.326)	2.052 (2.376)	2.272 (2.693)
CTC_1h	0.214 (0.220)	0.980 (0.220)	1.074 (0.871)
CTC_8h	0.633 (0.726)	0.924 (0.746)	1.289 (1.177)

Table 2: RMSE of original model outputs and the modified models (shown in the parenthesis) using sparse datasets.

As shown in Table 2, the model modification is successful without losing accuracy. In fact, the NB modification performs better in the sparse dataset as the NB distributions are more suitable for the nature of discrete data. Full implementation details and comparison of the modified ST-GNN models can be found in Appendix A.1.

Method / Dataset	Full observations: Before calibration	Histogram binning	Isotonic regression	Temp. scaling	Platt scaling	QR	SAUC	Zero-only targets: Before calibration	Histogram binning	Isotonic regression	Temp. scaling	Platt scaling	QR	SAUC
STGCN-NB Outputs														
CCR_1h	1.160	0.410	0.517	0.887	0.443	0.235	**0.198**	1.167	0.361	0.386	0.473	0.477	0.273	**0.267**
CCR_8h	1.389	0.344	0.417	0.385	0.373	0.392	**0.336**	1.065	0.568	0.578	0.427	0.162	0.175	**0.127**
CCR_1d	0.413	0.593	0.546	0.364	0.294	0.297	**0.192**	1.266	1.047	0.979	1.167	0.840	0.843	**0.738**
CCR_1w	0.855	0.784	0.773	0.765	0.579	0.272	**0.242**	0.354	0.362	0.332	0.287	0.345	0.155	**0.113**
CTC_1h	0.578	0.227	0.200	1.390	0.452	0.388	**0.165**	3.211	0.415	0.363	2.042	0.646	0.416	**0.078**
CTC_8h	0.323	0.320	0.255	0.570	0.297	0.241	**0.233**	0.330	0.247	0.276	0.551	0.230	0.196	**0.184**
CTC_1d	0.722	0.050	**0.011**	0.862	0.060	0.048	0.047	0.434	0.128	**0.027**	5.451	0.152	0.379	0.150
CTC_1w	2.279	1.227	1.527	0.914	2.265	0.378	**0.365**	1.823	2.384	2.009	1.160	1.343	0.767	**0.755**
GWN-NB Outputs														
CCR_1h	1.200	1.022	0.889	1.149	0.834	0.604	**0.493**	2.042	1.796	1.274	1.710	1.356	0.476	**0.375**
CCR_8h	0.670	0.636	0.612	0.696	0.655	0.597	**0.566**	1.398	0.389	0.492	1.179	0.478	0.308	**0.186**
CCR_1d	0.997	0.662	0.633	0.581	0.562	0.587	**0.504**	1.158	0.583	0.580	0.725	0.747	0.644	**0.563**
CCR_1w	0.858	0.938	0.935	**0.611**	0.931	0.857	0.829	0.335	0.218	0.219	**0.161**	0.219	0.250	0.172
CTC_1h	0.868	0.816	0.891	0.859	0.634	0.832	**0.290**	1.300	0.712	0.561	1.001	0.143	0.998	**0.139**
CTC_8h	0.931	0.416	0.388	1.381	0.169	0.226	**0.148**	2.213	0.720	0.659	1.995	0.421	0.776	**0.221**
CTC_1d	0.523	0.420	0.448	0.571	0.523	0.432	**0.139**	0.250	0.464	0.405	0.248	0.484	0.439	**0.210**
CTC_1w	0.408	0.115	**0.114**	0.259	0.236	0.241	0.239	0.466	0.479	**0.436**	0.457	0.451	0.476	0.450
DSTAGNN-NB Outputs														
CCR_1h	0.328	0.695	0.695	0.958	0.667	0.230	**0.107**	0.292	0.773	0.752	4.853	0.900	0.120	**0.102**
CCR_8h	0.895	0.941	0.940	0.604	0.928	0.443	**0.440**	0.811	0.905	0.901	0.419	0.930	0.572	**0.314**
CCR_1d	0.962	0.950	0.950	0.810	0.946	0.792	**0.709**	0.822	0.790	0.788	0.619	0.797	0.235	**0.171**
CCR_1w	0.999	1.000	1.000	0.987	1.000	0.977	**0.952**	0.260	0.140	0.140	**0.138**	0.139	0.150	0.149
CTC_1h	0.398	0.187	0.121	0.138	0.129	0.159	**0.119**	0.937	0.652	0.661	1.222	0.970	0.631	**0.229**
CTC_8h	1.462	0.666	0.669	0.473	0.767	0.505	**0.452**	0.694	0.575	0.608	0.586	0.950	0.612	**0.523**
CTC_1d	1.393	0.776	0.774	1.304	0.725	0.481	**0.402**	1.650	0.789	0.771	1.259	1.000	0.637	**0.503**
CTC_1w	0.890	0.876	0.874	**0.717**	0.840	0.824	0.833	0.980	0.757	0.743	**0.498**	0.827	0.679	0.550

Table 3: ENCE of the calibration methods. Bold fonts mark the best and underlines denote the second-best calibration results.

5.3 Baseline Calibration Methods

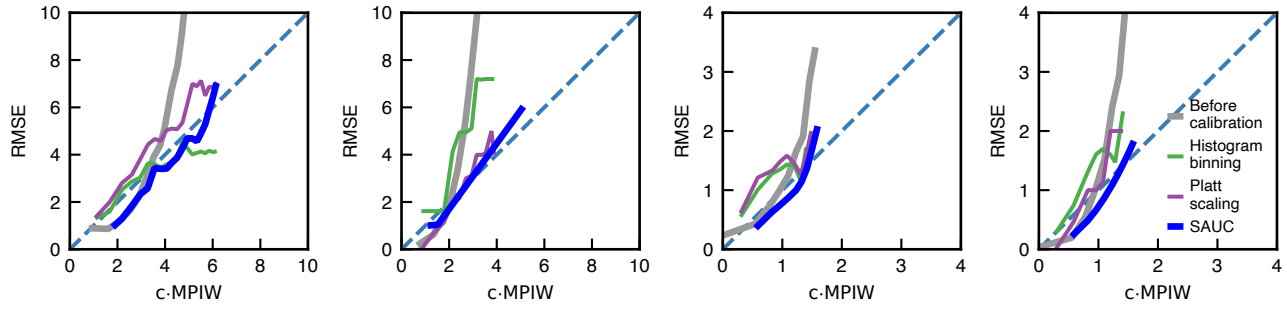
We compare the SAUC framework with existing post-hoc calibration methods proven to be effective in the regression tasks: (1) *Isotonic regression* [25], which ensures non-decreasing predictive probabilities, assuming higher model accuracy with increased predicted probabilities; (2) *Temperature scaling* [18], which scales the model's output using a learned parameter, whose efficacy for regression models depends on the output's monotonicity in relation to confidence; (3) *Histogram binning* [43], a model partitioning predicted values into bins, calibrating each independently according to observed frequencies, yet might lack flexibility for diverse prediction intervals; (4) *Platt scaling* [17], which applies a scaling transformation to the predicted values with the function derived from minimizing observed and predicted value discrepancies, providing linear adaptability but potentially inadequate for complex uncertainty patterns; (5) *Quantile regression* [8], see Section 4.1.

5.4 Calibration Results Comparison

5.4.1 ENCE Evaluation. Table 3 presents the calibration performance of baseline models and our proposed framework in different cases, measured by ENCE. The term "zero-only targets" denotes ENCE computation solely on true zero target values.

Table 3 highlights the superior performance of the SAUC framework across most datasets, with bold type indicating the best results and underlines denoting the second-best. Our findings reveal that the SAUC framework typically outperforms other models, though it might fall short in aggregated cases, such as those with a 1-week resolution. Notably, SAUC demonstrates approximately an overall 23% reduction in ENCE compared to the second-best model with full observations and a similar 20% reduction in sparsity entries. This reduction is assessed by computing the average percentage improvement relative to the second-best model across all scenarios. The separation of zero and non-zero data during calibration proved particularly beneficial, effectively addressing the issues of heteroscedasticity and zero inflation. However, some calibration methods may worsen errors due to overfitting, particularly evident in temperature scaling and isotonic regression.

Histogram binning and isotonic regression exhibit similar results among baseline models, facilitated by the non-decreasing nature of Equation 4. Temperature scaling performs well in coarse temporal resolutions, while Platt scaling excels in sparse scenarios. QR consistently ranks second-best, with our SAUC framework closely matching QR in aggregated non-sparse cases and outperforming notably in sparse scenarios.



(a) Full observations, CCR_8h (b) Zero-only targets, CCR_8h (c) Full observations, CCR_1h (d) Zero-only targets, CCR_1h

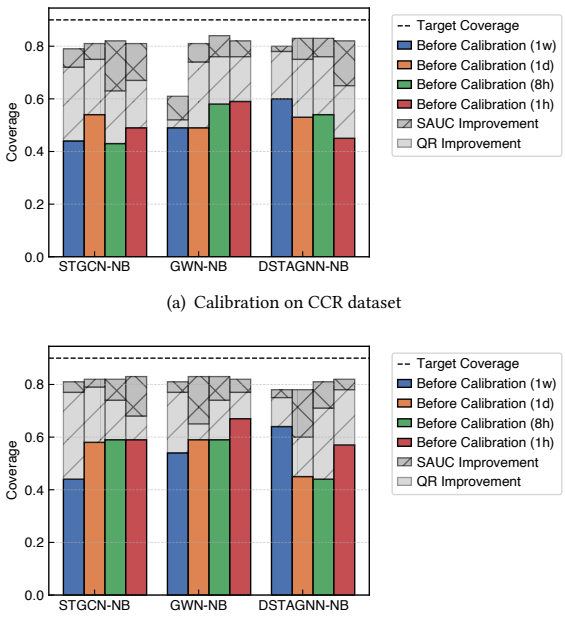
Figure 4: Reliability diagrams of different calibration methods applied to STGCN-NB outputs on different components of CCR_8h and CCR_1h data. Results closer to the diagonal dashed line are considered better.

5.4.2 Reliability Diagram Evaluation. Utilizing Equations 7 and 8, we refine the conventional reliability diagram to evaluate the calibration efficacy of a model [18]. This involves partitioning predictions into bins based on $\mathcal{I}$, after which we compute the RMSE and MPIW employing the calibrated μ^* and $|\mathcal{I}^*|$ for each respective bin.

Figure 4 presents a comparative reliability diagram of STGCN-NB outputs prior to calibration, alongside Isotonic regression, Platt scaling, and our SAUC framework, each applied to different segments of the CCR_8h dataset. We selectively highlight representative non-parametric and parametric baseline calibration approaches based on their performance. The diagonal dashed line symbolizes the calibration ideal: a closer alignment to this line indicates superior calibration.

A closer examination of Figure 4(a) reveals that both baseline models and our SAUC framework adeptly calibrate the model outputs across all observations. Yet, focusing solely on the reliability diagram of our calibrated outputs for ground-truth zero values, as depicted in Figure 4(b), there is a noticeable deviation from the ideal line. Conversely, our SAUC framework demonstrates commendable alignment with the ideal when authentic data are zeros. This observation remains consistent for the CCR_1h dataset, as seen in Figures 4(c) and 4(d). By contrasting Figures 4(a) and 4(c), it becomes evident that the calibration performance diminishes in scenarios with finer resolutions. Nevertheless, our SAUC framework retains commendable efficacy, particularly when predicting zero entries, attributed to our distinctive calibration approach coupled with QR's adeptness at managing heteroscedasticity, especially when predictions on zeros exhibit considerable variability. Contemporary probabilistic spatiotemporal prediction models predominantly prioritize refining MPIWs for enhanced accuracy, often unintentionally neglecting practical safety considerations. Our calibration strategy offers outputs with reinforced reliability.

5.4.3 Coverage Evaluation. We further evaluate the calibration performance through coverage, defined in Equation 11. The calibration improvements achieved using the SAUC framework and QR are illustrated in Figure 5, applied to various pre-calibrated models across different resolutions of the CCR and CTC datasets. The QR improvement in Figure 5 signifies the enhancement in coverage by applying QR to the pre-calibrated model, while SAUC improvement denotes the further enhancement over QR. Across all models and resolutions, calibration markedly enhances coverage. Initially, coverage levels were consistently below the target (0.9), exhibiting notable variations depending on the temporal resolution. Following calibration, all models and datasets achieve a similar level of coverage, closer to the target. Particularly on the sparser datasets (resolutions 1h and 8h), the improvements of SAUC compared with QR are generally more pronounced, notably for model performance on the CCR datasets. These findings underscore the efficacy of the SAUC framework in enhancing model calibration and ensuring improved prediction reliability across diverse temporal resolutions and models, particularly when handling sparser data inputs.



(a) Calibration on CCR dataset

(b) Calibration on CTC dataset

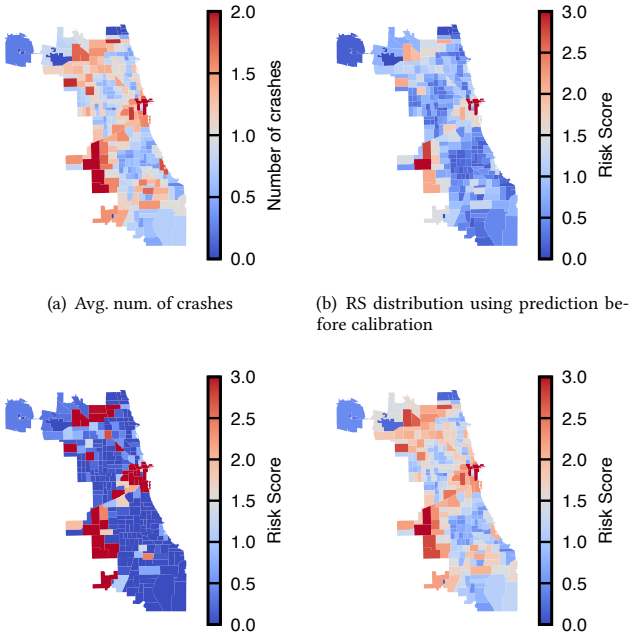
Figure 5: Coverage improvements by applying SAUC framework and QR on different data sets and models.

5.5 Risk Analysis

To evaluate the spatial effects of spatiotemporal prediction after SAUC calibration, we introduce Risk Score (RS), an enhancement of the traditional metric defined as the product of *event probability* and *potential loss magnitude* [24, 35]:

$$RS = \hat{\mu} \times |\hat{I}|. \tag{13}$$

This metric integrates anticipated risk ($\hat{\mu}$) and prediction uncertainty ($|\hat{I}|$), attributing higher RS to regions with both high incident frequency and uncertainty. This RS metric assigns higher risk to predictions that are characterized by both large predicted mean values and high uncertainty.



(a) Avg. num. of crashes

(b) RS distribution using prediction before calibration

(c) RS distribution using calibrated prediction via Platt scaling

(d) RS distribution using using calibrated prediction via SAUC

Figure 6: Traffic crash accident distributions and calibration using CTC_8h data. All the data are averaged over the temporal dimension.

Using the CTC_8h dataset and outputs from STGCN-NB, we demonstrate the importance of SAUC for evaluating the risks of traffic accidents. Figure 6 compares the average number of crashes and average RS values over time for each region, pre and post-calibration. Prior to calibration, as depicted in Figure 6(b), the RS in regions, especially Chicago's northern and southern sectors, did not provide actionable insights, given their frequent traffic incidents reported by Figure 6(a). This is due to the tight PIs from pre-calibration method outputs lower the RS. As shown in Figure 6(c), RS distribution from Platt scaling calibrated predictions could highlight certain risky areas, but still miss the major crash regions in southern Chicago. In contrast, SAUC-calibrated predictions, as seen in Figure 6(d), unveil coherent spatial distribution patterns highly consistent with Figure 6(a), demonstrating the effectiveness for urban safety monitoring.

Rather than just yielding exact numerical predictions with tight confidence intervals, it is imperative that models convey equivalent severity levels when integrating uncertainty. Such an approach ensures reliable outputs in crime and accident predictions. Notably, the reliable estimation of zero or non-zero incident counts is pivotal in risk management. Existing probabilistic prediction studies focus on exact values and restricted confidence intervals, sometimes neglecting the data variability, especially around zero incidences.

5.6 Sensitivity Analysis

The number of bins N in both the SAUC framework and ENCE discussion is important. We choose the bin number from the range $N = [1, 5, 10, 15, 20, 25, 30]$ and discuss the ENCE change with respect to the bin number. We conduct the sensitivity analysis using the four cases in the crime dataset using STGCN-NB model outputs.

Dataset/Bin	1	5	10	15	20	25
Full Observations						
CCR_1h	0.513	0.302	0.299	0.198	0.249	0.292
CCR_8h	0.096	0.113	0.185	0.336	0.123	0.115
CCR_1d	0.093	0.167	0.185	0.192	0.189	0.207
CCR_1w	0.074	0.216	0.323	0.242	0.254	0.272
Zero-only Observations						
CCR_1h	0.650	0.689	0.604	0.267	0.229	0.501
CCR_8h	0.642	0.583	0.504	0.127	0.172	0.345
CCR_1d	0.641	1.181	1.096	0.738	1.025	0.958
CCR_1w	1.445	0.950	0.361	0.113	0.363	0.567

Table 4: ENCE on the full observations and zero-only observations with respect to different bin numbers.

From Table 4, it is evident that the number of bins significantly impacts the SAUC framework and ENCE performance. Both very small and very large bin numbers may result in suboptimal ENCE values. Intuitively, with only one bin, all data points are grouped together, making it difficult to differentiate levels of variance as the comparison involves the overall data variability and the average MPIW. Conversely, having too many bins means each bin contains few data points, leading to fluctuations due to the small sample size. Therefore, an optimal bin number range between 10 and 20 is preferable for achieving reliable ENCE.

5.7 Discussion on Distribution Assumptions

A common implementation question arises: is the NB distribution flexible enough to capture the response distributions? The focus of this work lies in uncertainty calibration methods for sparse data, where specific selections of distributions are less discussed. Nevertheless, it is imperative to examine whether the presented distributions offer sufficient flexibility.

In comparison to the NB distribution, we have also implemented Poisson and Gaussian distributions as estimated distributions for our model. Models' modification remains unchanged following Section 3.1, with Equation 2 replaced by estimations of corresponding

parameters for Poisson and Gaussian distributions. The last layer is modified to output the distribution parameters of the Poisson distributions exclusively. For the Poisson distribution, only one parameter λ is outputted, and the loss function can be similarly defined as:

$$\mathcal{L}(\hat{\lambda}, y) = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{\lambda}_{s,i} - y_i \cdot \log(\lambda_{s,i})\right), \tag{14}$$

where $\hat{\lambda}_s = \hat{\lambda} + \epsilon$, with $\hat{\lambda}$ representing the predicted rate parameters, target representing the target counts, and ϵ a small constant (e.g., 10^{-10}) added for numerical stability. Equation 14 is derived from the Poisson likelihood function, ensuring robust and stable learning for our spatiotemporal prediction models.

Similarly, for Gaussian distribution, we estimate the location and scale parameters μ and σ with the loss function:

$$\mathcal{L}(\hat{\mu}, \hat{\sigma}, y) = \frac{1}{N}\sum_{i=1}^{N}\left(\frac{1}{2}\log(2\pi\hat{\sigma}_i^2) + \frac{(y_i - \hat{\mu}_i)^2}{2\hat{\sigma}_i^2}\right) + \lambda||\hat{\sigma}||_2, \tag{15}$$

where penalization for the variance is also added, similar to Equation 12. We discuss the prediction RMSE and ENCE performance of STGCN, STGCN-NB, STGCN-Poisson, and STGCN-Gaussian, as shown in Table 5.

Dataset	Metric	Original	NB	Poisson	Gaussian
CCR_1h	RMSE	0.637	0.703	0.581	0.741
	ENCE	/	1.160	1.692	3.292
CCR_8h	RMSE	2.386	2.326	2.419	2.303
	ENCE	/	1.389	0.913	30.286
CCR_1d	RMSE	5.167	4.286	6.296	6.701
	ENCE	/	0.413	0.999	67.267
CCR_1w	RMSE	23.333	20.242	23.907	31.044
	ENCE	/	0.855	0.100	168.889
CTC_1h	RMSE	0.214	0.220	0.282	0.321
	ENCE	/	0.578	3.079	0.962
CTC_8h	RMSE	0.633	0.726	0.962	0.972
	ENCE	/	0.323	0.960	2.320
CTC_1d	RMSE	1.148	1.786	1.138	1.130
	ENCE	/	0.722	1.509	3.351
CTC_1w	RMSE	3.317	3.929	3.450	3.321
	ENCE	/	2.279	0.145	5.867

Table 5: Comparison of RMSE and ENCE for the original model and its NB, Poisson, and Gaussian modification.

Based on Table 5, it is clear that the Poisson and Gaussian modifications exhibit similar performance to the NB distribution in terms of RMSE based on the experimental results. However, NB demonstrates better ENCE when capturing sparse data cases compared with the other two distributions, thus it is selected as the distribution predominantly discussed in this paper. More dedicated data distributions' assumptions can be explored, as demonstrated by the work of [4, 15, 47, 48].

5.8 Complexity Discussion

From Equation 5, we know that the complexity of the SAUC framework depends on the size of the data. Since our method is a post-hoc method, the added complexity is easier to address in terms of runtime. Same as the settings in sensitivity analysis, we visualize the model runtime on different bin numbers and data resolutions by running STGCN-NB, shown in Figure 7. It is clear that the selection of the bin number does not influence the complexity and runtime. Moreover, by reducing the resolution from 1 hour to 8 hours, 1 day, and 1 week, the runtime decreases nearly exponentially with the change in the temporal resolutions of the data.

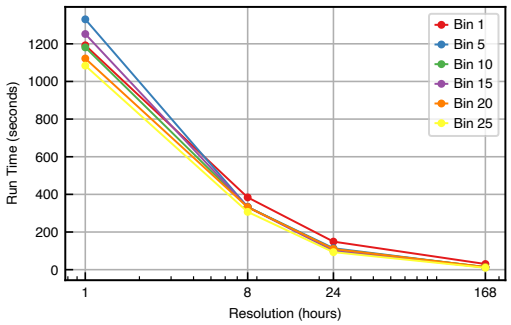


Figure 7: Runtime of the calibration methods on the full observations with different bin numbers and data resolutions of CCR dataset. The x-axis scale is logarithmic.

6 Conclusion

Current spatiotemporal GNN models mainly yield deterministic predictions, overlooking data uncertainties. Given the sparsity and asymmetry in high-resolution spatiotemporal data, quantifying uncertainty is challenging. Previous ST-GNN models covert the deterministic models to probabilistic ones by changing the output layers, without proper discussions in the reasonableness of learned distributions and UQ. We introduce the SAUC framework, which calibrates uncertainties for both zero and non-zero values using the zero-inflated nature of these datasets. As a post-hoc method, SAUC is available to be adapted to any existing ST-GNN model. We further introduce new calibration metrics suited for asymmetric distributions. Tests on real-world datasets show SAUC reduces calibration errors by 20% for zero-only targets. SAUC enhances GNN models and aids risk assessment in sparse datasets where accurate predictions are critical due to safety concerns.

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A Appendix

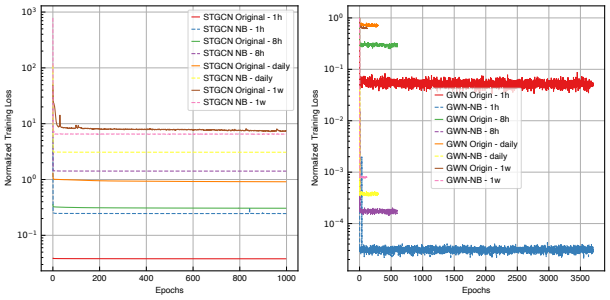
A.1 Modified Model Implementation Details

To implement the modified GNNs, we use the Github repositories for STGCN [4], GWN [5], and DSTAGNN [6] respectively. We basically keep all the hyper-parameters and important parameters as the default from the original repositories, which can be referred to in our Github repository [7]. Notice that we keep the same model parameters in both the original models and the modified models, which can be found in Table 6

Parameters/Models	STGCN	GWN	DSTAGNN
Input window length	12	12	12
Output window length	12	12	12
Learning rate	0.001	0.001	0.001
Batch size	24	24	24
Output layer	Linear	Conv2d	Linear
Training epochs	1000	100	500
Early stop	No	Yes	Yes

Table 6: Parameters of original and modified GNN models.

Training loss on CCR datasets can be found in Figure 8. All models are converged properly.



(a) Training Loss of STGCN on CCR dataset (b) Training Loss of GWN on CCR dataset

Figure 8: Training loss of STGCN, GWN, and their modifications on the CCR dataset.

The full implementation results can be found in Figure 7. It is clear that the model's performance in terms of numerical accuracy is close to the results using models before modification. In fact, the NB modification shows better performance in the sparse dataset as the NB distributions are more suitable for the nature of discrete data.

A.2 Baseline Model Implementation

We list the introductions and pseudo codes to implement the baseline models.

[4]https://github.com/FelixOpolka/STGCN-PyTorch
[5]https://github.com/nnzhan/Graph-WaveNet
[6]https://github.com/SYLan2019/DSTAGNN
[7]https://github.com/AnonymousSAUC/SAUC

Dataset	Error Metric	STGCN	STGCN-NB	GWN	GWN-NB	DSTAGNN	DSTAGNN-NB
CCR_1h	MAE	0.429	0.317	0.848	0.334	0.982	0.901
	RMSE	0.637	0.703	0.962	0.766	1.226	1.117
CCR_8h	MAE	1.499	1.452	1.395	1.478	1.475	1.198
	RMSE	2.386	2.326	2.052	2.376	2.272	2.693
CCR_daily	MAE	2.951	2.719	2.521	2.681	2.500	2.421
	RMSE	5.167	4.286	3.784	9.703	3.848	3.368
CCR_weekly	MAE	14.146	12.505	9.938	14.516	10.389	17.341
	RMSE	23.333	20.242	16.103	23.231	16.578	21.382
CTC_1h	MAE	0.084	0.044	0.960	0.044	1.049	0.832
	RMSE	0.214	0.220	0.980	0.220	1.074	0.871
CTC_8h	MAE	0.453	0.346	0.810	0.352	1.113	0.995
	RMSE	0.633	0.726	0.924	0.746	1.289	1.177
CTC_daily	MAE	0.861	1.433	0.895	1.019	1.299	0.986
	RMSE	1.148	1.786	1.154	1.550	1.596	1.353
CTC_weekly	MAE	2.474	3.618	2.423	3.279	2.502	2.956
	RMSE	3.317	3.929	3.178	3.689	3.256	3.876

Table 7: Comparisons between modified models and the original models.

A.2.1 Temperature Scaling. Temperature scaling is a parametric method that modifies the outputs of a regression model. This modification is controlled by a single parameter, the temperature T. The algorithms can be formulated as follows:

Algorithm 2 Temperature Scaling

1: Train a model on a training dataset, yielding point predictions $f(x)$.
2: Optimize the temperature T on the validation set by minimizing the mean squared error between the predicted and true values:

$$T = \arg\min_T \frac{1}{n}\sum_{i=1}^{n}(f(x_i) - y_i)^2$$

3: **for** each new input x **do**
4: Compute the model's prediction $f(x)$.
5: Apply temperature scaling with the learned T to the prediction $f(x)$, yielding scaled prediction:

$$g(f(x)) = \frac{f(x)}{T}$$

6: **end for**

Temperature scaling is an efficient and straightforward method for calibration that requires optimization of only a single parameter. By scaling the outputs, it can yield calibrated predictions without modifying the rank ordering of the model's predictions.

A.2.2 Isotonic Regression. Isotonic regression is a non-parametric method utilized for the calibration of a predictive model's outputs. In the context of regression, this method operates as shown below:

Isotonic regression makes no assumptions about the form of the function connecting the model's predictions to calibrated predictions. This flexibility allows it to fit complex, non-linear mappings, which can provide improved calibration performance when the model's outputs are not well-modeled by a simple function.

A.2.3 Histogram Binning. Histogram binning can also be applied in a regression setting to calibrate predictions. The idea is to split the range of your model's outputs into several bins, and then adjust

Algorithm 3 Isotonic Regression for Regression

Train a model on a training dataset, providing point predictions $f(x)$.
Fit an isotonic regression model on the validation set:
  Sort the predictions $f(x_i)$ and corresponding true values y_i in ascending order of $f(x_i)$.
  Solve the isotonic regression problem to find a non-decreasing sequence $g(f(x))$ that minimizes:

$$\min \sum_{i=1}^{n} (g(f(x_i)) - y_i)^2$$

for each new input x **do**
  Compute the model's prediction $f(x)$.
  Use the fitted isotonic regression model to convert the prediction $f(x)$ to a calibrated prediction: $g(f(x))$
end for

the predictions within each bin to match the average actual outcome within that bin. The specific algorithm is:

Algorithm 4 Histogram Binning

Train a model on a training dataset, yielding predictions $f(x)$.
Determine the bins for the predictions on the validation set.
for each bin B_i **do**
  Compute the average true value $\bar{y}_i$ for all examples in the bin:

$$\bar{y}_i = \frac{1}{|B_i|} \sum_{x_j \in B_i} y_j$$

  Adjust the prediction for each example in the bin to the average true value $\bar{y}_i$.
end for
for each new input x **do**
  Compute the model's prediction $f(x)$.
  Find the bin B_i that $f(x)$ falls into and adjust $f(x)$ to the average true value for that bin:

$$g(f(x)) = \bar{y}_i$$

end for

This method assumes that predictions within each bin are uniformly mis-calibrated. It is less flexible than isotonic regression, but it is simpler and less prone to overfitting, especially when the number of bins is small.

A.2.4 Platt Scaling. In a regression setting, Platt Scaling can be interpreted as applying a sigmoid function transformation to the model's outputs. The output is then considered as the mean of a Bernoulli distribution. While this might be useful in some contexts, it is not generally applicable to regression problems but can be useful in sparse datasets. A linear transformation is fitted to the model's predictions to minimize the mean squared error on the validation set. The parameters of this transformation are learned from the data, making this an instance of post-hoc calibration. Note that this adaptation might not be suitable for all regression tasks, especially those where the target variable has a non-linear relationship with the features.

Algorithm 5 Platt Scaling

Train a model on a training dataset, yielding predictions $f(x)$.
Minimize the mean squared error on the validation set between $g(f(x))$ and the true outcomes, where $g(f(x)) = af(x) + b$ is a linear transformation of the model's predictions: $a, b = \arg\min_{a,b} \frac{1}{n} \sum_{i=1}^{n} (af(x_i) + b - y_i)^2$
for each new input x **do**
  Compute the model's prediction $f(x)$.
  Apply the learned linear transformation to $f(x)$ to get the calibrated prediction: $g(f(x)) = af(x) + b$
end for

Uncertainty Quantification for Traffic Forecasting Using Deep-Ensemble-Based Spatiotemporal Graph Neural Networks

Tanwi Mallick, *Member, IEEE*, Jane Macfarlane, and Prasanna Balaprakash

***Abstract*— Deep-learning-based data-driven forecasting methods have achieved impressive results for traffic forecasting. Specifically, spatiotemporal graph neural networks have emerged as a promising class of approaches because of their ability to model both spatial and temporal patterns in traffic data. A major limitation of these methods, however, is that they provide forecasts without estimates of data and model uncertainty, which are critical for understanding inherent variations of the data and forecast limitations due to a lack of training data. We develop a scalable deep ensemble approach to quantify data and model uncertainties for spatiotemporal graph neural networks. Our approach consists of four stages: 1) using a Gaussian-assumption-free simultaneous quantile regression loss for training a spatiotemporal graph neural network to model the traffic distribution; 2) applying a scalable Bayesian optimization method to tune the hyperparameters of the spatiotemporal graph neural network; 3) fitting a Gaussian copula generative model to capture the joint distributions of the high-performing hyperparameter configurations, and training an ensemble of models by sampling a new set of hyperparameter configurations from the generative model; and 4) decomposing the data and model uncertainties from the spatiotemporal graph neural network ensembles. We illustrate the effectiveness of our approach on a diffusion convolutional recurrent neural network, a state-of-the-art method for short-term traffic forecasting. We demonstrate the efficacy of our ensemble-based uncertainty quantification method by comparing it with other uncertainty estimation techniques. We show that our generic and scalable approach outperforms the current state-of-the-art Bayesian and a number of other commonly used frequentist uncertainty estimation techniques. The code is available on GitHub: https://github.com/tanwimallick/DESQRUQ.**

***Index Terms*— Traffic forecasting, uncertainty quantification, deep ensemble, spatiotemporal graph neural network.**

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I. Introduction

TRAFFIC forecasting is a foundational component of intelligent transportation systems. Accurate forecasting across normal and extreme traffic conditions is crucial to improving traffic control, mitigating congestion, and resolving other traffic-related issues. For decades, traffic forecasting techniques have been extensively investigated by using integrated models [1], Kalman filters [2], support vector machines [3], [4], and shallow artificial neural networks [5], [6]. Recently, data-driven deep neural networks [7], [8], [9] have received increased attention in traffic forecasting because of their impressive accuracy. Examples include deep neural recurrent neural networks (RNNs) and long short-term memory networks [10], stacked autoencoder [11], convolutional neural networks [12], [13], and CNN-LSTM [14].

Spatiotemporal graph neural networks for traffic forecasting are a class of deep neural networks that learn the spatial and temporal correlations explicitly. In these methods, the spatial dependencies are modeled by using graph neural networks, and the temporal correlations of the traffic time series are learned by using sequence learning methods. These methods have achieved high forecasting accuracy when compared with other statistical and traditional machine learning approaches. Examples include spatiotemporal graph convolutional networks [7], [15], [16], [17], temporal graph convolutional networks [18], [19], attention-based Conv-LSTM [20], diffusion convolutional recurrent neural networks (DCRNNs) [8], and attention-based graph neural networks [9], [15], [21].

Given the black-box nature of the spatiotemporal graph neural networks, uncertainty quantification (UQ) is imperative for establishing the trustworthiness of the traffic forecasts [22], [23]. Uncertainty in traffic forecasting can be grouped into two major categories: (1) data or aleatoric uncertainty and (2) model or epistemic uncertainty [24]. The aleatoric uncertainty is attributed to inherent traffic variability and/or noise in the data. For example, at a given time of day and location, traffic can be dynamic, with no discernible historical pattern. Another source of data uncertainty stems from noisy or faulty sensors that are employed to collect the data. Signal processing techniques [25] can be useful to reduce the noise or variability in the measured signals. However, the inherent randomness or variability in the system will always be present to some degree. The aleatoric uncertainty arises due to data and cannot be reduced by gathering more data or training a

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large model. On the other hand, given a model with sufficient learning capacity, the epistemic uncertainty in traffic forecasts can be attributed to the lack of training data. Specifically, regions that are underrepresented in the training dataset can cause a mismatch between model estimation and data distribution (out-of-distribution generalization [26]). Thus, the epistemic uncertainty can be reduced by collecting more data.

Estimation of both the aleatoric and epistemic uncertainty for traffic forecasting is critical for understanding the limitations of the data and model, respectively. Estimating each uncertainty separately allows one to determine whether the predicted uncertainty comes from the model or is inherent in the data, as well as how to best utilize the model in the traffic management system. For example, the different quantiles of the aleatoric uncertainty can be used to characterize the traffic dynamics in the historical data, which is independent of the epistemic uncertainty. These quantiles (such as 95% and 99%) provide predictions for the worst traffic conditions in a given location. This information can then be used to inform the rerouting algorithms to plan for worst-case rare but critical traffic scenarios. Additionally, comparing a location's unusually high aleatoric uncertainty with its neighbor's very low aleatoric uncertainty can help us identify faulty sensors. On the other hand, consider the case when there is a lack of sufficient training data both spatially (locations that do not have data because of lack of data collection infrastructure) and temporally (representative data is lacking over longer periods of time to cover seasonal effects). Here, the model's lack of confidence in the forecasting results for those spatial and temporal regions increases the model's epistemic uncertainty. In the absence of additional data for training, the epistemic uncertainty can be used to establish trustworthiness and to know when and where the model predictions are reliable. The epistemic uncertainty can allow us to devise an adaptive forecasting horizon: For spatial locations characterized by low epistemic uncertainty, the forecasting horizon can be longer than for those with high epistemic uncertainty. Furthermore, epistemic uncertainty allows us to detect significant changes in the traffic pattern and targeted data collection.

Despite its importance, UQ in spatiotemporal graph neural networks for traffic forecasting has received limited attention from the research community. Existing UQ methods can be grouped into two classes: Bayesian and frequentist [23], [27]. The majority of the UQ methods for spatiotemporal graph neural networks (GNNs) belong to the former [28], [29], which assumes that the traffic follows a Gaussian distribution. This is a strong and often incorrect assumption for traffic and presents multiple modeling challenges. The Gaussian distribution assumes that the speed or flow can take values from minus to plus infinity, but these metrics are physically bounded within an interval (speed and flow will never be larger or smaller than certain values). The Gaussian distribution also assumes that the traffic metric is symmetric and unimodal with maximum density at the mean. This is also not the case since we can observe several modes in practice: peak and normal traffic patterns within a day; weekday and weekend patterns within a week; winter and non-winter traffic patterns within a given year. Consequently, the resulting distribution is often skewed, asymmetric, and multimodal. Furthermore, Bayesian UQ methods assume homoscedastic aleatoric noise in the data [30], where the variance is constant across spatial locations and temporal steps. This is not the case in traffic because aleatoric noise can change from one location to another location and from one sensor to another sensor and can vary temporally (normal vs peak hour, weekday vs weekend). The Bayesian UQ methods provide total uncertainty [31], which is a combination of aleatoric and epistemic uncertainty; they cannot separate these two uncertainties, which are critical for traffic forecasting applications. Moreover, training Bayesian UQ methods is computationally expensive [32] because they do not provide easy parallelization mechanisms for training.

The frequentist methods for UQ comprise bootstrap, quantile regression, batch ensemble, and deep ensembles. Of all these, ensemble methods have emerged as a high-performing approach for a wide range of applications compared with other frequentist and Bayesian counterparts [26]. They are relatively easier to train and parallelize than Bayesian neural networks, and they can provide decomposed aleatoric and epistemic uncertainty. Nevertheless, ensemble-based UQ methods also assume Gaussian distribution for the outputs, and consequently they suffer from the same limitations of Bayesian UQ methods for traffic forecasting: they cannot handle asymmetric, multimodal traffic distributions and heteroskedastic aleatoric noise. Moreover, a key difficulty in ensemble-based methods is the ensemble diversity. Research has shown that diverse neural network ensembles are required for robust uncertainty estimation [33]. Also, the effectiveness of ensemble-based UQ methods has not been investigated for multioutput spatiotemporal GNNs.

To address these limitations, we have developed a general scalable deep-ensemble-based UQ approach for a class of spatiotemporal GNNs originally developed for short-term traffic forecasting. Our approach comprises four steps. The first step is the adaptation of simultaneous quantile regression loss within the spatiotemporal GNN to model the quantiles of the forecasting traffic distribution. This addresses the key issue of handling asymmetric, multimodal traffic distributions and heteroskedastic aleatoric noise in traffic applications. The second step is the use of a scalable hyperparameter search for tuning the hyperparameters of the spatiotemporal GNN. The third step is generative modeling, where we model the joint probability distribution of the high-performing hyperparameter configurations from the previous step and use the generative model for generating more high-performing hyperparameter configurations to train an ensemble of GNN models. The fourth step is use of the ensemble of GNN models to forecast the traffic metrics and estimate the associated aleatoric and epistemic uncertainty using a new uncertainty decomposition for Gaussian-assumption-free simultaneous quantile regression loss. Using a diffusion convolution recurrent neural network as the target method, we demonstrate that the proposed UQ method is superior to previously proposed Bayesian UQ methods and other frequentist methods.

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The main contributions of the paper are as follows:

- A scalable approach to construct deep ensembles using Bayesian hyperparameter optimization and generative modeling
- A conceptually simple, yet effective, Gaussian assumption-free uncertainty quantification approach for spatiotemporal GNNs developed for traffic forecasting
- The first demonstration of aleatoric and epistemic uncertainty estimation for short-term traffic forecasting
- Demonstrated improvement over several state-of-the-art Bayesian UQ methods for spatiotemporal GNNs.

II. Related Work

Quantifying predictive uncertainty is important for deep learning models. Various methods for quantifying uncertainty have been investigated over the past decade (see, e.g., [27] and [34]. In this paper we look at the UQ techniques that are used specifically in traffic forecasting.

Despite their importance, UQ methods have not yet been extensively adopted or studied for traffic forecasting. Only a few attempts have been made to measure uncertainty in highway traffic forecasts. The generalized autoregressive conditional heteroscedasticity [35], [36], [37] model was used to estimate the uncertainty of traffic forecasting with a Kalman filter. A bootstrapping technique was used by Matas et al. [38] for uncertainty estimation. Quantile regression recently has been developed to forecast traffic volumes with confidence intervals [39]. Bayesian neural networks were used [40], [41], [42] to predict the travel time along with the confidence intervals. All of these methods, however, perform forecasting on individual locations. None of the methods consider the spatiotemporal dynamics of all the locations together in their models.

Recently, Wu et al. [23] developed and tested six uncertainty estimation techniques for diffusion convolution recurrent neural networks: five frequentist methods—bootstrap, mean interval score, quantile regression, spline quantile regression, and Monte Carlo dropout—and one Bayesian model trained through stochastic gradient Markov chain Monte Carlo (SG-MCMC). They compared these strategies empirically and reported on the effectiveness and computational trade-offs of various UQ methods. According to their experimental results, Bayesian DCRNN with SG-MCMC results in superior mean prediction when compared with the five frequentist methods, but it is computationally expensive. Frequentist approaches, on the other hand, are inexpensive and effective in covering data variations. The uncertainty calculated in Bayesian DCRNN is the total uncertainty, a combination of both epistemic and aleatoric; it cannot distinguish between the two types of uncertainty. Moreover, Bayesian neural networks are computationally more expensive to train than their deterministic counterparts and typically do not scale well for large datasets because of the sequential nature of the training procedures.

Compared with the state-of-the-art SG-MCMC UQ method, we show that our proposed method is robust in mean predictions and is effective in covering variations in the data. Moreover, the hyperparameter tuning and ensemble training can be scaled over multiple compute nodes. Overall, our method is more effective and more efficient than other Bayesian and frequentist approaches in terms of accuracy and scalability. Our approach was inspired by four recent works on uncertainty estimation through deep ensembles: deep ensemble [43], hyperensemble [44], neural ensemble search [45], and joint neural and hyperparameter search ensembles [33]. These methods assume a Gaussian distribution for the output variable, perform regression with a single output variable, do not make use of scale for hyperparameter search to generate ensembles, and do not adopt generative modeling. The uniqueness of our method stems from (1) its application to spatiotemporal GNNs, (2) inclusion of Gaussian-assumption-free simultaneous quantile regression loss, (3) multioutput spatiotemporal regression, (4) use of hyperparameter search at scale, and (5) Gaussian copula generative modeling for ensemble construction.

III. Methodology

Spatiotemporal GNNs consider a given traffic network as a weighted directed graph $G = (V, E, A)$, where the set V of nodes represents N traffic measurement (sensor) locations, the set E of edges represents the connectivity between the nodes, and the adjacency matrix A ($A \in R^{N\times N}$) represents the proximity between the nodes. Given the graph G and historical time series data of traffic metrics X, spatiotemporal GNNs seek to learn a function f(.) to forecast short-term traffic metrics for K time steps given the past K' traffic metrics as input $X(k-K'+1), \ldots, X(k); G \xrightarrow{\text{f(.)}} X(k+1), \ldots, X(k+K)$, where $X(k) \in \mathcal{R}^{N\times 1}$ is a vector of traffic metrics defined over N vertices at time step k.

We focus on uncertainty quantification in spatiotemporal GNNs. Our proposed method, deep ensemble with simultaneous quantile regression (DESQRUQ), seeks to estimate both aleatoric (data) and epistemic (model) uncertainties. Our methodology comprises four steps (see Figure 1 for a high-level overview): (1) using a Gaussian-assumption-free simultaneous quantile regression loss function for training spatiotemporal GNN to model the traffic distribution; (2) given the learning model and hyperparameter search space, applying a scalable Bayesian optimization method to tune the hyperparameters of the spatiotemporal graph neural network; (3) fitting a Gaussian copula generative model to capture the joint distributions of the high-performing hyperparameter configurations from the previous step and generate more high-performing hyperparameter configurations; and (4) training an ensemble of models by sampling a new set of hyperparameter configurations from the generative model and decomposing the aleatoric and epistemic uncertainties from the spatiotemporal GNN ensembles. We explain them in the following subsections.

A. Simultaneous Quantile Regression (SQR)

The first step of our approach is the adaptation of SQR in spatiotemporal GNNs to predict the quantiles of the forecasting traffic distribution.

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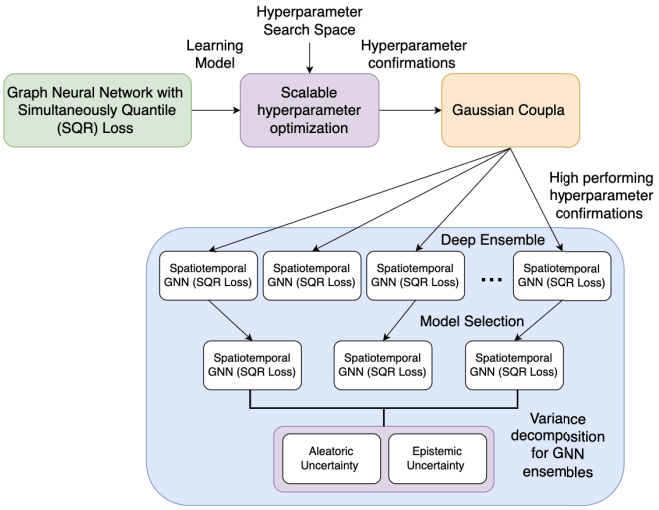

Fig. 1. Components of our proposed approach DESQRUQ: (1) simultaneous quantile regression (SQR) loss for spatiotemporal GNN, (2) scalable hyperparameter optimization, (3) high-performing hyperparameter generation using a Gaussian copula, and (4) uncertainty estimation using variance decomposition for the deep ensemble.

Consider a typical regression setting consisting of input features X and output variable Y taking real values y. Let $F(y) = P(Y \leq y)$ be the cumulative distribution function of Y; the qth quantile of Y can be calculated by the quantile distribution function $F^{-1}(q) = inf(y : F(y) \geq q)$, where $q \in (0, 1)$. The quantile regression approach seeks to model $\hat{y} = \hat{f}_q(x)$ that approximates the conditional quantile distribution function $y = F^{-1}(q \mid X = x)$. Specifically, the quantile regression estimates the value of Y at a given quantile q conditioned on the given x values of input features X. This estimation can be achieved by using the pinball loss function [46], [47]:

$$\ell_q(y, \hat{y}) = \begin{cases} q(y - \hat{y}) & \text{if } y - \hat{y} \geq 0 \\ (1 - q)(\hat{y} - y) & \text{else} \end{cases}$$

Building on this, the recently proposed SQR approach estimates all the quantiles simultaneously by solving the following optimization problem:

$$\hat{f} \in \arg\min \frac{1}{n} \sum_{i=1}^{n} \mathop{\mathbb{E}}_{q \sim U[0,1]} \left[\ell_q \left(f\left(x_i, q\right), y_i \right) \right].$$

This optimization problem is solved by using minibatch stochastic gradient descent, where random quantile levels $q \sim U[0, 1]$ (uniform distribution) are sampled afresh for each training point and for each minibatch.

In spatiotemporal GNNs, for each sensor location in the graph G and for each forecast time step, we seek to estimate the traffic metric at all quantiles conditioned on the past K' traffic metrics defined over N locations. This is achieved by replacing the MAE loss minimization with the SQR minimization. Applying the above equation on the graph, we get

$$\hat{f} \in \operatorname*{argmin}_{f} \frac{1}{N} \sum_{j=1}^{N} \frac{1}{n} \sum_{i=1}^{n} \mathop{\mathbb{E}}_{q \sim U[0,1]} [\ell_q(f(x_{ij}, q, G), y_{ij})]. \quad (1)$$

The key advantages of SQR include well-calibrated prediction intervals; joint estimation of the quantiles that can eliminate the unacceptable crossing quantiles [48]; and the ability to model non-Gaussian, skewed, asymmetric, multimodal, and heteroskedastic noise in the data. Note that SQR to estimate the quantiles of the distribution is generic for a wide range of sptaiotemporal GNNs; SQR loss does not depend on the specific model architecture. For example, in spatiotemporal graph convolutional networks [7], [15], [16], [17], temporal graph convolutional networks [18], [19], attention-based Conv-LSTM [20], dynamic graph neural networks [49], [50], and attention-based GNNs [9], [15], [21], one can simply replace the loss function from mean square error or mean absolute error to SQR loss given in Eq. 1.

B. Scalable Hyperparameter Optimization

The second step of our approach is the use of a scalable hyperparameter search for tuning the hyperparameters of the spatiotemporal GNNs.

Typically spatiotemporal GNNs have several hyperparameters that affect their forecasting accuracy values. They can be grouped into training hyperparameters, such as learning rate, learning rate decay, batch size, type of optimizer, and gradient norm clipping, and neural network architecture hyperparameters, such as the number of layers, units per layer, types of activation function, and regularization parameters per layer. All these hyperparameters often play a crucial role in achieving high forecasting accuracy. To tune the hyperparameters of a given spatiotemporal GNN model with SQR, we use a scalable asynchronous Bayesian optimization (BO) method [51]. BO optimizes an unknown function by iteratively evaluating it at different points, using a probabilistic surrogate model to guide the search toward promising regions of the search space.

For hyperparameter optimization, given the training and validation data, a neural network model, and the feasible set of values of the training hyperparameters, the asynchronous BO method seeks to minimize the validation error by repeatedly training the neural network model with various hyperparameter values. The asynchronous BO method uses an incrementally updated cheap surrogate model to learn the relationship between the hyperparameter values and their corresponding validation errors. The BO method uses the random forest model to predict the validation errors of the evaluated hyperparameter configurations along with the uncertainty associated with the predicted value. The BO search involves both evaluation of hyperparameter configurations, where the random forest model is most uncertain, and exploitation, focusing on hyperparameter configurations that are closer to the previously found high-performing configurations. The balance between exploration and exploitation is achieved by selecting an unevaluated hyperparameter configuration $\boldsymbol{x}^*$ that minimizes the lower confidence bound (LCB) acquisition function: $\boldsymbol{x}^* = \arg\min_{\boldsymbol{x} \in \mathcal{D}} \underbrace{\hat{\mu}(\boldsymbol{x}) - \lambda \times \hat{\sigma}(\boldsymbol{x})}_{\text{LCB}}$, where $\hat{\mu}(\boldsymbol{x})$ and $\hat{\sigma}(\boldsymbol{x})$ are respectively the predicted estimate and the standard deviation of that estimate from the random forest model of the given unevaluated hyperparameter configuration $\boldsymbol{x}$ sampled from

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the hyperparameter space $\mathcal{D}$ and where λ is a user-defined parameter that controls the trade-off between the exploration and exploitation.

A key reason for choosing the asynchronous BO method from [51] for the hyperparameter tuning is scalability. The asynchronous BO follows a single-manager multiple-workers parallelization scheme, where the manager runs the search, generates hyperparameter configurations, and sends them to workers for evaluation; each worker is responsible for evaluating a hyperparameter configuration by training the neural network model with the given configuration, computing the validation error, and returning that value to the manager. The manager generates hyperparameter configurations for evaluation in an asynchronous manner: as soon as a worker returns the validation error, the manager uses the returned value to retrain the surrogate model within the BO and generates the next hyperparameter configuration for the worker. To enable simultaneous multiworker evaluations, the BO uses a constant liar scheme, where configurations are generated sequentially by refitting the random forest with a lie for the validation error, given by a predicted value from the model itself, for the previously selected unevaluated hyperparameter configuration. For more details, we refer the reader to [51] and [52].

C. Generative Modeling and Ensemble Training

The third step of our approach is generative modeling. We model the joint probability distribution of the high-performing hyperparameter configurations from the previous step and use the distribution for generating more high-performing hyperparameter configurations to train an ensemble of DESQRUQ.

Copula is a statistical approach used to model the joint probability distribution of a given set of random variables by analyzing the dependencies between their marginal distributions. We use a multivariate Gaussian copula (GC) model [53] to learn the joint probability distribution between high-performing hyperparameter configurations. The copula model C takes multiple marginal distributions of the hyperparameters $u_1, u_2, \ldots, u_{hp}$ and returns a multivariate distribution as follows:

$$C(u_i, u_2, \ldots, u_{hp}) = \Phi_{\sum}(\Phi^{-1}(u_1), \Phi^{-1}(u_2), \ldots, \Phi^{-1}(u_{hp})), \tag{2}$$

where $\Phi_{\sum}$ represents the cumulative distribution function (CDF) of a multivariate normal, with covariance $\sum$ and mean 0, and Φ^{-1} is the inverse CDF for the standard normal. This captures the joint probability distribution of the high-performing hyperparameter values such as learning rate, batch size, number of layers, and objective function.

The rationale for generative modeling through a multivariate Gaussian copula is twofold. First, hyperparameter search problems characterized by integer, real, and categorical hyperparameters have multiple high-performing regions; while BO cannot sample these regions uniformly, it can sample a few configurations from different high-performing regions. By selecting these high-performing configurations and fitting a Gaussian copula, we can sample from the sparsely sampled high-performing regions as well. Second, if we have to find more high-performing configurations from different regions directly with BO, then we have to run the search multiple times with different initialization; this process will be both computationally expensive and resource intensive. Instead, once the copula model C is trained, we sample a number of hyperparameter configurations from C and run multiple GNN training sessions with the sampled hyperparameter configurations. From these trained GNN models, we select a set of high-performing models for ensemble construction.

D. Variance Decomposition for GNN Ensembles

The fourth step of our approach is to use the ensemble of trained models to estimate the traffic metrics: the aleatoric and epistemic uncertainty using variance decomposition.

Let $\mathcal{E} = \theta_i, i = 1, 2, \ldots, M$ be the ensemble of M models. Previous studies have showed that if each model is configured to predict the mean and variance of an output distribution that is assumed to be Gaussian and if the model is trained with negative log-likelihood loss [33], the empirical estimation of the mean and the total variance can be written as follows.

$$\begin{aligned} \mu_{\mathcal{E}} &= \frac{1}{M}\sum_{\theta\in\mathcal{E}}\mu_\theta \\ \sigma^2_{\mathcal{E}} &= \underbrace{\frac{1}{M}\sum_{\theta\in\mathcal{E}}\sigma^2_\theta}_{\text{Aleatoric Uncertainty}} + \underbrace{\frac{1}{M-1}\sum_{\theta\in\mathcal{E}}(\mu_\theta - \mu_{\mathcal{E}})^2}_{\text{Epistemic Uncertainty}} \end{aligned} \tag{3}$$

The total uncertainty quantified by $\sigma^2_{\mathcal{E}}$ is a combination of aleatoric and epistemic uncertainty, which is given by the mean of the predictive variance of each model in the ensemble and the predictive variance of the mean of each model in the ensemble.

Our proposed SQR loss seeks to predict quantiles of the distribution as opposed to the mean and variance of the distribution. Thus it is more generic and free from Gaussian assumptions. We derive the empirical estimate of the median ($q_{50\%}$) and the total variance of the ensemble as

$$\begin{aligned} q_{50\%\mathcal{E}} &= \frac{1}{M}\sum_{\theta\in\mathcal{E}} q_{50\%\theta} \\ \sigma^2_{\mathcal{E}} &= \underbrace{\frac{1}{M}\sum_{\theta\in\mathcal{E}}\left(\frac{q_{66\%\theta} - q_{33\%\theta}}{2}\right)^2}_{\text{Aleatoric Uncertainty}} \\ &\quad + \underbrace{\frac{1}{M-1}\sum_{\theta\in\mathcal{E}}(q_{50\%\theta} - q_{50\%\mathcal{E}})^2}_{\text{Epistemic Uncertainty}}, \end{aligned} \tag{4}$$

where $q_{33\%}$ and $q_{66\%}$ are the 33% and 66% quantiles of the output distribution, respectively. Note that Eq. 4 becomes Eq. 3 when the output distribution is Gaussian, where the mean and median will be the same ($q_{50\%\theta} = \mu_\theta$) and $(q_{66\%\theta} - q_{33\%\theta})/2$ will be equal to σ_θ. Equation 3 shows the aleatoric uncertainty for Gaussian distribution as σ^2_θ, where σ_θ corresponds to one standard deviation from the mean. In the symmetric Gaussian distribution, this corresponds to a 33%

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or 66% quantile, where the mean and median are the same. Therefore, the interval from 33% to 66% is $2 \times \sigma_\theta$. If we move to a non-Gaussian setting, this 33% to 66% quantile interval divided by 2 will correspond to the standard deviation equivalent in Gaussian.

IV. Experimental Results

We conducted experiments on the METR-LA [54] dataset, which contains speed profiles, given in miles per hour, of 207 sensor locations collected from the loop detector on the highways of Los Angeles County. It contains time series data from March 1, 2012, to June 30, 2012, aggregated at 5-minute intervals. The dataset includes various traffic dynamics such as rush hours, non-rush hours, incidents, weekdays, and weekend traffic. From the data, 70% is used for training, 10% for validation, and 20% for testing. This dataset is widely used for benchmarking traffic forecasting techniques [55], [56], [57], including the recent UQ estimation work in [23].

A. Diffusion Convolutional Recurrent Neural Network

We evaluate our proposed DESQRUQ method on DCRNN [8], one of the high-performing approaches to traffic forecasting, and compare it with state-of-the-art UQ techniques proposed for DCRNN. DCRNN is an encoder-decoder neural network architecture with RNN layers that operates on a graph. It models the temporal dependency of the traffic using GRUs, a variant of RNN. Specifically, the matrix multiplications of the GRU cell are replaced by a diffusion-convolutional operation to model the spatial dependency of the road network. The encoder takes the input adjacency matrix and K' time series data and maps the data into a latent representation. The decoder takes the latent representation and produces forecasts for future traffic for K timestamps. DCRNN has several training hyperparameters as follows: batch size – 64; filter type – dual-/bi-directional random walk; number of diffusion steps – 2; RNN layers – 2; RNN units per layer– 64; optimizer – Adam; threshold for gradient clipping – 5; initial learning rate – 0.01; and learning rate decay – 0.1. The adjacency matrix of the graph connectivity has been built by using the driving distance between the sensor locations. The DCRNN model uses the past 60 minutes of time series data at each node of the graph to forecast traffic for the next 60 minutes.

B. Evaluation Metrics

To evaluate the performance, we use the same metrics used in the current DCRNN UQ estimation work [23] such as mean absolute error, mean interval score (MIS), and interval score. MAE is defined as

$$\text{MAE} = \frac{1}{N}\sum_{j=1}^{N}\frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|,$$

where N is the total number of sensor locations and n is the number of timestamps for each sensor location. MIS [58] and interval scores are defined as follows:

$$\text{MIS}(q_u, q_l, q) = \frac{1}{N}\sum_{j=1}^{N}\frac{1}{n}\sum_{i=1}^{n}(q_{u_{ij}} - q_{l_{ij}})$$
$$+ \frac{2}{\rho}(y_{ij} - q_{u_{ij}})\mathbb{1}\{y_{ij} > q_{u_{ij}}\}$$
$$+ \frac{2}{\rho}(q_{l_{ij}} - y_{ij})\mathbb{1}\{y_{ij} < q_{l_{ij}}\} \quad \text{and}$$
$$\text{Interval} = \frac{1}{N}\sum_{j=1}^{N}\frac{1}{n}\sum_{i=1}^{n}(q_{u_{ij}} - q_{l_{ij}}),$$

where N is the total number of sensor locations and n is the number of timestamps for each sensor location. The estimated upper and lower quantiles q_u and q_l are the $(1 - \frac{\rho}{2})$ and $\frac{\rho}{2}$ quantiles for the confidence interval $(1 - \rho)$, and y is the observed value.

The experimental evaluation was conducted on a GPU-based cluster with 126 nodes. Each node contains two 2.4 GHz Intel Haswell E5-2620 v3 CPUs and one NVIDIA Tesla K80 dual-GPU card (two K40 GPUs), enabling two-model training per node. The code was implemented in TensorFlow 2.4 and used Python 3.7.0. We used the SDV package v0.14.0 [59] for fitting the GC models. For the hyperparameter search (HPS), we used the learning rate in the continuous range [0.0001, 0.02], learning rate decay ratio in the continuous range [0.001, 0.02], filter type in the categorical range [Laplacian, random walk, dual random walk], batch size [8, 16, ..., 256], maximum diffusion steps [1, 2, 3, 4, 5], number of layers for the encoder [1, 2, 3, 4, 5], number of layers for the decoder [1, 2, 3, 4, 5], number of RNN units per layer [4, 5,...,128], number of epoch [20,21,...,100], and threshold for gradient clipping [1,2,...,10] in the discrete range. The outcome of hyperparameter search can be influenced by the size of the search ranges and prior distribution. Smaller ranges with an informed prior distribution can significantly speed up the optimization process if they contain an optimal solution. This strategy is frequently employed as a form of transfer learning in hyperparameter optimization [60]. On the other hand, broader ranges lead to increased computational costs. However, without prior knowledge, it is better to extend the search range to prevent the exclusion of potential optimal solutions.

C. Impact of Scaling the Hyperparameter Search

Here we show that scaling the HPS results in an increased number of high-performing hyperparameter configurations, which can improve the quality of the Gaussian copula model and the overall ensemble accuracy.

For the scaling experiments, we ran DeepHyper HPS runs with 10, 25, 50, and 100 workers, each with 12 hours of wall time as the budget. We note that each compute node in the cluster can run up to 2 workers simultaneously. These runs resulted in 49, 108, 203, and 453 model evaluations, respectively. The default hyperparameter configuration achieved a validation SQR loss of 2.46. Therefore, we consider the hyperparameter configurations as high performing when they have a validation loss less than 2.46. Figure 2 shows the cumulative number of high-performing configurations obtained by DeepHyper as a function of search time. We observe that the number of high-performing configurations increases almost linearly with the increase in the number of workers. The

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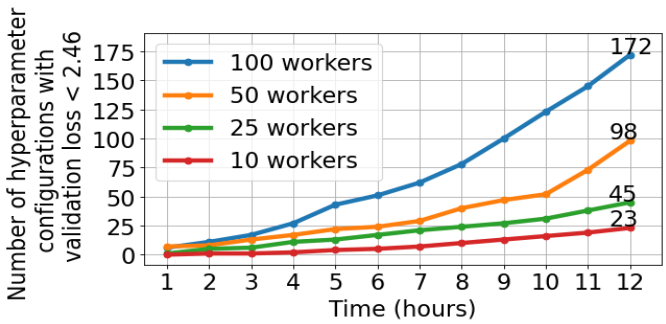

Fig. 2. Hyperparameter configurations with validation loss less than 2.46 for different numbers of workers used for HPS. The plot shows that scaling HPS is critical for finding a large number of high-performing configurations.

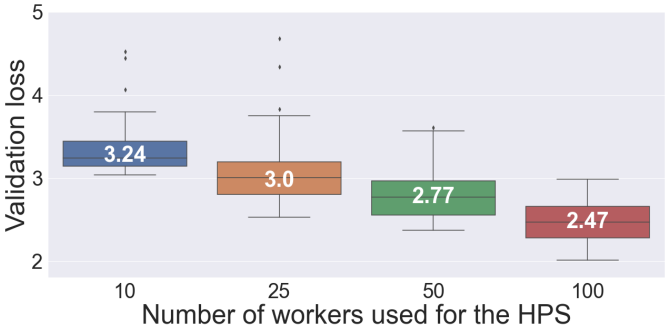

Fig. 3. Validation loss distribution of 100 models sampled from a GC trained with HPS results with different workers.

number of high-performing configurations obtained by the 10-, 25-, 50-, and 100-worker searches is 23, 45, 98, and 172, respectively.

Next, we selected the high-performing configurations from the DeepHyper HPS runs. For a given worker-count search run, we computed a 10% quantile of the validation loss values of all hyperparameter configurations and selected the configurations that were less than or equal to that cut-off value. These configurations were used to fit a Gaussian copula model. We applied this procedure for each worker-count search run and denoted the resulting four GC models as GC-10, GC-25, GC-50, and GC-100. We observe that the time required for fitting the GC model and sampling from the model is negligible. Specifically, the GC models GC-10, GC-25, GC-50, and GC-100 respectively take 2.57 sec, 2.82 sec, 2.89, and 2.96 sec to fit the configurations and 0.030 sec, 0.031 sec, 0.032 sec, and 0.038 sec to sample new hyperparameter configurations from the distribution. Once they were fitted, we sampled 100 new hyperparameter configurations from each GC model. The GC models' hyperparameter configurations were utilized to train a deep ensemble of DESQRUQ models. Ensembles of 100 models were simultaneously trained on 50 K80 nodes (two-model training per node). Figure 3 shows a box plot of the validation loss distribution of 100 trained models from GC-10, GC-25, GC-50, and GC-100. The results clearly show that the validation loss significantly decreases as we move to GC models trained with a larger number of workers. The median values of the validation loss distributions are 3.24, 3.0, 2.77, and 2.47 for GC models trained with 10-, 25-, 50-, and 100-worker HPS results, respectively. For model selection,

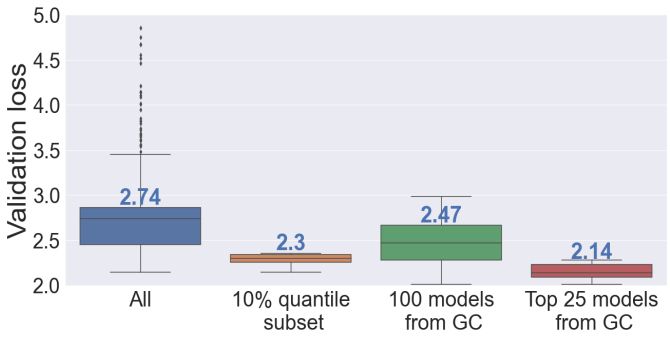

Fig. 4. Validation loss distributions of four settings: (1) all the hyperparameter configurations in the HPS run with 100 workers, (2) 10% quantile subset of the HPS run used for training the GC-100 model, (3) 100 models from GC-100, and (4) the top 25 of the 100 models sampled from GC-100.

we selected the top 25 models from the generated 100 based on the validation loss. The number 25 is motivated by the fact that the state-of-the-art SG-MCMC method used 25 posterior sample models from the trained model.

We also compared the validation loss distributions of (1) all the hyperparameter configurations in the HPS run, (2) 10% quantile subset of the HPS run used for training the GC-100 model, (3) 100 models from GC-100, and (4) the top 25 of 100 models from GC-100. The results, plotted in Figure 4, show that the validation loss distribution achieved by the top 25 models is superior to the hyperparameter values found by the search alone. Moreover, using all 100 models from the GC model results in a validation loss distribution that is better than that of the HPS run. The strategy of generating models from GC and selecting a subset from them seems a good strategy. The findings lead us to conclude that when performed on a large number of workers, HPS produces superior configurations, which can lead to higher ensemble prediction accuracy.

D. Comparison With the State-of-the-Art Methods

In this section we compare the uncertainty estimates from our proposed DESQRUQ with several other DCRNN UQ techniques that were used in a recent study [23], and we show that our method outperforms all these other methods. For fair experimental comparison, we adopt the same setup in terms of error metrics (MAE and MIS), forecasting horizons (15 minutes, 30 minutes, and 60 minutes), and confidence interval (2.5% and 97.5% quantiles) as described in [23].

The uncertainty estimation techniques reported in [23] are as follows. (1) Bootstrap [61]: This method randomly samples 50% of the training data points to train DCRNN model. The prediction is performed on the entire test dataset, and 25 samples are collected in order to calculate mean predictions and confidence intervals. (2) Quantile [62]: This method uses pinball loss for DCRNN, and the prediction is performed for three different quantiles 2.5%, 50%, and 97.5% (95% confidence interval). (3) Spline quantile regression (SQ) [63]: This method models the quantile functions using linear splines and optimizes the continuous ranked probability score as a loss function. (4) Mean interval score [23]: MIS is directly used in the loss function along with MAE, and a multiheaded model

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT UQ METHODS

METR-LA										
T	Metric	Point	Bootstrap	Quantile	SQ	MIS	MC Dropout	SG-MCMC	Ensemble	**DESQRUQ**
15 mins	MAE (mean)	2.38	2.63	2.43	2.67	2.63	2.47	2.32	2.28	**2.21**
	MIS		39.76	**18.32**	29.04	18.26	27.61	32.21	26.79	24.79
	Interval		**5.8**	12.42	8.38	12.46	8.99	8.73	10.74	8.29
30 mins	MAE (mean)	2.73	3.19	2.79	3.1	3.08	2.94	2.54	2.76	**2.44**
	MIS		38.48	21.54	40.93	**21.09**	33.38	31.87	31.28	27.4
	Interval		**7.86**	13.48	8.37	13.99	11.1	12.62	11.64	10.53
60 mins	MAE (mean)	3.14	3.99	3.19	3.75	3.65	3.7	3	3.26	**2.83**
	MIS		38.58	25.74	60.56	**24.33**	43.11	30.35	32.03	29.33
	Interval		11.65	14.5	**8.38**	15.55	14.46	18.79	13.30	12.34
PEMS-BAY										
T	Metric	Point	Bootstrap	Quantile	SQ	MIS	MC Dropout	SG-MCMC	Ensemble	**DESQRUQ**
15 mins	MAE	1.38	1.73	1.52	1.66	1.61	1.59	1.32	1.45	**1.26**
	MIS		30.61	15.02	31.17	**12.65**	29.31	23.77	22.54	20.29
	Interval		4.2	7.3	6.48	8.55	6.92	6.89	6.02	6.71
30 mins	MAE	1.74	2.36	1.85	2.05	1.97	1.87	1.65	1.72	**1.51**
	MIS		32.66	17.65	34.93	**15.58**	31.53	25.73	25.12	22.09
	Interval		4.9	8.9	7.43	10.21	7.43	7.33	6.93	7.16
60 mins	MAE	2.07	2.98	2.27	2.64	2.53	2.41	1.89	1.93	**1.76**
	MIS		33.57	21.23	37.41	**17.52**	34.67	28.12	27.34	24.11
	Interval		5.6	10.34	8.83	11.64	8.89	8.37	7.87	8.42

is used to jointly output the upper bound, lower bound, and prediction for a given input. (5) Monte Carlo Dropout [64]: This method uses the algorithm developed by Zhu et al. [42]. A 5% random drop rate is used during the testing time over 50 iterations to get a stable prediction. (6) Stochastic gradient Markov chain Monte Carlo (SG-MCMC) [65]: This is the current state-of-the-art Bayesian neural network for DCRNN and uses a subsampling technique to minimize the cost of Markov chain Monte Carlo per iteration. A Gaussian prior $\mathcal{N}(0,\ 0.4)$ with random initialization of $\mathcal{N}(0,\ 0.2)$ is used for the model parameters. The mean and confidence intervals are obtained by averaging from 25 posterior model samples. We also compare the performance of our method DeepSQRUQ with an ensemble of DCRNN models with different random initializations [66].

For our proposed DESQRUQ method we constructed a deep ensemble of 100 models trained using high-performing hyperparameter configurations generated from GC-100. The validation loss from the 100 models was used to select the top 25 models to perform a head-to-head comparison with the previous methods [23]. The comparison results of the above-mentioned methods are shown in Table I. The results indicate that our DESQRUQ method outperforms the state-of-the-art SG-MCMC method and several other UQ methods. The mean and the median MAE of our DESQRUQ show a better prediction accuracy in MAE than does SG-MCMC across all three forecasting horizons. Moreover, the MIS score and interval show that our method achieves a low MIS score and interval with a narrow interval compared with other techniques. An exception is for the MIS method that directly minimizes the MIS score in the loss function; as expected, it achieves the best MIS performance. The performance of the mean prediction using DESQRUQ has improved by 5.66% when compared with the SG-MCMC approach and by 9.87% when compared with the point estimate on the 60-minute forecasting horizon. We also compared our proposed method on the PEMS-BAY [54] dataset. The results show that our proposed DESQRUQ is more robust in prediction accuracy compared with other methods. Furthermore, we compared the computational time of the above-mentioned methods; see Table II. The wall time indicates the maximum time necessary for each method to complete, and the workers indicate the number of parallel workers employed to complete the calculation. The table indicates that, except for Bootstrap, the maximum wall time is the same for all other approaches. Moreover, Bootstrap, SG-MCMC, ensemble, and DESQRUQ required 25 workers, but all other methods required only one worker. The DESQRUQ wall time excludes HPS time since it is a one-time cost and, once it is completed, high-performing hyperparameters can be fit and sampled from GC in a small amount of time.

TABLE II
COMPARISON OF COMPUTATIONAL TIME OF DIFFERENT UNCERTAINTY QUANTIFICATION METHODS

Methods	**Wall time (Hours)**	**Workers**
Bootstrap	10	25
Quantile	14	1
SQ	14	1
MIS	14	1
MC Dropout	14	1
SG-MCMC	14	25
Ensembles	14	25
DESQRUQ	14	25

E. Uncertainty vs Traffic Dynamics

Here we estimate aleatoric and epistemic uncertainty using DESQRUQ and investigate the relationships between the traffic dynamics and the uncertainty measures. According to this analysis, fast changes in traffic behavior result in high aleatoric and epistemic uncertainty. Currently, none of the existing UQ methods developed for traffic forecasting separate aleatoric and epistemic uncertainties.

Traffic dynamics refers to the change in traffic behavior revealed by the variation in traffic speed. The traffic dynamics

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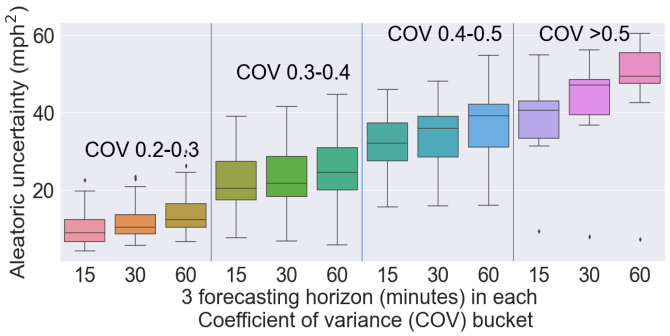


Fig. 5. COV of the sensors divided into four bins. Box plots depict the aleatoric uncertainty for each bin on the 15-, 30-, and 60-minute forecasting horizons using DESQRUQ.

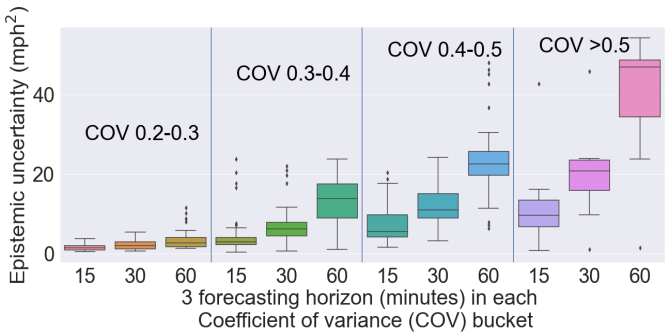
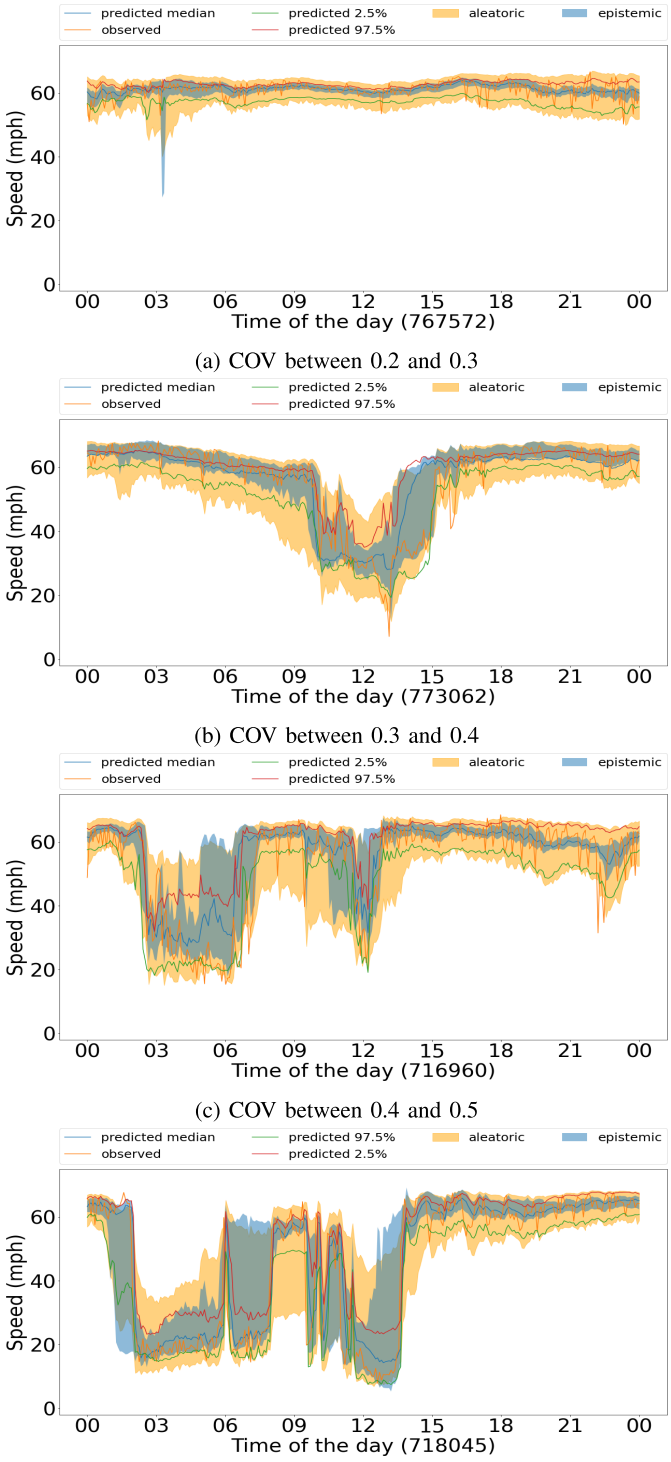


Fig. 6. COV of the sensors divided into four bins. Box plots depict the epistemic uncertainty for each bin on the 15-, 30-, and 60-minute forecasting horizons using DESQRUQ.

at a given location can be measured by the coefficient of variation (COV) [56] of the time series data (speed). The COV becomes large (small) when the variations or changes in the speed values are high (low). To measure COV, we calculated the mean (μ) and standard deviation (σ) of the speed of each location; the COV was calculated by σ/μ. Afterward, the COV values were grouped in four bins: (1) $0.2 < \text{COV} < 0.3$; (2) $0.3 < \text{COV} < 0.4$; (3) $0.4 < \text{COV} < 0.5$; and (4) $0.5 < \text{COV}$. The binning helps us identify the sensor location with low to high traffic dynamics. Of the 207 sensor locations, 34 have COV between 0.2 and 0.3, 120 have COV between 0.3 and 0.4, 42 have COV between 0.4 and 0.5, and 11 have COV more than 0.5.

We estimate the aleatoric and epistemic uncertainty using Eq. 4. The aleatoric uncertainty is the mean of the prediction variance of the 2.5% and 97.5% quantile over the ensemble of top 25 models. For each sensor location we now have the COV value and the aleatoric uncertainty measure. We separate the sensor locations based on the COV values into the four bins mentioned above and plot the distribution of aleatoric uncertainty across three forecasting horizons in a box-and-whisker plot in Figure 5. We have three forecasting horizons and four COV bins, for a total of 12 box plots per figure. The median of the distribution is shown as the horizontal line in the middle of the box. The ends of each box are 25% (bottom) and 75% (top) quantiles of the distribution. Across all the COV bins we can observe that 25%, median, and 75% quantiles of the aleatoric uncertainty increase with the increasing COV values. The box plots for all forecasting horizons show a similar trend. This result indicates that the aleatoric uncertainty



(a) COV between 0.2 and 0.3

(b) COV between 0.3 and 0.4

(c) COV between 0.4 and 0.5

(d) COV above 0.5

Fig. 7. 60-min-ahead forecasting on 4 sensors selected from 4 COV bins. The aleatoric uncertainty is shown in yellow, and the epistemic uncertainty is shown in blue.

is high in the sensor locations with high traffic dynamics or COV. In a proactive traffic management system, it can be used to estimate worst-case traffic management and planning.

Next, we estimate the epistemic uncertainty by using Eq. 4. Epistemic uncertainty measures the variance of the mean across ensemble of the top 25 models. Similar to the approach used for the aleatoric uncertainty, we separate the sensor locations based on COV values and plot the distribution of epistemic uncertainty across three forecasting horizons in

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Figure 6. Similar to the aleatoric uncertainty the epistemic uncertainty also increases for the locations having a high COV value. The epistemic uncertainty for 60-minute forecasting is high for COV $>$ 0.5 (9th box in Figure 6). The uncertainty of the model is high due to the high traffic dynamics. Hence, we recommend performing 15-minute forecasts for those sensor locations where COV $>$ 0.5 (11 sensors). The 15-minute forecast (7th box in Figure 6) has a median of the distribution of the epistemic uncertainty below 10. Similarly, uncertainty is high for 60-minute forecasting in the COV bin 0.4 to 0.5. Hence, we recommend performing 30-minute forecasting for those sensor locations (42 sensors). Limiting the forecasting horizon for highly dynamic sensors can stabilize the model prediction and minimize the risk when using the forecasting results for crucial decision-making.

Further, we plotted four sets of time series data taken from four different COV bins in Figure 7. Each plot contains the observed and predicted timeseries data and the epistemic and aleatoric uncertainty intervals estimated by our proposed DESQRUQ methods for the 60-minute forecasting horizon. The aleatoric uncertainty intervals are the medians of the 2.5% and 97.5% quantiles of the top 25 models, shown in yellow. The epistemic uncertainty interval is calculated by using the 2.5% and 97.5% quantiles of the median prediction of the top 25 models, shown in blue. Figure 7a shows that traffic behavior does not change rapidly in the location with the lowest COV value (between 0.2 and 0.3). Hence, the interval band is narrow for both of the uncertainty measures. Figure 7b illustrates that traffic dynamics are higher on the sensor location with COV values between 0.3 and 0.4. Both the aleatoric and the epistemic uncertainty measures are high around 9 am and 3 pm. The traffic dynamic increases more in Figure 7c, and the uncertainty band becomes even wider here. Epistemic uncertainty estimates are high during the drop/rise in traffic speed. In contrast, aleatoric uncertainty is high around 6 am, 9 am, 12 pm (noon), and 11 pm. Specifically, around 9 am, the aleatoric uncertainty band is high but the traffic is stable since this uncertainty is present in the data.

V. Summary and Conclusion

We developed a scalable ensemble-based uncertainty quantification approach for spatiotemporal graph neural networks. We demonstrated the effectiveness of our proposed approach on DCRNN, a state-of-the-art method developed exclusively for short-term traffic forecasting. We demonstrated that our approach outperforms the state-of-the-art Bayesian neural network method and a variety of other frequentist uncertainty techniques. We also investigated the relationship between aleatoric and epistemic uncertainty and traffic dynamics. The analysis showed that increased traffic dynamics caused by rapid changes in traffic behavior result in high aleatoric and epistemic uncertainty levels. Currently, no other UQ method can provide separate aleatoric and epistemic uncertainty estimates for traffic forecasting methods.

The key advantages of our approach are generality and scalability. Our proposed approach for building model ensembles is based on a scalable hyperparameter search and computationally inexpensive generative modeling. In principle, we can apply our UQ approach to any neural network model by using SQR loss, running HPS, training a generative model from the HPS runs, performing model selection, and decomposing the uncertainty estimates. The scalability of the approach is derived from the scalable hyperparameter search and scalable ensemble training.

Limitations of our approach stem from the ensemble modeling, which requires more compute resources for training, which involves increased complexity of handling multiple models in production. We will develop transfer learning methods to reduce the compute resources requirement and implement ensemble workflows for model management in production. Moreover, we will focus on (1) applying the proposed method for various spatiotemporal forecasting models; (2) reducing the epistemic uncertainty by penalizing the uncertainty cost in the loss function [67]; (3) comparing with other forms of Bayesian neural networks; and (4) using continual learning for spatial and temporal forecasting.

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Uncertainty Quantification over Graph with Conformalized Graph Neural Networks

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Abstract

Graph Neural Networks (GNNs) are powerful machine learning prediction models on graph-structured data. However, GNNs lack rigorous uncertainty estimates, limiting their reliable deployment in settings where the cost of errors is significant. We propose conformalized GNN (CF-GNN), extending conformal prediction (CP) to graph-based models for guaranteed uncertainty estimates. Given an entity in the graph, CF-GNN produces a prediction set/interval that provably contains the true label with pre-defined coverage probability (e.g. 90%). We establish a permutation invariance condition that enables the validity of CP on graph data and provide an exact characterization of the test-time coverage. Besides valid coverage, it is crucial to reduce the prediction set size/interval length for practical use. We observe a key connection between non-conformity scores and network structures, which motivates us to develop a topology-aware output correction model that learns to update the prediction and produces more efficient prediction sets/intervals. Extensive experiments show that CF-GNN achieves any pre-defined target marginal coverage while significantly reducing the prediction set/interval size by up to 74% over the baselines. It also empirically achieves satisfactory conditional coverage over various raw and network features.

1 Introduction

Graph Neural Networks (GNNs) have shown great potential in learning representations for graph-structured data, which has led to their widespread adoption in weather forecasting [29], drug discovery [31], and recommender systems [46], etc. As GNNs are increasingly deployed in high-stakes settings, it is important to understand the uncertainty in the predictions they produce. One prominent approach to uncertainty quantification is to construct a prediction set/interval that informs a plausible range of values the true outcome may take. A large number of methods have been proposed to achieve this goal [17, 49, 28, 44]. However, these methods often lack theoretical and empirical guarantees regarding their validity, i.e. the probability that the prediction set/interval covers the outcome [2]. This lack of rigor hinders their reliable deployment in situations where errors can be consequential.

Conformal prediction [43] (CP) is a framework for producing statistically guaranteed uncertainty estimates. Given a user-specified miscoverage level $\alpha \in (0, 1)$, it leverages a set of "calibration" data to output prediction sets/intervals for new test points that provably include the true outcome with probability at least $1 - \alpha$. Put another way, the conformal prediction sets provably only miss the test outcomes at most α fraction of the time. With its simple formulation, clear guarantee and

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distribution-free nature, it has been successfully applied to various problems in computer vision [2, 4], causal inference [30, 22, 47], time series forecasting [11, 48], and drug discovery [21].

Despite its success in numerous domains, conformal prediction has remained largely unexplored in the context of graph-structured data. One primary challenge is that it is unclear if the only, yet crucial, assumption for CP—exchangeability between the test and calibration samples—holds for graph data. When applying conformal prediction, exchangeability is usually ensured by independence among the trained model, the calibration data, and test samples (see Appendix B for more discussion). However, in the transductive setting, GNN training employs all nodes within the same graph–including test points–for message passing, creating intricate dependencies among them. Thus, to deploy conformal prediction for graph data, the first challenge is to identify situations where valid conformal prediction is possible given a fitted GNN model that already involves test information.

Efficiency is another crucial aspect of conformal prediction for practical use: a prediction set/interval with an enormous set size/interval length might not be practically desirable even though it achieves valid coverage. Therefore, the second major challenge is to develop a graph-specific approach to reduce the size of the prediction set or the length of the prediction interval (dubbed as *inefficiency* hereafter for brevity) while retaining the attractive coverage property of conformal prediction.

Present work. We propose conformalized GNN (CF-GNN),[1] extending conformal prediction to GNN for rigorous uncertainty quantification over graphs. We begin by establishing the validity of conformal prediction for graphs. We show that in the transductive setting, regardless of the dependence among calibration, test, and training nodes, standard conformal prediction [43] is valid as long as the score function (whose definition will be made clear in Section 2) is invariant to the ordering of calibration and test samples. This condition is easily satisfied by popular GNN models. Furthermore, we provide an exact characterization of the empirical test-time coverage.

Subsequently, we present a new approach that learns to optimize the inefficiencies of conformal prediction. We conduct an empirical analysis which reveals that inefficiencies are highly correlated along the network edges. Based on this observation, we add a topology-aware correction model that updates the node predictions based on their neighbors. This model is trained by minimizing a differentiable efficiency loss that simulates the CP set sizes/interval lengths. In this way, unlike the raw prediction that is often optimized for prediction accuracy, the corrected GNN prediction is optimized to yield smaller/shorter conformal prediction sets/intervals. Crucially, our approach aligns with the developed theory of graph exchangeability, ensuring valid coverage guarantees while simultaneously enhancing efficiency.

We conduct extensive experiments across 15 diverse datasets for both node classification and regression with 8 uncertainty quantification (UQ) baselines, covering a wide range of application domains. While all previous UQ methods fail to reach pre-defined target coverage, CF-GNN achieves the pre-defined empirical marginal coverage. It also significantly reduces the prediction set sizes/interval lengths by up to 74% compared with a direct application of conformal prediction to GNN. Such improvement in efficiency does not appear to sacrifice adaptivity: we show that CF-GNN achieves strong empirical conditional coverage over various network features.

2 Background and Problem Formulation

Let $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$ be a graph, where $\mathcal{V}$ is a set of nodes, $\mathcal{E}$ is a set of edges, and $\mathbf{X} = \{\mathbf{x}_v\}_{v\in\mathcal{V}}$ is the attributes, where $\mathbf{x}_v \in \mathbb{R}^d$ is a d-dimensional feature vector for node $v \in \mathcal{V}$. The label of node v is $y_v \in \mathcal{Y}$. For classification, $\mathcal{Y}$ is the discrete set of possible label classes. For regression, $\mathcal{Y} = \mathbb{R}$.

Transductive setting. We focus on transductive node classification/regression problems with random data split. In this setting, the graph G is fixed. At the beginning, we have access to $\{(\mathbf{x}_v, y_v)\}_{v\in\mathcal{D}}$ as the "training" data, as well as test data $\mathcal{D}_{\text{test}}$ with unknown labels $\{y_v\}_{v\in\mathcal{D}_{\text{test}}}$. Here $\mathcal{D}$ and $\mathcal{D}_{\text{test}}$ are disjoint subsets of $\mathcal{V}$. We work on the prevalent random split setting where nodes in $\mathcal{D}$ and $\mathcal{D}_{\text{test}}$ are randomly allocated from the entire graph, and the test sample size is $m = |\mathcal{D}_{\text{test}}|$. The training node set $\mathcal{D}$ is then randomly split into $\mathcal{D}_{\text{train}}/\mathcal{D}_{\text{valid}}/\mathcal{D}_{\text{calib}}$ of fixed sizes, the training/validation/calibration set, correspondingly. A perhaps nonstandard point here is that we withhold a subset $\mathcal{D}_{\text{calib}}$ as "calibration" data in order to apply conformal prediction later on. During the training step, the data

[1] See Figure 1 for an overview. The code is available at https://github.com/snap-stanford/conformalized-gnn.

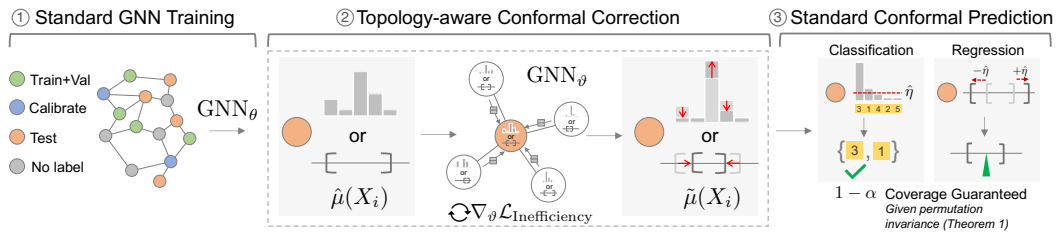

Figure 1: Conformal prediction for graph-structured data. (1) GNN training. We first use standard GNN training to obtain a base GNN model (GNN_θ) that produces prediction scores $\widehat{\mu}(X_i)$ for node i. It is fixed once trained. (2) Conformal correction. Since the training step is not aware of the conformal calibration step, the size/length of prediction sets/intervals (i.e. efficiency) are not optimized. We propose a novel correction step that learns to correct the prediction to achieve desirable properties such as efficiency. We use a topology-aware correction model GNN_ϑ that takes $\widehat{\mu}(X_i)$ as the input node feature and aggregates information from its local subgraph to produce an updated prediction $\tilde{\mu}(X_i)$. ϑ is trained by simulating the conformal prediction step and optimizing a differentiable inefficiency loss. (3) Conformal prediction. We prove that in a transductive random split setting, graph exchangeability holds (Section 3) given permutation invariance. Thus, standard CP can be used to produce a prediction set/interval based on $\tilde{\mu}$ that includes true label with pre-specified coverage rate 1-α.

$\{(\mathbf{x}_v, y_v)\}_{v\in\mathcal{D}_{\text{train}}\cup\mathcal{D}_{\text{valid}}}$, the attribute information in $\{\mathbf{x}_v\}_{v\in\mathcal{D}_{\text{calib}}\cup\mathcal{D}_{\text{test}}}$ and the entire graph structure $(\mathcal{V}, \mathcal{E})$ are available to the GNN to compute training nodes representations, while $\{y_v\}_{v\in\mathcal{D}_{\text{calib}}\cup\mathcal{D}_{\text{test}}}$ are not seen.

Graph Neural Networks (GNNs). GNNs learn compact representations that capture network structure and node features. A GNN generates outputs through a series of propagation layers [12], where propagation at layer l consists of the following three steps: (1) Neural message passing. GNN computes a message $\mathbf{m}_{uv}^{(l)} = \text{MSG}(\mathbf{h}_u^{(l-1)}, \mathbf{h}_v^{(l-1)})$ for every linked nodes u, v based on their embeddings from the previous layer $\mathbf{h}_u^{(l-1)}$ and $\mathbf{h}_v^{(l-1)}$. (2) Neighborhood aggregation. The messages between node u and its neighbors $\mathcal{N}_u$ are aggregated as $\widehat{\mathbf{m}}_u^{(l)} = \text{AGG}(\mathbf{m}_{uv}^{(l)} | v \in \mathcal{N}_u)$. (3) Update. Finally, GNN uses a non-linear function to update node embeddings as $\mathbf{h}_u^{(l)} = \text{UPD}(\widehat{\mathbf{m}}_u^{(l)}, \mathbf{h}_u^{(l-1)})$ using the aggregated message and the embedding from the previous layer. The obtained final node representation is then fed to a classifier or regressor to obtain a prediction $\widehat{\mu}(X)$.

Conformal prediction. In this work, we focus on the computationally efficient split conformal prediction method [43].[2] Given a predefined miscoverage rate $\alpha \in [0, 1]$, it proceeds in three steps: (1) non-conformity scores. CP first obtains any heuristic notion of uncertainty called non-conformity score $V\colon \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$. Intuitively, $V(x, y)$ measures how y "conforms" to the prediction at x. An example is the predicted probability of a class y in classification or the residual value $V(x, y) = |y - \widehat{\mu}(x)|$ in regression for a predictor $\widehat{\mu}\colon \mathcal{X} \to \mathcal{Y}$. (2) Quantile computation. CP then takes the $1-\alpha$ quantile of the non-conformity scores computed on the calibration set. Let $\{(X_i, Y_i)\}_{i=1}^n$ be the calibration data, where $n = |\mathcal{D}_{\text{calib}}|$, and compute $\widehat{\eta} = \text{quantile}(\{V(X_1, Y_1), \cdots, V(X_n, Y_n)\}, (1-\alpha)(1+\frac{1}{n}))$. (3) Prediction set/interval construction. Given a new test point X_{n+1}, CP constructs a prediction set/interval $C(X_{n+1}) = \{y \in \mathcal{Y} : V(X_{n+1}, y) \leq \widehat{\eta}\}$. If $\{Z_i\}_{i=1}^{n+1} := \{(X_i, Y_i)\}_{i=1}^{n+1}$ are exchangeable,[3] then $V_{n+1} := V(X_{n+1}, Y_{n+1})$ is exchangeable with $\{V_i\}_{i=1}^n$ since $\widehat{\mu}$ is given. Thus, $\widehat{C}(X_{n+1})$ contains the true label with predefined coverage rate [43]: $P\{Y_{n+1} \in C(X_{n+1})\} = \mathbb{P}\{V_{n+1} \geq \text{Quantile}(\{V_1, \ldots, V_{n+1}\}, 1-\alpha) \geq 1-\alpha$ due to exchangeability of $\{V_i\}_{i=1}^{n+1}$. This framework works for any non-conformity score. CF-GNN is similarly non-conformity score-agnostic. However, for demonstration, we focus on two popular scores, described in detail below.

Adaptive Prediction Set (APS). For the classification task, we use the non-conformity score in APS proposed by [37]. It takes the cumulative sum of ordered class probabilities till the true class. Formally, given any estimator $\widehat{\mu}_j(x)$ for the conditional probability of Y being class j at $X = x$, $j = 1, \ldots, |\mathcal{Y}|$, we denote the cumulative probability till the k-th most promising class as $V(x, k) = \sum_{j=1}^k \widehat{\mu}_{\pi_{(j)}}(x)$,

[2] See Appendix B for discussion on full and split conformal prediction. We refer to the split conformal prediction when we describe conformal prediction throughout the paper.

[3] Exchangeability definition: for any $z_1, \ldots, z_{n+1}$ and any permutation π of $\{1, \ldots, n+1\}$, it holds that $\mathbb{P}((Z_{\pi(1)}, \ldots, Z_{\pi(n+1)}) = (z_1, \ldots, z_{n+1})) = \mathbb{P}((Z_1, \ldots, Z_{n+1}) = (z_1, \ldots, z_{n+1}))$.

where π is a permutation of $\mathcal{Y}$ so that $\widehat{\mu}_{\pi(1)}(x) \geq \widehat{\mu}_{\pi(2)}(x) \geq \cdots \geq \widehat{\mu}_{\pi(|\mathcal{Y}|)}(x)$. Then, the prediction set is constructed as $C(x) = \{\pi(1), \cdots, \pi(k^*)\}$, where $k^* = \inf\{k : \sum_{j=1}^{k} \widehat{\mu}_{\pi(j)}(x) \geq \widehat{\eta}\}$.

Conformalized Quantile Regression (CQR). For the regression task, we use CQR in [36]. CQR is based on quantile regression (QR). QR obtains heuristic estimates $\widehat{\mu}_{\alpha/2}(x)$ and $\widehat{\mu}_{1-\alpha/2}(x)$ for the $\alpha/2$-th and $1-\alpha/2$-th conditional quantile functions of Y given $X = x$. The non-conformity score is $V(x, y) = \max\{\widehat{\mu}_{\alpha/2}(x) - y, y - \widehat{\mu}_{1-\alpha/2}(x)\}$, interpreted as the residual of true label projected to the closest quantile. The prediction interval is then $C(x) = [\widehat{\mu}_{\alpha/2}(x) - \widehat{\eta}, \widehat{\mu}_{1-\alpha/2}(x) + \widehat{\eta}]$.

In its vanilla form, the non-conformity score (including APS and CQR) in CP does not depend on the calibration and test data. That means, $\{\mathbf{x}_v\}_{v \in \mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}}$ are not revealed in the training process of V, which is the key to exchangeability. In contrast, GNN training typically leverages the entire graph, and hence the learned model depends on the calibration and test attributes in a complicated way. In the following, for clarity, we denote any non-conformity score built on a GNN-trained model as

$$V(x, y; \{z_v\}_{v \in \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{valid}}}, \{\mathbf{x}_v\}_{v \in \mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}}, \mathcal{V}, \mathcal{E})$$

to emphasize its dependence on the entire graph, where $z_v = (\mathbf{x}_v, Y_v)$ for $v \in \mathcal{V}$.

Evaluation metrics. The goal is to ensure valid marginal coverage while decreasing the inefficiency as much as possible. Given the test set $\mathcal{D}_{\text{test}}$, the empirical marginal coverage is defined as $\text{Coverage} := \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i \in \mathcal{D}_{\text{test}}} \mathbb{I}(Y_i \in C(X_i))$. For the regression task, inefficiency is measured as the interval length while for the classification task, the inefficiency is the size of the prediction set: $\text{Ineff} := \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{i \in \mathcal{D}_{\text{test}}} |C(X_i)|$. The larger the length/size, the more inefficient. Note that inefficiency of conformal prediction is different from accuracy of the original predictions. Our method does not change the trained prediction but modifies the prediction sets from conformal prediction.

3 Exchangeability and Validity of Conformal Prediction on Graph

To deploy CP for graph-structured data, we first study the exchangeability of node information under the transductive setting. We show that under a general permutation invariant condition (Assumption 1), exchangeability of the non-conformity scores is still valid even though GNN training uses the calibration and test information; this paves the way for applying conformal prediction to GNN models. We develop an exact characterization of the test-time coverage of conformal prediction in such settings. Proofs of these results are in Appendix A.1.

Assumption 1. *For any permutation π of $\mathcal{D}_{calib} \cup \mathcal{D}_{test}$, the non-conformity score V obeys*

$$\begin{aligned} &V(x, y; \{z_v\}_{v \in \mathcal{D}_{train} \cup \mathcal{D}_{valid}}, \{\mathbf{x}_v\}_{v \in \mathcal{D}_{calib} \cup \mathcal{D}_{test}}, \mathcal{V}, \mathcal{E}) \\ &= V(x, y; \{z_v\}_{v \in \mathcal{D}_{train} \cup \mathcal{D}_{valid}}, \{\mathbf{x}_{\pi(v)}\}_{v \in \mathcal{D}_{calib} \cup \mathcal{D}_{test}}, \mathcal{V}_\pi, \mathcal{E}_\pi), \end{aligned}$$

where $(\mathcal{V}_\pi, \mathcal{E}_\pi)$ represents a graph where $\mathcal{D}_{calib} \cup \mathcal{D}_{test}$ nodes (indices) are permuted according to π.

Assumption 1 imposes a permutation invariance condition for the GNN training, i.e., model output/non-conformity score is invariant to permuting the ordering of the calibration and test nodes (with their edges permuted accordingly) on the graph. To put it differently, different selections of calibration sets do not modify the non-conformity scores for any node in the graph. GNN models (including those evaluated in our experiments) typically obey Assumption 1, because they only use the structures and attributes in the graph without information on the ordering of the nodes [24, 16, 12].

For clarity, we write the calibration data as $\{(X_i, Y_i)\}_{i=1}^{n}$, where $X_i = X_{v_i}$, and $v_i \in \mathcal{D}_{\text{calib}}$ is the i-th node in the calibration data under some pre-defined ordering. Similarly, the test data are $\{(X_{n+j}, Y_{n+j})\}_{j=1}^{m}$, where $X_{n+j} = X_{v_j}$, and $v_j \in \mathcal{D}_{\text{test}}$ is the j-th node in the test data. We write

$$V_i = V(X_i, Y_i; \{z_i\}_{i \in \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{valid}}}, \{\mathbf{x}_v\}_{v \in \mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}}, \mathcal{V}, \mathcal{E}), \quad i = 1, \ldots, n, n+1, \ldots, n+m.$$

V_i is a random variable that depends on the training process and the split of calibration and test data. The next lemma shows that under Assumption 1, the non-conformity scores are still exchangeable.

Lemma 2. *In the transductive setting described in Section 2, conditional on the entire unordered graph $(\mathcal{V}, \mathcal{E})$, all the attribute and label information $\{(\mathbf{x}_v, y_v)\}_{v \in \mathcal{V}}$, and the index sets $\mathcal{D}_{train}$ and $\mathcal{D}_{ct} := \mathcal{D}_{calib} \cup \mathcal{D}_{test}$, the unordered set of the scores $[V_i]_{i=1}^{n+m}$ are fixed. Also, the calibration scores $\{V_i\}_{i=1}^{n}$ are a simple random sample from $\{V_i\}_{i=1}^{n+m}$. That is, for any subset $\{v_1, \ldots, v_n\} \subseteq \{V_i\}_{i=1}^{n+m}$ of size n, $P(\{V_i\}_{i=1}^{n} = \{v_1, \ldots, v_n\} \,\big|\, \{V_i\}_{i=1}^{n+m}) = 1/\binom{n}{|\mathcal{D}_{ct}|}$.*

Based on Lemma 2, we next show that any permutation-invariant non-conformity score leads to valid prediction sets, and provide an exact characterization of the distribution of test-time coverage.

Theorem 3. *Given any score V obeying Assumption 1 and any confidence level $\alpha \in (0,1)$, we define the split conformal prediction set as $\widehat{C}(x) = \{y\colon V(x,y) \le \widehat{\eta}\}$, where*

$$\widehat{\eta} = \inf\Big\{\eta\colon \tfrac{1}{n}\textstyle\sum_{i=1}^{n} \mathbb{1}\{V(X_i,Y_i) \le \eta\} \ge (1-\alpha)(1+1/n)\Big\}.$$

Then $\mathbb{P}(Y_{n+j} \in \widehat{C}(X_{n+j})) \ge 1-\alpha$, $\forall\, j = 1,\dots,m$. Moreover, define $\widehat{\text{Cover}} = \frac{1}{m}\sum_{j=1}^{m} \mathbb{1}\{Y_{n+j} \in \widehat{C}(X_{n+j})\}$. If the V_i's, $i \in \mathcal{D}_{calib} \cup \mathcal{D}_{test}$, have no ties almost surely, then for any $t \in (0,1)$,

$$\mathbb{P}\big(\widehat{\text{Cover}} \le t\big) = 1 - \Phi_{\text{HG}}\big(\lceil (n+1)(1-q)\rceil - 1; m+n, n, \lceil (1-q)(n+1)\rceil + \lceil mt\rceil\big),$$

where $\Phi_{\text{HG}}(\cdot; N, n, k)$ denotes the cumulative distribution function of a hyper-geometric distribution with parameters N, n, k (drawing k balls from an urn wherein n out of N balls are white balls).

Figure 2 plots the probability density functions (p.d.f.) of $\widehat{\text{Cover}}$ at a sequence of $t \in [0,1]$ fixing $n = 1000$ while varying m. The exact distribution described in Theorem 3 is useful in determining the size of the calibration data in order for the test-time coverage to concentrate sufficiently tightly around $1-\alpha$. More discussion and visualization of $\widehat{\text{Cover}}$ under different values of (n,m) are in Appendix A.2. Note that similar exchangeability and validity results are obtained in several concurrent works [15, 33], yet without exact characterization of the test-time coverage.

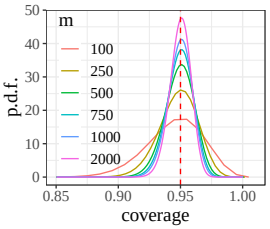

Figure 2: P.d.f. of $\widehat{\text{Cover}}$ for $n = 1000$ and $\alpha = 0.05$; curves represent different values of test sample size m.

4 CF-GNN: Conformalized Graph Neural Networks

We now propose a new method called CF-GNN to reduce inefficiency while maintaining valid coverage. The key idea of CF-GNN is to boost any given non-conformity score with graphical information. The method illustration is in Figure 1 and pseudo-code is in Appendix C.

Efficiency-aware boosting. Standard CP takes any pre-trained predictor to construct the prediction set/interval (see Section 2). A key observation is that the training stage is not aware of the post-hoc stage of conformal prediction set/interval construction. Thus, it is not optimized for efficiency. Our high-level idea is to include an additional correction step that boosts the non-conformity score, which happens after the model training and before conformal prediction. To ensure flexibility and practicality, our framework is designed as a generic wrapper that works for any pre-trained GNN model, without changing the base model training process.

Motivation: Inefficiency correlation. Our approach to boosting the scores is based on exploiting the correlation among connected nodes. Since the connected nodes usually represent entities that interact in the real world, there can be strong correlation between them. To be more specific, it is well established in network science that prediction residuals are correlated along edges [20]. Such a result implies a similar phenomenon for inefficiency: taking CQR for regression as an example, the prediction interval largely depends on the residual of the true outcome from the predicted quantiles. Hence, the prediction interval lengths are also highly correlated for connected nodes. We empirically verify this intuition in Figure 3, where we plot the difference in the prediction interval lengths for connected/unconnected node pairs in the Anaheim dataset using vanilla CQR for GCN.[4] In Figure 3, we observe that inefficiency has a topological root: connected nodes usually have similar residual scales, suggesting the existence of rich neighborhood information for the residuals. This motivates us to utilize such information to correct the scores and achieve better efficiency.

Topology-aware correction model. Based on the motivation above, we update the model predictions using the neighbor predictions. However, the relationship in neighborhood predictions could be complex (i.e. beyond homophily), making heuristic aggregation such as averaging/summing/etc. overly simplistic and not generalizable to all types of graphs. Ideally, we want to design a general and powerful mechanism that flexibly aggregates graph information. This requirement could be perfectly

[4] That is, we take the predicted quantiles from GCN and directly builds prediction intervals using CQR.

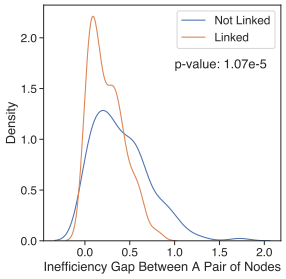

Figure 3: Inefficiency is correlated in the network. Connected nodes have significantly smaller gaps in prediction interval length compared to unconnected nodes.

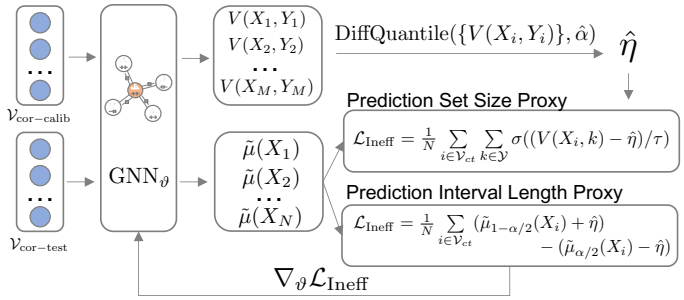

Figure 4: We simulate the downstream conformal step and optimize for inefficiency directly. We first produce differentiable quantile $\widehat{\eta}$ using $V(X_i, Y_i)$ from $\mathcal{V}_{\text{cor-cal}}$. We then construct a prediction set size/interval length proxy on $\mathcal{V}_{\text{cor-test}}$ and directly minimize inefficiency loss by updating GNN_ϑ.

fulfilled by GNN message passing as it represents a class of learnable aggregation functions over graphs. Therefore, we use a separate GNN learner parameterized by ϑ for the same network G but with modified input node features; specifically, we use the base GNN prediction ($\mathbf{X}_0 = \widehat{\mu}(X)$) as input, and output $\tilde{\mu}(X) = \text{GNN}_\vartheta(\widehat{\mu}(X), G)$. We will then use $\tilde{\mu}(X)$ as the input for constructing conformal prediction sets/intervals. Note that this second GNN model is a post-hoc process and only requires base GNN predictions, instead of access to the base GNN model.

Training with conformal-aware inefficiency loss. Given the hypothesis class, it remains to devise a concrete recipe for training the correction model GNN_ϑ parameters. Recall that as with many other prediction models, a GNN model is typically trained to optimize prediction loss (i.e.cross-entropy loss or mean squared error) but not geared towards efficiency for the post-hoc conformal prediction step. We design GNN_ϑ to be efficiency-aware by proposing a differentiable inefficiency loss that GNN_ϑ optimizes over; this allows integration of GNN message passing to exploit the neighborhood information and also ensures a good $\tilde{\mu}(\cdot)$ that leads to efficient prediction sets in downstream steps.

We first withhold a small fraction γ of the calibration dataset and use it for the correction procedure. The remaining data is used as the "usual" calibration data for building the conformal prediction set. We then further split the withheld data into a correction calibration set $\mathcal{V}_{\text{cor-cal}}$ and correction testing set $\mathcal{V}_{\text{cor-test}}$, to simulate the downstream conformal inference step. Given $\tilde{\mu}(X) = \text{GNN}_\vartheta(\widehat{\mu}(X), G)$ and a target miscoverage rate α, the framework follows three steps:

(1) Differentiable quantile: we compute a smooth quantile $\widehat{\eta}$ based on the $\mathcal{V}_{\text{cor-cal}}$ by

$$\widehat{\eta} = \text{DiffQuantile}(\{V(X_i, Y_i) | i \in \mathcal{V}_{\text{cor-cal}}\}, (1-\alpha)(1 + 1/|\mathcal{V}_{\text{cor-cal}}|).$$

Since the non-conformity score is usually differentiable, it only requires differentiable quantile calculation where there are well-established methods available [6, 5].

(2) Differentiable inefficiency proxy: we then construct a differentiable proxy $\mathbf{c}$ of the inefficiency on $\mathcal{V}_{\text{cor-test}}$ by simulating the downstream conformal prediction procedures. We propose general formulas to construct $\mathbf{c}$ that are applicable for any classification and regression tasks respectively:

a. Inefficiency loss instantiation for Classification: The desirable proxy is to simulate the prediction set size using $\mathcal{D}_{\text{cor-test}}$ as the "test" data and $\mathcal{D}_{\text{cor-calib}}$ as the "calibration" data. For class k and node i in $\mathcal{D}_{\text{cor-test}}$, the non-conformity score is $V(X_i, k)$ for class k, where $V(\cdot)$, for instance, is the APS score in Section 2. Then, we define the inefficiency proxy as

$$\mathbf{c}_{i,k} = \sigma(\frac{V(X_i, k) - \widehat{\eta}}{\tau}),$$

where $\sigma(x) = \frac{1}{1+e^{-x}}$ is the sigmoid function and τ is a temperature hyper-parameter [40]. It can be interpreted as a soft assignment of class k to the prediction set. When $\tau \to 0$, it becomes a hard assignment. The per-sample inefficiency proxy is then readily constructed as $\mathbf{c}_i = \frac{1}{|\mathcal{Y}|} \sum_{k \in \mathcal{Y}} \mathbf{c}_{i,k}$.

b. Inefficiency loss instantiation for Regression: The desirable proxy is to simulate the prediction interval length. For node i in $\mathcal{V}_{\text{cor-test}}$, the conformal prediction interval is $[\tilde{\mu}_{\alpha/2}(X_i) - \widehat{\eta}, \tilde{\mu}_{1-\alpha/2}(X_i) + \widehat{\eta}]$. Thus, the per-sample prediction interval length could be directly calculated as

$$\mathbf{c}_i = (\tilde{\mu}_{1-\alpha/2}(X_i) + \widehat{\eta}) - (\tilde{\mu}_{\alpha/2}(X_i) - \widehat{\eta}).$$

Since GNN_ϑ maps intervals to intervals and do not pose a constraint on the prediction, it may incur a trivial optimized solution where $\tilde{\mu}_{1-\alpha/2}(X) < \tilde{\mu}_{\alpha/2}(X)$. Thus, we pose an additional consistency regularization term: $(\tilde{\mu}_{1-\alpha/2}(X) - \widehat{\mu}_{1-\alpha/2}(X))^2 + (\tilde{\mu}_{\alpha/2}(X) - \widehat{\mu}_{\alpha/2}(X))^2$. This regularizes the updated intervals to not deviate significantly to reach the trivial solution.

(3) Inefficiency loss: finally, the inefficiency loss is an average of inefficiency proxies $L_{\text{ineff}} = \frac{1}{|\mathcal{V}_{\text{cor-test}}|} \sum_i \mathbf{c}_i$. The GNN_ϑ is optimized using backpropagation in an end-to-end fashion.

Conditional coverage. A natural question is whether optimizing the efficiency of conformal prediction may hurt its conditional validity.[5] In Section 5, we empirically demonstrate satisfactory conditional coverage across various graph features, which even improves upon the direct application of APS and CQR to graph data. We conjecture that it is because we correct for the correlation among nodes. However, theoretical understanding is left for future investigation.

Graph exchangeability. The post-hoc correction model is GNN-based, thus, it is permutation-invariant. Thus, it satisfies the exchangeability condition laid out in our theory in Section 3. Empirically, we demonstrate in Section 5 that CF-GNN achieves target empirical marginal coverage.

Computational cost. We remark that CF-GNN scales similarly as base GNN training since the correction step follows a standard GNN training procedure but with a modified input attribute and loss function. Notably, as the input to the correction model usually has a smaller attribute size (the number of classes for classification and 2 for regression), it has smaller parameter size than standard GNN training. In addition, it is also compatible with standard GNN mini-batching techniques.

General loss functions. Finally, we note that the choice of our loss function can be quite general. For instance, one may directly optimize for conditional validity by choosing a proper loss function.

5 Experiments

We conduct experiments to demonstrate the advantages of CF-GNN over other UQ methods in achieving empirical marginal coverage for graph data, as well as the efficiency improvement with CF-GNN. We also evaluate conditional coverage of CF-GNN and conduct systematic ablation and parameter analysis to show the robustness of CF-GNN.

Evaluation setup. For node classification, we follow a standard semi-supervised learning evaluation procedure [24], where we randomly split data into folds with 20%/10%/70% nodes as $\mathcal{D}_{\text{train}}/\mathcal{D}_{\text{valid}}/\mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}$. For the node regression task, we follow a previous evaluation procedure from [20] and randomly split the data into folds with 50%/10%/40% nodes as $\mathcal{D}_{\text{train}}/\mathcal{D}_{\text{valid}}/\mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}$. We conduct 100 random splits of calibration/testing sets to estimate the empirical coverage. Using the test-time coverage distribution in Figure 2 to ensure that coverage is concentrated tightly around 1-α, we modify the calibration set size to $\min\{1000, |\mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}}|/2\}$, and use the rest as the test sample. For a fair comparison, we first train 10 runs of the base GNN model and then fix the predictions (i.e. the input to UQ baselines and CF-GNN). In this way, we ensure that the gain is not from randomness in base model training. The hyperparameter search strategy and configurations for CF-GNN and baselines can be found in Appendix D.1.

Models & baselines to evaluate coverage. For classification, we first use general statistical calibration approaches including temperate scaling [13], vector scaling [13], ensemble temperate scaling [49]. We also use SOTA GNN-specific calibration learners including CaGCN [44] and GATS [17]. The prediction set is the set of classes from highest to lowest scores until accumulative scores exceed 1-α. For regression, we construct prediction intervals using quantile regression (QR) [25], Monte Carlo dropouts (MC dropout) [9], and Bayesian loss to model both aleatoric and epistemic uncertainty [23]. More information about baselines can be found in Appendix D.2.

Models & baselines to evaluate efficiency. As smaller coverage always leads to higher efficiency, for a fair comparison, we can only compare methods on efficiency that achieve the same coverage. Thus, we do not evaluate UQ baselines here since they do not produce exact coverage and are thus not comparable. While any CP-based methods produce exact coverage, to the best of our knowledge,

[5] Conditional coverage asks for $\mathbb{P}(Y_{n+j} \in \widehat{C}(X_{n+j}) \,|\, X_{n+j} = x) \approx 1 - \alpha$ for all $x \in \mathcal{X}$. Although exact conditional validity is statistically impossible [8], approximate conditional validity is a practically important property that APS and CQR are designed for. See Section 5 for common ways to assess conditional coverage.

Table 1: Empirical marginal coverage of node classification(upper table) and node regression tasks(lower table). The result takes the average and standard deviation across 10 GNN runs with 100 calib/test splits. ✔ means that the UQ method reaches the target coverage (i.e. coverage ≥ 0.95) while ✗ means that it fails to reach it. The last column "Covered" becomes ✔ if a UQ method reaches target coverage for all datasets and ✗ vice versus.

Task	UQ Model	Cora	DBLP	CiteSeer	PubMed	Computers	Photo	CS	Physics	Covered?
Node classif.	Temp. Scale.	$0.946_{\pm.003}$ ✗	$0.920_{\pm.009}$ ✗	$0.952_{\pm.004}$ ✔	$0.899_{\pm.002}$ ✗	$0.929_{\pm.002}$ ✗	$0.962_{\pm.002}$ ✔	$0.957_{\pm.001}$ ✔	$0.969_{\pm.000}$ ✔	✗
	Vector Scale.	$0.944_{\pm.004}$ ✗	$0.921_{\pm.009}$ ✗	$0.951_{\pm.004}$ ✔	$0.899_{\pm.003}$ ✗	$0.932_{\pm.002}$ ✗	$0.963_{\pm.002}$ ✔	$0.958_{\pm.001}$ ✔	$0.969_{\pm.000}$ ✔	✗
	Ensemble TS	$0.947_{\pm.003}$ ✗	$0.920_{\pm.008}$ ✗	$0.953_{\pm.003}$ ✔	$0.899_{\pm.002}$ ✗	$0.930_{\pm.002}$ ✗	$0.964_{\pm.002}$ ✔	$0.958_{\pm.001}$ ✔	$0.969_{\pm.000}$ ✔	✗
	CaGCN	$0.939_{\pm.005}$ ✗	$0.922_{\pm.004}$ ✗	$0.949_{\pm.005}$ ✗	$0.898_{\pm.003}$ ✗	$0.926_{\pm.003}$ ✗	$0.956_{\pm.002}$ ✔	$0.954_{\pm.003}$ ✔	$0.968_{\pm.001}$ ✔	✗
	GATS	$0.939_{\pm.005}$ ✗	$0.921_{\pm.004}$ ✗	$0.951_{\pm.005}$ ✔	$0.898_{\pm.002}$ ✗	$0.925_{\pm.002}$ ✗	$0.957_{\pm.002}$ ✔	$0.957_{\pm.001}$ ✔	$0.968_{\pm.000}$ ✔	✗
	CF-GNN	$0.952_{\pm.001}$ ✔	$0.952_{\pm.001}$ ✔	$0.953_{\pm.001}$ ✔	$0.953_{\pm.001}$ ✔	$0.952_{\pm.001}$ ✔	$0.953_{\pm.001}$ ✔	$0.952_{\pm.001}$ ✔	$0.952_{\pm.001}$ ✔	✔

Task	UQ Model	Anaheim	Chicago	Education	Election	Income	Unemploy.	Twitch	Covered?
Node regress.	QR	$0.943_{\pm.031}$ ✗	$0.950_{\pm.007}$ ✗	$0.959_{\pm.001}$ ✔	$0.956_{\pm.004}$ ✔	$0.960_{\pm.005}$ ✔	$0.954_{\pm.004}$ ✔	$0.900_{\pm.015}$ ✗	✗
	MC dropout	$0.553_{\pm.022}$ ✗	$0.427_{\pm.015}$ ✗	$0.423_{\pm.013}$ ✗	$0.417_{\pm.008}$ ✗	$0.532_{\pm.022}$ ✗	$0.489_{\pm.016}$ ✗	$0.448_{\pm.017}$ ✗	✗
	BayesianNN	$0.967_{\pm.001}$ ✔	$0.955_{\pm.003}$ ✔	$0.957_{\pm.002}$ ✔	$0.958_{\pm.009}$ ✔	$0.970_{\pm.004}$ ✔	$0.960_{\pm.001}$ ✔	$0.923_{\pm.006}$ ✗	✗
	CF-GNN	$0.957_{\pm.003}$ ✔	$0.954_{\pm.002}$ ✔	$0.951_{\pm.001}$ ✔	$0.950_{\pm.001}$ ✔	$0.951_{\pm.001}$ ✔	$0.951_{\pm.001}$ ✔	$0.954_{\pm.001}$ ✔	✔

there are no existing graph-based conformal prediction methods for transductive settings. Thus, we can only compare with the direct application of conformal prediction (CP) to base GNN. In the main text, we only show results for GCN [24] as the base model; results of three additional popular GNN models (GraphSAGE [16], GAT [42], and SGC [45]) are deferred to Appendix D.4.

Datasets. We evaluate CF-GNN on 8 node classification datasets and 7 node regression datasets with diverse network types such as geographical network, transportation network, social network, citation network, and financial network. Dataset statistics are in Appendix D.3.

5.1 Results

CF-GNN achieves empirical marginal coverage while existing UQ methods do not. We report marginal coverage of various UQ methods with target coverage at 95% (Table 1). There are three key takeaways. Firstly, none of these UQ methods achieves the target coverage for all datasets while CF-GNN does, highlighting the lack of statistical rigor in those methods and the necessity for a guaranteed UQ method. Secondly, it validates our theory from Section 3 that CF-GNN achieves designated coverage in transductive GNN predictions. Lastly, CF-GNN achieves empirical coverage that is close to the target coverage while baseline UQ methods are not. This controllable feature of CF-GNN is practically useful for practitioners that aim for a specified coverage in settings such as planning and selection.

CF-GNN significantly reduces inefficiency. We report empirical inefficiency for 8 classification and 7 regression datasets (Table 2). We observe that we achieve consistent improvement across datasets with up to 74% reduction in the prediction set size/interval length. We additionally conduct the same experiments for 3 other GNN models including GAT, GraphSAGE, and SGC in Appendix D.4 and we observe that performance gain is generalizable to diverse architecture choices. Furthermore, CF-GNN yields more efficient prediction sets than existing UQ methods even if we manually adjust the nominal level of them to achieve 95% empirical coverage (it is however impossible to do so in practice, here we do this for evaluation). For instance, the best calibration method GATS yields an average prediction size of 1.82 on Cora when the nominal level is tuned to achieve 95% empirical coverage. In contrast, CF-GNN has an average size of 1.76, smaller than GATS. In Appendix D.5, we also observe that CF-GNN also reduces inefficiency for advanced conformal predictor RAPS for classification task. In addition, we find that CF-GNN yields little changes to the prediction accuracy of the original GNN model (Appendix D.7).

CF-GNN empirically maintains conditional coverage. While CF-GNN achieves marginal coverage, it is highly desirable to have a method that achieves reasonable conditional coverage, which was the motivation of APS and CQR. We follow [37] to evaluate conditional coverage via the Worst-Slice (WS) coverage, which takes the worst coverage across slices in the feature space (i.e. node input features). We observe that CF-GNN achieves a WS coverage close to $1 - \alpha$, indicating satisfactory conditional coverage (Cond. Cov. (Input Feat.) row in Table 3). Besides the raw features, for each node, we also construct several network features (which are label agnostic) including clustering

Table 2: Empirical inefficiency measured by the size/length of the prediction set/interval for node classification (left table)/regression(right table). A smaller number has better efficiency. We show the relative improvement (%) of CF-GNN over CP on top of the $\rightarrow$. The result uses APS for classification and CQR for regression with GCN as the base model. Additional results on other GNN models are at Appendix D.4. We report the average and standard deviation of prediction sizes/lengths calculated from 10 GNN runs, each with 100 calibration/test splits.

Task	Dataset	CP $\longrightarrow$ CF-GNN
Node classif.	Cora	$3.80_{\pm.28} \xrightarrow{-53.61\%} 1.76_{\pm.27}$
	DBLP	$2.43_{\pm.03} \xrightarrow{-49.13\%} 1.23_{\pm.01}$
	CiteSeer	$3.86_{\pm.11} \xrightarrow{-74.27\%} 0.99_{\pm.02}$
	PubMed	$1.60_{\pm.02} \xrightarrow{-19.05\%} 1.29_{\pm.03}$
	Computers	$3.56_{\pm.13} \xrightarrow{-49.05\%} 1.81_{\pm.12}$
	Photo	$3.79_{\pm.13} \xrightarrow{-56.28\%} 1.66_{\pm.21}$
	CS	$7.79_{\pm.29} \xrightarrow{-62.16\%} 2.95_{\pm.49}$
	Physics	$3.11_{\pm.07} \xrightarrow{-62.81\%} 1.16_{\pm.13}$
Average Improvement		-53.75%

Task	Dataset	CP $\longrightarrow$ CF-GNN
Node regress.	Anaheim	$2.89_{\pm.39} \xrightarrow{-25.00\%} 2.17_{\pm.11}$
	Chicago	$2.05_{\pm.07} \xrightarrow{-0.48\%} 2.04_{\pm.17}$
	Education	$2.56_{\pm.02} \xrightarrow{-5.07\%} 2.43_{\pm.05}$
	Election	$0.90_{\pm.01} \xrightarrow{+0.21\%} 0.90_{\pm.02}$
	Income	$2.51_{\pm.12} \xrightarrow{-4.58\%} 2.40_{\pm.05}$
	Unemploy	$2.72_{\pm.03} \xrightarrow{-10.83\%} 2.43_{\pm.04}$
	Twitch	$2.43_{\pm.10} \xrightarrow{-1.36\%} 2.39_{\pm.07}$
Average Improvement		-6.73%

Table 3: CF-GNN achieves conditional coverage, measured by Worse-Slice Coverage [37]. We use Cora/Twitch as an example classification/regression dataset. Results on other network features and results on target coverage of 0.9 can be found in Appendix D.6.

Target: 0.95	Classification		Regression	
Model	CP	CF-GNN	CP	CF-GNN
Marginal Cov.	$0.95_{\pm.01}$	$0.95_{\pm.01}$	$0.96_{\pm.02}$	$0.96_{\pm.02}$
Cond. Cov. (Input Feat.)	$0.94_{\pm.02}$	$0.94_{\pm.03}$	$0.95_{\pm.04}$	$0.94_{\pm.05}$
Cond. Cov. (Cluster)	$0.89_{\pm.06}$	$0.93_{\pm.04}$	$0.96_{\pm.03}$	$0.96_{\pm.03}$
Cond. Cov. (Between)	$0.81_{\pm.06}$	$0.95_{\pm.03}$	$0.94_{\pm.05}$	$0.94_{\pm.05}$
Cond. Cov. (PageRank)	$0.78_{\pm.06}$	$0.94_{\pm.03}$	$0.94_{\pm.05}$	$0.94_{\pm.05}$

Table 4: Ablation. For Size/length, we use Cora/Anaheim dataset with GCN backbone. Each experiment is with 10 independent base model runs with 100 conformal split runs.

Topology -aware	Ineff. Loss	Size	Length
✔	✔	$1.76_{\pm.27}$	$2.17_{\pm.11}$
✔	✗	$2.42_{\pm.35}$	$2.23_{\pm.10}$
✗	✔	$2.35_{\pm.47}$	$2.32_{\pm.18}$
✗	✗	$3.80_{\pm.28}$	$2.89_{\pm.39}$

coefficients, betweenness, PageRank, closeness, load, and harmonic centrality, and then calculate the WS coverage over the network feature space. We observe close to 95% WS coverage for various network features, suggesting CF-GNN also achieves robust conditional coverage over network properties. We also see that the direct application of CP (i.e. without graph correction) has much smaller WS coverage for classification, suggesting that adjusting for neigborhood information in CF-GNN implicitly improves conditional coverage.

Ablation. We conduct ablations in Table 4 to test two main components in CF-GNN, topology-aware correction model, and inefficiency loss. We first remove the inefficiency loss and replace it with standard prediction loss. The performance drops as expected, showing the power of directly modeling inefficiency loss in the correction step. Secondly, we replace the GNN correction model with an MLP correction model. The performance drops significantly, showing the importance of the design choice of correction model and justifying our motivation on inefficiency correlation over networks.

Parameter analysis. We conduct additional parameter analysis to test the robustness of CF-GNN. We first adjust the target coverage rate and calculate the inefficiency (Figure 5(1)). CF-GNN consistently beats the vanilla CP across all target coverages. Moreover, we adjust the fraction γ of the holdout calibration data in building the inefficiency loss, and observe that CF-GNN achieves consistent improvement in inefficiency (Figure 5(2)). We also observe a small fraction (10%) leads to excellent performance, showing that our model only requires a small amount of data for the inefficiency loss and leaves the majority of the calibration data for downstream conformal prediction.

6 Related Works

We discuss here related works that are closest to the ideas in CF-GNN and provide extended discussion on other related works in Appendix E.

(1) Uncertainty quantification (UQ) for GNN: Many UQ methods have been proposed to construct model-agnostic uncertain estimates for both classification [13, 49, 14, 27, 1] and regression [25, 41,

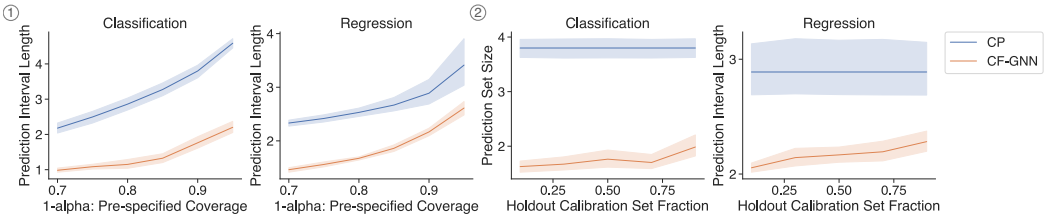

Figure 5: (1) Parameter analysis on inefficiency given different target coverage rate 1-α. (2) Parameter analysis on inefficiency given calibration set holdout fraction. Analyses use Cora/Anaheim for classification/regression.

38, 9, 28, 26, 35, 23, 19]. Recently, specialized calibration methods for GNNs that leverage network principles such as homophily have been developed [44, 17]. However, these UQ methods can fail to provide a statistically rigorous and empirically valid coverage guarantee (see Table 1). In contrast, CF-GNN achieves valid marginal coverage in both theory and practice.

(2) Conformal prediction for GNN: The application of CP to graph-structured data remains largely unexplored. [7] claims that nodes in the graph are not exchangeable in the inductive setting and employs the framework of [3] to construct prediction sets using neighborhood nodes as the calibration data for mitigating the miscoverage due to non-exchangeability. In contrast, we study the transductive setting where certain exchangeability property holds; thus, the method from [3] are not comparable to ours. Concurrent with our work, [15] studies the exchangeability under transductive setting and proposes a diffusion-based method for improving efficiency, which can be viewed as an instantiation of our approach where the GNN correction learns an identity mapping; [33] studies exchangeability in network regression for of non-conformity scores based on various network structures, with similar observations as our Theorem 3. Other recent efforts in conformal prediction for graphs include [32, 34] which focus on distinct problem settings.

(3) Efficiency of conformal prediction: How to achieve desirable properties beyond validity is an active topic in the CP community; we focus on the efficiency aspect here. One line of work designs "good" non-conformity scores in theory such as APS [37] and CQR [36]. More recent works take another approach, by modifying the training process of the prediction model. CF-GNN falls into the latter case, although our idea applies to any non-conformity score. ConfTr [40] also modifies training for improved efficiency. Our approach differs from theirs in significant ways. First, we develop a theory on CP validity on the graph data and leverage topological principles that are specialized to graph to improve efficiency while ConfTr focuses on i.i.d. vision image data. Also, ConfTr happens during base model training using the training set, while CF-GNN conducts post-hoc correction using withheld calibration data without assuming access to base model training, making ConfTr not comparable to us. Finally, we also propose a novel loss for efficiency in regression tasks.

7 Conclusion

In this work, we extend conformal prediction to GNNs by laying out the theoretical conditions for finite-sample validity and proposing a flexible graph-based CP framework to improve efficiency. Potential directions for future work include generalizing the inefficiency loss to other desirable CP properties such as robustness and conditional coverage; extensions to inductive settings or transductive but non-random split settings; extensions to other graph tasks such as link prediction, community detection, and so on.

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A Deferred details for Section 3

A.1 Technical proofs for theoretical results

Proof of Lemma 2. Hereafter, we condition on the entire unordered graph, all the attribute and label information, and the index sets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ct}}$. We define the scores evaluated at the original node indices as

$$v_v = V(\mathbf{x}_v, y_v; \{z_i\}_{i\in\mathcal{D}_{\text{train}}\cup\mathcal{D}_{\text{valid}}}, \{\mathbf{x}_v\}_{v\in\mathcal{D}_{\text{calib}}\cup\mathcal{D}_{\text{test}}}, \mathcal{V}, \mathcal{E}), \quad v \in \mathcal{D}_{\text{calib}} \cup \mathcal{D}_{\text{test}} \subseteq \mathcal{V}.$$

By Assumption 1, for any permutation π of $\mathcal{D}_{\text{ct}}$, we always have

$$v_v = V(\mathbf{x}_v, y_v; \{z_i\}_{i\in\mathcal{D}_{\text{train}}\cup\mathcal{D}_{\text{valid}}}, \{\mathbf{x}_{\pi(v)}\}_{v\in\mathcal{D}_{\text{calib}}\cup\mathcal{D}_{\text{test}}}, \mathcal{V}_\pi, \mathcal{E}_\pi).$$

That is, given $\mathcal{D}_{\text{ct}}$, the evaluated score at any $v \in \mathcal{D}_{\text{ct}}$ remains invariant no matter which subset of $\mathcal{D}_{\text{ct}}$ is designated as $\mathcal{D}_{\text{calib}}$. This implies that the scores are fixed after conditioning:

$$[V_1, \ldots, V_{n+m}] = [v_v]_{v\in\mathcal{D}_{\text{ct}}},$$

where we use [] to emphasize unordered sets. Thus, the calibration scores $\{V_i\}_{i=1}^n$ is a subset of size n of $[v_v]_{v\in\mathcal{D}_{\text{ct}}}$. Note that under random splitting in the transductive setting, any permutation π of $\mathcal{D}_{\text{ct}}$ occurs with the same probability, which gives the conditional probability in Lemma 2. □

Proof of Theorem 3. Throughout this proof, we condition on the entire unordered graph, all the attribute and label information, and the index sets $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{ct}}$. By Lemma 2, after the conditioning, the unordered set of $\{V_i\}_{i=1}^{n+m}$ is fixed as $[v_v]_{v\in\mathcal{D}_{\text{ct}}}$, and $\{V_i\}_{i=1}^n$ is a simple random sample from $[v_v]_{v\in\mathcal{D}_{\text{ct}}}$. As a result, any test sample $V(X_{n+j}, Y_{n+j})$, $j = 1, \ldots, m$ is exchangeable with $\{V_i\}_{i=1}^n$. By standard theory for conformal prediction [43], this ensures *valid marginal coverage*, i.e., $\mathbb{P}(Y_{n+j} \in \widehat{C}(X_{n+j})) \geq 1 - \alpha$, where the expectation is over all the randomness.

We now proceed to analyze the distribution of $\widehat{\text{Cover}}$. For notational convenience, we write $N = m + n$, and view $\mathcal{D}_{\text{ct}}$ as the 'population'. In this way, $\{V_i\}_{i=1}^n$ is a simple random sample from $[v_v]_{v\in\mathcal{D}_{\text{ct}}}$. For every $\eta \in \mathbb{R}$, we define the 'population' cumulative distribution function (c.d.f.)

$$F(\eta) = \frac{1}{N}\sum_{v\in\mathcal{D}_{\text{ct}}} \mathbb{1}\{v_v \leq \eta\},$$

which is a deterministic function. We also define the calibration c.d.f. as

$$\widehat{F}_n(\eta) = \frac{1}{n}\sum_{v\in\mathcal{D}_{\text{calib}}} \mathbb{1}\{v_v \leq \eta\} = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{V_i \leq \eta\},$$

which is random, and its randomness comes from which subset of $\mathcal{D}_{\text{ct}}$ is $\mathcal{D}_{\text{calib}}$. By definition,

$$\widehat{\eta} = \inf\{\eta \colon \widehat{F}_n(\eta) \geq (1-q)(1+1/n)\}.$$

Since the scores have no ties, we know

$$\widehat{F}_n(\widehat{\eta}) = \lceil (1-q)(n+1)\rceil / n.$$

The test-time coverage can be written as

$$\begin{aligned}
\widehat{\text{Cover}} &= \frac{1}{m}\sum_{j=1}^m \mathbb{1}\{V_{n+j} \leq \widehat{\eta}\} \\
&= \frac{1}{N-n}\Big(\sum_{v\in\mathcal{D}_{\text{ct}}} \mathbb{1}\{v_v \leq \widehat{\eta}) - \sum_{v\in\mathcal{D}_{\text{calib}}} \mathbb{1}\{v_v \leq \widehat{\eta})\Big) \\
&= \frac{N}{N-n}F(\widehat{\eta}) - \frac{n}{N-n}\widehat{F}_n(\widehat{\eta}) = \frac{N}{N-n}F(\widehat{\eta}) - \frac{\lceil (1-q)(n+1)\rceil}{N-n}.
\end{aligned}$$

Now we characterize the distribution of $\widehat{\eta}$. For any $\eta \in \mathbb{R}$, by the definition of $\widehat{\eta}$,

$$\mathbb{P}(\widehat{\eta} \leq \eta) = \mathbb{P}\big(n\widehat{F}_n(\eta) \geq (n+1)(1-q)\big) = \mathbb{P}\big(n\widehat{F}_n(\eta) \geq \lceil (n+1)(1-q)\rceil\big).$$

Note that $n\widehat{F}_n(\eta) = \sum_{v\in\mathcal{D}_{\text{calib}}} \mathbb{1}\{v_v \le \eta\}$ is the count of data in $\mathcal{D}_{\text{calib}}$ such that the score is below η. By the simple random sample (i.e., sampling without replacement), $n\widehat{F}_n(\eta)$ follows a hyper-geometric distribution with parameter $N, n, NF(\eta)$. That is,

$$\mathbb{P}(n\widehat{F}_n(\eta) = k) = \frac{\binom{NF(\eta)}{k}\binom{N-NF(\eta)}{n-k}}{\binom{N}{n}}, \quad 0 \le k \le NF(\eta).$$

Denoting the c.d.f. of hypergeometric distribution as $\Phi_{\text{HG}}(k; N, n, NF(\eta))$, we have

$$\mathbb{P}(\widehat{\eta} \le \eta) = 1 - \Phi_{\text{HG}}\big(\lceil (n+1)(1-q)\rceil - 1; N, n, NF(\eta)\big).$$

Then, for any $t \in [0,1]$,

$$\begin{aligned}\mathbb{P}\big(\widehat{\text{Cover}} \le t\big) &= \mathbb{P}\left(\frac{N}{N-n}F(\widehat{\eta}) - \frac{\lceil (1-q)(n+1)\rceil}{N-n} \le t\right) \\ &= \mathbb{P}\left(F(\widehat{\eta}) \le \frac{\lceil (1-q)(n+1)\rceil + (N-n)t}{N}\right).\end{aligned}$$

Since $F(\cdot)$ is monotonely increasing,

$$\mathbb{P}\big(\widehat{\text{Cover}} \le t\big) = \mathbb{P}\left(\widehat{\eta} \le F^{-1}\left(\frac{\lceil (1-q)(n+1)\rceil + (N-n)t}{N}\right)\right),$$

where $F^{-1}(s) = \inf\{\eta\colon F(\eta) \ge s\}$ for any $s \in [0,1]$. Plugging in the previous results on the distribution of $\widehat{\eta}$, we have

$$\begin{aligned}\mathbb{P}\big(\widehat{\text{Cover}} \le t\big) &= 1 - \Phi_{\text{HG}}\left(\lceil (n+1)(1-q)\rceil - 1; N, n, NF\left(F^{-1}\big(\tfrac{\lceil (1-q)(n+1)\rceil + (N-n)t}{N}\big)\right)\right) \\ &= 1 - \Phi_{\text{HG}}\left(\lceil (n+1)(1-q)\rceil - 1; N, n, N\tfrac{\lceil \lceil (1-q)(n+1)\rceil + (N-n)t\rceil}{N}\right) \\ &= 1 - \Phi_{\text{HG}}\left(\lceil (n+1)(1-q)\rceil - 1; N, n, \lceil \lceil (1-q)(n+1)\rceil + (N-n)t\rceil\right) \\ &= 1 - \Phi_{\text{HG}}\left(\lceil (n+1)(1-q)\rceil - 1; N, n, \lceil (1-q)(n+1)\rceil + \lceil (N-n)t\rceil\right)\end{aligned}$$

where the second equality uses the fact that $F(\eta) \in \{0, 1/N, \ldots, (N-1)/N, 1\}$, hence $F(F^{-1}(s)) = \lceil Ns\rceil/N$ for any $s \in [0,1]$. By tower property, such an equation also holds for the unconditional distribution, marginalized over all the randomness. This completes the proof of Theorem 3. □

A.2 Additional visualization of test-time coverage

In this part, we provide more visualization of the distributions of test time coverage $\widehat{\text{Cover}}$ under various sample size configurations. We note that such results also apply to standard application of split conformal prediction when the non-conformity score function V is independent of calibration and test samples, so that Assumption 1 is satisfied.

Figures 6 and 7 plot the p.d.f. of $\widehat{\text{Cover}}$ for $\alpha = 0.05$ and $\alpha = 0.1$, respectively, when fixing n and varying the test sample size m. The y-axis is obtained by computing $\mathbb{P}(t_{k-1} < \widehat{\text{Cover}} \le t_k)/(t_k - t_{k-1})$ at $x = (t_{k-1} + t_k)/2$ for a sequence of evenly spaced $\{t_k\} \in [0,1]$. All figures in this paper for p.d.f.s are obtained in the same way. We see that $\widehat{\text{Cover}}$ concentrates more tightly around the target value $1-\alpha$ as m and n increases.

Figures 8 and 9 plot the p.d.f. of $\widehat{\text{Cover}}$ for $\alpha = 0.05$ and $\alpha = 0.1$, respectively, where we fix $N = m + n$ but vary the calibration sample size n. This mimics the situation where the total number of nodes on the graph is fixed, while we may have flexibility in collecting data as the calibration set. We observe a tradeoff between the calibration accuracy determined by n and the test-sample concentration determined by n. The distribution of $\widehat{\text{Cover}}$ is more concentrated around $1-\alpha$ when m and n are relatively balanced.

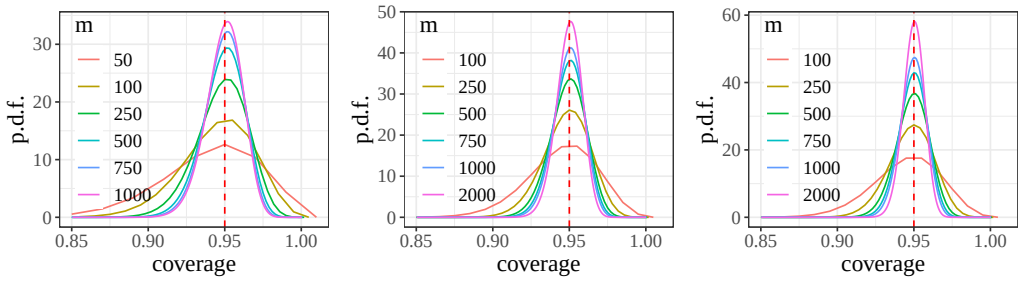

Figure 6: P.d.f. of test-time coverage $\widehat{\text{Cover}}$ for $n = 500$ (left), 1000 (middle), 2000 (right) and $\alpha = 0.05$ with curves representing different values of m, the test sample size.

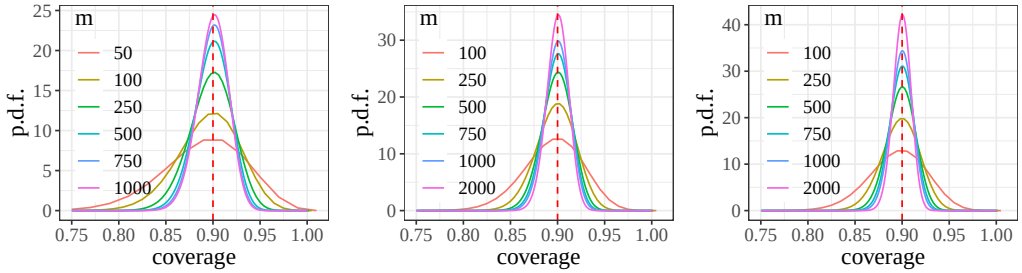

Figure 7: P.d.f. of test-time coverage $\widehat{\text{Cover}}$ for $n = 500$ (left), 1000 (middle), 2000 (right) and $\alpha = 0.1$ with curves representing different values of m, the test sample size.

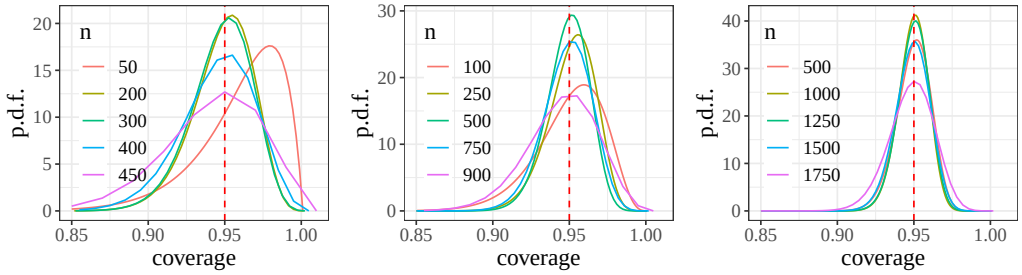

Figure 8: P.d.f. of test-time coverage $\widehat{\text{Cover}}$ for $N = m + n = 500$ (left), 1000 (middle), 2000 (right) and $\alpha = 0.05$ with curves representing different values of n, the calibration sample size.

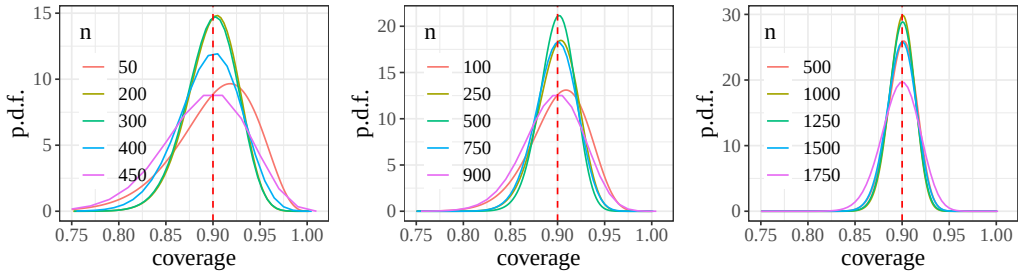

Figure 9: P.d.f. of test-time coverage $\widehat{\text{Cover}}$ for $N = m + n = 500$ (left), 1000 (middle), 2000 (right) and $\alpha = 0.1$ with curves representing different values of n, the calibration sample size.

B Discussion on full conformal prediction, split conformal prediction

In this part, we discuss the relation of our application of conformal prediction to full conformal prediction and split conformal prediction, two prominent conformal prediction methods proposed by Vovk and his coauthors in [43]. Split conformal prediction is mostly widely used due to its computational efficiency, where exchangeability is usually ensured by independence (which is not obvious for graph data) as we discussed briefly in the introduction.

Full conformal prediction (FCP) is arguably the most versatile form of conformal prediction. Given calibration data $Z_i = (X_i, Y_i) \in \mathcal{X} \times \mathcal{Y}$, $i = 1, \ldots, n$, and given a test point $X_{n+1} \in \mathcal{X}$ whose outcome $Y_{n+1} \in \mathcal{Y}$ is unknown, at every hypothesized value $y \in \mathcal{Y}$, FCP uses any algorithm S to train the following scores

$$S_i^y = S(X_i, Y_i; Z_1, \ldots, Z_{i-1}, Z_{i+1}, \ldots, Z_n, (X_{n+1}, y)), \quad i = 1, \ldots, n,$$

where S is symmetric in the arguments $Z_1, \ldots, Z_{i-1}, Z_{i+1}, \ldots, Z_n, (X_{n+1}, y)$, as well as

$$S_{n+1}^y = S(X_{n+1}, y; Z_1, \ldots, Z_n).$$

Here, for $1 \le i \le n$, S_i^y intuitively measures how well the observation (X_i, Y_i) conforms to the observations $Z_1, \ldots, Z_{i-1}, Z_{i+1}, \ldots, Z_n, (X_{n+1}, y)$ with the hypothesized value of y. For instance, when using a linear prediction model, it can be chosen as the prediction residual

$$S_i^y = |Y_i - X_i^\top \widehat{\theta}^y|,$$

where $\widehat{\theta}^y$ is the ordinary least squares coefficient by a linear regression of $Y_1, \ldots, Y_{i-1}, Y_{i+1}, \ldots, Y_n, y$ over $X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n, X_{n+1}$. More generally, one may train a prediction model $\widehat{\mu}^y \colon \mathcal{X} \to \mathcal{Y}$ using $Z_1, \ldots, Z_{i-1}, Z_{i+1}, \ldots, Z_n, (X_{n+1}, y)$, and set $S_i^y = |Y_i - \widehat{\mu}^y(X_i)|$. For a confidence level $\alpha \in (0, 1)$, the FCP prediction set is then

$$\widehat{C}(X_{n+1}) := \Big\{ y \colon \tfrac{1 + \mathbb{1}\{S_i^y > S_{n+1}^y\}}{n+1} \le \alpha \Big\}.$$

Since the original form of FCP involves training $n + 1$ models at each hypothesized value y, its computation can be very intense. It is thus impractical to directly apply FCP to GNN models (i.e., imagining S as the GNN training process on the entire graph with a hypothesized outcome y).

Split conformal prediction (SCP) is a computationally-efficient special case of FCP that is most widely used for i.i.d. data. The idea is to set aside an independent fold of data to output a single trained model. To be specific, we assume access to a given non-conformity score $V \colon \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, i.i.d. calibration data $Z_i = (X_i, Y_i)_{i=1}^n$, and an independent test sample (X_{n+1}, Y_{n+1}) from the same distribution with Y_{n+1} unobserved. Here by a "given" score, we mean that it is obtained without knowing the calibration and test sample; usually, it is trained on an independent set of data $\{(X_j, Y_j)\}_{j \in \mathcal{D}_{\text{train}}}$ before seeing the calibration and test sample. Then define $V_i = V(X_i, Y_i)$ for $i = 1, \ldots, n$. The SCP prediction set is

$$\widehat{C}(X_{n+1}) = \Big\{ y \colon \tfrac{1 + \mathbb{1}\{V_i > V(X_{n+1}, y)\}}{n+1} \le \alpha \Big\}.$$

The above set is usually convenient to compute, because we only need one single model to obtain V. The validity of SCP usually relies on the independence of V to calibration and test data as we mentioned in the introduction. However, the application of SCP to GNN model is also not straightforward: as we discussed in the main text, the model training step already uses the calibration and test samples, and the nodes are correlated.

Indeed, our method can be seen as a middle ground between FCP and SCP: it only requires one single prediction model as SCP does, but allows to use calibration and test data in the training step as FCP does. In our method introduced in the main text, there exists a fixed function $V \colon \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ (provided by APS and CQR) such that

$$S_i^y = V(\widehat{\mu}(X_i), Y_i), \quad S_{n+1}^y = V(\widehat{\mu}(X_{n+1}, y),$$

where $\widehat{\mu}$ is the final output from the second GNN model whose training process does not utilize the outcomes $Y_1, \ldots, Y_n$ and y, but uses the features $X_1, \ldots, X_n$ and X_{n+1}.

Algorithm 1: Pseudo-code for CF-GNN algorithm.

Input: Graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$; a trained base GNN model GNN_θ; non-conformity score function $V(X, Y)$; pre-specified mis-coverage rate α, Randomly initialized ϑ for the conformal correction model GNN_ϑ.

while *not done* **do**

  for *i in {1, ..., $|\mathcal{V}_{\text{cor-calib}} \cup \mathcal{V}_{\text{cor-test}}|$}* **do**

    $\widehat{\mu}(X_i) = \text{GNN}_\theta(\mathbf{X_i}, G)$ `// Base GNN output scores`

    $\tilde{\mu}(X_i) = \text{GNN}_\vartheta(\widehat{\mu}(X_i), G)$ `// Correction model output scores`

  end

  $n, m = |\mathcal{V}_{\text{cor-calib}}|, |\mathcal{V}_{\text{cor-test}}|$ `// Size of correction calib/test set`

  $\widehat{\alpha} = \frac{1}{n+1} * \alpha$ `// Finite-sample correction`

  $\widehat{\eta} = \text{DiffQuantile}(\{V(X_i, Y_i) | i \in \mathcal{V}_{\text{cor-calib}}\})$ `// Compute non-conformity scores`

  if Classification **then**

    $\mathcal{L}_{\text{Ineff}} = \frac{1}{m} \sum_{i \in \mathcal{V}_{\text{cor-test}}} \frac{1}{|\mathcal{Y}|} \sum_{k \in \mathcal{Y}} \sigma(\frac{V(X_i, k) - \widehat{\eta}}{\tau})$

    `// Inefficiency proxy for classification tasks`

  end

  if Regression **then**

    $\mathcal{L}_{\text{Ineff}} = \frac{1}{m} \sum_{i \in \mathcal{V}_{\text{cor-test}}} (\tilde{\mu}_{1-\alpha/2}(X)_i + \widehat{\eta}) - (\tilde{\mu}_{\alpha/2}(X)_i - \widehat{\eta})$

    `// Inefficiency proxy for regression tasks`

    $\mathcal{L}_{\text{Ineff}} += \gamma \frac{1}{m} \sum_{i \in \mathcal{V}_{\text{cor-test}}} (\tilde{\mu}_{1-\alpha/2}(X)_i - \widehat{\mu}_{1-\alpha/2}(X)_i)^2 + (\tilde{\mu}_{\alpha/2}(X)_i - \widehat{\mu}_{\alpha/2}(X)_i)^2$

    `// Consistency regularization term`

  end

  $\vartheta \leftarrow \vartheta - \nabla_\vartheta \mathcal{L}_{\text{Ineff}}$ `// Optimizing` ϑ `to reduce inefficiency`

end

C Algorithm overview

We describe the pseudo-code of CF-GNN in Algorithm 1.

D Deferred details for experiments

D.1 Hyperparameters

Table 5 reports our set of hyperparameter ranges. We conduct 100 iterations of Bayesian Optimization for CF-GNN with the validation set inefficiency proxy as the optimization metric. To avoid overfitting, each iteration only uses the first GNN run. The optimized hyperparameters are then used for all 10 GNN runs and we then reported the average and standard deviation across runs. Each experiment is done with a single NVIDIA 2080 Ti RTX 11GB GPU.

Table 5: Hyperparameter range for CF-GNN.

Task	Param.	Range
Classification	GNN_ϑ Hidden dimension	[16,32,64,128,256]
	Learning rate	[1e-1, 1e-2, 1e-3, 1e-4]
	GNN_ϑ Number of GNN Layers	[1,2,3,4]
	GNN_ϑ Base Model	[GCN, GAT, GraphSAGE, SGC]
	τ	[10, 1, 1e-1, 1e-2, 1e-3]
Regression	GNN_ϑ Hidden dimension	[16,32,64,128,256]
	Learning rate	[1e-1, 1e-2, 1e-3, 1e-4]
	GNN_ϑ Number of GNN Layers	[1, 2, 3, 4]
	GNN_ϑ Base Model	[GCN, GAT, GraphSAGE, SGC]
	Reg. loss coeff. γ	[1, 1e-1]

D.2 Baseline Details

We report the details about baselines below and the hyperparameter range in Table 6.

1. Temperature Scaling [13] divides the logits with a learnable scalar. It is optimized over NLL loss in the validation set.
2. Vector Scaling [13] has a scalar to scale the logits for each class dimension and adds an additional classwide bias. It is optimized over NLL loss in the validation set.
3. Ensemble Temperature Scaling [49] learns an ensemble of uncalibrated, temperature-scaled calibrated calibrators.
4. CaGCN [44] uses an additional GCN model that learns a temperature scalar for each node based on its neighborhood information.
5. GATS [17] identifies five factors that affect GNN calibration and designs a model that accounts for these five factors by using per-node temperature scaling and attentive aggregation from the local neighborhood.
6. QR [25] uses a pinball loss to produce quantile scores. It is CQR without the conformal prediction adjustment.
7. MC dropout [9] turns on dropout during evaluation and produces K predictions. We then take the 95% quantile of the predicted distribution. We also experimented with taking a 95% confidence interval but 95% quantile has better coverage, thus we adopt the quantile approach.
8. BayesianNN [23] model the label with normal distribution and the model produces two heads, where one corresponds to the mean and the second log variance. We then calculate the standard deviation as the square root of the exponent of log variance. Then we take the [mean-1.96*standard deviation, mean+1.96*standard deviation] for the 95% interval.

Table 6: Hyperparameter range for baselines.

Baseline	Param.	Range
Temperature Scaling	No hyperparameter	Not Applicable
Vector Scaling	No hyperparameter	Not Applicable
Ensemble Temp Scaling	No hyperparameter	Not Applicable
CaGCN	Dropout Hidden dimension Number of GNN Layers Weight Decay	[0.3, 0.5, 0.7] [16, 32, 64, 128, 256] [1,2,3,4] [0, 1e-3, 1e-2, 1e-1]
GATS	Dropout Hidden dimension Number of GNN Layers Weight Decay	[0.3, 0.5, 0.7] [16, 32, 64, 128, 256] [1,2,3,4] [0, 1e-3, 1e-2, 1e-1]
MC Dropout	Number of Predictions	[100, 500, 1,000]
BayesianNN	No hyperparameter	Not Applicable

D.3 Dataset

For node classification, we use the common node classification datasets in Pytorch Geometric package. For node regression, we use datasets in [20]. We report the dataset statistics at Table 7.

D.4 Marginal coverage and inefficiency across GNN architectures

We additionally conduct marginal coverage and inefficiency comparisons of CF-GNN over the vanilla CP across 4 different GNN architectures: GCN, GAT, GraphSAGE, and SGC. The result for marginal

Table 7: Dataset statistics.

Domain	Dataset	Task	# Nodes	# Edges	# Features	# Labels
Citation	Cora	Classification	2,995	16,346	2,879	7
	DBLP	Classification	17,716	105,734	1,639	4
	CiteSeer	Classification	4,230	10,674	602	6
	PubMed	Classification	19,717	88,648	500	3
Co-purchase	Computers	Classification	13,752	491,722	767	10
	Photos	Classification	7,650	238,162	745	8
Co-author	CS	Classification	18,333	163,788	6,805	15
	Physics	Classification	34,493	495,924	8,415	5
Transportation	Anaheim	Regression	914	3,881	4	–
	Chicago	Regression	2,176	15,104	4	–
Geography	Education	Regression	3,234	12,717	6	–
	Election	Regression	3,234	12,717	6	–
	Income	Regression	3,234	12,717	6	–
	Unemployment	Regression	3,234	12,717	6	–
Social	Twitch	Regression	1,912	31,299	3,170	–

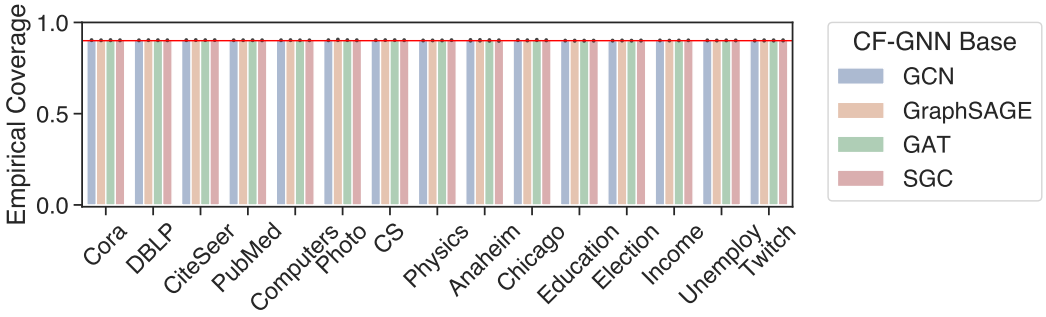

Figure 10: Empirical coverage across 15 datasets with 10 independent runs of GNN, using CF-GNN.

coverage is in Figure 10. The result for inefficiency is in Table 8. We observe consistent improvement in inefficiency reduction across these architectures, suggesting CF-GNN is a GNN-agnostic efficiency improvement approach.

D.5 CF-GNN with Regularized Adaptive Prediction Sets

To further showcase that CF-GNN is a versatile framework that adapts to any advancement in non-conformity scores, we experiment on RAPS [2], which regularizes APS to produce a smaller prediction set size. We report the performance using the GCN backbond in Table 9. We observe that CF-GNN still obtains impressive inefficiency reduction compared to the vanilla application of RAPS to GNN.

D.6 Conditional coverage on full set of network features

We report the full set of network features and calculate the worse-slice coverage in Table 10 for target coverage of 0.9 and Table 11 for a target coverage of 0.95. We observe that CF-GNN achieves satisfactory conditional coverage across a wide range of diverse network features.

D.7 Prediction accuracy versus uncertainty calibration

As we discussed in the main text, the original GNN trained towards optimal prediction accuracy does not necessarily yield the most efficient prediction model; this is corrected by the second GNN in CF-GNN which improves the efficiency of conformal prediction sets/intervals. With our approach, one can use the output of the original GNN for point prediction while that of the second GNN for efficient uncertainty quantification, without necessarily overwriting the first accurate prediction model. However, a natural question here still remains, which is that whether applying the second

Table 8: Empirical inefficiency measured by the size/length of the prediction set/interval for node classification/regression. The result uses APS for classification and CQR for regression. We report the average and standard deviation calculated from 10 GNN runs with each run of 100 conformal splits.

Task	GNN Model	GCN	GraphSAGE	GAT	SGC
	Dataset	CP ⟶ CF-GNN	CP ⟶ CF-GNN	CP ⟶ CF-GNN	CP ⟶ CF-GNN
Node classif.	Cora	$3.80_{\pm.28} \xrightarrow{-58.44\%} 1.58_{\pm.22}$	$6.73_{\pm.19} \xrightarrow{-76.50\%} 1.58_{\pm.15}$	$4.14_{\pm.16} \xrightarrow{-62.53\%} 1.55_{\pm.10}$	$3.88_{\pm.19} \xrightarrow{-62.23\%} 1.47_{\pm.10}$
	DBLP	$2.43_{\pm.03} \xrightarrow{-49.20\%} 1.23_{\pm.01}$	$3.91_{\pm.01} \xrightarrow{-68.27\%} 1.24_{\pm.01}$	$2.02_{\pm.06} \xrightarrow{-37.51\%} 1.26_{\pm.01}$	$2.44_{\pm.04} \xrightarrow{-48.93\%} 1.24_{\pm.02}$
	CiteSeer	$3.86_{\pm.11} \xrightarrow{-74.50\%} 0.99_{\pm.01}$	$5.88_{\pm.02} \xrightarrow{-83.07\%} 1.00_{\pm.01}$	$3.18_{\pm.25} \xrightarrow{-68.56\%} 1.00_{\pm.01}$	$3.79_{\pm.14} \xrightarrow{-73.43\%} 1.01_{\pm.02}$
	PubMed	$1.60_{\pm.02} \xrightarrow{-19.44\%} 1.29_{\pm.04}$	$1.93_{\pm.28} \xrightarrow{-36.95\%} 1.22_{\pm.03}$	$1.37_{\pm.02} \xrightarrow{-10.54\%} 1.23_{\pm.02}$	$1.60_{\pm.02} \xrightarrow{-24.27\%} 1.21_{\pm.02}$
	Computers	$3.56_{\pm.13} \xrightarrow{-50.22\%} 1.77_{\pm.11}$	$6.00_{\pm.10} \xrightarrow{-45.74\%} 3.26_{\pm.48}$	$2.33_{\pm.11} \xrightarrow{-8.18\%} 2.14_{\pm.12}$	$3.44_{\pm.14} \xrightarrow{-45.69\%} 1.87_{\pm.15}$
	Photo	$3.79_{\pm.13} \xrightarrow{-57.03\%} 1.63_{\pm.17}$	$4.52_{\pm.47} \xrightarrow{-37.22\%} 2.84_{\pm.67}$	$2.24_{\pm.21} \xrightarrow{-19.38\%} 1.81_{\pm.16}$	$3.81_{\pm.14} \xrightarrow{-62.86\%} 1.41_{\pm.05}$
	CS	$7.79_{\pm.29} \xrightarrow{-55.83\%} 3.44_{\pm.33}$	$14.68_{\pm.02} \xrightarrow{-88.63\%} 1.67_{\pm.14}$	$6.87_{\pm.48} \xrightarrow{-73.08\%} 1.85_{\pm.10}$	$7.76_{\pm.25} \xrightarrow{-73.92\%} 2.02_{\pm.22}$
	Physics	$3.11_{\pm.07} \xrightarrow{-65.36\%} 1.08_{\pm.10}$	$4.91_{\pm.01} \xrightarrow{-72.97\%} 1.33_{\pm.08}$	$2.00_{\pm.19} \xrightarrow{-45.23\%} 1.09_{\pm.06}$	$3.10_{\pm.08} \xrightarrow{-57.22\%} 1.32_{\pm.12}$
Average Improvement		-53.75%	-63.75%	-40.63%	-56.07%
Node regress.	Anaheim	$2.89_{\pm.39} \xrightarrow{-25.00\%} 2.17_{\pm.11}$	$2.37_{\pm.05} \xrightarrow{-23.12\%} 1.82_{\pm.07}$	$3.12_{\pm.38} \xrightarrow{-31.27\%} 2.14_{\pm.11}$	$2.94_{\pm.24} \xrightarrow{-24.90\%} 2.21_{\pm.16}$
	Chicago	$2.05_{\pm.07} \xrightarrow{-0.48\%} 2.04_{\pm.17}$	$2.08_{\pm.05} \xrightarrow{-7.90\%} 1.92_{\pm.09}$	$1.95_{\pm.04} \xrightarrow{-68.15\%} 0.62_{\pm.93}$	$2.02_{\pm.03} \xrightarrow{-1.37\%} 1.99_{\pm.07}$
	Education	$2.56_{\pm.02} \xrightarrow{-5.07\%} 2.43_{\pm.05}$	$2.20_{\pm.04} \xrightarrow{+8.44\%} 2.38_{\pm.08}$	$2.48_{\pm.05} \xrightarrow{-2.76\%} 2.41_{\pm.04}$	$2.55_{\pm.02} \xrightarrow{-2.80\%} 2.48_{\pm.04}$
	Election	$0.90_{\pm.01} \xrightarrow{+0.21\%} 0.90_{\pm.02}$	$0.87_{\pm.01} \xrightarrow{-0.80\%} 0.86_{\pm.02}$	$0.89_{\pm.00} \xrightarrow{-1.23\%} 0.88_{\pm.02}$	$0.90_{\pm.00} \xrightarrow{-0.42\%} 0.90_{\pm.02}$
	Income	$2.51_{\pm.12} \xrightarrow{-4.58\%} 2.40_{\pm.05}$	$2.08_{\pm.04} \xrightarrow{+32.23\%} 2.75_{\pm.23}$	$2.35_{\pm.03} \xrightarrow{-0.23\%} 2.34_{\pm.07}$	$2.42_{\pm.01} \xrightarrow{+3.04\%} 2.49_{\pm.04}$
	Unemploy.	$2.72_{\pm.03} \xrightarrow{-10.83\%} 2.43_{\pm.04}$	$2.75_{\pm.06} \xrightarrow{-12.90\%} 2.39_{\pm.05}$	$2.80_{\pm.08} \xrightarrow{-14.56\%} 2.40_{\pm.04}$	$2.72_{\pm.02} \xrightarrow{-11.05\%} 2.42_{\pm.04}$
	Twitch	$2.43_{\pm.10} \xrightarrow{-1.36\%} 2.39_{\pm.07}$	$2.48_{\pm.09} \xrightarrow{-3.06\%} 2.40_{\pm.07}$	$2.50_{\pm.14} \xrightarrow{-5.53\%} 2.36_{\pm.07}$	$2.42_{\pm.08} \xrightarrow{-1.43\%} 2.38_{\pm.06}$
Average Improvement		-6.73%	-1.02%	-17.68%	-5.56%

Table 9: Comparison with other non-conformity scores that reduce inefficiency.

Size	CP ⟶ CF-GNN
Cora	$1.67_{\pm.11} \xrightarrow{-15.35\%} 1.42_{\pm.05}$
DBLP	$1.39_{\pm.02} \xrightarrow{-5.00\%} 1.32_{\pm.01}$
CiteSeer	$1.30_{\pm.07} \xrightarrow{-19.85\%} 1.04_{\pm.04}$
PubMed	$1.23_{\pm.01} \xrightarrow{+2.40\%} 1.26_{\pm.02}$
Computers	$1.58_{\pm.02} \xrightarrow{-4.59\%} 1.51_{\pm.05}$
Photo	$1.34_{\pm.01} \xrightarrow{-10.47\%} 1.20_{\pm.01}$
CS	$1.29_{\pm.04} \xrightarrow{-6.13\%} 1.21_{\pm.02}$

GNN drastically changes the prediction accuracy. This question is more relevant to the classification problem since for regression our method only adjusts the confidence band. For classification, we consider top-1 class prediction as the "point prediction". We present its accuracy "Before" and "After" the correction in Table 12, which shows that this correction typically does not result in a visible change in accuracy. In addition, in a new experiment on Cora, we find that 100% of the top-1 class from the base GNN are in CF-GNN's prediction sets. The potential to develop steps that explicitly consider point prediction accuracy is an exciting avenue for future research.

E Extended Related Works

Uncertainty quantification for graph neural networks. Uncertainty quantification (UQ) is a well-studied subject in general machine learning and also recently in GNNs. For multi-class classification, the raw prediction scores are often under/over-confident and thus various calibration methods are proposed for valid uncertainty estimation such as temperate scaling [13], vector scaling [13], ensemble temperate scaling [49], and so on [14, 27, 39, 1]. Recently, specialized calibration methods that leverage network principles such as homophily have been developed: examples include CaGCN [44] and GATS [17]. In regression, various methods have been proposed to construct prediction intervals, such as quantile regression [25, 41, 38], bootstrapping with subsampling, model ensembles, and dropout initialization [9, 28, 26, 35], and bayesian approaches with strong modeling assumptions on parameter and data distributions [23, 19]. However, these UQ methods can fail to provide statistically

Table 10: CF-GNN achieves conditional coverage. We use Cora/Twitch as an example classification/regression dataset.

Target: 0.9	Classification		Regression	
Model	CP	CF-GNN	CP	CF-GNN
Marginal Cov.	0.90±.02	0.90±.01	0.91±.02	0.91±.03
Cond. Cov. (Input Feat.)	0.89±.04	0.90±.03	0.90±.07	0.86±.08
Cond. Cov. (Cluster)	0.82±.07	0.89±.03	0.90±.06	0.88±.07
Cond. Cov. (Between)	0.82±.06	0.89±.03	0.86±.08	0.88±.07
Cond. Cov. (PageRank)	0.71±.08	0.87±.05	0.87±.09	0.89±.07
Cond. Cov. (Load)	0.83±.05	0.90±.03	0.86±.08	0.88±.07
Cond. Cov. (Harmonic)	0.89±.04	0.87±.05	0.88±.08	0.91±.06
Cond. Cov. (Degree)	0.79±.05	0.89±.04	0.86±.08	0.89±.06

Table 11: CF-GNN achieves conditional coverage. We use Cora/Twitch as an example classification/regression dataset.

Target: 0.95	Classification		Regression	
Model	CP	CF-GNN	CP	CF-GNN
Marginal Cov.	0.95±.01	0.95±.01	0.96±.02	0.96±.02
Cond. Cov. (Input Feat.)	0.94±.02	0.94±.03	0.95±.04	0.94±.05
Cond. Cov. (Cluster)	0.89±.06	0.93±.04	0.96±.03	0.96±.03
Cond. Cov. (Between)	0.81±.06	0.95±.03	0.94±.05	0.94±.05
Cond. Cov. (PageRank)	0.78±.06	0.94±.03	0.94±.05	0.94±.05
Cond. Cov. (Load)	0.81±.06	0.94±.03	0.94±.05	0.95±.05
Cond. Cov. (Harmonic)	0.88±.04	0.95±.03	0.96±.04	0.95±.04
Cond. Cov. (Degree)	0.83±.05	0.88±.06	0.94±.04	0.94±.04

rigorous and empirically valid coverage guarantee (see Table 1). In contrast, CF-GNN achieves valid marginal coverage in both theory and practice. Uncertainty quantification has also been leveraged to deal with out-of-distribution detection and imbalanced data in graph neural networks [50, 10]. While it is not the focus here, we remark that conformal prediction can also be extended to tackle such issues [18], and it would be interesting to explore such applications for graph data.

Conformal prediction for graph neural networks. As we discussed, the application of conformal prediction to graph-structured data remains largely unexplored. At the time of submission, the only work we awared of is [7], who claims that nodes in the graph are not exchangeable in the inductive setting and employs the framework of [3] to construct conformal prediction sets using neighborhood nodes as the calibration data. In contrast, we study the transductive setting where certain exchangeablility property holds and allows for flexibility in the training step. We also study the efficiency aspect that is absent in [7]. In addition, there have been concurrent works [15, 33] that observe similar exchangeability and validity of conformal prediction in either transductive setting or other network models. In particular, [15] proposes a diffusion-based method that aggregates non-conformity scores of neighbor nodes to improve efficiency, while our approach learns the aggregation of neighbor scores, which is more general than their approach. [32] studies the exchangeability for node regression under certain network models instead of our transductive setting with GNNs, and without considering the efficiency aspect. With a growing recent interest in conformal prediction for graphs, there are even more recent works that focus on validity [32] and link prediction [34].

Efficiency of conformal prediction. While conformal prediction enjoys distribution-free coverage for any non-conformity score based on any prediction model, its efficiency (i.e., size of prediction sets or length of prediction intervals) varies with specific choice of the scores and models. How to achieve desirable properties such as efficiency is a topic under intense research in conformal prediction. To this end, one major thread designs good non-conformity scores such as APS [37] and CQR [36]. More recent works take another approach, by modifying the training process of the prediction model to

Table 12: CF-GNN does not change the top-1 class prediction accuracy for classification tasks.

Dataset	Before	After
Cora	$0.844_{\pm 0.004}$	$0.843_{\pm 0.016}$
DBLP	$0.835_{\pm 0.001}$	$0.832_{\pm 0.002}$
CiteSeer	$0.913_{\pm 0.002}$	$0.911_{\pm 0.002}$

further improve efficiency. This work falls into the latter case. Our idea applies to any non-conformity scores, as demonstrated with APS and CQR, two prominent examples of the former case. Related to our work, ConfTr [40] also simulates conformal prediction so as to train a prediction model that eventually leads to more efficient conformal prediction sets. However, our approach differs from theirs in significant ways. First, ConfTr modifies model training, while CF-GNN conducts post-hoc correction without changing the original prediction. Second, ConfTr uses the training set to simultaneously optimize model prediction and efficiency of conformal prediction, while we withhold a fraction of calibration data to optimize the efficiency. Third, our approach specifically leverages the rich topological information in graph-structured data to achieve more improvement in efficiency. Finally, we also propose a novel loss for efficiency in regression tasks.

Conformal Prediction: A Data Perspective

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Conformal prediction (CP), a distribution-free uncertainty quantification (UQ) framework, reliably provides valid predictive inference for black-box models. CP constructs prediction sets or intervals that contain the true output with a specified probability. However, modern data science's diverse modalities, along with increasing data and model complexity, challenge traditional CP methods. These developments have spurred novel approaches to address evolving scenarios. This survey reviews the foundational concepts of CP and recent advancements from a data-centric perspective, including applications to structured, unstructured, and dynamic data. We also discuss the challenges and opportunities CP faces in large-scale data and models.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Uncertainty quantification;**

Additional Key Words and Phrases: Natural language processing, Computer vision, Time series

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1 Introduction

The recent rapid development of well-designed and powerful **machine learning** (**ML**) models has significantly transformed our lives. However, the success of these models is often evaluated based on the accuracy of their predictions, which, while important, is not sufficient in many real-world scenarios. In high-stakes applications, it is equally critical to assess the *uncertainty* of model outputs. **Uncertainty quantification** (**UQ**) has long been a central problem in fields like statistics and ML. Several well-established methods, such as Bayesian inference and resampling techniques, have been widely adopted to address UQ. However, Bayesian posterior intervals are only valid if the distributional assumptions of the model are correctly specified, which may not always be the case in practical applications. Similarly, when constructing prediction sets for unobserved outputs,

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resampling approaches based on the residuals of observed data can result in overly narrow intervals since the estimate of variance is too small [19], compromising the validity of the prediction set. This raises a fundamental question:

Is it possible to provide a valid confidence or prediction set,
regardless of the model in use and with minimal assumptions about the data distribution?

CP [186] is model-agnostic and can be applied to any ML model, generating prediction sets/intervals based on non-conformity scores derived from the model's outputs. CP offers several desirable features: (1) *Finite-sample validity*: CP generates calibrated prediction sets/intervals that ensure high-probability coverage with finite-sample theoretical guarantees, without simplifying the models' complexity or relying on strong assumptions about the data generating process. More concretely, the validity of CP only requires the *exchangeability* assumption of data. (2) *Lightweight:* As a wrapper method, CP does not require retraining the underlying model, making it computationally efficient and easy to apply to existing models. (3) *Versatility:* CP is model-agnostic, meaning it can be used with any model, regardless of its architecture or complexity. (4) *Informativeness:* When appropriately designed, CP's prediction set or interval can be sufficiently small to provide meaningful information about the original outputs.

1.1 CP from a Data Perspective

The increasing prevalence of big data, particularly with richer and more complex modalities, presents unique challenges for CP beyond traditional statistical settings. For instance: (1) **Generative models in natural language and image processing**: In these contexts, the target output can be ultra-high-dimensional or even infinite. For example, the output of a language model can be a permutation of any tokens in the pre-training corpus with arbitrary length , unlike the univariate real-valued responses in classical regression or classification tasks. For complex tasks like open-domain **question answering** (**QA**), the output space is often unbounded [60, 170] (i.e., all possible text sequences). Adapting CP to handle these multi-dimensional and even infinite-dimensional outputs is essential to improve its applicability and predictive accuracy. (2) **Time series and streaming data**: These types of data are becoming increasingly important in fields such as economics, meteorology, geosciences, urban planning, and healthcare. However, time-dependent data poses significant challenges for CP, particularly because the assumption of data exchangeability is often violated in the presence of distribution shifts or temporal dependencies. (3) **Structured data** (e.g., groups, trees, and graphs): Structured data is typically quantitative and organized in predefined formats like rows and columns in relational databases while unstructured data (like text) is often qualitative and lacking a fixed schema [176, 210]. In ML applications involving structured data, canonical CP methods may fail to capture the intrinsic dependencies when the feature structure is ignored. Traditional inferential methods often rely on distributional assumptions specific to structured data. Thus, developing model-agnostic UQ tools that leverage implicit (or weighted) exchangeability within the data structure is crucial.

These challenges motivate this survey on CP *from a data perspective*, which introduces CP methods tailored to different data categories, as outlined in Figure 1. We categorize data into two main types based on the collection process: static data and dynamic data. Static data is further divided into structured data—including matrix/tensor data, hierarchical data, tree-structured data, and graph data—and unstructured data including image, text, and multi-modal data. Dynamic data is focused on spatio-temporal data, which is further subdivided into streaming, one-dimensional, and multi-dimensional categories. Note that dynamic data can be also structured, e.g., time-series data. This hierarchical organization provides a clear and systematic framework for understanding various data types, allowing for the appropriate application of CP techniques.

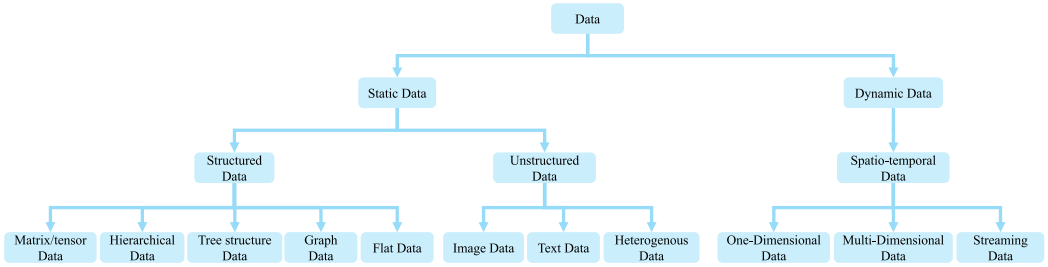

Fig. 1. The proposed taxonomy for CP from a data perspective.

1.2 Comparison with Existing Surveys

There are several existing surveys that either cover the theoretical foundations of CP with a review of more advanced algorithm developments (e.g., [9]) or review CP methods for a specific data type (e.g., [173]). Particularly, [9, 61] and [8] introduce theoretical foundations of CP including split CP, full CP, conformalized quantile regression, as well as recent advanced applications of CP. The survey paper [173] focuses on the application of CP in the specific field with spatio-temporal data and [26] centers around the field of natural language processing (NLP). [115] focuses CP's application in autonomous systems, dealing with dynamic data by having trajectory calibration data. However, in light of the increasingly richer data types and modalities, a comprehensive and structured overview of CP from a data perspective is urgently needed to comprehensively understand the unique challenges of CP adapted to various data types. In this survey, we propose a novel taxonomy of CP from a data perspective (Figure 1) and review CP approaches and applications for structured data, unstructured data, and spatio-tempoal data. We will start with a background introduction to basic CP and then provide a thorough discussion of CP across different tasks and data types, examining how the nature of the data influences the effectiveness of CP methods. Lastly, we summarize several open challenges and promising future directions that hope to spark future research in CP.

2 Background of CP

In this section, we first introduce basic CP approaches and important variants and then summarize common metrics in evaluating UQ methods. The major notations used throughout the survey are summarized in Table 2 in the supplementary.

2.1 Full CP

Let X and Y denote features and label, respectively. Taking the regression problem as an example, consider observed data $(X_1, Y_1), \ldots, (X_n, Y_n)$ and the test data (X_{n+1}, Y_{n+1}) with Y_{n+1} unobserved. Our goal is to construct a prediction set for Y_{n+1}. Suppose that (X_i, Y_i), $i = 1, \ldots, n+1$, are exchangeable (a relaxed assumption of i.i.d.[1]), i.e., the joint distribution remains unchanged under any permutation, we have the property such that for any score function $V : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$,

$$\left\{V(X_{n+1}, Y_{n+1}) \mid \{V(X_i, Y_i)\}_{i=1}^{n+1} = \{v_i\}_{i=1}^{n+1}\right\} \sim \text{Uniform}(\{v_i\}_{i=1}^{n+1}), \tag{1}$$

which implies that conditioning on the realizations of the $n+1$ scores $\{v_i\}_{i=1}^{n+1}$, the test score $V(X_{n+1}, Y_{n+1})$ can be viewed as a draw from the uniform distribution over all the realized scores. The score function, also referred to as the *non-conformity score*, quantifies how "a typical" a

[1]Note that exchangeability does not imply independence. For example, consider variables $X_i+\varepsilon, i \in [n]$, where $X_1, \ldots, X_n$ are i.i.d. and ε is a shared additive term with $\text{Var}(\varepsilon) > 0$. Then, $X_i + \varepsilon$'s are exchangeable but the dependence is not zero.

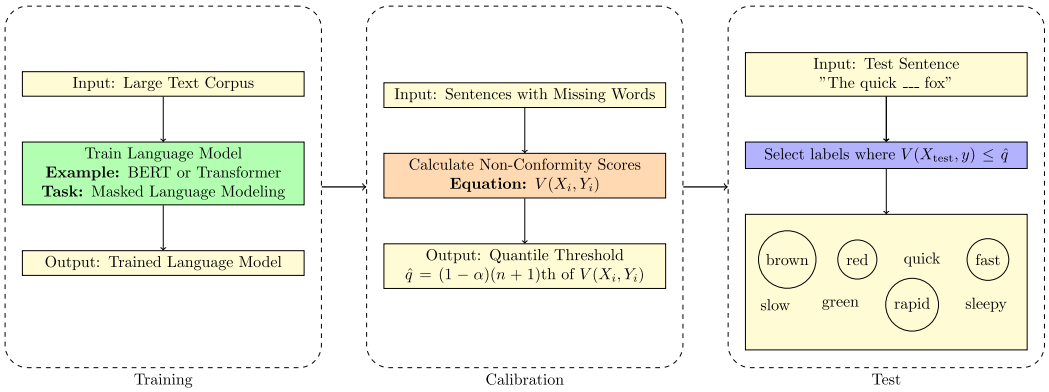

Fig. 2. Example of Split CP for Text Infilling. In this task, the model predicts missing words in a sentence, and Split CP is used to create prediction sets that quantify the uncertainty of these predictions. $V(X_i, Y_i) = 1 - P(Y_i|X_i)$. where P represents the predictive probability output by the trained base model.

prediction is relative to the true value. Then, denoting the $(1-\alpha)$th-quantile of the uniform distribution over $\{V(X_i, Y_i))\}_{i=1}^{n+1}$ by $Q_{1-\alpha}$, based on the property of the uniform distribution, we have

$$\mathbb{P}\left(V(X_{n+1}, Y_{n+1}) \leq Q_{1-\alpha}\right) \geq 1 - \alpha. \tag{2}$$

However, as Y_{n+1} is not observed, consider plugging in $Y_{n+1} = y$ in model fitting. Then, we denote $V^y(X, Y)$ as the score function trained with $Y_{n+1} = y$ and $Q^y_{1-\alpha}$ as the $(1-\alpha)$th-quantile of the uniform distribution over $\{V^y(X_{n+1}, y)\} \cup \{V^y(X_i, Y_i))\}_{i=1}^{n}$. Consider the following set

$$C_{1-\alpha}(X_{n+1}) = \{y \in \mathcal{Y} : \ V^y(X_{n+1}, y) \leq Q^y_{1-\alpha}\}, \tag{3}$$

which, according to Equation (2), satisfies that $\mathbb{P}(Y_{n+1} \in C_{1-\alpha}(X_{n+1})) = \mathbb{P}\left(V^{Y_{n+1}}(X_{n+1}, Y_{n+1}) \leq Q_{1-\alpha}\right) \geq 1 - \alpha$. This indicates that with a probability no less than $1-\alpha$, the response Y_{n+1} is included in the constructed set. A common choice of the score function is $V^y(X, Y) = |Y - f^y(X)|$, where $f^y(X)$ is some fitted estimate of $\mathbb{E}[Y \mid X]$ trained with $Y_{n+1} = y$. In Figure 2, this process is demonstrated for a text infilling task without calibration. Here, the score function measures the model's uncertainty in predicting missing words on training data itself, and the resulting prediction set captures multiple plausible completions while maintaining the required coverage guarantee.

2.2 Split CP and Variants

However, as we may have seen from above, full CP generally requires model re-fitting for all the values $y \in \mathcal{Y}$, which is infeasible in general scenarios with $|\mathcal{Y}| = \infty$. For computational efficiency, various variants of full CP are proposed, as detailed below.

Split CP One of the most common approaches in CP is *split CP* [105, 139], which begins by pre-fitting a model f and subsequently assessing its efficacy using a separate calibration set of labeled examples $\{(X_i, Y_i)\}_{i=1}^{n}$, with which we obtain non-conformity scores $\{V_i = V(X_i, Y_i)\}_{i=1}^{n}$.

Central to this method is the foundational assumption of exchangeability and the independence between the training and calibration sets. Based on the scores, we can construct the prediction set

$$C(X_{n+1}) = \left\{y \in \mathcal{Y} : \ V(X_{n+1}, y) \leq \widehat{Q}_{1-\alpha}\right\}, \tag{4}$$

where the quantile $\widehat{Q}_{1-\alpha}$ can be directly obtained from the non-conformity scores:

$$\widehat{Q}_{1-\alpha} = \text{the } \lceil (1-\alpha)(n+1) \rceil \text{th smallest } V_i. \tag{5}$$

The coverage guarantee of split CP ensures that the true value of a new observation falls within the predicted interval with high probability, i.e., $\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$. To illustrate this process, Figure 2 provides a concrete example of Split CP applied to a text infilling task. The figure highlights how the calibration process enables a trained model to compute non-conformity scores and subsequently generate prediction sets with coverage guarantees.

Split CP involves just a single model fitting step, which, while computationally efficient, can lead to lower accuracy. Due to the reduction of sample size. In contrast, full CP achieves high statistical efficiency by requiring an extensive number of model fittings [9, 105]. However, these two methods represent extremes on a spectrum of possibilities. There are intermediate techniques that balance statistical and computational efficiency in different ways.

CP with cross-validation As an extension of split CP, [184, 187] proposed cross-CP, which involves splitting the training dataset into K disjoint subsets, $S_1, \ldots, S_K$, each of equal size $m = n/K$, where n is divisible by K. For each subset S_k, a regression model f_{-S_k} is trained on the complement of S_k in the dataset. These models are then used to predict the outcomes of the samples within their respective subsets, resulting in a set of cross-validation residuals, e.g.

$$V_i^{\mathrm{CV}} = |Y_i - f_{-S_{k(i)}}(X_i)|, \tag{6}$$

where $k(i)$ identifies the subset containing instance i. Consider the text infilling task, the text corpus is split into k disjoint subsets of sentences with missing words. For each subset, a language model is trained on the other $k - 1$ subsets and then used to predict the missing words in the held-out subset. However, the finite-sample validity of the cross-conformal predictors [184, 187] at the level of $1 - \alpha$ may not hold. To address this issue, [15] proposed CV+ prediction intervals unsymmetrical around the prediction for X_{n+1} but have the finite-sample coverage at least $1 - 2\alpha$.

CP with jackknife Jackknife-based CP can be viewed as a specific example of cross-validated CP with $K = n$. Barber et al. [15] propose CP with jackknife+, which relieves the overfitting of canonical jackknife intervals and exhibits a finite-sample guarantee. Concretely, with **leave-one-out** (**LOO**) predictors f_{-i}, $i \in [n]$, one can calculate the residuals $V_i^{\mathrm{LOO}} = |Y_i - f_{-i}(X_i)|$ and construct the prediction interval similar with Equation (6) by defining $k(i) = i$. In the text infilling task, Jackknife+ trains a language model by leaving out one sentence at a time, computes residuals from the model's predictions, and then uses these residuals to generate prediction sets for new inputs.

This modification is crucial in scenarios where predictions are sensitive to slight data variations, providing intervals that more accurately reflect the model's uncertainty. The Jackknife+ prediction interval satisfies the following coverage guarantee:

$$\mathbb{P}\{Y_{n+1} \in \widehat{C}_{n,1-\alpha}^{\text{jackknife+}}(X_{n+1})\} \geq 1 - 2\alpha. \tag{7}$$

Moreover, jackknife-minimax builds on the foundation of Jackknife+ by adopting a more conservative approach, aiming to remove the factor of two from the theoretical error bound to achieve tighter coverage control as is shown in [15]. Tigher coverage guarantees around $1 - \alpha$ are also presented in [15] when algorithmic stability is assumed. Alternatives to relieve the computational costs of full CP are also explored in the literature. For example, [127] proposes approximate full CP and utilizes influence functions to construct easier-to-compute non-conformity scores.

2.3 Weighted CP

The exchangeability assumption is crucial in ensuring the finite-sample guarantee of CP. However, in many practical scenarios with population heterogeneity or distribution shift, this assumption can be violated. Tibshirani et al. [180] extended the principle of exchangeability to a weighted setting, providing a theoretical foundation for weighted CP when the data are weighted exchangeable.

More concretely, consider the setting with covariate shift:

$$(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} P = P_X \times P_{Y|X}, \quad i \in [n], \qquad (X_{n+1}, Y_{n+1}) \sim \widetilde{P} = \widetilde{P}_X \times P_{Y|X}. \tag{8}$$

The data are weighted exchangeable in the sense that, with the weight function $w(x) = \frac{d\widetilde{P}_X}{dP_X}(x)$,

$$\left(V(X_{n+1}, Y_{n+1}) \mid \{V(X_i, Y_i)\}_{i\in[n+1]}\right) \sim \sum_{i=1}^{n+1} \frac{w(X_i)}{\sum_{j=1}^{n+1} w(X_j)} \delta_{V(X_i,Y_i)}, \tag{9}$$

where δ is the point mass function. This means that given the observed values $\{V(X_i, Y_i)\}_{i\in[n+1]}$, the new data point (X_{n+1}, Y_{n+1}) is considered to be drawn from a weighted combination of the empirical distribution of the observed data. Weighted CP has been used across various fields such as experimental design [54], survival analysis [27], and causal inference [108]. Subsequent studies have primarily focused on tackling the challenge of accurately estimating high-dimensional likelihood ratios to ensure effective coverage in complex data settings. For instance, Barber et al. [16] extend it to general non-exchangeable settings, Bhattacharyya and Barber [22] enhance coverage guarantees in scenarios where observations belong to distinct groups, and Liu et al. [116], Lu and Kalpathy-Cramer [119], Ying et al. [202] bolster the informativeness of weighted CP when facing covariate shifts across multiple data sources. To address the problem with distribution shift, there is another line of research focusing on calibrating the confidence level α to maintain validity for the worst-case distribution [1, 28, 76], which is known as distributionally robust methods.

2.4 Conformal Risk Control

Conformal risk control [10] is a more general framework compared to traditional CP. While CP focuses on constructing prediction sets with guaranteed coverage probability, conformal risk control expands this idea to allow for more flexible risk management, considering different risk functions. Given a parametrized family of prediction sets $\{C_\lambda(x) : \lambda \in \Lambda\}$, e.g., $C_\lambda(x) = \{y \in \mathcal{Y} : V(x, y) \le \lambda\}$ with a given non-conformity score function V, the problem of constructing a valid prediction set boils down to the calibration of the parameter λ. The main idea is to choose λ such that the average miscoverage in the calibration set is controlled, i.e.,

$$\widehat{\lambda} = \inf\left\{\lambda \in \Lambda : \frac{1}{n_{\text{cal}}+1} \sum_{i=1}^{n_{\text{cal}}} \mathbb{1}\{Y_i \notin C_\lambda(X_i)\} + \frac{1}{n_{\text{cal}}+1} \le \alpha\right\}. \tag{10}$$

Then, the prediction set $C_{\widehat{\lambda}}(x)$ satisfies that $\mathbb{P}(Y_{n+1} \in C_{\widehat{\lambda}}(X_{n+1}) \ge 1 - \alpha$. Extensions to the generic loss function $\ell(x, y)$ are also presented in [10]. Similar ideas are also explored in [141–143], in which PAC-type control of coverage is established.

2.5 Evaluation Metrics

In this section, we present a summary of common validity and accuracy guarantees for CP under a supervised setting. We denote $\widehat{C}_{n,1-\alpha}$ as the prediction set based on the previous n samples $\{(X_i, Y_i)\}_{i\le n}$ at the level of $1 - \alpha$ and consider the following metrics.

2.5.1 Marginal Coverage. The most basic guarantee for a prediction interval is the finite-sample marginal coverage such that for the test sample X_{n+1}, one has

$$\mathbb{P}(Y_{n+1} \in \widehat{C}_{n,1-\alpha}) \ge 1 - \alpha, \tag{11}$$

where the probability is with respect to the randomness of $\{(X_i, Y_i)\}_{i\le n+1}$. Furthermore, to obtain tight coverage, it usually requires that $|\mathbb{P}(Y_{n+1} \in \widehat{C}_{n,1-\alpha}) - (1 - \alpha)| = O(n^{-1})$. CP is shown to meet both requirements with exchangeable data [3, 106, 162, 186].

2.5.2 Adversarial and Instantaneous Coverage in Online Setting. In the online setting with spatio-temporal data, where data arrive sequentially and are not guaranteed to have the same distribution (see more details in Section 5), the exchangeability assumption is violated, thus the worst-case or adversarial coverage is considered in the literature.

Adversarial marginal coverage. The worst-case coverage is the *mean prediction interval coverage* averaged over all time points, which can be written as $T^{-1}\sum_{t=1}^{T}\mathbb{1}\{Y_t \in \widehat{C}_t(X_t)\}$, where T is the total number of time points and $\widehat{C}_t(\cdot)$ is the prediction interval constructed at time t, and 0 otherwise [16, 21, 68, 88, 112, 166, 169, 174, 192, 194, 196, 200, 204].

Instantaneous coverage. The worst-case marginal validity holds without distributional assumption for **Adaptive conformal Inference** (**ACI**)-type approaches, or alternatively, can be viewed as the property of a fixed sequence. To show the adaptivity of online CP approaches, [7] also validates that the ACI-type has instantaneous validity under the i.i.d. assumption with sufficiently large t. With a fitted non-conformity score s_t, [7] consider the asymptotic control of the coverage rate at time t: $\text{Coverage}_t(q) = \mathbb{P}(V_t(X, Y) \leq q \mid V_t)$.

2.5.3 Conditional Coverage. Conditional coverage is a stronger validity criterion. For example, if we are interested in a subpopulation determined by a function $g(x, y)$, it leads us to prediction intervals satisfying the conditional coverage guarantee such that for all $\widetilde{g}$,

$$\mathbb{P}\left(Y_{n+1} \in \widehat{C}_{n,1-\alpha}(X_{n+1}) \mid g(X_{n+1}, Y_{n+1}) = \widetilde{g}\right) \geq 1 - \alpha. \tag{12}$$

In the following subsections, we consider different choices of the function g in practice.

Covariate-conditional coverage. A natural partition of the entire space is according to the realization of the test feature, i.e., $g(x, y) = x$, in which case we seek the covariate-conditional (or local) coverage. Unfortunately, when the marginal distribution of X has no point mass, this guarantee is shown to be impossible in [107, 186]. Various kinds of relaxation of the exact conditional coverage are explored in the literature (see [62, 73, 84, 105, 152]).

Class-conditional coverage. In the task of classification, it is of interest to provide valid prediction interval conditioning on each class, i.e., $g(x, y) = y$, in which we consider the label conditional guarantee. In addition to the Mondrian CP in [45, 186] archives class-conditional validity in the setting where the number of classes is large such that effective samples in each class may not be sufficient. Empirical studies are conducted in [74, 82, 117, 156, 164, 172].

Training-conditional coverage. Since the coverage probably has high variability in the training data, it drives us to consider the coverage guarantee conditional on the training set $\mathcal{D}_n$:

$$\mathbb{P}\left(\alpha(\mathcal{D}_n) \geq \alpha + \varepsilon_n\right) \leq \delta_n, \text{ where } \alpha(\mathcal{D}_n) = \mathbb{P}\left(Y_{n+1} \notin \widehat{C}_{n,1-\alpha}(X_{n+1}) \mid \mathcal{D}_n\right), \tag{13}$$

and $\varepsilon_n, \delta_n \to 0$ as $n \to +\infty$. Specifically, the marginal coverage can be written as $\mathbb{E}[\alpha(\mathcal{D}_n)] \leq \alpha$.

2.5.4 Prediction Efficiency. The efficiency of the prediction set/interval measures its size or length $\|\widehat{C}_{n,1-\alpha}\|$. It is also common to compare with an oracle prediction set $C^*_{1-\alpha}$ under some specified models, see e.g., [105]. More criteria to measure prediction efficiency are discussed in [185]. For example, one can evaluate the second-highest p-values or the sum of all p-values excluding the largest one, as these metrics reflect the distribution of confidence across potential labels. In the online setting, the mean prediction interval width, rolling width, and average median length of the prediction intervals are considered. To address the issue of infinite-length intervals, it is suggested to measure the *fraction of infinite-length prediction intervals* [200] and analyze the *75%, 90%, and 95% quantiles of the set sizes* [6] to understand the distribution of interval lengths better.

2.5.5 Hybrid Metric. There are also several metrics that consider validity and efficiency together. [113] introduces the *inverse coverage efficiency*, which is the average prediction interval length divided by the marginal coverage rate (the smaller, the better). [88] considers the ***coverage width-based criterion (CWC)*** that penalizes under—and overcoverage in a similar way. Additional metrics that aim at balancing validity and accuracy are explored in [14, 166].

3 Static Data–Structured Data

3.1 Flat Data

We start with the scenario where data are numeric and assumed to be independently and identically distributed (i.i.d or *flat*), which is common in regression/classification problems.

3.1.1 Regression. Consider observed covariates $\{X_i\}_{i=1}^n \subseteq \mathbb{R}^d$ and response $\{Y_i\}_{i=1}^n \subseteq \mathbb{R}$, where we assume $\{(X_i, Y_i)\}_{i=1}^n$ are i.i.d. Then, according to the previous section, for any non-conformity score $V(x, y)$, the prediction set defined in Equation (4) is valid. Particularly, for the regression problem, the score $V(x, y) = |y - \widehat{f}(x)|$ is widely adopted [105, 162, 186] with a fitted function f, with which the prediction interval takes the form $C_{1-\alpha}(X_{n+1}) = f(X_{n+1}) \pm \widehat{Q}_{1-\alpha}$. Note that the prediction interval has an equalized width for different X_{n+1}. To incorporate the local uncertainty of covariates when the noise distributions are heteroscedastic, a locally weighted score is proposed [105, 139]:

$$V(x, y) = \frac{|y - \widehat{f}(x)|}{\widehat{\sigma}(x)}, \tag{14}$$

where the local uncertainty score $\widehat{\sigma}(x) > 0$ is an estimated standard error of the residual. Variants of the local uncertainty score are proposed and applied to various scenarios [39, 83, 138].

However, as the distribution of $(Y \mid X)$ may not lie in the location-scale family such that $Y = f(X) + \sigma(X) \cdot e$, the prediction interval based on the locally weighted score can lose efficiency when dealing with more complicated distributions. Another notable line of research focuses on non-conformity scores based on more general distributional quantities such as estimated quantiles and **cumulative density functions** (**CDF**), which are introduced as follows.

Conformalized quantile regression(CQR). Even when the distribution does not belong to the location-scale family, the quantile function can capture the intrinsic characteristics (e.g., the shape) of the distribution. [152] proposes CQR where the prediction interval is defined as

$$\left[\widehat{Q}_{\alpha/2}(X_{n+1}) - \eta_{1-\alpha},\ \widehat{Q}_{1-\alpha/2}(X_{n+1}) + \eta_{1-\alpha}\right], \tag{15}$$

where $\widehat{Q}$ is the estimated quantile based on data splitting and on the calibration set with sample size n_{cal}, $\eta_{1-\alpha}$ is the $(1-\alpha)(1+1/n_{\text{cal}})$th-quantile of

$$V_i = \max\left\{\widehat{Q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \widehat{Q}_{1-\alpha/2}(X_i)\right\}, \qquad 1 \le i \le n_{\text{cal}}. \tag{16}$$

When the estimated quantile is accurate, CQR also exhibits good local performance and to improve local coverage, further extensions are proposed in [4, 154, 160, 167].

Distributional CP. In comparison, with the same motivation to capture the characteristics of the distribution beyond the location-scale family and achieve better conditional performance, distributional CP [35] considers the scores based on an estimated CDF. Consider the data splitting case where $\widehat{F}(x, y)$ is an estimated CDF using a hold-out set. [35] defines the scores as

$$V_i = \psi\left(\widehat{F}(X_i, Y_i)\right), \qquad 1 \le i \le n_{\text{cal}}, \tag{17}$$

where ψ is a user-specified scalar function to enhance accuracy and the default choice is $\psi(t) = |t - 1/2|$. When the CDF is estimated accurately, distributional CP is also shown to have an asymptotic conditional coverage guarantee.

Discussion. CP with flat data in the regression setting mainly focuses on designing non-conformity scores to achieve better local coverage. Further discussion is presented in Section 2.5.3.

3.1.2 Classification. To apply CP to classification problems with K classes such that $Y \in [K]$, a widely adopted non-conformity score is the estimated likelihood [156, 162]. Taking split CP as an example where we have $\widehat{p}(x;k)$ as an estimate for $\mathbb{P}(Y = k \mid X = x)$, then with the non-conformity score $V_i = \widehat{p}(X_i; Y_i)$, the prediction set takes the form

$$C_{1-\alpha}(X_{n+1}) = \left\{k : \widehat{p}(X_{n+1};k) \geq \widehat{Q}_{1-\alpha}\right\}, \tag{18}$$

where $\widehat{Q}_{1-\alpha}$ is the $\lfloor \alpha \cdot (1 + n_{\text{cal}}) \rfloor$th-smallest of $\{V_i\}_{i=1}^{n_{\text{cal}}}$.

A modified approach is proposed in [153] where the non-conformity score takes the form

$$\widetilde{V}_i = \sum_{k \in \mathcal{S}_i} \widehat{p}(X_i;k), \qquad \mathcal{S}_i = \{k : \widehat{p}(X_i;k) \geq \widehat{p}(X_i;Y_i)\}, \tag{19}$$

which is a score that tends to be small for the more "likely" labels for a certain x. Let $\widehat{Q}_{1-\alpha}$ be the $\lceil (1-\alpha)(1+n_{\text{cal}}) \rceil$th-smallest of $\{\widetilde{V}_i\}_{i=1}^{n_{\text{cal}}}$ and $\{\pi(j;x)\}_{j=1}^{K}$ be permutation of $[K]$ such that

$$\widehat{p}(x;\pi(1;x)) \geq \widehat{p}(x;\pi(2;x)) \geq \cdots \geq \widehat{p}(x;\pi(K;x)). \tag{20}$$

Then, the prediction set takes the form

$$C_{1-\alpha}(X_{n+1}) = \{\pi(k;X_{n+1}) : 1 \leq k \leq k(X_{n+1})\}, \tag{21}$$

where $k(x) = \min\{k : \sum_{j=1}^{k} \widehat{p}(x;\pi(j;x)) \leq \widetilde{Q}_{1-\alpha}\}$ and $\widetilde{Q}_{1-\alpha}$ is the $\lceil (1-\alpha)(1+n_{\text{cal}}) \rceil$th-smallest of $\{\widetilde{V}_i\}_{i=1}^{n_{\text{cal}}}$. In [29], a non-conformity score based on estimated quantiles, which is similar to CQR, is adopted for better local coverage performance. More recent work [85] further improves the proposed non-conformity score via label ranking. [52] calibrates the output probability of deep neural networks for more informative prediction sets.

Ordinal classification. When the labels are ordinal, the prediction set is required to be contiguous. However, the prediction sets constructed above for general classification problems do not have this property unless the estimated conditional distribution $\widehat{P}_{Y|X}$ is unimodal. Motivated by this property, [43] proposed a procedure that constructs unimodal classifiers. In addition, [30] adopts conformal p-values to determine the lower and upper bounds for the contiguous prediction set for the ordinal responses, and [198] applies the conformal risk control framework to this setting. Conformalized ordinal classification has applications in disease severity rating [30, 43, 118], age recognition [43, 198], and historical image dating [43], and so on.

Multi-label classification. Maxime Cauchois et al. [29] apply CP on multi-class and multi-label image classification. One key challenge here is that typical CP methods, which give marginal validity (coverage) guarantees–provide uneven coverage, in that they address easy examples at the expense of essentially ignoring difficult examples. They build tree-structured classifiers that efficently account for interactions between labels. To asymptotically get conditional coverage for both multi-class and multi-label prediction, they adopt the idea from quantile regression [152].

Discussion. Conditional coverage is also of concern in classification problems, which motivates a series of works [29, 153] in constructing scores similar to conditional quantiles. In addition, UQ for ordinal classification with additional rank constraints on the labels, which lies between regression and classification with unordered labels, is still an active field. Selective classification

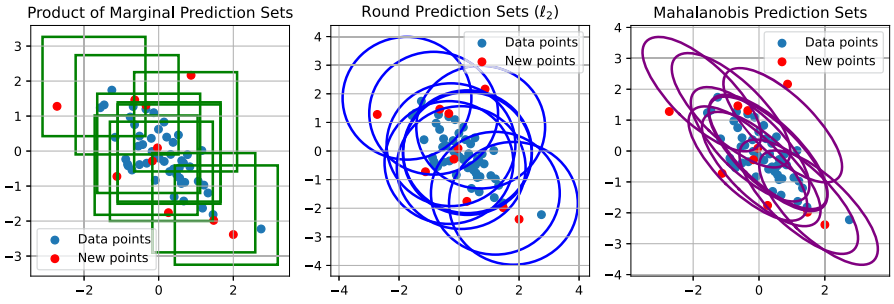

Fig. 3. Illustration of **conformal prediction (CP)** sets with bivariate response.

is also of interest in ML, where more cautious prediction sets with the ability to abstain are constructed [82]

3.1.3 Multivariate Response. As an extension to CP with univariate response Y, the setting with a multivariate response is also of interest in the scenario where $Y \in \mathbb{R}^k$ is a k-dimensional series for each data point. Although the validity of CP still holds with properly defined non-conformity scores, the specification of the scores is a crucial question to determine the shape of the prediction set for better accuracy. A naive approach is to construct the CP prediction interval $\widehat{C}_j(X_{n+1})$ at the level of $1-\alpha_j$ for $Y_{n+1,j}$ for each $j \in [k]$ as is depicted in Figure 3 (left panel). Then, by the union bound, one has $\mathbb{P}\{Y_{n+1} \in \widehat{C}_1(X_{n+1}) \times \cdots \times \widehat{C}_k(X_{n+1})\} \geq 1 - \sum_{j=1}^k \alpha_j$. Specifically, when coordinates in Y are independent, we can further have $\mathbb{P}\{Y_{n+1} \in \widehat{C}_1(X_{n+1}) \times \cdots \times \widehat{C}_k(X_{n+1})\} \geq \prod_{j=1}^k (1-\alpha_j)$, which has much tighter overage. To achieve a middle-ground between the independent and adversarially dependent settings, [129] and [174] further consider the copula transformation to achieve independent coordinates and apply an inverse map to obtain the predict set for the multivariate response. In comparison, [95] extends the score $|Y - \widehat{f}(X)|$ to $\|Y - \widehat{f}(X)\|$ with different choices of the norm $\|\cdot\|$ in $\mathbb{R}^k$ and define the prediction set by $\{y \in \mathbb{R}^k : \|y - \widehat{f}(X_{n+1})\| \leq \widehat{q}_{1-\alpha}\}$ with calibrated quantile $\widehat{q}_{1-\alpha}$ for the scores. See Figure 3 (middle and right panels) for prediction sets with two different choices of norms: $\|Y - \widehat{f}(X)\|_2$ and $\|Y - \widehat{f}(X)\|_\Omega = (Y - \widehat{f}(X))^\top \Omega^{-1} (Y - \widehat{f}(X))$, where Ω is an estimate of the covariance matrix of residual vectors. Other extensions to the setting with multivariate responses are discussed in [44, 47, 56, 192].

3.1.4 Functional Data. CP with functional data can be viewed an a further extension to the previous section with multivariate responses since the response here is a function in an infinite-dimensional space. Consider the setting with random functions $X_1(\cdot), \ldots, X_n(\cdot) \in \Omega$ with domain $\mathcal{T}$. The goal is to find a prediction set $C_{1-\alpha} \subseteq \Omega$ such that $\mathbb{P}(X_{n+1} \in C_{1-\alpha}) \geq 1-\alpha$. For brevity, we consider the split CP. To solve this problem, a naive trial is to find a predefined function $g : \Omega \to \mathbb{R}$, e.g., $g(X) = -\sup_{t \in \mathcal{T}} |X(t)|$, and then define $C_{1-\alpha} = \{x \in \Omega : g(x) \geq \lambda\}$. Here λ is $(\lceil \alpha(n_{\text{cal}}+1) \rceil - 1)$th-smallest among $\{g(X_i)\}_{i=1}^{n_{\text{cal}}}$. However, this criterion is too ambitious and may lose efficiency. In [106], a modified criterion is considered: $\mathbb{P}(\Pi(X_{n+1}) \in C_{1-\alpha}) \geq 1-\alpha$, where Π maps functions in Ω to a finite dimensional space $\Omega_p \subset \Omega$. More concretely, [106] considers the prediction bands $B(t) = [l(t), u(t)]$, $t \in \mathcal{T}$, such that $\mathbb{P}(l(t) \leq X_{n+1}(t) \leq u(t),\ \forall t \in \mathcal{T}) \geq 1-\alpha$. The proposed approach projects functions to vectors $\{\xi_i\}_{i \leq n} \subseteq \mathbb{R}^p$ in the Euclidean space via a basis $\{\phi_j\}_{j=1}^p$. Then, with non-conformity scores $V(\xi_i)$, $i \in [n_{\text{cal}}]$ and

$$\mathcal{E}_n = \left\{\zeta \in \mathbb{R}^p : V(\zeta) \geq V_{(\lceil \alpha(n_{\text{cal}}+1) \rceil - 1)}\right\}, \tag{22}$$

where $V_{(k)}$ is the kth order statistic of $\{V(\xi_i)\}_{i=1}^{n_{\text{cal}}}$, one can construct the prediction band by $B(t) = \{\sum_{j=1}^{p} \zeta_j \phi_j : \zeta \in \mathcal{E}_n\}$. In practice, the non-conformity score $V(\cdot)$ is chosen to be the Gaussian mixture density estimator with K components. [46] constructs prediction bands for X_{n+1} without projection by considering the non-conformity score

$$V(X_i) = \sup_{t \in \mathcal{T}} \left| \frac{X_i(t) - \widehat{G}(t)}{\widehat{s}(t)} \right|, \quad i = 1, \ldots, n_{\text{cal}}, \tag{23}$$

where $\widehat{G}(t)$ is the mean of $X_i(t)$ in the hold-out training set and $\widehat{s}(t)$ is designed to penalize extreme values in [46]. This procedure is further extended to the multivariate functional setting [47].

3.1.5 Data with Noise, Missingness, or Censoring. The ideal exchangeability assumption can be violated in many scenarios, among which data corruption is a common setting in statistics. In this section, we summarize related works on CP with flat data but either the covariates or the response is not directly observed, which can be missing or perturbed.

Covariate noise. The simplest setting with data corruption is the case with noisy covariates for the test data, which may happen when measurement errors are taken into account [38] or perturbations are applied to enhance privacy [51]. Consider the case where we observe $\{(X_i, Y_i)\}_{i=1}^{n}$ and a new sample with noisy covariates $\widetilde{X}_{n+1}$. Gendler et al. [65] adopt randomized smoothing to construct a non-conformity score $\widetilde{V}(x, y)$ which satisfies that

$$V(\widetilde{X}_{n+1}, y) \leq V(X_{n+1}, y) + M_\delta, \qquad \forall y \in \mathcal{Y}, \quad \text{almost surely}, \tag{24}$$

where M_δ is a constant. Concretely, given any $V(x, y)$, the score $\widetilde{V}(x, y)$ can be constructed by

$$\widetilde{V}(x, y) = \Phi^{-1}\left(\mathbb{E}_{v \sim \mathcal{N}(0_d, I_d)}[V(x + v, y)]\right), \tag{25}$$

where Φ is the CDF of $\mathcal{N}(0_d, I_d)$. A more recent work [67] only requires $V(\widetilde{X}_{n+1}, y) \leq V(X_{n+1}, y) + M_\delta$ to hold with a high probability and proposes an adaptive approach to achieve improved trade-offs between performance on clean data and robustness to adversarial examples based on the principle of quantile-of-quantile design. [205] further considers the performance of CP under poisoning/evasion attack.

Label noise. [57] focuses on the setting where we observe $\{(X_i, g(Y_i))\}_{i=1}^{n}$ with a corruption function $g(\cdot)$ and the goal is to construct a valid prediction interval for Y_{n+1}. For any $u \in (0, \alpha - 1/(n+1))$, if the test score $V_{n+1} = V(X_{n+1}, Y_{n+1})$ and the corrupted score $\widetilde{V}_{n+1} = V(X_{n+1}, \widetilde{Y}_{n+1})$ satisfy

$$\mathbb{P}(\widetilde{V}_{n+1} \leq t) + u \leq \mathbb{P}(V_{n+1} \leq t), \tag{26}$$

then with the protection interval $\widetilde{C}_{1-\alpha}(\cdot)$ based on $(X_i, g(Y_i))$'s, it holds that

$$\mathbb{P}(Y_{n+1} \in \widetilde{C}_{1-\alpha}(X_{n+1})) \geq 1 - \alpha + \frac{1}{n+1} + u. \tag{27}$$

Moreover, if the noisy distribution of $\widetilde{Y} = g(Y)$ satisfies that $d_{\text{TV}}(Y, \widetilde{Y}) \leq \varepsilon$, one has

$$\mathbb{P}(Y_{n+1} \in \widetilde{C}_{1-\alpha}(X_{n+1})) \geq 1 - \alpha + \frac{1}{n+1} + \frac{n}{n+1}\varepsilon, \tag{28}$$

which shows CP sets are robust when the strength of label noise is mild. In the study [161], they propose a novel CP method that adjusts for label contamination through a precise characterization of the effective coverage inflation or deflation caused by label contamination. Recently, a non-conformity score that interpolates noisy and clean data to achieve robustness is proposed [144].

Covariate missingness. In [203], the setting with missing coordinates in X_i's is considered, which can be viewed as a specific case of noisy covariates. For example, the test data can take the form $X_{n+1} = (X_{n+1,1}, \cdot, \cdot, X_{n+1,4})$, where the second and third entries are unobserved. Solutions

are proposed in [203] and the main idea is to artificially create missingness in the calibration dataset such that the second and third entries in calibration data are also missing. In the first approach, calibration data with additional missingness (e.g., $X_j = (X_{j,1}, \cdot, \cdot, \cdot)$ will be removed, in which case the exchangeability is recovered. To relieve the issue of sample size insufficiency, [203] also suggests that we could impute X_j to obtain $X_j = (X_{j,1}, \cdot, \cdot, \widehat{X}_{j,4})$ for CP. Another line of work focusing on data (e.g., matrix or tensor) completion will be introduced in Section 3.2.

Causal inference. Causal inference under the potential outcome model [155] can be viewed as an example of the missing response. Consider covariates X_i's and potential outcomes $(Y_i(0), Y_i(1))$ for each unit. A treatment $T_i \in \{0, 1\}$ is assigned to unit i and the strong ignorability assumption is commonly considered: $(Y_i(0), Y_i(1)) \perp\!\!\!\perp T_i \mid X_i$, i.e., no unobserved hidden confounders. The two major challenges of applying CP in causal inference are the covariate shift and inductive bias due to counterfactuals. In [108], under the strong ignorability assumption, prediction intervals are constructed for random variables, such as $Y_{n+1}(1)$ and $Y_{n+1}(1) - Y_{n+1}(0)$, via weighted CP. To see this, for $Y_{n+1}(1)$, the density ratio between the population of interest $P_{Y(1),X|T=1}$ and $P_{Y(1),X}$ only depends on X, which lies in the covariate shift setting. Different choices of weight functions are presented in [108] for various inferential targets. In comparison, [98] adopts CP without reweighting to obtain prediction intervals that are valid conditioning on the treatment T. [33, 89] propose robust CP that maintains valid coverage when the assumption is slightly violated. More concretely, for example, with the bounds constraint of the density ratios, the weight function can be replaced by a lower or an upper bound function in calculating the weighted quantile for weighted CP. In addition, [32, 146, 199] further propose doubly robust results by incorporating the missingness/covariate shift in the framework of non-parametric/semi-parametric statistics and Alaa et al. [3] propose conformal meta learner which combines CP with meta-learner [100] to estimate causal effects.

Censored data. The setting with the censored response is commonly known as survival analysis where X consists of covariates, T is the survival time of interest and C is the censoring time. In practice, only the censored time $\widetilde{T} = \min\{C, T\}$ can be observed. With the goal of constructing a **lower prediction bound** (**LPB**) for T such that $\mathbb{P}(L_{1-\alpha}(X) \leq T) \geq 1 - \alpha$, if we directly treat $\widetilde{T}$ as the response, the LPB for $\widetilde{T}$ is also valid for T. However, when $C < T$ with a high probability, the LPB for $\widetilde{T}$ will be too conservative for T. The main idea in [27, 77] is to find a cutoff $q(X)$ for C and focus on a subpopulation with $C \geq q(X)$, in which the censoring rate is lower. The difference lies in that [27] adopts a fixed cutoff c_0 while [77] considers an adaptive cutoff as a function of X. We note that, under the conditionally independent censoring assumption that $T \perp\!\!\!\perp C \mid X$, the distribution shift from (T, C, X) to $(T, C, X \mid C \geq q(X))$ is covariate shift. Thus, by weighted CP, the LPB for $\widetilde{T}$ constrained on the subpopulation will gain more accuracy.

Discussion. UQ for a wide range of statistical problems with missing/noisy observations can be addressed via (weighted) CP. However, its validity usually depends on the covariate shift condition, which has different variants in specific problems (e.g., the strong ignorability assumption in causal inference and the conditionally independent censoring assumption in survival analysis). Motivated by [16, 89], there is ongoing interest in robustness evaluation or sensitive analysis of (weighted) CP when the covariate shift assumption is violated.

3.2 Other Structured Data

The validity of CP holds with flat data. However, in practice, the i.i.d. assumption is too strong and does not hold in general. This section provides an overview of applications of CP with hierarchical, matrix/tensor, and graph data and additional assumptions required for the exchangeability to hold.

3.2.1 Data with Hierarchical Structure. When data are generated via a hierarchical sampling procedure, it creates a simple form of heterogeneity within data. Consider the setting with K many

groups of observations and $\Pi_k \sim P_\Pi, \quad N_k \mid \Pi_k \sim P_{N|\Pi}$, independently for $k \in [K]$, where N_k is the observed sample size for the kth group. For each group k, we have observations $\{(X_{k,i}, Y_{k,i})\}_{i=1}^{N_k}$. As mentioned in [49], there are two tasks of interest: (1) constructing a prediction set for the new sample $(X_{K+1,1}, Y_{K+1,1})$ drawn from a new group $\Pi_{K+1} \sim P_\Pi$; (2) constructing a prediction set for the new sample $(X_{\text{test}}, Y_{\text{test}})$ from one of existing K groups. Let $\widetilde{Z}_k = (Z_{k,i})_{i=1}^{N_k} = ((X_{k,i}, Y_{k,i}))_{i=1}^{N_k}$ for $k \in [K+1]$. As shown in [49], in this setting, the hierarchical exchangeability condition is assumed as an extension of the generic exchangeability assumption, where not only are observations within each group exchangeable, K groups are also assumed to be exchangeable.

For task (1), four methods are proposed in [49]. The first approach, *pooled CDF*, utilizes the estimated CDF from each group to construct asymptotically valid prediction sets. The second approach, *double conformal*, is similar to the quantile-of-quantile approach and achieves finite-sample validity under the hierarchical exchangeability assumption but tends to be conservative in practice. The third approach, *subsampling once*, samples one sample from each group and constructs the prediction set following the generic CP procedure without the requirement of hierarchical exchangeability. The shortcoming is that this approach does not utilize all the available data and may exhibit high variance. In light of this, *repeated subsampling* repeats the procedure B times to derandomize the subsampling procedure via p-values aggregation.

Another work [22] considers the setting with training and test data drawn from different multinomial distributions, and to account for the heterogeneity in the sampling probabilities, weighted CP is adopted to adapt the test sample to "similar" observations.

3.2.2 Matrix/tensor Data. An important line of research in unsupervised learning focuses on matrix estimation/prediction, e.g., forecasting in panel data with spatio-temporal correlations. In this setting, we have an underlying matrix $M \in \mathbb{R}^{d_1 \times d_2}$ of interest, and each entry (i, j) is missing following a certain pattern. With the random set of observed items $\mathcal{S} \subseteq [d_1] \times [d_2]$, the task is to construct intervals $\{C(i^*, j^*) : (i^*, j^*) \in \mathcal{S}^c\}$ such that $M_{i^*j^*} \in C(i^*, j^*)$ with a high probability.

In [75], it is assumed that the matrix M is deterministic and each entry (i, j) is observed with probability $p_{ij} \in (0, 1)$ independently and weighted exchangeability of indices in $\mathcal{S}_{\text{cal}} \cup \{(i^*, j^*)\}$ conditioning on $\mathcal{S} \setminus \mathcal{S}_{\text{cal}}$ is established for (i^*, j^*) uniformly drawn from $\mathcal{S}^c$, which enables the splitting approach that uses $\mathcal{S} \setminus \mathcal{S}_{\text{cal}}$ for estimating quantities of interest. More concretely, with $\widehat{M}$, $\widehat{p}$, and the local uncertainty score $\widehat{s}$ estimated with $M_{\mathcal{S} \setminus \mathcal{S}_{\text{cal}}}$, we have $V_{ij} = (|M_{ij} - \widehat{M}_{ij}|)/\widehat{s}_{ij}$ for all $(i, j) \in \mathcal{S}_{\text{cal}}$. Then, by weighted CP with $\widehat{w}_{ij}$ as the normalized odds ratio, the prediction interval for each (i^*, j^*) can be constructed by $C(i^*, j^*) = \widehat{M}_{i^*j^*} \pm \widehat{s}_{i^*j^*} \widehat{Q}$ where

$$\widehat{Q} = \text{Quantile}_{1-\alpha}\left(\sum_{(i,j) \in \mathcal{S}_{\text{cal}}} \widehat{w}_{ij} \delta_{V_{ij}} + \widehat{w}_{i^*j^*} \delta_{+\infty} \right). \tag{29}$$

This approach archives the validity in $\mathcal{S}^c$ on average such that

$$\frac{1}{|\mathcal{S}^c|} \sum_{(i^*, j^*) \in \mathcal{S}^c} \mathbb{1}\left\{ M_{i^*j^*} \in C(i^*, j^*) \right\} \geq 1 - \alpha - \Delta, \tag{30}$$

where Δ measures the estimation accuracy of $\widehat{p}$.

In comparison, [163] considers the setting with a random matrix and fixed missing patterns. Valid intervals are constructed under the row/column exchangeability assumption on the random matrix. Since the coverage guarantee in [75] is on average for $\mathcal{S}^c$, a recent work [110] considers a different criterion of coverage, *simutaneous coverage*, which constructs a joint confidence region for M. [171] further extends the problem to the scenarios with a noisy and partially observed tensor.

3.2.3 Data with Graph (Tree) Structure. Graph data is widely used across various fields, such as recommendation systems [81] and drug discovery [63]. CP for graph data presents both unique challenges and opportunities. One key challenge arises from the reliance of standard CP on the exchangeability assumption, which may not hold for graph data due to its inherent dependencies. Several studies [36, 86, 207] have explored the conditions under which the exchangeability assumption applies to graph data. A common consensus is that in inductive settings, where test data nodes/links are not present in the calibration data, the exchangeability assumption typically does not hold. However, in transductive settings, where all nodes from both the training and testing datasets are included in the graph, the assumption is more likely to be valid. Despite this challenge, the unique structure of graph data also presents opportunities to enhance the performance of CP methods, offering untapped potential for improvement.

Inductive setting In the inductive setting, models are trained on one graph (or a subset of nodes and edges) and then applied to unseen data, such as new nodes or entirely new graphs, during inference. As a result, the exchangeability assumption does not hold, causing CP to fail to provide coverage guarantees. To address this challenge, Clarkson [36] propose the Neighbourhood Adaptive Prediction Set. By leveraging the adjacency matrix, they generate prediction sets for test data, using the neighborhoods of test nodes as calibration data and introduce a weighting scheme based on the path length between nodes to adjust the importance of nodes in calibration, inspired by the homophily principle. In a different approach, Zargarbashi and Bojchevski [207] note that the previous method may not work well with realistic sparse datasets, such as social network interactions. To tackle this issue, they suggest recalculating the non-conformity score and quantile each time new nodes are added to the graph. For edge classification, the same strategy is employed, but with a weighted non-conformity score defined as $1/\text{degree}(v)$, where v represents a node, to account for covariate shift. In Calissano et al. [25], they proposed a CP method applicable to both labeled and unlabeled graphs. For unlabeled graphs, they embed the prediction set into a quotient space, where isomorphic graphs are grouped into equivalence classes. As a result, this approach effectively handles graph structures without reliance on specific node labels.

Transductive setting In this setting, research has shown that exchangeability holds for both node and link prediction tasks [86, 211]. As a result, most works focus on improving efficiency. In [206], the authors update the non-conformity score by incorporating the non-conformity scores of neighboring nodes, effectively leveraging the graph structure to improve prediction efficiency. Specifically, they propose a diffused score as the non-conformity score:

$$\widehat{V}(v, y) = (1 - \lambda)V(x, y) + \frac{\lambda}{|\mathcal{N}_x|} \sum_{u \in \mathcal{N}_x} V(u, y), \tag{31}$$

where $\mathcal{N}_x$ is the set of neighboring nodes of node v, and λ is a hyperparameter. Huang et al. [86] further propose a more general method, highlighting that current approaches mainly focus on designing non-conformity scores while overlooking the direct training for efficiency. Based on the empirical finding that nodes linked together often produce similar sizes of prediction sets, they employ a second GNN model to help aggregate neighborhood information and introduce an inefficiency loss, defined as the size of the prediction set.

In addition to the aforementioned node classification tasks, there are some works that focus on applications of CP on link prediction. For example, extending previous exchangeability results in [86], Zhao et al. [211] prove that the assumption also holds for link prediction tasks in GNNs, and introduce conformalized link prediction. The key finding is that graphs with node degrees following the power law distribution more closely tend to display smaller average prediction intervals, indicating higher efficiency. They then propose a sampling-based CP method to align the node degree distribution of the graph with a power-law distribution to enhance the efficiency of

the CP process. Another study [126] develops a conformal link prediction method that provides a set of true edges while controlling the **False Discovery Rate** (**FDR**). The motivation is that most methods consider link prediction as a ranking problem, which is satisfying when the application constrains the number of pairs of nodes to be declared as true edges to be fixed. Alternatively, other practices may require a classification of missing links into true and false edges with a control of false positives, e.g., reconstruction of biological networks. They collect a set of observed false links (those confirmed to not exist) and calculate p-values on these. Then, they apply the Benjamini-Hochberg procedure to the conformal p-values to establish a threshold for determining whether a link is a true connection. In a follow-up study, Blanchard et al. [23] validate the effectiveness of this approach and extend it to include False Discovery Proportion bounds.

Another area of link prediction is recommendation systems, which aim to predict interactions between users and items. For example, Angelopoulos et al. [12] focus on controlling FDR for evaluating variable-sized sets in drug recommendation. To achieve this, they propose a method that calibrates **learning-to-rank** (**L2R**) models to maintain FDR control. Additionally, in [188], the authors introduce two novel CP-based losses: **CP Set Size** (**CPS**) and **CP Set Distance** (**CPD**). CPS is designed to improve the efficiency of CP, while CPD minimizes the cosine similarity distance between the top-K closest items in the CP set and the ground truth. This enhances recommendation precision by ensuring that items are not only selected with confidence (as with CPS) but also closely match user preferences. They train the model using joint learning to achieve these goals.

Discussion CP for graphs is a relatively underexplored area, and several challenges remain. First, the graph structure itself can be more thoroughly leveraged. Current research primarily focuses on neighborhood information, but other relationships—such as nodes with similar degrees or shared topological properties—could also be incorporated into CP. Second, there is limited research on applying CP to weighted graphs, such as transportation networks, or other graph variants, which presents a significant gap in the literature. Additionally, optimizing the efficiency of CP in settings beyond the current focus remains an open area for exploration.

4 Static Data–Unstructured Data

4.1 Text Data

The rapid expansion of **large language models** (**LLMs**) and NLP applications has highlighted the critical need for UQ to mitigate risks such as hallucinations and to improve decision-making reliability in high-stakes scenarios. Text and LLMs possess unique characteristics that engender new challenges in CP due to their complex nature and unbounded output space. These LLMs are often trained on vast, inaccessible datasets and parameters, making re-training or even fine-tuning nearly impossible. CP addresses uncertainty without oversimplifying the complexity of LLMs and avoids the need for retraining and can be applied to any LLM. Primarily used in classification and regression tasks, harnessing the full potential of CP for text data and generative models such as LLMs is not trivial. In this section, we will explore CP in unstructured text data. We illustrate the structure of the CP method applied to NLP tasks in Figure 4. For a more comprehensive review of CP for NLP, please refer to the survey [26].

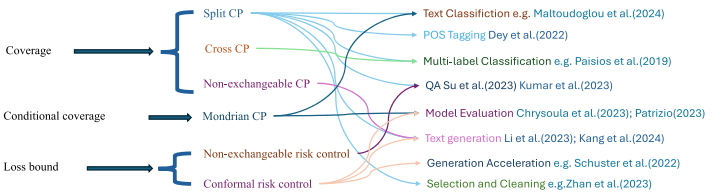

Fig. 4. A taxonomy of CP methods for text data. Loss bound refers to a formal guarantee on the maximum expected loss of a prediction set generated by a conformal method.

4.1.1 Text Classification. Text classification is often applied in high-stakes areas such as medical diagnosis, security, and legal document review. For example, Maltoudoglou et al. [124] build CP [140] upon the transformer method for sentiment analysis to address the problem of heavy computation of CP. Another work [70] similarly applied inductive Venn—ABERS predictor to transformers and constructed prediction sets based on conformal p-values calculated from estimated probabilities. Recent work focuses on the conditional coverage of CP for text classification, which focuses on the coverage for certain subpopulations (e.g., conditional on labels) and is particularly essential for unbalanced datasets and respective tasks. Giovannotti and Gammerman [72] compare several transformer-based conformal predictors and found that Mondrian CP demonstrated the smallest theoretically expected efficiency drop for conditional coverage of CP.

Another thread of work focuses on multi-label text classification, where each instance may have multiple labels, requiring CP to output a combination of labels. For example, [136] applies the power-set approach [137] to multi-label classification, which treats each label set as a candidate classification and is assigned a p-value. They also propose to only use the maximum observed label cardinality of the corpus to reduce the combination of labels. In the study [123], they extend the **Label Powerset** (**LP**) Inductive (i.e., Split) CP by reducing computational complexity, specifically by eliminating a significant number of label sets that are likely to have p-values below the specified miscoverate rate. [59] proposes a modified CP algorithm that produces multi-label prediction sets with a strict limit on false positives. By using a greedy algorithm based on the non-conformity scores, it optimizes true positive rates while controlling false positives below a user-specified level.

*4.1.2 CP for **Natural Language Understanding (NLU)**.* NLU focuses on enabling machines to comprehend and interpret the structure and semantics of human language, it is becoming increasingly pivotal in various commercial settings. NLU involves analyzing text to extract meaning, intent, and context, handling ambiguity, and recognizing entities or relationships.

One application of NLU is *QA* tasks, which can be categorized into **Multiple Choice Question Answering** (**MCQA**) and open-domain QA. For the MCQA task, a common approach is to simplify it into a classification task for the given choices. Current approaches to applying CP to this task primarily rely on the standard split CP and developing various non-conformity scores for different needs. For example, one line of research [99, 201] involves appending the input with "The correct answer is the option:", prompting the LLMs to predict the next word to acquire the logits of the option. Subsequently, a softmax operation is applied to these logits and then used for CP algorithm. In open-domain QA, an LLM receives a question and is asked to generate an answer from a vast, unbounded space based on its comprehension of the question. In a black-box setting, when sampled answers as the LLM output are available, CP can be used to construct a reliable set of potential answers. For example, Su et al. [170] employs sampling to assess the model's uncertainty and calculates non-conformity scores based on frequency to construct prediction sets. They also distinguish the results using normalized entropy and semantic similarity as an extension. Another application of NLU in open-domain QA is to detect LLM hallucinations and verify the facts. Li et al. [109] construct a prediction set by sequentially building the retriever and response set with a coverage guarantee and then combining them into an aggregated set. Then it employs Bayesian optimization to enhance efficiency by finding the best hyper-parameters. An earlier work [60] first expands the CP correctness criterion to allow for additional, inferred "admissible" answers, which can substantially reduce the size of the predicted set while still providing valid performance guarantees. It then amortizes costs by conformalizing prediction cascades, in which it aggressively prunes implausible labels early on by using progressively stronger classifiers–again, while still providing valid performance guarantees.

Another three common NLU tasks where CP has been studied are *machine translation evaluation*, text infilling, and ***part-of-speech (POS)*** *tagging*. Machine translation evaluation involves assessing the quality of generations in a target language based on the input provided in a source language. Some studies apply the CP algorithm to machine translation evaluation to address uncertainty [71, 208]. For example, Zerva and Martins [208] show that split CP adjusts confidence intervals to achieve the desired coverage and identify biases in these intervals based on translation language pairs and translation quality. Using conditional CP, they create calibration subsets for each data subgroup, ensuring equalized coverage. In the POS tagging and text infilling, Dey et al. [42] propose CP-enhanced BERT and BiLSTM algorithms to ensure their statistical reliability. The non-conformity scores are designed based on multi-label classification tasks.

4.1.3 CP for ***Natural Language Generation (NLG)****.* NLG is the process of transforming structured information or text into natural language output. It focuses on producing coherent, contextually appropriate language, such as generating summaries or translations.

In the field of NLP, transformer-based generative LLMs play a significant role [183]. These models generate text by predicting the next word based on previous inputs. When applying CP to text generation tasks, these tasks are categorized by the length of the generated content, distinguishing between next-word prediction and long sentence generation tasks. We note that different from MCQA, in the next-word prediction task, the label set encompasses the entire vocabulary, from which CP generates a prediction set containing potential next words.

To assess whether a top-p sampling set is indeed aligned with its probabilistic meaning in various linguistic contexts, Ravfogel et al. [150] leverage CP and customize the **Adaptive Prediction Sets (APS)** score [153], which aims to provide prediction sets that adaptively include all possible labels that meet a certain confidence level, ensuring that the true label is contained within the prediction set with a high probability. Another challenge in next-word prediction is that the complex intrinsic relationships within the data may violate the exchangeability assumption, leading to non-i.i.d. test and calibration data. To address this, Dennis Ulmer et al. [182] combine APS scores with non-exchangeable CP for next-word prediction.

Another increasingly popular NLG task is *code generation*, where LLMs are tasked with producing a piece of code based on input. One challenge lies in structured prediction problems, where the space of labels is exponential in size. To address this issue, Adam Khakhar et al. [97] propose generating code based on the abstract syntax tree structure and partial programs. They utilize partial programs for prediction [165] and establish a pre-selected 1D search space over the prediction sets by mapping features to labels in order to construct **Probably Approximately Correct (PAC)** prediction sets. ***Retrieval Augmented Generation (RAG)*** combines retrieval-based methods with LLMs for text generation. Integrating CP in its unique generation process is challenging. Kang et al. [96] propose to apply conformal risk control with RAG, in which they show that using RAG can lead to lower risk than normal LLM with or without covariate shift. Mohri et al. [131] address the problem of factual correctness in LLM outputs. During calibration, each output is broken into subclaims, scored by GPT-4 for confidence, and filtered using a threshold based on a preset confidence rate α. In the prediction stage, subclaims with confidence scores above this threshold are seen as correct factual and concatenated to form the final output.

Another line of research adopts CP to control the quality of LLM generations. To deploy the outputs of foundation models reliably, Gui et al. [78] present conformal alignment, which aims to provide guarantees that a prescribed fraction of selected model outputs are trustworthy and meet the alignment criterion. This is achieved by training an alignment predictor on reference data and selecting outputs whose predicted alignment scores exceed a calibrated threshold, thus controlling the FDR of untrustworthy outputs. Victor et al. [147] introduce a conformal sampling framework

for LLMs to generate diagnoses based on input content. Its goal is to generate prediction sets with provable guarantees of containing at least one acceptable response, while filtering out low-quality or redundant outputs. This is achieved through a calibrated sampling process that stops when confidence criteria are met, using rejection rules to eliminate undesirable samples, ensuring efficient, and reliable output.

4.1.4 Inference-time Efficiency. Besides generating prediction sets, CP can also improve inference-time efficiency by reducing the need for multiple samples or complex generation and leveraging efficient calibration strategies. Some works focus on achieving early stopping using CP, enabling the model to produce reliable predictions with reduced inference time. In [102], the authors propose training an LLM using minimal computational resources while maintaining accuracy. By leveraging the **Learn then Test** (**LTT**) framework, they identify optimal risk-controlling hyperparameter configurations. Their key contribution, Pareto Testing, further improves efficiency by strategically guiding the number and order of tested configurations when searching for valid settings. Schuster et al. [159] propose a novel method called **Confident Adaptive Language Modeling** (**CALM**), which allows the transformer to exit early at a specific layer of the decoder, thereby significantly reducing inference time. This method employs various confidence measures to determine when to exit early, such as softmax response, hidden-state saturation, and a trained early exit classifier, then they uses a calibration dataset to threshold confidence scores by grid search.

Discussion Although CP has recently achieved success in NLP, several questions remain to be addressed in the future: (1) Computational efficiency. The use of CP can be computationally expensive, especially with large datasets or packages like FAISS [93]. Optimizing CP for efficiency is essential. (2) Human and LLMs Interaction. As human-LLM interactions generate continuous new data, further research is needed to adapt CP to effectively and efficiently handle these unbounded output space and dynamic environments. (3) Ethical Concerns. CP's ability to produce multiple predictions increases the risk of generating unethical outputs, particularly in LLMs, and addressing these concerns remains an open challenge.

4.2 Image Data

Deep learning with images can achieve high predictive accuracy, but quantifying uncertainty remains a significant challenge, hindering its deployment in critical settings. Applying CP for image data involves navigating challenges related to high dimensionality and feature complexity. The following sections will introduce the development of CP in various **computer vision** (**CV**) tasks.

4.2.1 Image Classification. Some applications of CP for image classification are built upon the framework of canonical classification tasks, but further exploit the specific characteristics of image data across various domains to design powerful non-conformity score functions. For example, in [13], they aim to ensure the reliability of AI models in diverse clinical environments by introducing the conformal triage algorithm, which outputs labels {true, false, and unknown} to aid clinical decision-making. Another line of research utilizes learned deep (underlying) features of image data in constructing more efficient prediction sets. Teng et al. [179] extend the application of CP to semantic feature spaces, in which they leverage the difference between pre-trained and fine-tuned features to generate the prediction set via band estimation [197]. Ghosh et al. [66] further show that if learned image representation satisfies some mild conditions, their proposed NCP (neighborhood CP) approach will produce smaller prediction sets than traditional CP algorithms. They utilize the learned representation to identify the K-nearest neighbors of a given test input and assign important weights based on their distances to create adaptive prediction sets.

4.2.2 Image Segmentation. Given an image x as the input, image segmentation aims to find a map $f : x \mapsto y$, where y is the segmentation result as the output. Here x and y have the same dimension and for each pixel x_i, the corresponding value y_i represents the class for this pixel. As the softmax function has been shown to produce overconfident predictions with uncalibrated outputs, CP can be beneficial in calibrating the segmentation approaches. [191] combines deep learning with CP for tissue sub-region prediction with confidence. They define the non-conformity score based on the output softmax score for each pixel. They further use the conformal p-values to guide prediction: they first choose the class with the largest p-value as the predicted output. Then, confidence in the prediction is defined as the gap between the largest and the second largest p-values, with which the pixel with confidence lower than a given confidence level is excluded. In addition, a recent work [132] utilizes the framework of conformal risk control, where the nested sets $\{\widehat{C}_\lambda(\cdot)\}_{\lambda\in\Lambda}$ is defined by thresholding the output softmax scores by λ. The scarcity of images for pixel-level calibration makes it hard to obtain sufficient images for pixel-level calibration. Brunekreef et al. [24] introduce the concept of Kandinsky calibration, which leverages the inherent spatial structure within natural image distributions, to cluster pixels.

4.2.3 Image-to-image Tasks. In image-to-image tasks, including inpainting, super-resolution, and colorization, the objective of CP is to construct a prediction interval around each pixel such that the true pixel value falls within this interval with a user-defined probability.

Canonical approaches to this task define uncertainty regions based on probable values per pixel, while ignoring spatial correlations within the image, resulting in an exaggerated volume of uncertainty. To address this, one study [157] quantifies the uncertainty in a latent space, thus taking spatial dependencies into account. Specifically, they aim to develop a method that can provide principled uncertainty intervals for semantic factors within a disentangled latent space of a generative model. This method has a fixed decoder that can predict images based on latent code with conformal quantile regression and uses a decoder to generate restored images from the prediction set. However, by relying on a non-linear, non-invertible and uncertainty-oblivious transformation, this method suffers from interpretability limitations. Belhasin et al. [20] improve this by using spatial relationships between pixels for UQ in inverse imaging problems. They employ a diffusion model with three phases: (1) approximation, where sampling generates principal components and weights using SVD; (2) calibration, where three variants of **Principal UQ** (**PUQ**) are proposed: exact PUQ, Dimension-Adaptive PUQ, and Reduced Dimension-Adaptive PUQ, with the latter two improving computational efficiency; and (3) using the previous result for prediction.

There are several works that focus on conformal risk control for image-to-image tasks. For example, [11] proposes a distribution-free pipeline for image-to-image regression based on risk control [18]. [101] proposes a method for conformal risk control based on masking, which guarantee the masked reconstructed image and masked true image have a distance under threshold in a given probability. [168] proposes a novel method for applying diffusion models to CP. Their approach involves generating multiple reconstructed variations for each original image and then determining the interval bounds by taking the upper and lower quantiles for each pixel across these variations. To deal with the problem that classical risk control can only be applied to scalar hyper-parameters, in the study [178], they propose entrywise calibrated intervals and multi-dimensional risk control.

Discussion CP for image classification focuses on high efficiency, handling data imbalance or noise, and achieving the performance of full CP. These methods perform well because they address various challenges associated with CP. In image-to-image tasks, encoder-decoder structures like diffusion models are commonly used. However, there are still some problems that can be better

addressed. For example, in image-to-image tasks, leveraging the relationship between the prediction sets of different pixels to improve predictions is an area for further exploration.

4.3 Heterogeneous Data

Heterogeneous data consists of diverse types and formats, often originating from different sources. This diversity presents challenges for traditional CP methods, which are generally designed for more uniform data types. The variability in distributions, complexity in feature interactions, and the need to handle different data structures simultaneously make it difficult for traditional CP to provide accurate and reliable prediction intervals. Therefore, new CP methods are required to effectively manage heterogeneous data, ensuring that predictions remain reliable and maintain confidence guarantees across all types of data involved. For heterogeneous data, the objective is to find a function that can map $\mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X} = \mathcal{X}_1 \cup \ldots \cup \mathcal{X}_K$. Then in a standard CP method for heterogeneous data, for each sample x, a prediction interval/set is generated by:

$$C(x) = [\widehat{l}, \widehat{u}] \text{ or } [c_1, c_2, .., c_n], \tag{32}$$

where $\widehat{l}$ and $\widehat{u}$ are the lower and upper bounds of the predicted interval, respectively, c_i is a label.

One study [181] works on multi-modal data and highlights that previous methods for generating multi-modal prediction regions often result in overly complex solutions. To address this, it proposes using shape templates—such as ellipsoids, convex hulls, and hyperrectangles—over the residuals of calibration data, which simplify the non-conformity score functions, leading to more efficient and accurate prediction regions. When there is noise in the multi-modal data, Zhan et al. [209] propose a reliability-based training data cleaning method using split CP, which leverages well-labeled data as a calibration set to detect and correct mislabeled data and outliers. In another study [50] for image caption, they introduce a novel heuristic for zero-shot CP that leverages CLIP-based foundation models [148] calibrated by web data. Some work develops CP for limited data scenarios with multi-modal training, e.g., zero-shot, one-shot, and few-shot learning. For example, Fisch et al. [58] point out that when the dataset is small, the prediction generated by CP will be unusually large. They address it by casting CP as a meta-learning paradigm over exchangeable collections of auxiliary tasks. Another type of heterogeneous data is multi-view data which refers to datasets where the same underlying object or phenomenon is described by multiple different sets of modes or "views." **Multi-View CP** (**MVCP**) [151] addresses the challenge that the naive method for late fusion fails to leverage the uncertainty information of individual predictors, leading to less efficient and adaptive prediction intervals. Specifically, they define the score function for the predictor in each mode and use quantiles along projection directions. The final prediction region is an intersection of half-planes, ensuring efficient, distribution-free coverage.

Another application of heterogeneous data is in federated or transfer learning, where the model processes multi-source inputs. In [121], the authors highlight that data in federated learning is typically non-IID, as data from different sources often follow distinct distributions. This characteristic poses challenges to the non-exchangeability assumption of CP. To address this issue, they propose the concept of an ϵ-approximate β-quantile, which allows the quantile to deviate slightly from the target value while maintaining similar coverage properties. Within this framework, non-conformity scores are computed locally for calibration before communication occurs. Further, in transfer learning, Liu et al. [116] study covariate shifts when data originates from multiple sources. The proposed data-adaptive strategy begins by calculating site-specific quantiles for each source. For the target distribution, it then computes a discrepancy measure between the target site's quantile and each source site's quantile. This discrepancy is used to assign weights and minimize the discrepancy for each source site and the target site.

Table 1. A Summary of Representative CP Methods for Spatio-temporal Data Described in Section 5

Representative Work	Data	Distribution Assumption	Coverage Guarantee
NexCP [16]	None	None	$\geq 1-\alpha-\sum_{i=1}^{n}\widetilde{w}_i \cdot \mathrm{d_{TV}}\left(R(Z), R(Z^i)\right)$
EnbPI [194]	single time series	strongly mixing errors	$\approx 1-\alpha$
SPCI [196]	single time series	None	$\approx 1-\alpha$
ACI [68]	single time series	None (online)	Asymptotically $\approx 1-\alpha$
CF-RNN [169]	independent time series	exchangeable time series	$1-\alpha$
Copula-RNN [174]	independent time series	exchangeable time series	$1-\alpha$
MultiDimSPCI [192]	dependent time series	non-exchangeable time series	$\approx 1-\alpha$

Discussion Existing CP methods for heterogeneous data are relatively limited, primarily focusing on multi-modal and multi-view data, as well as scenarios like covariate shift and limited data availability. Future research could explore leveraging contrastive learning frameworks, similar to CLIP for multi-modal image captioning, to enhance conditional probability CP efficiency. Another promising direction is addressing covariate shifts across different modalities in multi-modal tasks, which remains largely unexplored. Further, managing uncertainty in multi-modal outputs, which often arises from the complexity of such data, presents a compelling challenge that warrants further investigation. Finally, more efforts are needed to explore CP in the semantic space.

5 Dynamice Data—Spatio-temporal Data

Spatio-temporal data captures phenomena that evolve over time and across locations, making it crucial in fields like meteorology, geosciences, urban planning, and public health. As prediction models for spatio-temporal data are often setting-specific, prediction sets instead of point predictions can be preferred in this scenario. Applying CP to this type of data helps manage risks associated with spatial and temporal variability by providing confidence measures for predictions. However, standard CP assumes data exchangeability, an assumption often violated by spatio-temporal data due to its structured dependencies and evolving nature. This can lead to misleading confidence measures if not properly addressed. This section discusses challenges and methodologies associated with applying CP to spatio-temporal data. We begin by discussing one-dimensional spatio-temporal data, which essentially constitutes time series data collected from a single geographic location. Then, we explore strategies for tackling multi-location (i.e., multi-dimensional) spatio-temporal data, where each entire series is treated as a single observation, exhibiting temporal dependencies both within and between the sequences. Therefore, the violation of exchangeability in multi-dimensional settings can be more severe and requires robust analytical approaches to ensure the accuracy and reliability of predictive models. In addition, we will cover the special case of streaming time series data, highlighting the need for new methods and identifying additional areas for the application of CP. We summarize representative CP methods in this section in Table 1 and Figure 5.

5.1 One-Dimensional Spatio-temporal Data

For one-dimensional spatio-temporal data, there are three primary lines of research:

— **Reweighting** adjusts the influence of historical samples based on their age or relevance, addressing the non-stationarity. By altering the weights assigned to each sample, sample reweighting helps to align the distribution of the weighted calibration data more closely with that of the test data, ultimately addressing the issue of distribution shift.
— **Updating non-conformity scores** updates non-conformity scores continuously as new data becomes available, ensuring that the model adapts to recent trends and patterns without being overly influenced by outdated information.

— **Updating α:** Instead of using a fixed miscoverate rate for all predictions, this method adjusts the miscoverate rate based on recent prediction errors and variability in the data. Such adaptivity allows for more flexible and context-sensitive prediction intervals, crucial for applications where sudden changes in volatility or trend are common.

5.1.1 Reweighting. Reweighting is one of the most common frameworks for non-exchangeable settings, including time-series data. As we show in Section 2.3, weighted CP can compute prediction intervals for problems with covariate shifts between training and test data [180]. Below, we discuss several reweighting approaches that can be applied to one-dimensional time series data.

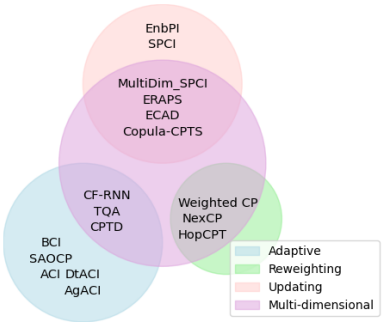

Fig. 5. Venn diagram for CP methods in spatio-temporal data.

Pre-defined weights To relax the covariate shift assumption and address the problem that the high-dimensional likelihood ratio is known beforehand or well approximated for correct coverage in weighted CP, Barber et al. [16] extended the weighted method in [180] to the scenario with arbitrary pre-defined weights, showing that the coverage gap can be bounded by the total variation between the calibration and test data distributions. The weights $\{w_1, \ldots, w_t \in [0,1]\}$ for non-conformity scores are normalized as follows:

$$\widetilde{w}_i = \frac{w_i}{w_1 + \ldots + w_t + 1}, \quad i = 1, \ldots, t, \quad \text{and} \quad \widetilde{w}_{t+1} = \frac{1}{w_1 + \ldots + w_t + 1}. \tag{33}$$

The weighted non-conformity scores for split and full CP are then:

$$\sum_{i=1}^{t} \widetilde{w}_i \cdot \delta_{V_i} + \widetilde{w}_{t+1} \cdot \delta_\infty \quad \text{and} \quad \sum_{i=1}^{t+1} \widetilde{w}_i \cdot \delta_{V_i}^{y}, \text{ respectively.} \tag{34}$$

Here, $V_i = |Y_i - \widehat{f}(X_i)|$ is the non-conformity score for the trained model $\widehat{f}$. Let $\mathrm{d_{TV}}$ denote the total variation distance between distributions, theoretically, [16] provided the lower bounds for coverage in non-exchangeable CP:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_n(X_{n+1})\right\} \geq 1 - \alpha - \sum_{i=1}^{n} \widetilde{w}_i \cdot \mathrm{d_{TV}}\left(R(Z), R(Z^i)\right). \tag{35}$$

This approach can be applied to non-symmetric algorithms like weighted regression and autoregressive models, which are common in time series forecasting. Building upon [16], subsequent research broadened the framework's applications. For example, Schlembach et al. [158] expanded the methodology for multi-step-ahead time series forecasting to the multivariate setting and conformal risk control measures were incorporated into one-dimensional spatio-temporal data [55].

Learned weights The seminal work [16] uses pre-defined weights, leading to poor generalization. HopCPT [14] addresses this by using a **Modern Hopfield Network** (**MHN**) for similarity-based reweighting, which enhances accuracy by prioritizing past instances that closely resemble current conditions [14, 149]. HopCPT assigns weights in time series forecasting, where each weight $a_{t,i}$ signifies the relevance of a past time step i to the current time step t. The weights and corresponding normalized weights are computed as follows:

$$a_{t,i} = \beta m(z_t)^\top W_q W_k m(z_i), \quad \widetilde{a}_{t,i} = \frac{e^{a_{t,i}}}{\sum_{j \in M} e^{a_{t,j}}}, \quad \forall i \in M, \tag{36}$$

where z_i is the encoded input for time step i, m is an encoding function, and W_q, W_k are matrices of learned weights. The hyperparameter β adjusts the focus of the probabilistic distribution generated by the softmax function. Using these weights, HopCPT generates the prediction interval by discounting the quantiles of relative errors:

$$\widehat{C}_\alpha(x_t, M) = \left[\widehat{f}(X_t) + Q_{\frac{\alpha}{2}}\left(\sum_{i\in M} \widetilde{a}_{t,i}\delta_{\epsilon_i}\right), \widehat{f}(X_t) + Q_{1-\frac{\alpha}{2}}\left(\sum_{i\in M} \widetilde{a}_{t,i}\delta_{\epsilon_i}\right)\right], \tag{37}$$

where Q_τ denotes the τ-quantile and δ_{ϵ_i} is a point mass distribution of the error ϵ_i at time step i. To train the MHN, an auxiliary task using absolute errors as proxy values is employed, aligning errors from comparable regime time steps to predict the current absolute error.

Discussion HopCPT exemplifies cutting-edge CP techniques for time series data, offering remarkable forecasting precision [14]. Future research can focus on refining weight selection mechanisms to better capture temporal dependencies in data (e.g., [31]). Advancing this aspect of CP can lead to significant theoretical innovations and enhance practical forecasting performance.

5.1.2 Updating Non-conformity Scores. In contrast to reweighting methods that adjust weights based on a predefined strategy, this section introduces updating non-conformity scores to address non-stationary time-series data. By leveraging the most recent T samples and continuously updating prediction intervals as new data becomes available, this approach adapts the intervals to reflect the latest data trends. For example, when data is continuously streamed, updating non-conformity scores uses only the latest data points to calculate prediction intervals, quickly adapting to new trends or shifts in the data. Therefore, updating non-conformity scores dynamically integrates new information without segregating data into calibration and test subsets.

A pioneering work proposes EnbPI [194], which uniquely updates the non-conformity score with sequential error patterns to adapt the intervals dynamically. This approach starts with bootstrap training data to create effective "**leave-one-out**" (**LOO**) ensemble prediction models with input $\widehat{X}$: $f_{\widehat{X}}(\cdot) = \phi(\{f_b(\cdot) : t \notin S_b\})$, where ϕ denotes an arbitrary aggregation function (e.g., mean, median) over a set of scalars, and $S_b \subset [T]$ is the bootstrap index set used to train the b-th bootstrap estimator f_b. The point predictor on test data is defined as $f_{\widehat{X}}(X_t)$, which aggregates all bootstrap predictions, reducing the possibility of overfitting. EnbPI then updates the past residuals during predictions to ensure the prediction intervals have adaptive width. For a fixed $h \geq 1$, define $\epsilon_t^h := \{\widehat{\epsilon}_1, \ldots, \widehat{\epsilon}_{t-h}\}$, where the residuals are $\widehat{\epsilon}_t = Y_t - f_{\widehat{X}}(X_t)$. The prediction intervals $C_{t-1}(\widehat{X})$ are given by:

$$\left[f_{\widehat{X}}(X_t) + Q_{\alpha/2}(\epsilon_t^T), f_{\widehat{X}}(X_t) + Q_{1-\alpha/2}(\epsilon_t^T)\right], \tag{38}$$

which utilize the past $h = T$ residuals and resemble traditional CP intervals due to the use of the empirical quantile function $Q_{1-\alpha/2}$ to compute interval width. This method offers computational efficiency, robustness against overfitting, and adaptability by avoiding the need to retrain models, instead using LOO ensemble models [194].

The approximately valid coverage guarantee of the EnbPI method is contingent on the assumption that the error terms $\{\epsilon_t\}_{t\geq 1}$ are both stationary and strongly mixing. However, this assumption may not hold in time-series models. SPCI [198] advances beyond EnbPI by relaxing the strict assumptions on error distributions. SPCI incorporates quantile regression to effectively address serial dependencies among residuals in sequential analysis. It replaces the empirical quantile with an estimate by a conditional quantile estimator and adjusts the prediction interval dynamically by optimizing its width. This is done by selecting a threshold parameter that minimizes the gap between two conditional quantile estimates while ensuring the desired coverage level.

Follow-up studies on SPCI have introduced more sophisticated methods to handle the dependent structure of the time-series data. For example, Lee et al. [103] applies a reweighted Nadaraya-Watson estimator within the realm of nonparametric kernel regression, thus enhancing the selection of data-dependent optimal weights. Additionally, Lee et al. [104] adopts a Transformer decoder as a conditional quantile estimator in the SPCI framework, recognizing its effectiveness in sequential modeling tasks. Both methods facilitate the computation of weights in a data-driven manner, eliminating the need for prior knowledge about the data.

Discussion The "updating" approach for time series is akin to reweighting, but it uniquely utilizes data-adaptive weights. By incorporating the most recent T samples into the calibration set, this method ensures that the weights adjust dynamically to reflect the latest data trends and patterns. This similarity to reweighting, yet with a focus on adaptive recalibration, maintains the model's relevance and accuracy amidst evolving data. Future enhancements could involve developing more sophisticated algorithms to determine optimal T values are based on real-time changes in data volatility or distribution patterns. Integrating predictive analytics to anticipate the need for updates could further refine its effectiveness.

5.1.3 Updating the Miscoverate Rate. Another direction for CP in time series focuses on adaptively adjusting the miscoverate rate α during test time to account for mis-coverage. This method can dynamically adjust the size of prediction sets in an online setting where the data-generating distribution varies over time in an unknown fashion. Such an approach is particularly advantageous for time series data. The continuous recalibration during the testing phase ensures that the prediction intervals maintain an appropriate level of coverage over time.

ACI [68] is designed for constructing prediction intervals for time series data in an online learning framework. The approach is simple in that it requires only the tracking of a single parameter that models the shift. The update rule for the significance level α is: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, where γ is a step size parameter, and err_t indicates if Y_t is excluded from $\widehat{C}_t(\alpha_t)$. The approach ensures that the prediction intervals adjust over time to account for shifts in the data distribution, maintaining the desired coverage probability.

The choice of γ gives a tradeoff between adaptability and stability. While raising the value of γ will make the method more adaptive to distribution shifts, it will induce greater volatility in α_t. Zaffran et al. [204] theoretically analyzed the influence of γ on ACI efficiency and proposes AgACI which runs multiple ACI instances with different learning rates and combines their outputs using an online aggregation of experts approach. Alternatively, Gibbs and Candès [69] dynamically adjusts γ based on expert performance. Each expert operates with a specific γ, and ACI is performed in parallel for all experts. The final α_t is selected based on the historical performance of each expert, effectively learning the optimal γ over time to adapt to shifts in the data distribution.

Another alternative to ACI is the **multi-valid CP** (**MVP**) method of [17] which accounts for worst-case adversarial guarantees to ensure the methods remain reliable even when data distributions are manipulated or shift unexpectedly, especially in adversarial environments. MVP achieves this by selecting prediction thresholds that control the long-term miscoverage rate. Similarly, Angelopoulos et al. [7] address worst-case scenarios by examining the effect of a decaying step size on the prediction threshold update rule. Their approach ensures strong long-term coverage in adversarial settings and achieves convergence to an optimal threshold.

Another follow-up work of ACI focuses on refining the regret condition within the online learning framework. SAOCP [17] targets minimizing ***Strongly Adaptive Regret*** (**SAReg**), which evaluates performance over all intervals of specific lengths rather than over the entire length. In [6] , control theory is used to dynamically manage uncertainties and adapt to changing data. It integrates ACI with traditional PID control mechanisms to enhance predictions. Additionally, Yang et al. [200]

proposes BCI to addresse ACI's limitations by incorporating **Model Predictive Control** (**MPC**) techniques to optimize prediction intervals. Together, these advancements extend ACI's capabilities by enhancing adaptability and robustness in dynamic and adversarial environments.

Discussion Approaches that dynamically adjust the miscoverate rate based on incoming data, akin to how online learning updates model parameters in response to new information. Future research could explore more nuanced methods for adjusting miscoverate rate, potentially incorporating reinforcement learning to optimize decision-making processes based on performance feedback. Importantly, these methods can be synergistically combined; for instance, updating may be viewed as a specialized form of reweighting that places greater emphasis on recent samples. Moreover, the integration of reweighting with adaptive strategies has been shown to enhance performance significantly [14, 79]. While the strategies outlined are broadly applicable to spatio-temporal data, research exists for specific types of time series, such as AR process [204], Hidden Markov Model [133], and β-mixing process [134], which further refine these general techniques.

5.1.4 Other Methods. Chernozhukov et al. [34] develops a randomization method and accounts for potential serial dependence by including block structures in the permutation scheme, which can be either overlapping or non-overlapping. The non-overlapping blocking scheme divides the data into separate blocks to reduce dependencies within each block. In contrast, the overlapping blocking scheme shifts the starting point of each block slightly to create overlapping sections between them. This overlap allows the scheme to capture more variations and dependencies between observations by including samples that are close together in multiple blocks. Follow-up work such as [2] employs similar blocking techniques to account for dependencies in a probabilistic forecasting scheme for two-dimensional functional time series. Cleaveland et al. [37] propose an optimization-based method that reduces the conservatism of CP for time series predictions. Instead of treating prediction errors independently at each time step, they consider a parameterized prediction error over multiple time steps and optimize the parameters over an additional dataset. Prinzhorn et al. [145] further analyzes different temporal components using time series decomposition. This method decomposes the original time series into remainder, trend, and seasonal components and leverage previous methods such as EnbPI [194] and ACI [68] to handle these different components. This strategy better captures the inherent characteristics of time series and improves performance.

A very recent work by [135] extends split CP to handle non-exchangeable data by introducing a novel framework based on concentration inequalities and data decoupling properties. The method assumes that the calibration and test data distributions are reasonably similar and stable, ensuring reliable prediction intervals with guarantees for both marginal coverage and conditional coverage.

5.2 Multi-Dimensional

Multi-dimensional spatio-temporal data spans multiple dimensions of both time and space, introducing additional complexity and opportunities for predictive modeling. The dependencies between spatial and temporal dimensions further undermine the exchangeability assumption inherent in traditional methods. Previous works on multi-dimensional data primarily focus on exchangeable data [56, 67, 94, 130], overlooking the temporal and spatial dependencies. This section mainly focuses on advanced techniques that adapt CP for multi-dimensional spatio-temporal data in regression problems.

A simple approach is to treat each time step as a separate prediction task. CF-RNN [169] constructs prediction intervals using Bonferroni correction, selecting the $(1 - \alpha/k)$ th quantile for a confidence level of $1 - \alpha$. This method results in conservative intervals and ignores spatial dependencies. Similarly, Mao et al. [125] achieves theoretical validity by assuming that spatial locations are exchangeable. Without making this assumption, they focus on specific locations and propose

a localized approach that utilizes nearby locations to address spatial dependence. Zhou et al. [212] also assume that each time series trajectory is exchangeable, combining with ACI to produce more informative prediction intervals for motion planning. Sun and Yu [174] considers spatial dependence by using copulas to model uncertainty over future time steps, reducing confidence interval widths while maintaining validity. Copulas, which model dependencies between variables, apply the empirical **cumulative distribution function** (**CDF**) to estimate non-conformity scores, ensuring the predicted trajectory falls within the confidence intervals at a level of $1-\alpha$.

The theory behind [174] assumes that each data sample in a time series is i.i.d., thus ignoring temporal dependencies. Xu et al. [192] relax this assumption by constructing episodic prediction sets that are adaptively and efficiently calibrated during test time to account for temporal dependencies. This method starts with a training dataset $Z_t = (X_t, Y_t)$ for $t = 1, \ldots, T$, and a prediction algorithm $\widehat{f}$ that forecasts Y. Key to this method is the computation of prediction residuals $\widehat{\sigma}_t = Y_t - \widehat{f}(X_t)$, which are used to estimate the covariance $\widehat{\Sigma}$, because it captures the dependencies and relationships between different dimensions of the data. The covariance is:

$$\widehat{\Sigma} = \frac{1}{T-1}\sum_{t=1}^{T}(\widehat{\sigma}_t - \bar{\sigma})(\widehat{\sigma}_t - \bar{\sigma})^T, \tag{39}$$

where $\bar{\sigma}$ is the mean of $\widehat{\sigma}_t$. They ensure the invertibility of $\widehat{\Sigma}$ via a low-rank approximation, crucial for defining the ellipsoidal (prediction) region $B(r, \bar{\sigma}, \widehat{\Sigma}_\rho)$ with a radius r. This radius is calibrated through CP by calculating a non-conformity score for a new sample (X, Y):

$$\widehat{e}(Y) = (\widehat{\sigma} - \bar{\sigma})^T \widehat{\Sigma}_\rho^{-1}(\widehat{\sigma} - \bar{\sigma}), \tag{40}$$

where $\widehat{\sigma} = Y - \widehat{f}(X)$. The method then determines r to ensure predictions fall within the ellipsoid at a specified confidence level. The prediction sets are then constructed via SPCI [192, 196].

5.2.1 Cross-sectional Coverage. In the multi-dimensional time series domain, an interesting branch considers both cross-sectional and longitudinal validity simultaneously. Two works [112, 113] enhance prediction intervals in time series forecasting by ensuring both cross-sectional and longitudinal coverage. Lin et al. [113] introduce **Temporal Quantile Adjustment** (**TQA**), which dynamically adjusts quantiles using temporal information. TQA employs a two-step strategy: predicting future quantiles based on non-conformity scores and refining them through a budgeting strategy (TQA-B) or error-based adjustment (TQA-E). By contrast, Lin et al. [112] focus on normalization to account for distribution changes. It calculates these scores based on historical data discrepancies and normalizes them with the median absolute deviation to adjust for shifts in data distribution over time.

5.2.2 Classificaion and Anomaly Detection. Classification and anomaly detection within spatio-temporal data contexts also hold significant importance. Building upon their foundational research, Xu and Xie [195] developed the **Ensemble Regularized Adaptive Prediction Set** (**ERAPS**), which adapts the EnbPI framework [194] for classification problems by applying the **Regularized Adaptive Prediction Set** (**RAPS**) method [5] from image classification to construct non-conformity scores. Additionally, [193] further developed EnbPI for anomaly detection in multi-dimensional data with ensembling to obviate the need for data splitting and enhance prediction accuracy and robustness.

Discussion This section discusses methods for handling dependencies in time and space, highlighting the challenge of solving non-exchangeability issues. Additionally, simultaneously considering longitudinal and cross-sectional coverage is an area that requires further exploration. Finally,

the use of graphical models for multi-dimensional time series remains underexplored, presenting a promising direction for future research.

5.3 Streaming Data

Streaming data is characterized by continuous flow, high volume, and rapid updates. **Adaptive CP** (**ACP**) methods [68, 69] for streaming data dynamically adjust α to maintain desired miscoverage rate as data characteristics evolve. In this section, we first introduce the application of ACP in motion planning. Next, we discuss the use of CP for detecting concept drift and enhancing interpretability in streaming data and finally extend CP to online learning beyond numerical data streams.

5.3.1 Streaming Numerical Data. Motion planning involves navigating agents from an origin to a destination without collisions, despite uncertainties from environmental factors, sensor errors, and model inaccuracies. The environment includes N dynamic agents with states Y_t following an unknown distribution $\mathcal{D}$. ACP manages streaming data to forecast future states over a prediction horizon H, updating the adaptive threshold δ_τ based on prediction accuracy:

$$\delta_{\tau+1} := \delta_\tau + \gamma(\delta - e_\tau), \tag{41}$$

where e_τ indicates prediction accuracy, and γ is a tuning parameter. Using ACP, an Uncertainty-Informed MPC optimizes a control sequence to ensure safe navigation by integrating real-time data and adapting to dynamic environments [48, 114].

CP can also help monitor online data streams, particularly in tasks like concept drift detection [53, 122, 177] and regression trees [90–92]. For instance, **Inductive CP** (**ICP**) assigns nonconformity scores to new samples based on a model trained on initial data. If the p-value for a new sample falls below a certain threshold, indicating a potential concept drift, a new classifier is trained; otherwise, the model is updated with the current labeled data. Additionally, CP enhances both interpretability and efficiency in regression trees by applying a Mondrian conformal predictor at each leaf, focusing on localized updates to maintain accuracy and adapt to changes in the data stream.

5.3.2 Beyond Numerical Data. Other studied problems with streaming data include image classification and document retrieval in online learning. Ge et al. [64] explore the use of stochastic online CP with semi-bandit feedback in real-time data streams, where the system only receives information about the true label if it is included in the prediction set. Stochastic online CP updates its prediction thresholds using the **Dvoretzky-Kiefer-Wolfowitz** (**DKW**) inequality to ensure the empirical distribution of non-conformity scores up to time t is not overly optimistic about the true distribution of the data by adding an adjustment factor. The adjusted threshold ensures that the prediction intervals remain conservative enough to maintain the desired coverage.

Discussion The seminal work ACI can be a general strategy to handle high-volume, non-stationary streaming data. When applying CP for streaming data, it is crucial to consider computational efficiency. Additionally, there is a significant need for more research on scalability, particularly in distributed systems where data streams are processed in parallel across multiple nodes. Developing efficient algorithms that can maintain statistical guarantees while operating in such environments would be highly beneficial. Finally, exploring the application of CP in multi-modal data streams, where data comes from various sources and types, presents another exciting research frontier.

6 Open Challenges and Future Directions

Non-exchangeability: distribution heterogeneity and temporal dependence. The exchangeability assumption often fails in real-world scenarios. While existing methods like nonexchangeable CP [16] assess the coverage gap of weighted CP based on the distance between distributions, developing an adaptive approach to minimize this gap remains an open challenge. Moreover, works such as ACI and related studies [7, 68] provide adversarial coverage guarantees for sequences with arbitrary distributions, but they do not ensure validity at each step. Given this, it would be valuable to explore how light distributional assumptions on the sequence could enhance adaptivity.

Robust CP with more general data imperfections. Ensuring the reliability of CP methods under data imperfections and adversarial conditions is crucial. As highlighted in the survey, critical scenarios include observations with missing values, noisy entries, or adversarial attacks. The validity of some existing methods hinges on specific distributional assumptions, such as the missing at-random assumption. However, real-world missingness can follow more complex patterns, like block-wise or non-random missingness. Extending CP to handle such general data imperfections, while maintaining prediction set validity with tractable coverage gaps, remains an open problem.

Handling emerging data types. As data evolves, new data types bring challenges and opportunities for CP. For example, multi-modal data, which integrates information from various sources such as text, images, and sensor data, presents the challenge of combining heterogeneous data sources with differing noise characteristics and distributions. A key open question is how to "optimally" aggregate data from multiple modalities to enhance CP performance. Similarly, the rise of streaming data underscores the need for adaptive and online CP techniques that can provide valid predictions in real time. This requires maintaining computational efficiency and prediction validity as new data arrives and as the underlying data distribution potentially shifts over time.

CP for responsible AI. CP offers a promising approach to responsible AI [118]. CP can generate clear, quantifiable confidence measures, aiding users in understanding and trusting model predictions, which promotes interpretability [87, 90, 91]. For fairness, CP can help equalize coverage across diverse population groups, correcting biases without the need for expensive retraining [120, 153, 189]. Privacy is another critical area, especially in federated learning [119, 121]; CP can be adapted to produce confidence intervals without accessing sensitive or personally identifiable information, thus preserving privacy while ensuring reliable predictions. By addressing these aspects, CP can significantly contribute to the development of AI systems that are not only accurate but also fair, transparent, and resilient across various applications.

CP in aiding decision-making. Research indicates that when humans are provided with CP sets as *side information*, their task accuracy improves compared to using fixed-size prediction sets, thereby enhancing human-in-the-loop decision-making and human-AI collaboration [40]. CP also aids in early stopping for iterative algorithms by providing robust uncertainty measures, which helps prevent overfitting and reduces computational costs [111]. In online auctions, CP can predict final prices and dynamically adjust bidding strategies, leading to improved efficiency and profitability [80]. These applications highlight CP's potential to transform various fields through reliable and adaptive predictions. Thus, it is essential to explore additional scenarios where CP can be applied to further unlock its potential and broaden its impact.

CP for cross-disciplinary research. Cross-disciplinary research offers significant opportunities to advance CP. For example, insights from statistics can contribute to developing new

theoretical foundations and methods for CP. Collaborations with domain experts in areas like healthcare, finance, and environmental science can lead to the creation of application-specific CP methods that address unique challenges in these fields. Additionally, leveraging advances in computer science, particularly in optimization and high-performance computing, can improve the scalability and efficiency of CP techniques. Interdisciplinary research that bridges theory and practice is also crucial for translating CP methods into real-world applications, thereby maximizing their impact.

Open source tools for CP. Despite progress in developing open-source tools for CP, challenges remain in expanding their usability. Current tools like PUNCC [128] and TorchCP [190] address traditional tasks such as regression, classification, and graph-based learning, while Fortuna [41] takes a broader approach, incorporating UQ techniques such as Bayesian inference. Additionally, Susmann et al. [175] introduced an open-source R package for ACI. While many tools effectively handle traditional data types, there is a notable lack of support for modern data types, such as unstructured data and streaming data, and for addressing the rapid growth of language modeling applications. Improving compatibility, scalability, and ease of use across ML platforms is crucial to foster wider adoption and innovation in CP.

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A Appendix

Table 2. Major Notations Used Throughout the Survey

Notation	Description
$X_i, \mathcal{X}$	feature of ith instance and feature domain
$Y_i, \mathcal{Y}$	label of ith instance and label domain
$V(X_i, Y_i), V_i$	non-conformity score of ith instance
$Q_{1-\alpha}, \widehat{Q}_{1-\alpha}$	quantile or estimated quantile at $1-\alpha$
α	miscoverage rate
$C_{1-\alpha}(X_i)$	predicted set of ith instance
D, n	dataset and its size
σ_i	residuals or estimated residual of ith instance
$P, P_X, P_{Y\|X}$	distribution, distribution of X, distribution of Y given X
$\widehat{F}(X_i, Y_i)$	estimated CDF
π	permutation
θ_Z	parameter of model get from data Z
z_i, Z	pair of label and feature, equal to (X_i, Y_i) or (X, Y)
ℓ	loss function
$\widehat{\tau}$	calibration threshold
t, T	time stamp
f, g	neural network or predictor
$\widehat{l}, \widehat{u}$	lower and upper bound of prediction interval
w_i	weight for ith instance
δ_{R_i}	point mass of ith instance

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ORIGINAL RESEARCH

A comparative study of machine learning models for taxi-demand prediction using a big data framework

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Abstract

The increase in urban vehicle numbers and ownership extends traffic congestion, with road expansion often escalating the problem, while growing individual car usage further contributes to the issue. To address these challenges, the development of efficient and accurate demand predictors, particularly for taxi-passenger demand, has attracted a significant attention from both industry and academia. This research aims to address the question of how predictive models, evaluated within a big data framework, can be effectively utilized to forecast taxi-passenger demand and enhance urban mobility. The study is structured around two key objectives: first, to evaluate existing predictive models for analyzing traffic data and forecasting taxi-passenger demand using a dataset from the New York City Taxi and Limousine Commission (TLC), and second, to provide practical recommendations for improving demand prediction through a comparative analysis of machine learning models—such as multinomial logistic regression, generalized linear regression, random forest, and decision tree algorithms. Our results demonstrate that the GLR outperforms both random forest and decision tree models in predicting taxi-passenger demand, achieving approximately 90% R^2 score. The multinomial logistic regression classifier predicts demand levels with over 70% accuracy. The lower performance of random forest and decision tree algorithms is attributed to the highly imbalanced nature of the demand, leading to overfitting and local optimization issues. The application of these predictive models can contribute to optimizing fleet management, reducing passenger waiting times, and improving taxi distribution, potentially alleviating congestion and supporting a better integration with other transport modes.

Keywords Traffic prediction · Machine learning · Predictive models · Urban mobility · Big data framework · Machine learning · Classification

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1 Introduction

In recent years, the proliferation of vehicles in urban areas has been a notable trend, driven by the rapid development of urban mobility and the insufficient expansion of road infrastructure, which has struggled to keep pace with the growing demand and has often increased congestion. There is considerable interest in transportation planning and urban data analytics for accurately characterizing mobility patterns and improving lifestyle by making smart cities a reality (Briand et al. 2016; Wang et al. 2022). Taxis represent a substantial share of urban road use, often playing a crucial role in urban public transportation systems (Liu et al. 2022). For example, data from the New York City Taxi and Limousine Commission show that the number of licensed taxi drivers in the city increased by 11.10% between 2011 and 2022 (De Blasio and Joshi 2016). This growth raises important considerations in urban mobility planning. It should be noted that urban road infrastructure, often already stressed by high car ownership and usage rates, faces challenges in expanding indefinitely to accommodate the growing number of cars on the road. Gridlock and congestion are potential consequences, even if other infrastructure and city environments are restructured for car mobility. Recent literature has examined the spatial impact of automobile infrastructure in urban areas, providing insight into the challenges associated with accommodating the increase in car use (Bwire and Zengo 2020). The total number of taxi licenses in New York City, as reported by the Taxi and Limousine Commission (TLC) (Taxi and Commission 2011–2021), grew steadily by 106% from 166,713 in 2011 to a peak of 343,786 in 2018. A decline followed in 2019, with a 2.8% decrease to 334,201, and a sharper drop of 12% to 291,936 in 2020, coinciding with the COVID-19 pandemic's impact on urban mobility. Post 2020, the trend stabilized at around 285,000 licenses in 2021 and 2022, before recovering by 4% to 294,483 in 2023. By 2024, licenses rebounded further to 302,000, highlighting recovery and adaptation within the transportation industry.

Accurate predictions of taxi-passenger demand, hereafter referred to as "demand," can be used to mitigate issues, such as excess vehicles on the road and traffic congestion (Liao et al. 2018; Veloso et al. 2011; Tang et al. 2018). The integration of diverse transportation modes, such as transport network companies with existing public systems, has been explored to improve urban mobility frameworks (Medeiros et al. 2024). The advent of sensor and GPS technologies has facilitated the accumulation of vast amounts of taxi trajectory data, making it possible to analyze travel patterns and forecast demand (Castro et al. 2012; Laha and Putatunda 2018). However, the uncertain nature of demand, influenced by factors, such as time, location, season, and weather, presents challenges for accurate prediction (Moreira-Matias et al. 2013a). Furthermore, the high frequency at which taxi trajectory data are generated, often every minute or even every second, poses difficulties for extracting valuable knowledge from the data (Yang et al. 2018). As a result, developing efficient and accurate predictors of demand has garnered a significant attention from industry and academia.

Multiple studies have been conducted to develop generic traffic demand predictors using traffic trajectory data, such as predicting travel destinations and modeling origin–destination routes (Laha and Putatunda 2018; Li et al. 2018; De Brébisson et al. 2015; Zhang et al. 2018, 2017), forecasting travel time (You and Kim 2000; Lee et al. 2009; Wang et al. 2016), and predicting transportation demand (Jing and Fan 2014; Jiao and Ge 2013; Chang et al. 2010; Zhao et al. 2016; Moreira-Matias et al. 2013b; Al-Ayyash et al. 2016). In addition, various data mining algorithms and machine learning models, such as k-means clustering (Ladino et al. 2017), linear regression (LR) (Tong et al. 2017), support vector machines (SVM) (Sun et al. 2015; Vanajakshi and Rilett 2007), and neural networks (Wang et al. 2016; Zhou et al. 2018), have been applied to build traffic predictors. Among these studies, predicting transportation demand is one of the most active research areas. Although many valuable insights have been presented, the most relevant articles focus on individual models or algorithms. In other words, comparative studies on the performance of different models remain limited and need further investigation.

This study explores how predictive models, within a big data framework, can be used to forecast taxi-passenger demand and inform urban mobility planning. It focuses on evaluating and comparing forecasting models using real-world data to derive practical insights that could lead to comprehensive solutions for mobility optimization. Specifically, the study aims to:

1. Evaluate existing predictive models for analyzing traffic data and predicting taxi-passenger demand using a dataset from the New York City Taxi and Limousine Commission (TLC).
2. Provide practical recommendations for enhancing taxi-passenger demand prediction through a comparative analysis of machine learning models assessed within a big data framework.

To achieve these objectives, we examine existing predictive models for forecasting taxi-passenger demand, evaluate their performance, and identify the most effective models. Furthermore, we offer practical recommendations for the deployment of these models in real-world scenarios. To comprehensively assess their effectiveness, we utilize the Amazon EMR (Elastic MapReduce) big data framework, which enables us to evaluate their capability to handle large volumes of traffic data efficiently. The findings and recommendations emerging from our research serve as valuable resources for taxi service providers, urban traffic management, and data analysts, empowering them to enhance traffic prediction and improve urban mobility. In essence, this study makes the following contributions:

1. **Evaluation of predictive models**: We evaluate existing predictive models for analyzing traffic data and forecasting taxi-passenger demand, providing insights into their performance in real-world scenarios.
2. **Performance in big data environments**: We evaluate the models' ability to handle large-scale traffic datasets within a big data framework, demonstrating their applicability in real-world settings.

3. **Practical recommendations**: Based on our evaluations, we offer recommendations to improve the use of these models, benefiting stakeholders such as taxi providers, urban traffic management, and data analysts.

This research adopts experimental quantitative analysis as its primary methodology, utilizing publicly available secondary data sources to address the research questions. The study conducts experiments to evaluate the performance of classification and regression models using the Spark Machine Learning library (MLlib), implemented in the R programming language, on the Amazon EMR (Elastic MapReduce) cloud computing service.

The paper is organized into seven sections. After the introduction section, Sect. 2 outlines the background, key concepts, machine learning techniques, models, and the criteria for evaluating these models. The next Sect. 3 reviews the existing literature and identifies research gaps addressed by this study. Section 4 details the research methodology, covering the experimental design and the steps involved. In Sect. 5, the paper presents the research findings and presents a thorough analysis, while Sect. 6 discusses their significance within the context of the research objectives and the broader field. Section 7 concludes the paper with a summary, the implications of the findings, and potential avenues for future research.

2 Background

This section explains the key-theoretical concepts used in the research, encompassing taxi-passenger demand metrics, an introduction to machine learning techniques and models, as well as performance evaluation methods and criteria for assessing machine learning models.

2.1 Taxi-passenger demand metrics

Previous research has studied several ways to conceptualize taxi-passenger demand. Yang et al. (2018) analyzed the demand as the total pick-up and drop-off events within one selected traffic analysis zone. Similarly, Liao et al. (2018) described demand in terms of time-specific and location-specific variation, i.e., the number of pick-ups and drop-offs occurring within a given region during a specific time interval. Generally, taxi-passenger demand is analyzed through both spatial and temporal dimensions. In this study, taxi-passenger demand is articulated as the accumulated number of passengers *(P)* at a specific location *(l)* over a defined time interval *(t)*

$$demand = sum(P_{(l,t)}). \tag{1}$$

Furthermore, recent studies reveal similarities between traditional taxi services and ride-hailing systems, particularly in their reliance on technology to optimize supply–demand matching and improve service efficiency (Medeiros et al. 2024). These findings provide a valuable foundation for understanding the complexities of passenger demand in urban transportation systems.

2.2 Machine learning techniques and algorithms

Machine learning techniques encompass the design and analysis of algorithms that enable computers to learn automatically (Witten and Frank 2005). These algorithms analyze and extract patterns from data to make predictions, which are essential in various domains, including data mining, natural language processing, computer vision, and traffic pattern discovery. Machine learning broadly categorizes its models into three types: supervised, unsupervised, and reinforcement learning (Caruana and Niculescu-Mizil 2006; Jordan and Mitchell 2015).

2.2.1 Supervised learning models

Supervised learning models involve the use of labeled data to train algorithms, allowing them to make predictions. These models include classification and regression. **Classification models** predict the class to which instances or objects belong. They can be binary, providing predictions for two classes (e.g., 0 or 1), or multi-class, generating predictions for multiple classes within a given problem context. A variety of algorithms have been developed for classification tasks, including logistic regression, random forest, decision trees, Naive Bayes, and support vector machines (SVM) (Kotsiantis 2007). It is important to note that data must undergo a discretization process and receive appropriate labels before classification, which significantly impacts prediction accuracy. **Regression models**, on the other hand, are used to predict a continuous numeric value by analyzing the quantitative relationships between multiple variables involved in the task. In our study, we employed classification techniques to predict the level of demand for taxi services and used regression analysis to forecast the specific value of this demand (Bishop 2006). Popular machine learning algorithms for both classification and regression problems include regression, random forest, and decision trees.

2.2.2 Regression

Regression is a statistical process used to explore the relationship between variables. It offers various techniques for modeling and analyzing the relationship between multiple independent (x) and dependent (y) variables (Rasmussen 1997; Dreiseitl and Ohno-Machado 2002). Regression aims to find a function that relates the dependent variable to one or more independent variables. A simple linear regression model is represented as

$$y = \beta_0 + \beta_1 x + \epsilon. \tag{2}$$

Here, y represents the dependent variable, x is the independent variable, and β_0 and β_1 collectively denote the regression parameters showing the intercept and slope. In the context of simple linear regression, there is one independent variable. In more

complex scenarios with multiple independent variables, a multiple regression model can be expressed as follows:

$$y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i + \epsilon \tag{3}$$

Here, x_i represents the i-th independent variable, and β_i represents the corresponding regression parameter. Simple linear regression is used to predict numerical values, where the goal is to estimate a numeric outcome based on one or more predictor variables. In contrast, logistic regression is used to predict categorical values based on independent variables and is a classification technique. It models the probability that a given input belongs to a particular category (Bishop 2006).[1]

2.2.3 Decision trees

Decision trees are a fundamental concept in machine learning. They build a model based on the properties of the data in a tree-like structure (Dietterich and Kong 1995). Decision trees are versatile and can be applied to both classification and regression problems. The core idea behind decision trees is to create a model that can predict the probability of unknown data based on the observed probabilities of different situations. Decision tree algorithms, such as the commonly used DT algorithm, construct a tree structure based on the attribute values of labeled instances. These trees are designed for classification tasks, where the goal is to predict the class label of an unlabeled instance based on its attribute values. A decision tree consists of nodes and branches, forming a hierarchical structure. The key components of a decision tree include the following: Root Node: The top node of the tree, representing the initial decision. Internal Nodes: Nodes in the middle of the tree that make decisions based on attribute values. Leaf Nodes: Terminal nodes that provide the final prediction or classification. The path from the root to a leaf node is determined by the attribute values of an unlabeled instance. This path specifies the label or prediction for that particular instance. The decision tree is constructed through a recursive process, where at each internal node, a decision is made based on a selected attribute. This decision guides the traversal of the tree structure until a leaf node is reached, which assigns the final class label or prediction. The decision tree model is expressed through a set of rules that can be used for classification tasks. Here is a simple representation of a decision tree rule for a binary classification problem:

$$\begin{aligned} &\text{IF Attribute A} \leq \text{Threshold T} \\ &\qquad\qquad \text{THEN Class} = \text{Class 1} \\ &\text{ELSE IF Attribute B} > \text{Threshold S} \\ &\qquad\qquad \text{THEN Class} = \text{Class 2} \end{aligned}$$

[1] For more comprehensive mathematical details on linear and logistic regression, you can refer to chapters 10 to 12 of "Pattern Recognition and Machine Learning" by Bishop (2006).

In this example, the decision tree checks two attributes, A and B, against specific thresholds, T and S, and assigns the instance to one of two classes, Class 1 or Class 2, based on the outcomes. Decision trees are powerful tools for understanding and modeling complex decision-making processes in a structured and interpretable way. They are particularly useful for tasks where transparency and explainability are important, and they can serve as a foundation for more advanced techniques like random forest and gradient boosting.

2.2.4 Random forest

Random forest is a classification algorithm that employs an ensemble of multiple decision trees constructed randomly. It is represented as

$$\text{Random Forest} = \{T_1, T_2, \ldots, T_N\}.$$

Here, each decision tree T_i is constructed using a bootstrapped data subset D_i and a random subset of features F_i.

For prediction, each tree T_i outputs a category prediction C_i. The final prediction is determined by selecting the category with the most votes from all trees

$$\text{Final Prediction} = \text{Mode}(C_1, C_2, \ldots, C_N).$$

Random forest is known for its robustness and superior predictive accuracy compared to traditional decision tree models. This improvement stems from its ensemble approach, which combines multiple decision trees to reduce overfitting and enhance generalization, making it a widely favored technique in machine learning (Breiman 2001; Svetnik et al. 2003).

2.3 Performance evaluation measures of machine learning models

The performance of a machine learning model is evaluated by training the model on a training dataset, making predictions with the trained model, and comparing the predictions with observations to score the model's predictive ability, using evaluation indicators such as percent correct classification and a confusion matrix.

Generally, the evaluation of the performance of a machine learning model involves three steps: 1) training the model using a training dataset; 2) utilizing the trained model to generate predictions; 3) comparing the predictions with the observations to assess the predictive capability of the model (Willmott 1982). Performance metrics, also known as evaluation indicators, are the criteria used to evaluate the generalization ability of machine learning models. Different indicators provide different perspectives on the evaluation, and it is important to note that different machine learning models should be evaluated using appropriate performance metrics (Sheiner and Beal 1981).

2.3.1 Performance metrics of classification models

For classification problems, the most common metrics are percent correct classification (PCC) and the confusion matrix (Chaudhri et al. 2022; Sokolova and Lapalme 2009; Ferri et al. 2009). The PCC measures overall accuracy without considering the type of errors made. In contrast, the confusion matrix provides a detailed summary of errors, such as false positives and false negatives. The specific metrics used are as follows (TP: true positive, TN: true negative, FP: false positive, FN: false negative):

1. Accuracy represents the proportion of successful predictions to total cases

$$\text{Accuracy} = \frac{(\text{TP} + \text{TN})}{N}. \tag{4}$$

2. Kappa score is similar to the accuracy score, but it considers the accuracy that would have happened in any case through random predictions (Ben-David 2008)

$$\text{Kappa} = \frac{\text{Accuracy} - RP\text{Accuracy}}{1 - RP\text{Accuracy}}, \tag{5}$$

   where Accuracy is the observed accuracy and RPAccuracy represents the expected accuracy by random prediction. RPAccuracy is calculated as the sum of squared proportions of each class in the dataset, i.e., $RP\text{Accuracy} = \sum p_i^2$, where p_i is the proportion of class i (Siemers and Bajorath 2023).
3. Precision represents the proportion of true results over all positive results

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{6}$$

4. Recall, also called sensitivity, represents the proportion of true results over all positive observations

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{7}$$

5. F-measure represents the weighted average of precision and recall in the [0,1] range. Generally, the F1-measure, a specific form of the F-measure, is employed, which is calculated as follows:

$$\begin{aligned} F1 =& \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \\ =& \frac{2 * \text{TP}}{2 * \text{TP} + \text{FP} + \text{FN}}. \end{aligned} \tag{8}$$

6. Balanced accuracy (BalancedAcc) is calculated as the arithmetic mean of the sensitivity (true positive rate) and specificity (true negative rate). It is useful when the classes are imbalanced in the data

$$\begin{aligned}\text{Balanced Acc} &= \frac{\text{Sensitivity} + \text{Specificity}}{2} \\ &= \frac{1}{2}\left(\frac{\text{TP}}{\text{TP} + \text{FN}} + \frac{\text{TN}}{\text{TN} + \text{FP}}\right).\end{aligned} \tag{9}$$

The precision, recall, and F1-measure indicators are calculated based on binary classification. For multi-class prediction, these indicators can be averaged in two ways, i.e., macro-average and micro-average (Ballabio et al. 2018; Kautz et al. 2017). The macro-average takes all classes as having the same weights, whereas the micro-average weights the classes with the number of observations that the class consists of.

2.3.2 Performance metrics of regression models

The regression models are usually assessed by the amount of error, which means the differences between the observation and the predictions (Verbeek and Nijman 1993; Huang and Xu 2015). In this study, some common performance metrics of regression models are applied as follows:

1. Mean absolute error (MAE):

$$\text{MAE} = {}^{1}/_{n} \sum_{i=1}^{n} |d_i - p_i| \tag{10}$$

2. Mean square error (MSE):

$$\text{MSE} = {}^{1}/_{n} \sum_{i=1}^{n} (d_i - p_i)^2 \tag{11}$$

3. Root-mean-squared error (RMSE):

$$\text{RMSE} = \sqrt{{}^{1}/_{n} \sum_{i=1}^{n} (d_i - p_i)^2} \tag{12}$$

4. *Relative squared error* (RSE):

$$\text{RSE} = \frac{\text{MSE}}{Var(d)} = \frac{{}^{1}/_{n} * \sum_{i=1}^{n} (d_i - p_i)^2}{{}^{1}/_{n} * \sum_{i=1}^{n} (mu - d_i)^2} \tag{13}$$

5. Coefficient of determination (R^2):

$$R^2 = 1 - \text{RSE} \tag{14}$$

3 Related work

Multiple studies have been conducted on generating predictions on traffic data, such as predicting travel destinations, modeling origin–destination routes, predicting travel time, and forecasting transportation demands. These studies have focused on building location predictors for forecasting travel destinations, taxi distributions, and origin–destination routes, forecasting travel time, and predicting traffic demands. These predictions are important for transportation planning, traffic management, and providing better service to drivers and passengers. The predictive models used for these studies include time-series models and machine learning models, such as classification, regression, clustering, and neural networks. Below are some of the most important works.

3.1 Predictions on traffic data

Multiple works have been carried out to generate predictions on traffic data, such as predicting travel destinations, modeling origin–destination routes, predicting travel time, and forecasting transportation demands.

Some studies focused on building location predictors, which can be used to forecast travel destinations, taxi distributions, and origin–destination routes. He et al. (2014) argued that predicting the taxi destination can benefit car-sharing services. Zhang et al. (2018) forecasted the taxi destination to match passengers with the nearest available taxi to reduce the waiting time. De Brébisson et al. (2015) tried to predict the taxi locations for planning the best path and providing intelligent navigation. Polson and Sokolov (2017) stated that understanding the origin–destination distribution of taxi trips can improve the effects of transportation planning and enhance the quality of taxi services. Li et al. (2018) studied estimating the origin–destination flow for smart transportation. Medeiros et al. (2024) explored the merging of transport network companies and taxis within the Bus Rapid Transit (BRT) system, demonstrating the potential of integrated systems to enhance urban mobility. Similarly, Veloso et al. (2011) uncovered urban human mobility for city and transportation planning.

In addition, forecasting travel time is another key topic when mining traffic data. Bin et al. (2006) proposed that the bus arrival time can be predicted based on the travel time of the current sector and the next sector, which can benefit the public transportation management system. Häll et al. (2015) proposed a simulation-based approach for evaluating the performance of dial-a-ride services and found that factors, such as vehicle capacity, passenger waiting times, and the size of the service area, can significantly affect the service's performance. Moccia et al. (2022) presented that urban transport systems are influenced by headway, capacity, and operating speed. Ladino et al. (2017) stated that accurate travel time predictions are crucial for traffic management and essential for drivers, passengers, and traffic management. In similar studies, a neural network model is suggested to forecast travel time (Wang et al. 2016) and service quality (Islam et al. 2016).

Furthermore, predicting traffic demands has attracted great attention from researchers. Tong et al. (2017) stated that making accurate predictions on the transportation demands can balance the supplies and the requests, which benefits traditional taxi service providers and brings advantages to modern on-demand taxi services. Therefore, they researched the large-scale online platform to propose an effective predictive model for forecasting taxi demand. Yang et al. (2018) analyzed the taxi demand with GPS trajectories. They suggested that understanding the impact factor of taxi demand is crucial for managing urban traffic. Similarly, Liao et al. (2018) forecast urban taxi demand with taxi trip records. Zaheer Khan et al. (2025) demonstrated that the placement of public transit stations significantly impacts accessibility and urban mobility through GIS-based analysis. For calculating the demand, Ke et al. (2017) proposed that the traffic demand at a specific region can be calculated using the total number of pick-ups and drop-offs over a given time interval at the location.

3.2 Predictive models for traffic data

Various predictive models have been proposed for analyzing traffic data, which can be broadly classified into two categories: time-series models and machine learning models. These models utilize various techniques such as classification, regression, clustering, and neural networks (Karlaftis and Vlahogianni 2011).

Moreira-Matias et al. (2012) utilized time-series techniques to predict taxi-passenger demand using GPS data. Three time-series techniques were integrated to predict the spatial and temporal distribution of taxi passengers. The proposed framework can make accurate predictions on taxi demand in the short term. Tong et al. (2017) proposed that complex non-linear models, while accurate, can be inflexible when features are high-dimensional and frequently changing. As a solution, they suggested a simple linear regression model to maintain flexibility in predictors and utilized optimization techniques to improve predictions. Iimi (2023) estimated the demand for informal public transport and found it highly elastic and sensitive to price and service quality.

In another study, Hung et al. (2015) applied clustering algorithms to identify patterns in the trajectories of taxi passengers. They suggested that the mobility of taxi passengers can be inferred from both spatial and temporal perspectives, and thus, similar trajectories can be grouped and analyzed to uncover movement patterns. De Brébisson et al. (2015) proposed a multi-layer perceptron (MLP) framework based on an artificial neural network (ANN) in which inputs were processed in hidden layers and predictions were produced in the output layer. The proposed method is capable of generating predictions with very little human intervention. Similarly, Zhang et al. (2018) presented a model based on a combination of the convolutional neural network (CNN) and the recurrent neural network (RNN) techniques. They argued that CNN has the ability to extract features from trajectory data, and the RNN can account for dependencies between input data, resulting in improved predictions. Golshanrad et al. (2025) proposed a model for predicting passenger origin–destination flows in online taxi-hailing systems. Their approach combines

K-means clustering with non-negative matrix factorization to identify travel patterns and employs a stacked recurrent neural network to predict travel counts, demonstrating significant improvements in accuracy compared to existing methods.

3.3 Performance of traffic predictive models

Diverse metrics were applied to evaluate the performance of different predictive models. The RMSE and mean percentage absolute error (MPAE) are widely used for assessing the performance of travel time predictors and traffic flow predictors, and the indicators of the confusion matrix, precision, recall, and the F1-measure are generally used to evaluate the accuracy of frequent route predictors.

Lee et al. (2009) proposed three knowledge-based predictors, the historical predictor, the current-time predictor, and the weight combination predictor, to forecast travel time and applied the RMSE and relative mean errors (RME) to compare the performance of the predictors. It showed that the weight combination predictor outperformed the others with lower RMSE and RME. Bin et al. (2006) also used the RMSE indicator to evaluate the accuracy of the SVM model and ANN model. According to their experimental results, the SVM outperformed the ANN model in predicting travel time. Similarly, You and Kim (2000) used the RMSE and MAPE indicators to assess the proposed model, which was designed for forecasting travel time.

Tang et al. (2018) used the Entropy-Maximizing model for modeling taxi trip distribution and applied MAE, RMSE, and MPAE to demonstrate that the Entropy-Maximizing model is superior to the Gravity model in performance comparison. Lv et al. (2014), Sun et al. (2015), and Li et al. (2018) studied traffic flow prediction and used the RMSE and MAPE indicators to evaluate the proposed models. He et al. (2014) used the indicators of the confusion matrix to assess the performance of the frequent route predictor proposed to plan the ride-sharing routing based on GPS data. Similarly, Niu et al. (2015) used precision, recall, and the F1-measure to measure the performance of O-Sense, a predictive model to find the fastest driving routes.

In the domain of taxi-demand prediction, a rich body of literature has explored various facets of urban transportation and ride-sharing dynamics. Tachet et al. (2017) provided a foundational understanding of the scalability of urban ride-sharing, revealing a universal scaling law that underscores the potential for ride-sharing across diverse urban landscapes. This work is complemented by Xie and Olszewski (2011), who delve into the impacts of road pricing on traffic, offering insights into demand elasticity that are crucial for predicting taxi demand under different pricing schemes. Santi et al. (2014) further quantified the benefits of vehicle pooling, providing a methodological framework that could enhance taxi-demand forecasting models by integrating the concept of shareability networks.

The empirical analysis of Nguyen-Phuoc et al. (2020) shifts the focus to customer loyalty and its determinants in the context of ride-hailing services, a factor that can significantly influence demand patterns. Noulas et al. (2015) leveraged open datasets to bring transparency to taxi transport, suggesting that data-driven approaches can lead to more accurate demand predictions. Wen et al. (2025) reviewed the resilience of urban transportation systems, which is a critical factor

for robust demand forecasting. Jin et al. (2019) examine the interplay between ride-hailing services like Uber and public transit, highlighting the socio-economic dimensions of urban transportation equity that can affect demand.

Clewlow and Mishra (2017) investigated the disruptive nature of ride-hailing services, providing a broader understanding of their adoption and utilization, which is essential for anticipating shifts in taxi demand. Grahn et al. (2021) analyzed the substitution patterns between transportation network companies (TNCs) and public buses, offering empirical evidence to refine demand prediction models. Hensher et al. (2021) critically assessed the concept of Mobility as a Service, probing into its potential to reshape travel behavior, a critical component of demand forecasting. Collectively, these studies provide a multi-dimensional view of the factors influencing taxi demand, from policy-induced changes to technological disruptions and evolving consumer preferences.

Each of these works contributes to a holistic understanding of taxi-demand prediction, providing both direct and indirect insights into the factors that drive taxi usage. Our research builds upon these foundations, seeking to integrate these diverse perspectives into a comprehensive predictive model that accounts for economic, social, and network-related variables.

In line with findings from related work, the utilization of multinomial logistic regression (MLR), generalized linear regression (GLR), *random forest* (RF), and *decision tree* (DT) algorithms in our experiments emerges as a judicious and well-supported choice for modeling taxi-passenger demand through both classification and regression tasks. A wide body of research highlights these algorithms as some of the most commonly adopted and effective methods to predict demand patterns, estimate travel time, and forecast pick-up/drop-off locations in urban mobility studies. In addition, these four models were driven by a combination of their popularity in the literature, their proven performance in similar contexts, and their suitability for the computational constraints of real-world deployments. In particular, MLR and GLR offer robust regression capabilities, while RF and DT provide flexibility for both classification and regression with high interpretability, an important consideration for practical applications where model transparency is essential. This selection balances model complexity with practical feasibility. While more advanced algorithms exist, our focus is on widely recognized, interpretable, and efficient models. By using these models, our study contributes to a targeted and replicable understanding of taxi-demand prediction in realistic conditions.

4 Case study

In this section, we describe the dataset, experimental settings, types of performed experiments, and the quality measures used to compare three machine learning approaches: regression, decision tree, and random forest.

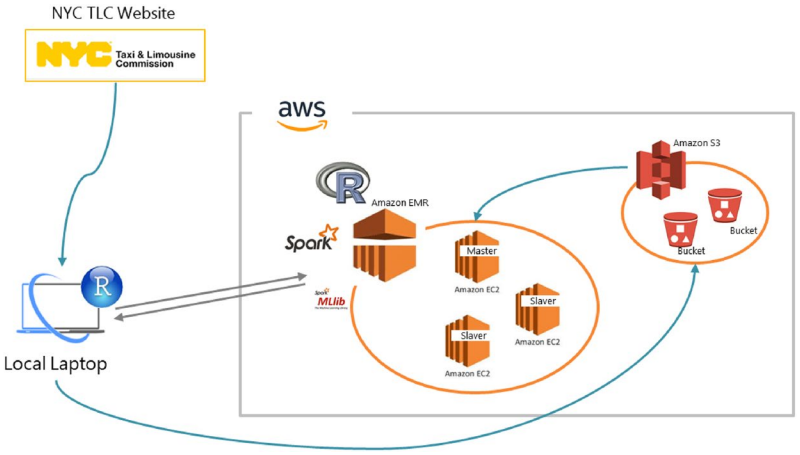

Fig. 1 Experimental instruments

4.1 Dataset description

The dataset for this study consists of taxi trip records from the New York City Taxi and Limousine Commission (NYC TLC). The dataset covers taxi trips from January 1, 2017, to March 31, 2017, with a total of 29,173,577 trip records. Each trip record contains information on the number of passengers, pick-up and drop-off times, locations of pick-up and drop-off, trip distance, fare, and payment type. This dataset provides sufficient coverage of different granularities, including six sub-datasets (with time intervals of 1, 2, 3, and 4 h, plus two peak-period subsets from 7-8 a.m. and 7-9 a.m.), and incorporates key temporal (hour, day) and spatial (pick-up location) features, making it well-suited for the forecasting tasks we aimed to evaluate. While some studies use longer timeframes, our 3-month period captures representative demand patterns while maintaining computational efficiency, as demonstrated in similar urban mobility studies (Ke et al. 2017; Zhao et al. 2016). The dataset's magnitude and comprehensive nature ensure a robust foundation for testing predictive models, with each instance representing aggregated demand for specific location–time pairs.

4.2 Experimental setup

The experimental instruments of this study are shown in Fig. 1. The dataset was uploaded to Amazon Simple Storage Service (S3), and the Amazon cloud computing service was employed to train the models and make predictions. Three Elastic Compute Cloud (EC2) instances were clustered by the Elastic MapReduce (EMR), one instance as the master node and the other two instances as the slave nodes. The Apache Spark and R server were deployed on EMR.

Three groups of experiments were conducted to assess and compare the performance of the classification and regression models of the Spark Machine Learning

library (MLlib)[2] using R language[3] based on Amazon cloud computing service.[4] The R packages-, such as "ggplot2", "e1071", "caret", and "dplyr", were used in the experiments.

While there is no single universally accepted big data framework, Apache Spark was selected for this study due to its high performance in distributed computing and strong support for iterative machine learning tasks. Integrated within Amazon EMR, it offers scalable and cost-effective processing, making it a widely adopted solution in data-intensive applications (Kaplunovich and Yesha 2018). The decision to use Spark was also guided by its compatibility with R and its popularity in the literature for similar time-series and demand forecasting problems. We implemented both classification (using multinomial logistic regression, random forest, and decision tree classifiers) and regression (using generalized linear regression and corresponding tree-based methods) approaches to provide comprehensive insights into model performance for different application scenarios. This dual methodology allows comparison with prior work while addressing both discrete demand-level prediction (useful for operational decisions) and continuous demand estimation (valuable for quantitative analysis). We recognize that other frameworks, such as Hadoop, Flink, or Dask, are also viable, but Spark provided the best balance for our computational needs, cloud integration, and experimental design (Amazon Web Services 2025).

4.3 Experimental design

The experiments were designed following the process of knowledge discovery in databases (KDD) (Fayyad et al. 1996), as shown in Fig. 2. The raw dataset was pre-processed, analyzed, and divided into training and prediction datasets. The selected predictive models were trained on the training dataset and used to make predictions on the test dataset. The predictions were then compared to the observed demand to evaluate the performance of the models.

Figure 3 illustrates the detailed design of the experiments. The raw data were grouped by departure location and departure hour to calculate the demand. Six datasets were generated with different time intervals, four of which included whole-day records and two of which included selected period records. For each dataset, the records of the first 89 days were split into a training dataset, and the records of the last day (March 31, 2017) were used as a predicting dataset, which

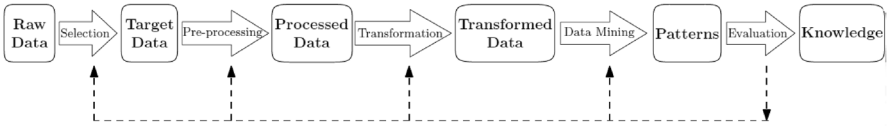

Fig. 2 The process of knowledge discovery in databases KDD for experimental design

[2] https://spark.apache.org/mllib/
[3] https://www.r-project.org/
[4] https://aws.amazon.com/

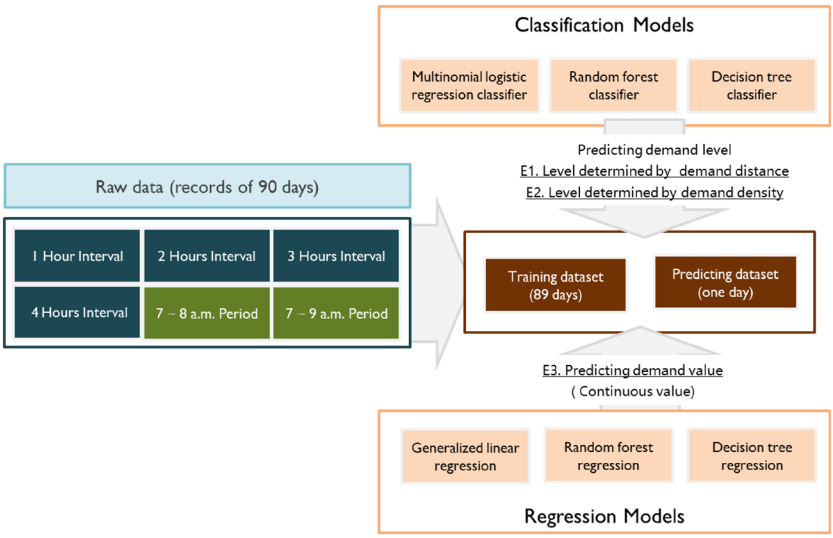

Fig. 3 Experimental design

Table 1 Descriptive statistics of taxi-passenger demand in the 1-h interval

Min	Max	Mean	Median	Mode	Variance	SD
0	3664	180	26	1	93,916	306

simulated the scenario of predicting 1-day demand using the taxi trip records of the past three months.

The datasets were labeled according to the level of demand for classification. Two methods were applied to nominalize the data. The first method involved labeling the demand based on equal distance, where the range of demand was the same for each class. For example, the demand from 0 to 360 was categorized as level one, 361 to 720 as level two, and the same for other levels. The second method involved labeling the demand based on equal density, where the number of records was the same for each class.

Three classification models were selected, including a multinomial logistic regression (MLR) classifier, a random forest (RF) classifier, and a decision tree (DT) classifier. Similarly, three regression models, generalized linear regression (GLR), RF regression, and DT regression, were chosen. Three groups of experiments were designed. The classification models were evaluated on the same distance classes datasets and the same density classes datasets in the first and second groups, respectively, and the regression models were assessed in the third group. All the models are tested and compared on the same datasets and test bed to ensure fairness.

As discussed in Sect. 2.3, the classification models are assessed via the accuracy, Kappa score, precision, recall, and the F1-measure. The macro-average

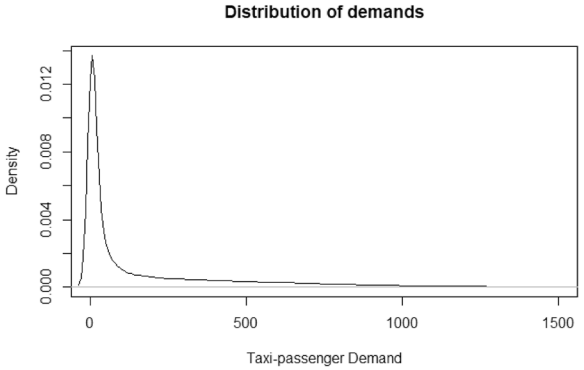

Fig. 4 Distribution of demands

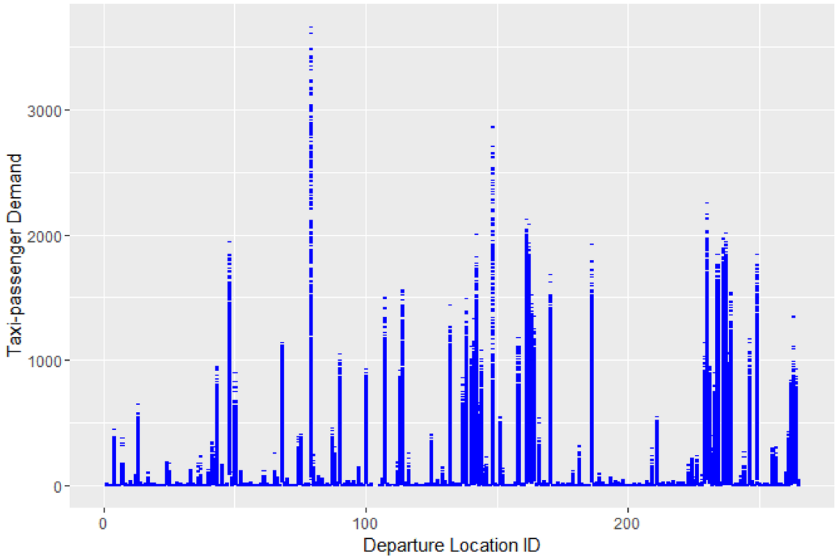

Fig. 5 Demand distribution by departure locations

value is applied to evaluate multi-class classification. The regression models are evaluated via the metrics of errors, MAE, MSE, RMSE, RSE, R^2.

4.4 Descriptive statistics

The experimental data should be understood deeply before being analyzed to mine valuable knowledge. Descriptive statistics demonstrate the central tendency and dispersion of taxi-passenger demand concerning temporal and spatial factors.

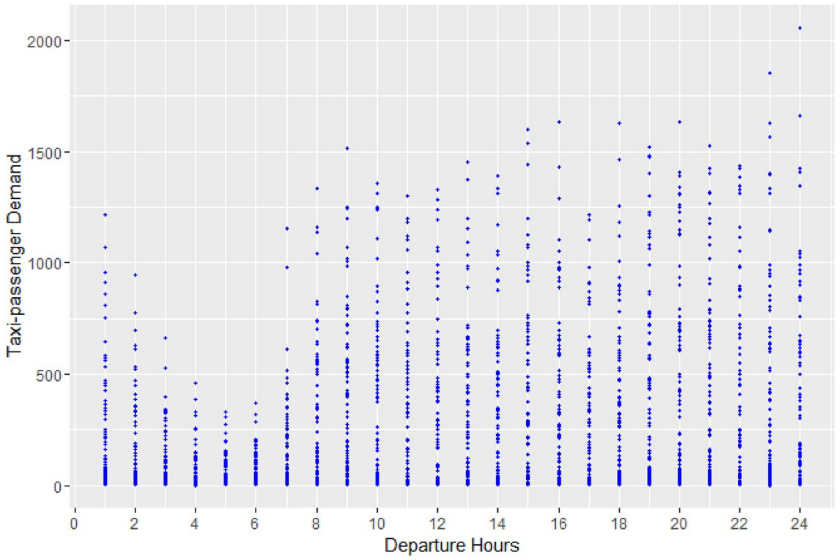

Fig. 6 Demand distribution by departure hours

4.4.1 Exploratory data analysis

We perform exploratory data analysis (EDA) to study the descriptive statistics of the dataset. The values for central tendency measures of 1-h interval demand are shown in Table 1, and the density of the demand is illustrated in Fig. 4. EDA shows that the range of the demand in the 1-h interval is from 0 to 3,664; however, the median is only 26, and the standard deviation (SD) is 306. The range is 12 times the SD; thus, the datasets were categorized into 12 classes in the following experiments.

Figures 5 and 6 expose the relationship between the demand and departure locations/departure hours, respectively. The demand varies considerably among different places. Meanwhile, the highest demand occurs at midnight, decreases dramatically after midnight, and reaches the lowest point at 5 a.m. The demand remains stable at approximately 1,500 per hour during work hours and increases again after 10 p.m. The figures indicate a solid relationship between the demand and temporal/spatial factors.

4.4.2 Findings from descriptive statistics

The density of the demand is absolutely uneven, as shown in Fig. 4. The demand under 100 accounts for most of the proportion, while the demand above 500 accounts for only a small percentage. The statistical data in Table 1 also support this imbalanced distribution. Zhao et al. (2016) and Ke et al. (2017) analyzed that this imbalance feature of the traffic data could be highly challenging to make predictions on traffic data. This opinion has been verified in the experimental

Table 2 Performance of the MLR classifier on the same distance classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	71.89	74.57	76.49	77.36	85.42	83.12
Kappa	28.01	31.30	33.87	37.11	72.42	70.38
Precision	13.51	14.07	15.35	14.88	44.68	61.02
Recall	18.01	19.23	20.71	26.40	53.46	66.68
F1-measure	14.29	15.09	16.63	16.11	46.88	62.20
BalancedAcc	37.10	42.00	42.70	45.63	67.76	82.61

Table 3 Performance of the RF classifier on the same distance classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	68.55	71.25%	72.99	73.84	65.97	62.34
Kappa	0.00	0.00	0.00	0.00	0.00	0.00
Precision	8.33	8.33	8.33	8.33	8.33	8.33
Recall	5.71	5.94	6.08	6.15	5.50	5.19
F1-measure	6.78	6.93	7.03	7.08	6.62	6.40
BalancedAcc	0.00	0.00	0.00	0.00	0.00	0.00

results of this study. Moreover, as illustrated in Fig. 5, the solid linear correlation between the demand and the departure location can be observed, which can also impact the evaluation results of different algorithms.

5 Performance evaluation of machine learning models

In this study, we evaluate the performance of three machine learning algorithms—multinomial logistic regression (MLR), random forest (RF), and decision tree (DT)—for both classification and regression tasks, as we aim to forecast taxi-passenger demand. The target feature for forecasting is initially processed to transform it into a categorical feature for the classification models and a continuous feature for the regression models.

For the classification models, we assess performance using a variety of quality measures, including accuracy, Kappa score, precision, recall, and the F1-measure. The macro-average value is applied to evaluate multi-class classification performance. For the regression models, we evaluate accuracy through error metrics, such as MAE, MSE, RMSE, RSE, and R^2, with a focus on forecasting accuracy. The evaluation ensures that both the classification and regression approaches are adequately assessed for their ability to predict taxi demand across different granularities.

5.1 Performance of classification models on the same distance classes

In our approach, taxi-passenger demand forecasting is treated as a classification problem by converting continuous demand values into categorical classes based on travel distance. The performance of the MLR classifier on six different datasets with the same distance classes is illustrated in Table 2. On the whole-day datasets, whether it concerns 1-h, 2-h, 3-h, or 4-h intervals, the accuracy of the MLR classifier is higher than 70%; however, the Kappa score and the F1-measure are relatively low. With the time interval extension, the Kappa score increases considerably, and the F1-measure still remains around 16%. The MLR classifier performs well on the period datasets, 7 a.m. to 8 a.m. and 7 a.m. to 9 a.m. The accuracy, Kappa, and F1-measure scores are better than those on the whole-day datasets.

The scores of the RF classifier on the same distance test datasets are demonstrated in Table 3. Although the accuracy of the RF classifier on all datasets is higher than

Table 4 Performance of the DT classifier on the same distance classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	69.89	72.22	74.10	74.65	68.75	65.58
Kappa	9.47	9.08	9.89	8.79	13.65	12.94
Precision	10.06	9.83	10.11	9.51	16.67	14.53
Recall	17.50	15.76	24.36	19.80	27.92	22.04
F1-measure	9.89	9.63	10.16	9.22	18.00	14.39
BalancedAcc	28.66	23.82	32.27	30.05	30.25	23.14

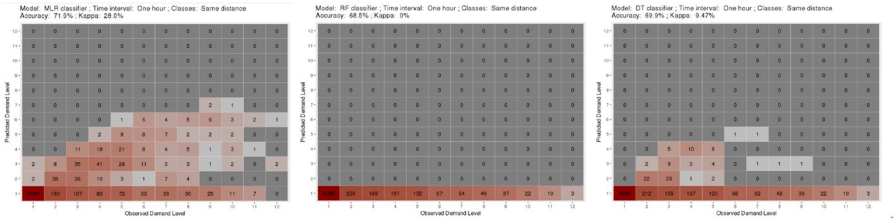

Fig. 7 Confusion matrix of the MLR, the RF, and the DT classifiers on 1-h interval dataset with the same distance classes

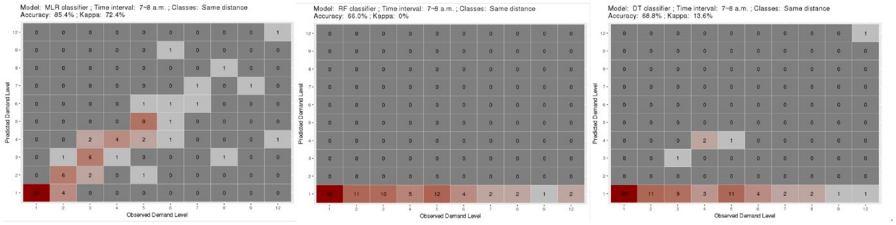

Fig. 8 Confusion matrix of the MLR, the RF, and the DT classifiers on 7-8 a.m. period dataset with the same distance classes

Table 5 Performance of multinomial logistic regression classifier on the same density classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	37.78	40.32	41.27	41.65	58.33	61.04
Kappa	32.11	34.86	35.97	36.39	54.38	57.33
Precision	37.31	39.70	40.94	41.56	56.91	60.29
Recall	37.30	40.68	42.00	42.28	57.61	60.06
F1-measure	36.46	39.27	40.63	41.32	56.26	59.46
BalancedAcc	65.83	67.63	68.34	68.49	76.91	78.26

Table 6 Performance of random forest classifier on the same density classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	19.46	18.16	18.33	19.11	25.69	20.78
Kappa	11.88	10.50	10.95	10.87	18.28	16.22
Precision	18.55	17.48	17.81	17.90	24.15	23.83
Recall	26.33	21.63	26.59	23.57	43.62	49.22
F1-measure	14.82	14.19	15.49	13.30	24.44	23.59
BalancedAcc	48.18	38.06	44.40	39.17	60.98	56.05

60%, the Kappa score is zero, and the F1-measure is only about 6.5%. The interesting finding is that the precision remains at 8.33% on all datasets. As mentioned in the previous section, there are 12 classes for the demand since the probability of random selection on 12 classes is 8.33%, which indicates that the precision of the RF classifier equals the result of the random selection.

The experimental results of the DT classifier of the first experiments group are shown in Table 4. It can be noted that the performance of the DT classifier is better than the performance of the RF classifier; nevertheless, it is still much lower than the performance of the MLR classifier.

Figures 7 and 8 illustrate the confusion matrices of three classifiers on the 1-h interval dataset and the 7-8 a.m. period dataset, respectively. It can be seen that the RF classifier predicts all the demand as "level-1", and the DT classifier makes the predictions in a similar way, whereas the MLR classifier makes over 60% true predictions on the 7-8 a.m. period dataset.

5.2 Performance of classification models on the same density classes

As with distance-based grouping, demand forecasting is approached as a classification task by categorizing continuous demand values into density-based classes. Table 5 illustrates the evaluating results of the MLR classifier on the same density classes. On the whole-day datasets, the accuracy of the MLR classifier is about 40%, while it is about 60% on the period datasets, which is much lower than the results

Table 7 Performance of decision tree classifier on the same density classes

Time interval	1 h (%)	2 h (%)	3 h (%)	4 h (%)	7–8 a.m (%)	7–9 a.m (%)
Accuracy	13.40	12.22	11.39	14.19	8.33	7.79
Kappa	5.26	4.60	3.53	5.30	2.82	3.32
Precision	12.74	12.30	11.38	12.68	11.40	12.26
Recall	15.13	24.70	21.85	14.16	33.81	17.06
F1-measure	6.52	6.60	6.50	4.55	6.43	6.62
BalancedAcc	27.01	35.57	34.02	22.78	40.16	20.51

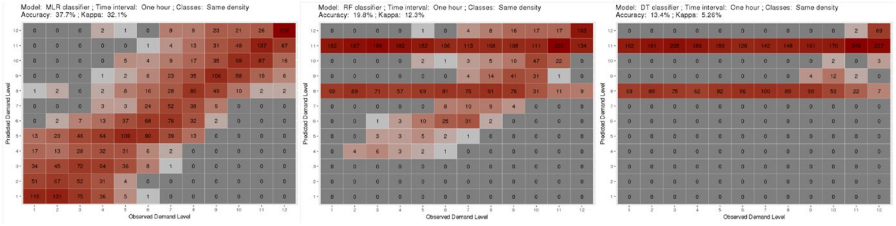

Fig. 9 Confusion matrix of the MLR, RF, and DT classifiers on a 1-h interval dataset with the same density classes

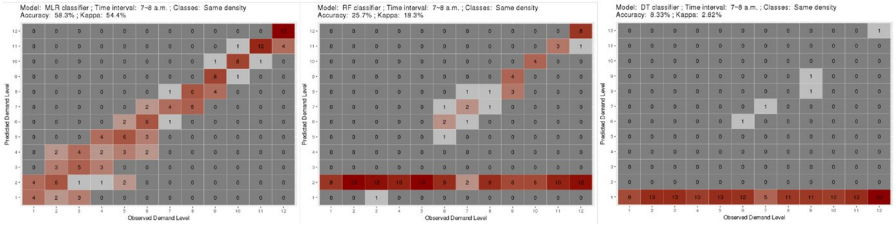

Fig. 10 Confusion matrix of MLR, RF and DT classifiers on the 7-8 a.m. period dataset with the same density classes

of the previous test group. By contrast, the Kappa score and F1-measure are much higher than the results of the last group. With the extension of the time interval, both the Kappa and the F1-measure scores increase.

Table 6 demonstrates the assessment results of the RF classifier on the same density test datasets. The accuracy of the RF classifier on the whole-day datasets is lower than 20%. Nevertheless, the Kappa score and the F1-measure in this test group are higher than the results of the last group, similar to the experimental results of the MLR classifier.

Table 7 shows the performance of the DT classifier on the same distance classes. It can be found that the indicators of the DT classifier in this test group are much lower than the performance evaluation results of the MLR and RF classifiers. Another

Table 8 Performance of generalized linear regression

Time interval	1 h	2 h	3 h	4 h	7–8 a.m	7–9 a.m
Mean of demand	200.12	349.81	486.66	614.45	279.77	472.65
Mean of prediction	185.34	323.78	450.07	570.10	221.75	370.52
MAE	105.88	187.48	259.54	321.72	71.89	124.52
MSE	28,848.59	96,975.36	189,002.80	289,077.70	21,840.54	62,856.54
RMSE	169.85	311.41	434.74	537.66	147.79	250.71
RSE	0.27	0.25	0.23	0.21	0.11	0.11
R^2	73.13%	74.94%	76.68%	78.64%	89.04%	89.22%

Table 9 Performance of random forest regression

Time interval	1 h	2 h	3 h	4 h	7–8 a.m	7–9 a.m
Mean of demand	200.12	349.81	486.66	614.45	279.77	472.65
Mean of prediction	179.85	313.34	435.47	547.02	210.90	349.42
MAE	200.40	375.00	534.30	675.66	256.96	449.67
MSE	77,140.80	281,780.90	584,524.90	951,307.10	121,046.60	390,830.30
RMSE	277.74	530.83	764.54	975.35	347.92	625.16
RSE	0.72	0.73	0.72	0.70	0.61	0.67
R^2	28.15%	27.20%	27.88%	29.70%	39.28%	32.96%

interesting finding is that the accuracy of the period datasets is lower than that of the whole-day datasets, which is different from the results of other experiments.

The confusion matrices of three classifiers on the 1-h interval dataset and 7-8 a.m. period dataset are illustrated in Figs. 9 and 10, respectively. On the 1-h interval dataset, the DT classifier predicted most demand as "Level-8" and "Level-11". The RF classifier made similar predictions with more true predictions. On the period dataset, the DT classifier classified most demand as "Level −1", while the RF classifier put them to "Level - 2". Most of the MLR predictions fell to the observed level or near the observed level, especially on the period dataset.

While comparing the scores of three selected classifiers in Tables 2, 3, and 4, it can be seen that the MLR classifier outperforms the RF and DT classifiers on all the test datasets. The same type of results can be found in Tables 5, 6, and 7. On the same distance datasets, the DT classifier performs better than the RF classifier with a higher score of accuracy, Kappa, and the F1-measure. However, the RF classifier outperforms the DT classifier on the same density datasets.

Furthermore, while closely analyzing the confusion matrix of the selected algorithms on different datasets, as shown in Figs. 7, 8, 9, and 10, it is obvious that the RF and DT classifiers have severe overfitting problems; they predict most of the demand as just one or two classes. By contrast, the MLR classifier has better sensitivity and specificity; it predicts most of the demand on the observed level or near the observed level, especially on the period dataset.

Table 10 Performance of decision tree regression

Time interval	1 h	2 h	3 h	4 h	7–8 a.m	7–9 a.m
Mean of demand	200.12	349.81	486.66	614.45	279.77	472.65
Mean of prediction	179.30	313.78	435.07	547.88	217.30	358.19
MAE	200.25	373.44	530.51	675.13	251.52	436.62
MSE	80,860.37	292,493.30	609,353.10	1,011,804.00	123,602.40	376,651.80
RMSE	284.36	540.83	780.61	1,005.89	351.57	613.72
RSE	0.75	0.76	0.75	0.75	0.62	0.65
R^2	24.68%	24.43%	24.82%	25.23%	38.00%	35.39%

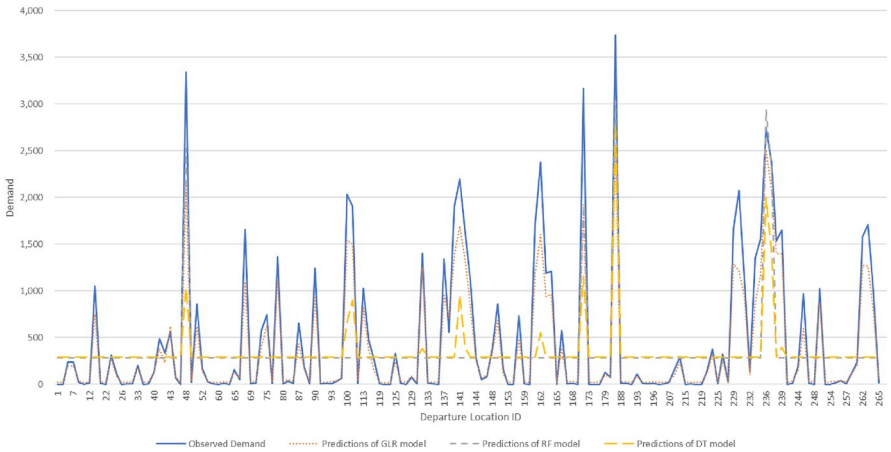

Fig. 11 Observed demand vs. predictions of the GLR model, the RF regression model, and the DT regression model by departure location

5.3 Performance of regression models

To evaluate the accuracy of taxi-passenger demand forecasting as a regression task, we used both linear and non-linear models, assessing performance based on standard error metrics. Table 8 illustrates the assessing results of the GLR model. It can be seen that the accuracy increases with the time interval extension. Furthermore, the coefficient on the period dataset is much higher than on the whole-day dataset. The R^2 indicator achieves nearly 90% over the period dataset, indicating high predictive accuracy.

Table 9 shows the performance of the RF regression. The changes in the scores on the different time intervals show the same trend as on the GLR model. However, the R^2 indicator is much lower than the score of the GLR model, at only about 30%.

The evaluation results of the DT regression are shown in Table 10. It can be noted that the DT regression has the lowest R^2 and the highest RMSE among the three selected regression models. Its scores are similar to the scores of RF regression.

The observed demand and predictions of three regression models are shown in Fig. 11. The predictions made by the GLR model are much closer to the observation; by contrast, the RF and DT regression predicted most of the demand around 300, which is near the standard deviation (SD) of the observed demand.

The GLR model outperformed the RF and DT regression with lower RMSE and higher R^2 score, as shown in Tables 8, 9 and 10. On the period dataset, the R^2 score of the GLR model is almost 90%, which indicates that the GLR model can predict the demand accurately with simple taxi trip records. Similarly, Tong et al. (2017) argued that the simple linear regression model could outperform the non-linear model in accuracy.

Figure 11 demonstrates that the predictions from the GLR model are better than the predictions from the RF and DT regression. It is noticeable that the GLR model is sensitive to outliers, whereas the RF and DT regression cannot make predictions with low values. Similar to the results of classification models, the RF and DT regressions overfit the demand at the SD level.

6 Discussion

Our discussion revolves around the exploration of predictive models for taxi-passenger demand, their performance evaluation in a big data framework, and the resulting recommendations.

In analyzing traffic data and predicting taxi-passenger demand, we assessed three primary models: generalized linear regression (GLR), Random Forest (RF) Regression, and decision tree (DT) Regression. The GLR model emerged as the top performer, excelling in terms of accuracy and reliability compared to RF and DT Regression models. The GLR model's lower Root-Mean-Square Error (RMSE) and higher R-squared (R^2) scores highlight its precision, especially with straightforward taxi trip records. However, the choice of model should align with dataset characteristics and task requirements, particularly when dealing with imbalanced datasets.

To assess predictive model performance in the context of big data, we employed a comprehensive set of metrics, including MAE, MSE, RMSE, RSE, and R_2. Emphasizing data preprocessing was vital due to dataset imbalances, ensuring a fair model evaluation. Our evaluation process was designed to address big data challenges, providing valuable insights for urban transportation management.

For demand classification, we compared multinomial logistic regression (MLR), Random Forest (RF), and decision tree (DT). MLR consistently outperformed the other models, exhibiting higher accuracy, Kappa scores, and F1-measures. MLR's strength in providing better sensitivity and specificity, especially for datasets with seasonal variations, was evident. In contrast, RF and DT models faced overfitting issues. When classifying distance vs. density-based classes, MLR excelled in distance-based scenarios, while RF performed better than DT in density-based cases. However, all models faced reduced accuracy in density-based classification. In the domain of demand prediction, GLR consistently outperformed Random Forest Regression and decision tree regression. GLR demonstrated higher accuracy and R-squared values, highlighting its effectiveness. In summary, the choice of a predictive model significantly impacts accuracy and reliability in predicting taxi-passenger demand. Data

preprocessing is vital for big data contexts, and our findings provide valuable guidance for researchers and practitioners in the urban transportation domain.

7 Conclusions and future work

The rapid increase in urban vehicles, particularly the growing number of licensed taxi drivers, has created an urgent need for better transportation planning and innovative approaches to urban data analysis. Simply expanding road infrastructure may not be enough to tackle the challenges this trend presents. This study examines predictive models for analyzing traffic data, beginning with a detailed review of existing literature that highlights how machine learning techniques like classification, regression, and clustering have been applied to forecast taxi demand.

The literature highlights that, among the various modeling techniques, classification and regression are the most commonly used for predicting taxi-passenger demand. Drawing from the literature review, we selected multinomial logistic regression (MLR), linear regression (LR), random forest (RF), and decision tree (DT) algorithms for further investigation. Performance evaluation of the classification models is based on metrics such as accuracy (Acc), Kappa score (κ), precision (Prec), recall (Rec), and F1-measure (F1). The regression models are assessed using critical indicators, specifically Mean Absolute Error (MAE), Mean Squared Error (MSE), Root-Mean-Squared Error (RMSE), and R-squared (R^2) to quantify their predictive capabilities.

Our experiments show that regression models, particularly the generalized linear regression (GLR) model, outperform RF and DT in predicting taxi-passenger demand. This is likely due to imbalanced demand data, which leads to local optimization and overfitting in RF and DT. These results highlight the importance of selecting models suited to the dataset and task, alongside robust evaluation metrics and data preprocessing, offering valuable insights for urban transportation planning.

Building on these findings, we provide practical implications for urban mobility through our research. Accurate demand predictions enable more efficient fleet management, reducing passenger waiting times and balancing taxi distribution to better meet demand. By optimizing taxi operations, these improvements in taxi services can help alleviate congestion and create opportunities for integrating taxis within a multi-modal transport framework, ultimately enhancing the overall efficiency of urban transportation systems.

The limitations of this work include time and resource constraints. Under ideal conditions, we would have conducted more extensive data collection and analysis, exploring additional factors like weather, holidays, and special events that could impact taxi-passenger demand. However, due to these constraints, our experiments primarily focused on whole-day and two-period datasets, leaving the correlations between demand and day-of-month and day-of-week unexplored. While we acknowledge that such constraints are common in academic research, our work lays the foundation for future research to dig deeper into these areas, enabling a more comprehensive understanding of taxi-passenger demand. Subsequent studies could prioritize gathering additional data on weather conditions, holidays, and events to

develop advanced predictive models. These limitations highlight the need for future work to address these constraints, potentially exploring more sophisticated models and incorporating additional variables.

In future research, exploring demand trends on different days of the week could provide insights into variations in taxi-passenger demand related to weekdays, weekends, holidays, and special events. Moreover, addressing imbalanced data challenges is essential. Researchers can focus on techniques like synthetic data generation, cost-sensitive learning, and advanced feature engineering methods. Additionally, investigating neural network architectures, anomaly detection methods, and ensemble learning may enhance prediction accuracy and robustness. These avenues of research aim to improve the field of taxi-passenger demand prediction by addressing various challenges and offering opportunities for better accuracy and reliability.

Author contributions S. Alam has presented the idea, updated the write-up for final submission, and supervised the project. MS. Ayub has improved the model and revised the write-up. H. Cui has performed the implementation and evaluation. MA. Khan has updated evaluations and reviewed the write-up.

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Data availability The datasets used during the current study are publicly available from the New York City Taxi and Limousine Commission (NYC TLC).

Declarations

Ethical approval Not applicable.

Informed consent Not applicable.

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

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EDITORIAL

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Recent advances in deep learning for traffic probabilistic prediction

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1. Introduction

Traffic prediction is an essential task in today's transportation management schemes, allowing the identification and prediction of crucial information such as the volume of traffic, speed, travel demand, time-of-travel, and movement trajectories (Ermagun & Levinson, 2018; Huang et al., 2023). In fact, these forecasts are a vital component in traffic planning and control, and ensuring efficient, safe and convenient travelling (Liu et al., 2022; Steentoft et al., in press). As has been identified, traffic prediction is of great importance in view of effective resource allocation, better traffic control, and better planning of travel and freight logistics (Cui et al., 2023). That will reduce the duration of travel, fuel consumption and result in lower environmental impacts (Sun et al., 2021). Now, in the context of autonomous vehicles and smart cities, traffic prediction has been subjected to the routing of individual vehicles for city-wide traffic management strategies (Pang et al., 2021; Xu, Di, Yang, et al., 2023).

Deep learning (DL) has attracted attention in the field of traffic prediction over the past decade and has far surpassed traditional forecasting methods by successfully learning the complex spatial and temporal dependencies in traffic data. Despite these, most deep learning-based models in the current literature make deterministic predictions, providing only single-point estimates about future traffic states. This paradigm ignores the intrinsic variability and uncertainty of traffic systems that can lead to suboptimal decisions and unexpected outcomes (C. Li et al., 2024). Recognising the deficiencies of deterministic models has driven interest in the recent application of deep probabilistic modelling in traffic prediction. Probabilistic models differ from deterministic methods because, instead of forecasting a single future state, they evaluate the probability of a number of possible outcomes and quantify the uncertainties associated with the forecast of these outcomes, which could benefit decision-making under uncertainty and equip traffic managers and systems to better predict and mitigate potential adverse conditions (Murça, 2021; Yu et al., 2020). Compared to conventional probabilistic prediction methods such as Gaussian mixture models (GMM) and Monte Carlo (MC) sampling, deep learning models have demonstrated superior accuracy and generalisability in probabilistic traffic prediction (Murça, 2021).

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This editorial provides a concise review focussing on the recent advancements in deep learning techniques for probabilistic traffic prediction. We will explore the methodologies used to quantify uncertainty, discuss their practical implications, and consider potential future directions in the field. By integrating probabilistic models, traffic management can progress towards more robust and resilient systems, adept at handling the dynamic and uncertain nature of real-world traffic environments (Fei et al., 2020; H. Li et al., 2023).

2. Deep learning methods in traffic probabilistic prediction

Probabilistic traffic prediction is a significant departure from previous deterministic prediction approaches in the sense that it provides a forecast with a range of possible traffic states and their corresponding probabilities. Probabilistic methods are crucial to mitigate the intrinsic uncertainties that exist in traffic systems, enhancing the dependability and precision of the results. Various techniques in this regard include Bayesian Deep Learning (BDL), Deep Gaussian Processes (DGPs), Quantile Regression (QR), and diffusion models. Bayesian deep learning views the model parameters as random variables instead of fixed elements; therefore, the model can express uncertainty regarding its predictions by giving probabilities for different traffic states (Liu et al., 2022; Pang et al., 2021; Sengupta et al., 2024). On the other hand, deep Gaussian processes represent a nonparametric approach that assumes the traffic flow follows a Gaussian distribution. This is particularly useful in approximating the expected traffic state and its associated uncertainties, especially when data is scarce (Steentoft et al., in press; Yang & Klabjan, 2021). In contrast, quantile regression predicts different percentiles of the traffic states, thus providing a range of possible solutions, rather than just an expected single mean value (H. Li et al., 2023). Furthermore, by noising the observed data in a specific way and training a neural network to denoise it, diffusion models are adept at generating plausible values for prediction targets, effectively estimating the probability distribution of potential outcomes rather than offering single-point estimates (Lei et al., 2024a).

The probabilistic approach can be further enriched by Ensemble Learning (EL) and Prediction Intervals (PIs) to integrate the predictions from various models for better prediction performance with uncertainty quantification (Mallick et al., 2024). In ensemble learning, several predictive models are combined to capture heterogeneous features of the data, providing more accurate forecasts using the mean or variance of the outputs from each predictive model (Noureldin et al., 2023; Olivier et al., 2023). PIs offer a range within which the true traffic condition is likely to fall, improving the prediction reliability (Y. Li et al., 2022, 2023). Additionally, Transformer models with attention mechanisms and Graph Convolutional Networks (GCNs) are often used in traffic probabilistic predictions to improve the model's focus and the learning of spatial dependency, respectively (Huang et al., 2023; Qian et al., 2024). Transformer-based models selectively focus on relevant data points, such as nearby road conditions or historical patterns, while GCNs represent road networks as graphs to learn traffic influence across connected road segments (Karim & Nower, 2024; Kong et al., 2023; Xu, Di, Yang, et al., 2023; Yu et al., 2020; Zhou et al., 2020).

Handling missing data and incorporating external factors are also critical in refining probabilistic traffic predictions. Models designed to impute missing values ensure reliability even with incomplete data sets (Xu, Di, Ding, et al., 2023). Moreover,

Table 1. A selection of recent deep learning applications in traffic probabilistic prediction.

Authors	Data	Method*	Target
Liu et al. (2022)	Freeway incident data, traffic sensor data	BDL	Incident detection
Steentoft et al. (in press)	Urban mobility flows, population data	DGPs	Mobility flow predictions
Pang et al. (2021)	Aircraft flight plans, weather data	BDL	Air traffic trajectory prediction
Xu, Di, Yang, et al. (2023)	Bike-sharing data	Transformer	Supply-demand prediction
Olivier et al. (2023)	Ambulance travel time data	BDL	Travel time prediction
G. Li et al. (2024)	Trajectory data	EL	Trajectory prediction
C. Li et al. (2024)	Ride-sourcing demand data	Transformer	Ride-sourcing demand prediction
Yu et al. (2020)	Road traffic speed data	GCNs	Traffic speed forecasting
Fei et al. (2020)	Vehicle trajectory data	EL	Vehicle trajectory prediction
H. Li et al. (2023)	Taxi GPS data	QR	Travel time prediction
S. Wang et al. (in press)	Vehicle trajectory data	EL	Trajectory prediction
Kong et al. (2023)	Traffic flow data	BDL	Traffic flow prediction
Qian et al. (2024)	Traffic speed data	GCNs	Traffic speed forecasting
Mallick et al. (2024)	Traffic flow data	DGPs	Traffic flow prediction
C. Li et al. (2020)	Public transit demand data	GCNs	Travel demand prediction
Y. Li et al. (2022)	Long-term traffic flow data	PIs	Traffic flow prediction
Y. Li et al. (2023)	Electric vehicle charging demand data	PIs	EV charging demand
Huang et al. (2023)	Electric vehicle charging data	Transformer	EV charging demand forecasting
Hu et al. (2022)	Traffic interaction data	GCNs	Traffic interactions prediction
Q. Wang et al. (2024)	Travel demand data	GCNs	Travel demand prediction
Tang et al. (2022)	Autonomous vehicle data	EL	Traffic flow prediction
Zhou et al. (2020)	Traffic speed data	GCNs	Traffic speed prediction
Sengupta et al. (2024)	Multi-source traffic data	BDL	Traffic flow prediction
Lei et al. (2024a)	Traffic flow with missing data	Diffusion model	Traffic flow prediction
de Nailly et al. (in press)	Pedestrian count data	PIs	Pedestrian count prediction
Zechin and Cybis (2023)	Traffic breakdown data	BDL	Traffic breakdown prediction
Karim and Nower (2024)	Traffic flow data	GCNs	Traffic flow prediction
Xu, Di, Ding, et al. (2023)	Traffic speed data	DGPs	Traffic speed prediction
Wu et al. (2021)	Spatiotemporal traffic data	BDL	Traffic flow prediction
Zhuang et al. (2022)	Sparse travel demand data	GCNs	Travel demand prediction

*BDL: Bayesian Deep Learning, DGPs: Deep Gaussian Processes, EL: Ensemble Learning, GCNs: Graph Convolutional Networks, QR: Quantile Regression, PIs: Prediction Intervals.

incorporating inputs like weather conditions, events, and road closures allows models to adjust predictions based on external influences (Y. Li et al., 2022; Pang et al., 2021). The evaluation of these probabilistic models necessitates metrics that account for the predictive uncertainty, using tools such as the Prediction Interval Coverage Probability (PICP), Mean Interval Score (MIS), and Continuous Ranked Probability Score (CRPS) (Y. Li et al., 2022; Zhang et al., 2020). By integrating these diverse techniques, researchers and practitioners can develop comprehensive and dependable probabilistic traffic prediction models, which are crucial for informed decision-making in traffic planning, management and control. In Table 1, we present a selection of recent deep learning applications in traffic probabilistic prediction.

3. Practical implications and future directions

3.1. Practical implications

Probabilistic traffic prediction using deep learning models could bring significant practical value to real-world applications in multiple aspects of Intelligent Transportation

Systems (ITS). Recognising the intrinsic uncertainty in traffic systems, the use of deep probabilistic models could provide a deeper understanding of traffic dynamics and guide better decision-making for transportation agencies (Fei et al., 2020). In particular, traffic management can use a probabilistic congestion forecast to take proactive control actions, such as dynamic speed limits, ramp metering, and re-routing, and therefore mitigate congestion before it actually happens (Chen et al., 2023). In public transport, the probabilistic prediction of the Estimated Time of Arrival (ETA) can be used to refine passenger information systems, allowing travellers to make better-informed decisions regarding departure times and route choices (Chen et al., 2023; Ricard et al., 2022). Moreover, the probabilistic predictions could facilitate improved risk assessment in safety-critical applications such as autonomous driving by providing probabilities for different traffic scenarios, allowing systems to anticipate and mitigate potential hazards more effectively (Liu et al., 2022; Tang et al., 2022). In addition, navigation applications can use probabilistic prediction outcomes to provide multiple routing solutions based on the computation of some confidence bounds so that the user can choose between shorter routes that might have delays and longer but more certain routes (Sun et al., 2021). Probabilistic traffic prediction could also help calibrate traffic simulation models to better represent the variability encountered in reality and test traffic management strategies under conditions of uncertainty (Olivier et al., 2023). Still, more widely, probabilistic models can be used to improve public transport planning by modelling the probability of delays; therefore, it helps transit agencies optimize schedules, allocate resources more effectively and give passengers estimates of arrival times that are more reliable (C. Li et al., 2020; Lyons & Marsden, 2021; Q. Wang et al., 2024).

3.2. Future directions

While significant advancements have been made in probabilistic traffic prediction using deep learning techniques, there remain several promising research directions that need further exploration. Efficient probabilistic deep models are expected to be developed while considering the interplay between various modes of transportation, including cars, public transportation, walking, and cycling, to facilitate intelligent multimodal transport systems (Lei et al., 2024b). The results of probabilistic prediction using deep learning should be extended toward long-term reliable predictions that could be used for effective strategic planning and infrastructure development. Furthermore, more scholarly focus is needed to improve the interpretability of deep probabilistic models, allowing a deeper understanding of the factors that influence predictions and the assessment of uncertainties (Noureldin et al., 2023). In addition, there is an emerging demand for deep probabilistic models capable of adapting in real time, with the ability to make updated predictions and reassess uncertainties based on newly obtained data – an essential feature for effective traffic management systems. Pang et al. (2021). Improving data efficiency with deep probabilistic models, despite having limited amounts of training data, is of great importance in some resource-limited environments. Integrating probabilistic traffic prediction using deep learning with other ITS applications, such as connected vehicles and autonomous driving systems, could also contribute to creating more efficient and safer transportation systems (Zechin & Cybis, 2023).

In addition, some deep Bayesian methods are quite computationally intensive under the large-scale datasets available now for uncertainty quantification in traffic predictions (G. Li et al., 2024). This creates an imperative need for developing more efficient and scalable deep Bayesian inference techniques for practical applications (Wu et al., 2021). Combining different sources of data such as loop detectors, GPS devices, cameras, and social media could also result in much more precise and reliable predictions in probabilistic traffic prediction using deep learning. Furthermore, it is essential to develop deep probabilistic modelling techniques that can capture the dynamics of traffic flow influenced by various factors, including weather conditions, accidents, special events, and driver behaviour (de Nailly et al., in press). Lastly, it is crucial to integrate uncertainty estimates into decision-making frameworks for traffic planning and management to ensure robust and optimal solutions (Hu et al., 2022). For example, real-time traffic management systems can leverage these uncertainty estimates to implement more resilient control strategies that proactively adapt to unforeseen events and minimize disruptions.

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Conformal multistep-ahead multivariate time-series forecasting

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Abstract
Time-series forecasts underpin decision-making processes in a wide range of application domains. Recently it has been shown that these processes can be strengthened by conformal prediction, a framework that allows adding prediction intervals to point forecasts. The prediction intervals quantify the uncertainty of a predictive model with mathematical coverage guarantees, giving the user a range of scenarios to consider. However, applying conformal prediction to time-series tasks is not trivial. This is either because the exchangeability condition the framework places on the data is violated, or because the framework only allows for one-step-ahead univariate forecasts. In this article we combine two existing methods derived from conformal prediction, one built for multi-target regression and one designed to handle non-exchangeable data. The resulting method, called non-exchangeable multi-target conformal prediction (nmtCP) produces provably robust prediction regions for multi-step ahead multidimensional time-series forecasts, meaning that the miscoverage rate is bound. Additionally, nmtCP is computationally efficient and easy to implement. Due to its model-agnostic nature, nmtCP can be used on top of any time-series model that produces point forecasts. A theoretical analysis proves the method's robustness while experiments on real-world data sets give insights into its practical behavior and performance.

Keywords Conformal time series forecasting · Conformal prediction · Multivariate multi-step ahead forecasting · Uncertainty quantification

1 Introduction

Time-series forecasting is an important part of the decision-making process in a wide range of application domains such as low-carbon electricity (Rolnick et al., 2019; Xu & Xie, 2021), stock price predictions, service demand forecasting, and medical prognoses (Stankevičiūtė et al., 2021). While modern time-series forecasting techniques have become increasingly accurate (Hossain & Mahmood, 2020; Lin et al., 2021, 2021) they

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usually produce point forecasts, providing little to no information about the uncertainty associated with their predictions. This limited quantification of the uncertainty constrains decision-making in practice, especially in high-stakes situations.

Recently, *conformal prediction* has been proposed for uncertainty quantification of models' predictions (Vovk et al., 2005; Barber et al., 2022). It is a modern, model agnostic framework capable of providing valid prediction sets or intervals without making assumptions about the distribution of the data. Conformal prediction relies however on two assumptions that are generally not met in time-series forecasting tasks. It assumes, that the examples in the data set are exchangeable and that the model fitting algorithm for the underlying model treats the examples in the data set symmetrically, i.e. that the model fitting algorithm produces the same model independently of the order of the examples in the training set.

Multiple attempts (Gibbs & Candès, 2021; Barber et al., 2022; Chernozhukov et al., 2018; Feldman et al., 2022; Stankevičiūtė et al., 2021) have been made to lift these requirements imposed by conformal prediction. While some of these techniques allow for non-exchangeable data and asymmetrical model-fitting algorithms, their applicability is generally restricted because they are limited to univariate, single-step-ahead time-series forecasting tasks.

Motivated by Rolnick et al. (2019), we set out to analyze if and how conformal prediction can be extended to the more complex multistep-ahead multivariate time-series forecasting tasks. The work of Stankevičiūtė et al. (2021) provides an approach that allows for the construction of valid prediction regions for multi-target regression by combining multiple conformal predictors. However, its application to time-series is limited by the exchangeability requirement. Barber et al. (2022) propose a method that is specifically designed to overcome the requirements of the original conformal predictor and render it applicable to time-series forecasting tasks. These properties are achieved under the assumption that the method's ideal parameters are known or can be estimated well enough. While powerful, their method is bound to single-target regression. We propose a new method, *non-exchangeable multi-target conformal prediction* (nmtCP), that is capable of producing robust prediction regions for multistep-ahead multivariate time-series forecasting tasks. nmtCP achieves this by combining the approaches taken by Barber et al. (2022) and Stankevičiūtė et al. (2021). This article describes the general concept of nmtCP, shows how it is implemented in practice, offers a theoretical analysis of its validity, and compares it to existing methods.

The development of nmtCP is preceded by an extensive literature review that spans form a standard conformal prediction method to the most recent adaptations for the time-series domain, describing their respective contributions. This includes a discussion of the difficulties that arise when applying conformal prediction to multi-target regression and some solutions proposed by other researchers in the field.

Following the theoretical analysis of nmtCP, a series of experiments validate its theoretical properties on four synthetic and two real-world data sets. This also serves to compare nmtCP to another state-of-the-art method and to discuss its behavior and performance. To conduct the practical experiments, we implement the method as one step in a modular pipeline that can easily accommodate different underlying models.

The contributions of this article are summarized as follows:

- An in-depth literature review of related works. It provides a comparison of existing methods with regard to their applicability to non-exchangeable data and to multi-target regression in the context of time-series forecasting.

- A new method, nmtCP, to generate prediction regions for multistep-ahead multivariate time-series forecasting tasks.
- A theoretical proof of the validity of nmtCP.
- Systematic experiments applying nmtCP that confirm the theoretical results.

While a subset of these results was included in Schlembach et al. (2022), that publication misses the context provided by the literature review and the discussion and, crucially, the theoretical proof provided in this article.

The remainder of this article is structured as follows. First, Sect. 2 introduces conformal prediction and discusses its subsequent extensions related to time-series forecasting and multi-target regression tasks. Next, Sect. 3 describes nmtCP and its theoretical properties. Subsequently, in Sect. 4, nmtCP's theoretical claims are tested experimentally on simulated and real-world data sets. Finally, Sect. 5 gives the conclusions and provides an outlook for future research.

2 Background and related work

This section provides the necessary background and reviews the relevant related work. Subsect. 2.1 introduces (inductive) conformal prediction by Vovk et al. (2005) and the learning settings in which it can be used. This is followed by a description of the extensions of ICPs for time-series forecasting in Subsect. 2.2. Subsection 2.3 introduces recent methods for conformal multi-target regression. Further analyses and discussions are provided in Subsection 2.4.

2.1 Conformal prediction

The upper bounds on the probability of error of machine learning algorithms' predictions provided by statistical learning theory are generally too loose to be useful in practice (Vovk et al. 2005, p.4). This is why Vovk et al. (2005) set out to quantify the confidence of machine learning algorithms' outputs derived from statistical learning theory. Their idea is to use a model's performance on previous samples as well as information about the current sample to estimate the confidence with which the label of a new object is predicted. *Conformal Prediction*, the resulting method, can be used on top of almost any machine-learning algorithm while retaining its theoretical guarantees (Vovk et al., 2005) under minimal assumptions (Toccaceli, 2022). It usually only requires that the examples in the training set are exchangeable and that the underlying model treats them symmetrically (Barber et al., 2022). Conformal Prediction also requires minimal to no modifications of the underlying model (Stankevičiūtė et al., 2021). In the next two subsections we provide a detailed description of computationally efficient conformal predictors used in the method presented in this article.

2.1.1 Inductive conformal predictors

Let $\mathcal{X}$ be a measurable space of objects and $\mathcal{Y}$ be a measurable space of labels [cf. Vovk et al. (2005)]. The example space $\mathcal{Z}$ is defined as the Cartesian product $\mathcal{X} \times \mathcal{Y}$. A sequence $z_{1:n}$ of n training examples $z_1, \ldots, z_n$ is generated by an unknown probability distribution Q

defined over $\mathcal{Z}$. We note that no assumptions are made about Q beyond the exchangeability assumption. In this context we can train two types of predictors: simple predictors and confidence predictors.

A *simple predictor* h is defined as $h : \mathcal{Z}^{(n)} \times \mathcal{X} \to \mathcal{Y}$ where $\mathcal{Z}^{(n)}$ is the set of all bags of size n of elements from $\mathcal{Z}$. Given a training bag $\{z_{1:n}\} \in \mathcal{Z}$ and object $x_{n+1} \in \mathcal{X}$, the predictor h assigns a label $\hat{y}_{n+1} \in \mathcal{Y}$ to x_{n+1} (Vovk et al., 2005, p.18). In contrast, $\Gamma : \mathcal{Z}^{(n)} \times \mathcal{X} \times (0,1) \to 2^{\mathcal{Y}}$ is a *confidence predictor*. Given training bag $\{z_{1:n}\} \in \mathcal{Z}$, object $x_{n+1} \in \mathcal{X}$, and *significance level* $\alpha \in (0,1)$, the predictor Γ assigns a region $\hat{\mathbf{y}}_{n+1}$ of $\mathcal{Y}$ to x_{n+1} (Vovk et al., 2005, p.19).

A confidence predictor Γ is said to be *exactly valid* or *exact* if for any example $z_{n+1} = (x_{n+1}, y_{n+1})$ the error probability $\mathsf{P}(y_{n+1} \notin \hat{\mathbf{y}}_{n+1}) = \alpha$, i.e. the probability of the true label y_{n+1} not being included in the prediction region $\hat{\mathbf{y}}_{n+1}$ equals significance level α (Vovk etal., 2005, p.20). If $\mathsf{P}(y_{n+1} \notin \hat{\mathbf{y}}_{n+1}) \leq \alpha$, confidence predictor Γ is said to be is *conservatively valid* (Vovk et al., 2005, p.20). Vovk et al. (2005, p.21) show that no confidence predictor achieves exact validity.

In addition to being valid, confidence predictors need to be *informationally efficient*. A confidence predictor Γ is said to be *informationally efficient* if the prediction regions $\hat{\mathbf{y}}_{n+1} \subseteq \mathcal{Y}$ are non-empty and as small as possible (Vovk et al., 2005, p.20).

Inductive conformal predictors (ICPs) are conservatively valid confidence predictors. They are usually a preferred choice due to their computational efficiency. This is achieved by splitting the training data $\mathcal{D} = \{z_{1:n}\}$ into two disjoint sets (bags), a proper training set $\mathcal{D}_{train} = \{z_{1:m_0}\}$ of size m_0 and a calibration set $\mathcal{D}_{cal} = \{z_{m_0+1:n}\}$ of size $m_c = n - m_0$ (Toccaceli, 2022)[1] The proper training set $\mathcal{D}_{train}$ is used to train a *nonconformity measure* (NCM) $A : \mathcal{Z}^{(m_0)} \times \mathcal{Z} \to \bar{\mathbb{R}}$. A is a measurable function that estimates how different any example in $\mathcal{D}_{cal}$ is from the examples in $\mathcal{D}_{train}$ (Vovk et al., 2005).

A common class of nonconformity measures is based on a simple predictor $\hat{\mu} = h(\mathcal{D}_{train})$. An example of such a nonconformity measure A are residuals:

$$r_i = \begin{cases} |y_i - \hat{\mu}(x_i)| & \forall i = m_0 + 1, \ldots, n \\ |\bar{y}_{n+1} - \hat{\mu}(x_{n+1})| & i = n+1 \end{cases} \tag{1}$$

Once the residuals have been computed for the validation set, an ICP determined by the NCM (1) produces the prediction region:

$$\hat{\mathbf{y}}_{n+1} = \Gamma^{\alpha}(\mathcal{D}, x_{n+1}) = \hat{\mu}(x_{n+1}) \pm \mathsf{Q}_{1-\alpha}\left(\sum_{i=m_0+1}^{n} \frac{1}{m_c+1} \cdot \delta_{r_i} + \frac{1}{m_c+1} \cdot \delta_{+\infty} \right) \tag{2}$$

where $\mathsf{Q}_{1-\alpha}$ denotes the quantile function and δ_x the Kronecker delta function. $\mathsf{Q}_{1-\alpha}$ returns the $\lceil (1-\alpha)(m_c+1) \rceil$th smallest value of the residuals $r_{m_0+1:n}$ (Barber et al., 2022; Stankevičiūtė et al., 2021).

We note that in contrast to other types of conformal predictors, ICPs split the training data D and use less data for training the nonconformity measures as well as for computing

[1] Due to the exchangeability of the examples z_i, this split is done randomly, even if the index might suggest otherwise.

prediction set. Thus, ICPs are relatively less informally efficient; i.e. they trade the informal efficiency for the computational efficiency.

For a more in-depth overview of conformal prediction, please refer to (Fontana et al., 2023; Toccaceli, 2022; Vovk et al., 2005). An excellent hands-on introduction to conformal prediction is provided by Angelopoulos and Bates (2023).

2.1.2 On-line, semi-on-line, semi-off-line and off-line settings

ICPs can be used in on-line, semi-on-line, semi-off-line and off-line settings. This subsection outlines how these situations are handled in practice.

In an *on-line setting* nature provides objects x_i sequentially and their corresponding labels y_i delayed by one time step (Vovk et al., 2005, p.5). This means that at the $n_0 + 1$st step when $\mathcal{D} = \{z_{1:n_0}\}$, an ICP as defined in Eq. (2) can generate a prediction region $\hat{\mathbf{y}}_{n_0+1} = \Gamma^{\alpha}(\mathcal{D}, x_{n_0+1})$ (Vovk et al., 2005, p.5). Then, the true label y_{n_0+1} and a new object x_{n_0+2} are revealed and the ICP can make the prediction $\hat{\mathbf{y}}_{n_0+2}$, and so on (Vovk et al., 2005, p.6). As a consequence the training data $\mathcal{D}$ grows at every time step. In this setting, the underlying model is re-trained at every time step for a new split of the updated training data $\mathcal{D}$ as shown in Algorithm 1.

Algorithm 1 On-line inductive conformal prediction for every new time step.

Require: $\mathcal{D} \in \mathcal{Z}^{(n-1)}$ ▷ bag of known examples
Require: $x_n \in \mathcal{X}$ ▷ last observed object
Require: $\alpha \in [0, 1]$ ▷ significance level
Require: $c \in [0, 1]$ ▷ relative calibration set size

$x_{n+1}, y_n \leftarrow$ observe reality
$\mathcal{D} \leftarrow \mathcal{D} \cup \{(x_n, y_n)\}$
$m_c = \lceil c \cdot (n) \rceil$ ▷ $|\mathcal{D}_{cal}|$
$m_0 = n - m_c$ ▷ $|\mathcal{D}_{train}|$
$\mathcal{D}_{train}, \mathcal{D}_{cal} \leftarrow$ split $\mathcal{D}$ into bags of size m_0 and m_c
$\hat{\mu} \leftarrow h(\mathcal{D}_{train})$ ▷ fit the model
for each $(x_j, y_j) \in \mathcal{D}_{cal}$ **do**
    $r_j \leftarrow |y_j - \hat{\mu}(x_j)|$
end for
$\hat{y}_{n+1} \leftarrow \hat{\mu}(x_{n+1})$
$\hat{\mathbf{y}}_{n+1} \leftarrow \hat{y}_{n+1} \pm \mathsf{Q}_{1-\alpha}\left(\sum_{j=1}^{m_c} \frac{1}{m_c+1} \cdot \delta_{r_j} + \frac{1}{m_c+1} \cdot \delta_{+\infty}\right)$
$n \leftarrow n + 1$
return $\hat{\mathbf{y}}_{n+1}$

ICPs in the online setting, such as presented in Algorithm 1, require the underlying model h to be fitted for every new object that is encountered. To fully utilize the computational improvements provided by ICPs, Vovk et al. (2005, p.98) propose to use sequence of numbers $m_1 < m_2 < \ldots$ with $m_i \in \mathbb{N}$. These numbers m_i act as thresholds that determine after how many new observations the underlying model is fitted to an updated training set. Once the number of known examples n surpasses a previously unsurpassed m_k, such that $m_k < n \leq m_{k+1}$, the underlying model is fitted to proper training examples $\mathcal{D}_{train} = \{z_{1:m_k}\}$. The remaining examples $z_{m_k+1:n}$ are used as calibration set $\mathcal{D}_{cal}$ to

compute the residuals (Vovk et al., 2005, p.99). Moving forward we call this situation the *semi-on-line setting*.

Vovk et al. (2005, p.110) call the semi-on-line setting with only a single m_1 *semi-off-line ICP*. The underlying model is only trained once while nature reveals new objects sequentially and their true labels with a delay. Once a label becomes known, it, together with its associated object, is added to the calibration set. This procedure ensures that the method retains the validity guarantees (Vovk et al., 2005, p.111).

Finally, in the *off-line setting*, nature does not provide the objects and labels sequentially. Instead, it provides all the labeled data $\mathcal{D} = \{z_{1:n}\}$ and a set of objects $x_{n+1:n+m_u}$ with unknown labels. In this situation the underlying model is fitted only once to the proper training set $\mathcal{D}_{train}$ and the ICP is calibrated on the calibration set $\mathcal{D}_{cal}$. Then the resulting ICP generates prediction regions for all $x_{n+1:n+m_u}$. We note that this setup weakens the validity guarantee. While $\mathsf{P}(y_i \notin \hat{\mathbf{y}}_i) \leq \alpha, \forall i \in n+1, \ldots, n+m_u$ still holds, the errors are no longer independent and $\sum_{i=1}^{n+m_u} \text{err}_i / m_u \leq \alpha$ may no longer be true (Vovk et al., 2005, p.112).

Note that we make the distinction between the on-line setting, the semi-on-line setting, the semi-off-line and the off-line setting based on the number of times the underlying model is fitted to a set of examples and the points in time when that happens. This classification is different from the one used by Toccaceli (2022) who uses *batch mode of operation* to denote what we call the off-line setting.

To summarize, we describe the different settings using the following terms:

- On-line setting: the underlying model is fitted for every new example that nature provides with a new training and calibration split;
- Semi-on-line: the underlying the model is fitted periodically, after $m_1, m_2, \cdots$ examples and the calibration set is updated with every new observed label;
- Semi-off-line: underlying model is only trained once and the calibration set is updated with every new observed label;
- Off-line: the underlying model is only trained once and the calibration set is not updated.

2.2 Conformal time-series forecasting

The definition of ICPs in an on-line setting provided in Eq. (2) and the subsequent variations lose their validity guarantee if applied to time-series. This is because the observations in time-series are generally not exchangeable. In this subsection, we present the work of some researchers that have sought to overcome this limitation. We group the presented methods broadly by the approach the authors took. This is preceded by an introduction to the notation we use for the representation of time-series data.

Going forward we use the following notation. $o_{t,j} \in \mathbb{R}$ denotes the value of feature j at time step t. $o_t = o_{t,1:F} \in \mathbb{R}^F$ represents the vector of observations for all F features at time step t and $o_{1:T} \in \mathbb{R}^{T \times F}$ stands for the entire multivariate time-series with T time steps and F features. In the case where only a single feature is observed, the second index is omitted and $o_{1:T} \in \mathbb{R}^T$ represents a univariate time-series with T time steps.

A *single-step ahead time-series forecast* predicts $y = o_{T+1}$ given a univariate time-series $x = o_{1:T}$. If $x = o_{1:T}$ is a multivariate time-series, the single-step ahead forecast predicts some or all features of $y = o_{T+1,1:F}$. In contrast, a *multi-step ahead time-series forecast* (called *multi-horizon time-series forecast* by some authors (Stankevičiūtė et al., 2021))

predicts the values $y = o_{T+1:T+T_h}$ for multiple future time steps in the univariate setting. In the multivariate setting, it predicts the value for some or all features of these future time steps $y = o_{T+1:T+T_h}$. The number of predicted future time steps T_h is called the *prediction horizon*.

2.2.1 Methods relying on block-wise exchangeability

Chernozhukov et al. (2018) and Stankevičiūtė et al. (2021) soften the exchangeability condition by requiring the data to be block-wise exchangeable. Chernozhukov et al. (2018) first split the time-series into blocks containing an equal number consecutive observations. These blocks are permuted while the values within the block retain their order to preserve the dependence structure of the time-series. They treat the elements between the last known observation and the prediction horizon as one such block. The authors prove that their method remains approximately valid under weak conditions on the nonconformity score, relaxing the exchangeability condition of the data. However, the method requires an infinite number of hypothetical labels $\bar{y}$ to be tested for regression tasks, and is not developed further to benefit from the improvements in computational efficiency presented in Subsect.. 2.1.2.[2] For exact validity, their method requires the examples to be exchangeable under the permutations π, a condition which might not be met for time-series with long term dependencies surpassing the block size.

Stankevičiūtė et al. (2021) base their CF-RNN method on blocks of consecutive observations that do not belong to the same time-series but are drawn from the same distribution. This makes it impractical for a single multivariate time-series. Subsection 2.3 contains a more detailed discussion of CF-RNN.

2.2.2 Adaptive methods

Instead of constraining the observations, Xu and Xie (2021) make assumptions about the time-series' forecasts' stochastic errors and the estimation quality of the underlying models to build distribution-free prediction intervals with approximate coverage guarantee. In practice, they create B bags $\mathcal{D}_b$ of size n by sampling with replacement from the known examples $z_{1:n}$. Next, they fit B underlying models $\hat{\mu}_b = h_{\mathcal{D}_b}, \forall b \in B$. The residuals

$$r_i = |y_i - \phi(\{\hat{\mu}_b(x_i)\ \forall b : z_i \notin \mathcal{D}_b\})| \tag{3}$$

are the absolute difference between the real label y_i and the aggregated predictions of all underlying models that were not trained on z_i. ϕ is the chosen aggregation function. Finally, they produce the prediction interval

$$\hat{\mathbf{y}}_{n+1} = \mathrm{Q}_{1-\alpha}(\{\hat{\mu}_b(x_{n+1})\ \forall b : z_i \notin \mathcal{D}_b\}_{i=1}^{n}) \pm \mathrm{Q}_{1-\alpha}(r_{1:n}) \tag{4}$$

for a new object x_{n+1}. In addition, they implement a mechanism to add the residual r_{n+1} when the label for a new object becomes available and to remove the oldest residual. This sliding window applied to the residuals used in the quantile function in Eq. (4) allows the method to adapt over time. Xu and Xie (2021) prove that EnbPI produces valid prediction intervals under two assumptions. The first assumption is that the $\{r_{1:n}\}$ are stationary and

[2] The reason for this is that these improvements only work for single target regression. For a more detailed analysis see Subsect. 2.3.

strongly mixing. The second assumption states that the average difference between the residuals and the stochastic part of the data generating process decreases over time. Xu and Xie (2021) test their EnbPI method on single-step ahead and multi-step ahead solar power predictions tasks. They compare it to the ARIMA method, which does not retain valid coverage in the experiments.

2.2.3 Weighted methods

Gibbs and Candès (2021) construct their *adaptive conformal inference* (ACI) method explicitly with a data distribution that shifts over time in mind, moving away from the exchangeability assumption. They do this by introducing two additions. The first addition is increasing or decreasing the significance level α over time, based on ACI's performance, making it an adaptive method. The second addition is weighing the nonconformity scores in the quantile function, laying the foundation for other weighted methods. Although Gibbs and Candès (2021) describe ACI in more general terms, we only show the situation where an underlying model is used to produce point forecasts in the semi-off-line setting to keep the notation consistent. In this situation initially n examples $z_{1:n}$ are known. We split them into two disjoint sets, a proper training set $\mathcal{D}_{train} = \{z_{1:m_0}\}$ of size m_0 and a calibration set $\mathcal{D}_{cal} = \{z_{m_0+1:n}\}$ of size $m_c = n - m_0$. After fitting the underlying model $\hat{\mu} = h_{\mathcal{D}_{train}}$ to the training set, we use it to compute the nonconformity scores $r_{m_0+1:n}$ on the calibration set. For future objects x_{n+i} that nature provides sequentially, ACI computes the prediction interval

$$\hat{\mathbf{y}}_{n+i} = \hat{\mu}(x_{n+i}) \pm \mathsf{Q}_{1-\alpha_{n+i}}\left(\sum_{j=m_0+1}^{n+i-1} \frac{1}{m_c+i} \cdot \delta_{r_j} + \frac{1}{m_c+i} \cdot \delta_{+\infty}\right) \tag{5}$$

with

$$\alpha_{n+i} = \alpha_{n+i-1} + \gamma(\alpha - \mathrm{err}_{n+i-1}) \tag{6}$$

starting at $\alpha_{n+1} = \alpha$. An alternative version of the method

$$\hat{\mathbf{y}}_{n+i} = \hat{\mu}(x_{n+i}) \pm \mathsf{Q}_{1-\alpha_{n+i}}\left(\sum_{j=m_0+1}^{n+i-1} \frac{\tilde{w}_j}{m_c+i} \cdot \delta_{r_j} + \frac{\tilde{w}_{n+i}}{m_c+i} \cdot \delta_{+\infty}\right) \tag{7}$$

Gibbs and Candès (2021) present under the same name introduces a series of increasing weights $\tilde{w}_j$ on the nonconformity scores r_j with $\sum \tilde{w}_j = 1$. It uses a weighted quantile function, giving more recent nonconformity scores a higher weight compared to older ones. In both cases, the *step size parameter* γ determines how quickly the method adapts to shifts in distribution. Gibbs and Candès (2021) define the

$$\text{coverage gap} = (1 - \alpha) - \mathsf{P}\{y_{n+i} \in \hat{\mathbf{y}}_{n+i}\} \tag{8}$$

as the loss in coverage compared to what would be achieved if the examples were exchangeable. They show that by correctly calibrating α_{n+i}, the coverage gap can be made arbitrarily small. In addition, they prove that using the on-line update scheme described in Eq. (6), ACI is guaranteed to be asymptotically valid. This stems from the observation that $\forall m_1 \in \mathbb{N}$

$$\left| \frac{1}{m_1} \sum_{i=n+1}^{n+m_1} \text{err}_i - \alpha \right| \leq \frac{\max\{\alpha_{n+1}, 1-\alpha_{n+1}\} + \gamma}{m_1 \gamma} \tag{9}$$

by taking $\lim_{m_1 \to +\infty}$ (Gibbs & Candès, 2021). To avoid discontinuity and prove these theoretical guarantees the quantile functions in Eqs. (5) and (7) use Dirac delta instead of the Kronecker delta introduced in Eq. (2).

Zaffran et al. (2022) provide an extensive analysis of the validity, the efficiency and the general behavior of ACI in various situations. They argue that forecasting of dependent examples, even in the absence of distribution shifts, can benefit from the theoretical framework created by Gibbs and Candès (2021). In addition, they propose two methods to automatically select a value for the learning rate γ. The first one consists in computing the quantiles for $K \in \mathbb{N}$ different values of γ and, for every new object, choosing the one that has produced the smallest intervals in the past while retaining validity. The second one is called *Online Expert Aggregation on ACI* (AgACI). Instead of just choosing the learning rate γ that performed best in the past, the intervals for different values of γ are aggregated using an on-line aggregation rule. Zaffran et al. (2022) compare their method to ACI, EnbPI and semi-off-line ICP. They show that simply choosing the value for γ that was most efficient in the past does not lead to valid results. In their experiments EnbPI and semi-off-line ICP lose validity, while AgACI is more efficient than ACI and (almost) retains validity.

Feldman et al. (2022) present a methods they call *rolling conformal inference* (Rolling CI) that aims to avoid splitting the known examples into a proper training set and a calibration set as is done for ICP. It builds directly upon ACI by removing the calibration set and and adapting the interval width in a fully on-line manner. Assuming n known examples $z_{1:n}$, Rolling CI uses all of them to fit the underlying model $\hat{\mu} = h_{\lfloor z_{1:n} \rfloor}$. It then constructs the prediction region

$$\hat{\mathbf{y}}_{n+i} = f(x_{n+i}, \theta_{n+i}, \hat{\mu}_{n+i}) \tag{10}$$

for every new object x_{n+i} that is observed sequentially using an interval construction function f that takes the new object x_{n+i}, a calibration parameter θ_{n+i} and the current underlying model $\hat{\mu}_{n+i}$ as inputs. The calibration parameter

$$\theta_{n+i+1} = \theta_{n+i} + \gamma(\text{err}_{n+i} - \alpha) \tag{11}$$

is updated after the label of y_{n+i} is revealed. This calibration parameter θ_{n+i} controls the size of the predicted region, similar to α_{n+i} in the ACI method. Finally, Rolling CI updates the underlying model to $\hat{\mu}_{n+i+1}$ using a single gradient step if the model supports it. Feldman et al. (2022) show, that for Rolling CI to produce valid intervals according to

$$\lim_{m_1 \to +\infty} \frac{1}{m_1} \sum_{i=1}^{m_1} \text{err}_{n+i} = \alpha \tag{12}$$

the only condition is the existence of two constants c_1, c_2 such that $f(x, \theta, \hat{\mu})$ returns an empty interval for $\theta < c_1$ and $\mathcal{Y}$ for $c_2 < \theta$. These requirements are fulfilled by the quantile functions in Eqs. (5) and (7) of ACI. They also show that

$$\left| \frac{1}{m_1} \sum_{i=1}^{m_1} (\text{err}_{n+i} - \alpha) \right| \leq \frac{\max\{\theta_{n+1} - c_1, c_2 - \theta_{n+1}\} + \gamma}{m_1 \gamma} \tag{13}$$

for a finite number of examples m_1. In contrast to the other methods, Rolling CI can directly be applied to a multi-output regression setting and therefore also to a multi-step ahead time-series prediction setting. In this case, the parameter θ inflates or deflates the predicted region in all dimensions.

Barber et al. (2022) directly address the two exchangeability conditions of the original conformal prediction method. Their *non-exchangeable conformal prediction* (nexCP) method does not require the examples to be exchangeable and does not require that they are treated symmetrically by the algorithm fitting the underlying model. A symmetric model fitting algorithm produces the same model[3] independently of the order of the examples in the training set. Barber et al. (2022) apply their developments to CPs, ICPs, and the jackknife+ method. Due to its computational efficiency, we focus on the adaptation of ICPs. In the semi-off-line setting, the method splits the $n = m_0 + m_c$ known examples into a proper training set $\mathcal{D}_{train}$ of size m_0 and a calibration set $\mathcal{D}_{cal}$ of size m_c. The underlying model $\hat{\mu} = h_{\mathcal{D}_{train}}$ is fitted to the proper training set and the calibration set is used to compute the residuals $r_{m_0+1:n}$. By using the weighted quantile function introduced in Eq. (7), the method adapts the generated prediction intervals to shifts in distribution without changing the value of α that is passed to the quantile function. This is achieved by assigning residuals of more recent examples higher weights than residuals of older ones. Assuming a new object x_{n+1} with the associated label y_{n+1}, Barber et al. (2022) show that the coverage gap as defined in Eq. (8) is bound by

$$\text{coverage gap} \leq \frac{\sum_{i=m_0+1}^{n} w_i \cdot \mathsf{d}_{\mathsf{TV}}(\mathcal{D}, \mathcal{D}^{[i]})}{1 + \sum_{i=m_0+1}^{n} w_i} \tag{14}$$

where $\mathcal{D} = \mathcal{D}_{cal} \cup \{z_{n+1}\}$ and $\mathcal{D}^{[i]}$ swaps elements i and $n+1$ in $\mathcal{D}$. d_{TV} is the total variation distance. This means more generally, that by reducing the weights w_i associated with residuals of examples that contribute to an increase in the coverage gap, the validity of the predicted intervals can be improved. While a careful selection of the weights w_i leads to a reduction in the coverage gap and thereby addresses the exchangeability of the examples it does not lift the requirement, that the model fitting algorithm for the underlying model treats the examples symmetrically. This can be ignored for ICPs as the solution to lift this restriction proposed by Barber et al. (2022) does not change the method beyond what has already been discussed.[4] Barber et al. (2022) show further that

$$\mathsf{P}(y_{n+1} \in \hat{\mathbf{y}}_{n+1}) \geq 1 - \alpha - \sum_{i=m_0+1}^{n} \tilde{w}_i \cdot \mathsf{d}_{\mathsf{TV}}(r_{1:n}, r_{1:n}^{[i]}) \tag{15}$$

where the weights $\tilde{w}_i = w_i / \left(\sum_{i=m_0+1}^{n} w_i \right)$ on the nonconformity scores are normalized weights, $r_{1:n}$ are the residuals computed on $\mathcal{D}$ and $r_{1:n}^{[i]}$ the residuals computed on $\mathcal{D}^{[i]}$. By setting $w_i = 1$, $\forall i \in \{m_0 + 1, \ldots, n\}$, these results also give an upper limit for the miscoverage rate of the original ICP in case the exchangeability condition is violated. Barber et al. (2022) test their method applied to a CP in three settings, comparing it to the standard CP. In the first setting, using i.i.d. examples, they show that their method performs

[3] The model fitting algorithm needs to produce models with the same distribution for randomized algorithms.

[4] Please refer to the original paper for the changes necessary for CP and jackknife+.

similarly to the standard CP. In the second and third setting, the examples present change points and distribution shift respectively, and nexCP retains approximately valid coverage while the standard CP does not.

2.3 Conformal multi-target regression

The work presented in Subsect. 2.2 addressed the non-exchangeability of time-series data and how conformal prediction can be adapted to produce valid prediction intervals in the presence of non-exchangeable data. This subsection treats the difficulties that arise when applying conformal prediction to multi-target regression and some proposed solutions.

In the case of single-target regression, the quantile function $\mathsf{Q}_{1-\alpha}$ in Eq. (2) returns the $\lceil (1-\alpha)(m_c+1) \rceil$th smallest value of the residuals, denoted as r_j here. For an object x_{n+1}, the prediction region is constructed by selecting all possible labels y such that the associated nonconformity score satisfies $r_{n+1} \leq r_j$. When using the absolute residuals as NCM, this is achieved by simply solving

$$|y - \hat{\mu}(x_{n+1})| \leq r_j. \tag{16}$$

Solving Eq. (16) for y produces exactly one region, the prediction interval $[\hat{\mu}(x_{n+1}) - r_j, \hat{\mu}(x_{n+1}) + r_j]$, as solution. This shows that there exists a bijective relationship between the upper and lower bound of the prediction interval and the nonconformity score r_{n+1} necessary to reach the desired coverage rate.

In the multi-target regression setting, i.e. for a label space $\mathcal{Y} \subset \mathbb{R}^F$,[5] replacing the absolute residuals by the Euclidean distance as NCM is a natural choice. This allows for a procedure that is similarly computationally efficient to the one outlined for the single-target regression case. Solving Eq. (16) for this generalized NCM results in a prediction region bound by a hyper-sphere centered around $\hat{\mu}(x_{n+1})$ and with a radius of r_j. While simple to compute, such a prediction region does not provide the user with any information about the error the underlying model makes in the various dimension of the label space. Instead, it assumes the underlying model is equally likely to make an error in any direction. This can lead to an inefficient prediction region, with a volume larger than necessary, if the underlying model's errors are generally larger in some directions than in others. In addition, a prediction region in the shape of a hypersphere is difficult to visualize if the label space has more than three dimensions. While Messoudi et al. (2020) note that "[...] conformal prediction for multi-target regression is still a largely unexplored area", some propositions have been put forward.

Vovk (2013) and later Stankevičiūtė et al. (2021) circumvent the issue of applying conformal prediction to multi-target regression by calibrating one CP for every dimension in the label space, reverting to single target regression and a one-dimensional label space for each. Equation (19) shows how they combine the individual prediction intervals generated by these CPs for every dimension into a multi-dimensional prediction region. Stankevičiūtė et al. (2021) explicitly extend ICP to univariate multi-horizon time-series forecasts. In this situation, the objects

[5] F stands for an arbitrary number of dimensions here and not necessarily the number of considered time-series.

Table 1 Comparison between the different methods discussed in Sect. 2. *Coverage guarantee* column displays the long-term theoretical coverage guarantee the original authors provide for their methods. *Multi-target* column displays if the method supports multi-target regression. *NE data* column displays if the method retains the coverage guarantee in the presence of non-exchangeable data. Symbol ✻ indicates methods that allow for short term dependencies, but lose their validity guarantee if applied to time-series with long term dependencies

Method	Coverage guarantee	Multi-target	NE data
CIbP Chernozhukov et al. (2018)	$1-\alpha$	✓	✻
CF-RNN Stankevičiūtė et al. (2021)	$1-\alpha$	✓	✻
EnbPI Xu and Xie (2021)	$1-\alpha$	✗	✗
ACI Gibbs and Candès (2021)	$1-\alpha$	✗	✓
AgACI Zaffran et al. (2022)	none	✗	✓
Rolling CI Feldman et al. (2022)	$1-\alpha$	✓	✓
nexCP Barber et al. (2022)	$1-\alpha$	✗	✓
Messoudi et al. (2020)	$1-\alpha$	✓	✗

$$x_i = o^{(i)}_{1:T_w} \in \mathbb{R}^{T_w} \tag{17}$$

are the observations made over T_w time steps and the labels

$$y_i = o^{(i)}_{T_w+1:T_w+T_h} \in \mathbb{R}^{T_h} \tag{18}$$

are the observations made during the following T_h time steps. The superscript (i) indicates that the observation $o^{(i)}$ belongs to the ith example. Given a set of exchangeable examples $z_{1:n}$, they follow the ICP procedure and split them into a proper training set $\mathcal{D}_{train}$ of size m_0 and a calibration set $\mathcal{D}_{cal}$ of size m_c. Then they fit an underlying model $\hat{\mu} = h_{\mathcal{D}_{train}} : \mathbb{R}^{T_w} \to \mathbb{R}^{T_h}$ to the training set and compute the residuals $r_i = |y_i - \hat{\mu}(x_i)| \in \mathbb{R}^{T_h}$ for the calibration set. For a new object x_{n+1} the predicted region

$$\hat{\mathbf{y}}_{n+1} = \left[\hat{\mu}(x_i)_j \pm \mathrm{Q}_{1-\alpha/T_h}(r_{m_0+1:n,j}),\ \forall j \in \{1,\dots,T_h\}\right] \in \mathbb{R}^{T_w \times 2} \tag{19}$$

is a T_h dimensional hyper-rectangle, where every dimension corresponds to one time step in the prediction horizon. T_h conformal predictors construct this hyper rectangle, one for each dimension. These conformal predictors use only the residuals associated with that particular time step in the prediction horizon from the examples in the calibration set in their quantile function. Stankevičiūtė et al. (2021) apply the Bonferroni Correction, setting the confidence level of the quantile function to $1-\alpha/T_h$ instead of $1-\alpha$. This guarantees, that the predicted region is valid in general. This approach allows for dependence within the $o^{(i)}_{1:T_w+T_h}$ that form an example z_i, as only the z_i have to be exchangeable. Designed with recurrent neural networks (RNNs) as underlying models in mind, Stankevičiūtė et al. (2021) call their method CF-RNN. They test their method for different RNN based underlying models on synthetic and real world data sets and show that the produced intervals are valid.

Messoudi et al. (2020) try to account for the interactions of errors in each dimension of the label space through the use of different NCMs. Similar to Vovk (2013) and Stankevičiūtė et al. (2021), the produced prediction region is a hyper-rectangle, however Messoudi et al. (2020) employ the Šidák correction, which assumes independence and

which is a little less stringent than the Bonferroni correction. They use the median volume of the produced prediction regions as metric for the efficiency of the compared NCMs. While the authors identify general trends, the best NCM depends on the data set.

2.4 Discussion

Methods generating prediction intervals for time-series should, as is the case for conformal predictors in general, be valid and efficient. Due to the nature of time-series, such a methods need to support non-exchangeable data and produce intervals for more than one feature and more than one future time step. Table 1 offers a comparison of the methods discussed in Sect. 2 according to these criteria. Ideally, it would also be computationally efficient (or at least feasible in practice) and user-friendly by providing guidance on how to set the method's parameters.

While other methods to convey uncertainty in the form of prediction intervals exist Lin et al. (2021), they do not come with the validity guarantees that some methods derived from CPs offer and that are desired in high stakes situations (Feldman et al., 2022). Gibbs and Candès (2021) have shown that these other methods can be used as underlying models for CP based methods. Because of this, we have chosen to focus on CP based methods.

As mentioned at the beginning of Subsect. 2.2, the lack of exchangeability in time-series data means that the conditions for the validity guaranteed by CP and ICP are not met. CIbP relaxes this condition by splitting the data into blocks consisting of values of consecutive time steps. Chernozhukov et al. (2018) assume that these blocks are exchangeable. By regarding each block as a point in a multidimensional space instead of a sequence of consecutive scalar values, with each value corresponding to a time step, the values within the block no longer need to be exchangeable. While some time-series may present such a block wise exchangeability, it is not given in general, especially if the time-series displays long term dependencies. Another drawback of the proposed method is that by evaluating all possible values for all dimensions of the label space, the method considers an infinite number of label candidates and is therefore not applicable in practice. CF-RNN overcomes the computational complexity of CIbP by estimating the intervals for every dimension in the label space individually and correcting for the family-wise error rate. CF-RNN relies on a block wise exchangeable structure in the same way as CIbP making it unsuitable for time-series with long term dependencies or a change in distribution over time.

EnbPI takes a different approach to relaxing the exchangeability condition for the examples by requiring the residuals to be stationary and strongly mixing. This assumption is only met if the underlying model maintains the quality of its predictions when the distribution of the data changes over time. Because this is difficult to prove in general, EnbPI is listed as not retaining coverage in presence of non-exchangeable data in Table 1. Xu and Xie (2021) apply EnbPI to the multi-step ahead setting by predicting the future time steps in the prediction horizon individually. If the method's conditions are met, this approach guarantees validity for every individual dimension of the label space, but it does not guarantee validity in general. This is why Table 1 lists EnbPI as not supporting multi-target regression.

By growing or shrinking the generated intervals according to the coverage errors it has made in the past, ACI achieves the desired coverage frequency in the long run without making any assumptions about the data or the errors the underlying model

makes. This achievement comes at the cost of a quantile function that can return infinite values resulting in a predicted region that covers the entire label space which is uninformative. The version of the method using the unweighted quantile function requires only a single additional parameter, the learning rate γ, adding to its ease of use. ACI is also computationally efficient as it takes advantage of all advancements made for ICPs. The method is developed for single-target forecasts, meaning it suffers from the challenges discussed in Subsect. 2.3 when applied to multivariate multistep-head time-series forecasts.

Multiple extensions for ACI are proposed. AgACI automatically estimates the learning rate γ thereby achieving better efficiency. This requires the choice of an on-line aggregation method and does not address any of the other shortcomings of ACI. Rolling CI is the second extension of ACI. It does not require a split of the known examples, allowing the underlying model to be fitted to all of them. Rolling CI also allows for multi-target forecasts by growing or shrinking the predicted region in all dimensions. If a shift in the distribution only occurs for a subset of features, this might lead to overly conservative intervals for the others.

nexCP could be seen as a third extension of ACI. It departs from the idea of adapting the α used in the quantile function based on the observed error rate of the method, and focuses instead on the weighted quantile function. This approach comes with strong theoretical guarantees. However, just as ACI, nexCP may produce infinite prediction intervals and faces the same the challenges when applied to multivariate multistep-head time-series forecasts. In addition, more research is required to determine the optimal weights for the weighted quantile function.

Applying conformal prediction to multivariate multistep-head time-series forecasts poses two challenges. The first being that time-series are generally non-exchangeable, and the second, that it is not sufficient to produce a prediction interval but instead a predictive region needs to cover all dimensions of the label space. Different methods have been proposed to address these challenges individually. And while Rolling CI addresses both simultaneously, we believe that it can further be improved as it adapts the predicted region in all dimensions over time without considering which features are responsible for a change in coverage rate.

3 nmtCP method

This section aims to address some of the gaps in previous methods we presented in Subsect. 2.4. Subsection 3.1 lays out our proposal on how to address these shortcomings. Subsection 3.2 takes this idea, turns it into a method and explains how to implement it in practice. This is followed by a theoretical analysis of the validity of the method in Subsection 3.3 and a discussion and a comparison with related methods in Subsection 3.4.

3.1 Idea

Applying the concepts of conformal prediction to multistep-ahead multivariate time-series forecasting faces two main obstacles. Subsection 2.2 shows the challenges that the nonexchangeability of the data in time-series poses and discuss methods to overcome them. Subsection 2.3 briefly explains the additional difficulty introduced when moving from

single target to multi-target regression, necessary to achieve multistep-ahead multivariate forecasts.

In this subsection, we propose a new method that addresses both of these obstacles by combining two existing methods. On the one hand, Barber et al. (2022) has provided a method that relaxes the exchangeability assumption while maintaining strong validity guarantees that is computationally efficient for single target regression. On the other hand, Stankevičiūtė et al. (2021) shows how multiple conformal predictors can be combined to produce prediction regions for multi-target regression. We merge these two approaches, replacing the CPs in the method of Stankevičiūtė et al. (2021) with nexCPs. This leads to a method that produces prediction intervals for multi-step ahead multivariate time-series forecasts for any underlying model producing point forecasts for every dimension in the label space. Henceforth, we refer to this method as non-exchangeable multi-target conformal prediction or nmtCP.

3.2 Implementation

This subsection explains how we put the idea outlined in Subsect. 3.1 into practice, starting with the representation and preprocessing of the time-series data in Subsect. 3.2.1. It also expands the notation introduced in Subsect. 2.2. Next, Subsect. 3.2.2 provides a detailed look into the generation of the predictive region in the semi-off-line setting. Finally, other settings are discussed in Subsect. 3.2.3.

3.2.1 Preprocessing

When loading the data, it is generally stored in a matrix o, where each line contains the values of the observed features for a specific time step and each column contains all measurements for a specific feature. This matrix can be represented as $o_{1:T,1:F}$, where T is the total number of time steps and F the total number of features.

$$\begin{matrix} o_{1,1} & o_{1,2} & \cdots & o_{1,F} \\ o_{2,1} & o_{2,2} & \cdots & o_{2,F} \\ \vdots & \vdots & \ddots & \vdots \\ o_{T,1} & o_{T,2} & \cdots & o_{T,F} \end{matrix} \tag{20}$$

We split this *raw data set* into examples $z_i = (x_i, y_i)$ consisting of objects x_i and their corresponding labels y_i. The objects x_i are created following a sliding window scheme. The ith object consists of

$$x_i = o_{i \cdot s+1 : i \cdot s+T_w, F_w} \tag{21}$$

starting at $i \cdot s + 1$ where s is the stride, describing the number of time steps between the starting points of two consecutive objects, i.e. windows[6]. T_w is the number of time steps each object contains and $F_w \subset F$ is the set of input features for the model. Consecutive objects overlap, unless $s \geq T_w$. Equation (23) visualizes the object x_i as a blue block. The label associated with the ith object is

[6] To avoid an overly convoluted notation we assume that i starts at 0.

$$y_i = o_{i \cdot s + T_o + 1 : i \cdot s + T_o + T_h, F_h} \tag{22}$$

where T_o is the temporal offset between the start of the object and its associated label, T_h is the number of time steps in the prediction horizon and $F_h \subset F$ is the set of features predicted by the model. The green block represents y_i in eq. (23).

$$\begin{array}{ccc|ccc}
o_{1,1} & \cdots & o_{1,|F_w|} & o_{1,|F_w|+1} & \cdots & o_{1,F} \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
o_{i\cdot s-1,1} & \cdots & o_{i\cdot s-1,|F_w|} & o_{i\cdot s-1,|F_w|+1} & \cdots & o_{i\cdot s-1,F} \\
o_{i\cdot s,1} & \cdots & o_{i\cdot s,|F_w|} & o_{i\cdot s,|F_w|+1} & \cdots & o_{i\cdot s,F} \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
o_{i\cdot s+T_w,1} & \cdots & o_{i\cdot s+T_w,|F_w|} & o_{i\cdot s+T_w,|F_w|+1} & \cdots & o_{i\cdot s+T_w,F} \\
o_{i\cdot s+T_w+1,1} & \cdots & o_{i\cdot s+T_w+1,|F_w|} & o_{i\cdot s+T_w+1,|F_w|+1} & \cdots & o_{i\cdot s+T_w+1,F} \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
o_{i\cdot s+T_w+T_h,1} & \cdots & o_{i\cdot s+T_w+T_h,|F_w|} & o_{i\cdot s+T_w+T_h,|F_w|+1} & \cdots & o_{i\cdot s+T_w+T_h,F} \\
o_{i\cdot s+T_w+T_h+1,1} & \cdots & o_{i\cdot s+T_w+T_h+1,|F_w|} & o_{i\cdot s+T_w+T_h+1,|F_w|+1} & \cdots & o_{i\cdot s+T_w+T_h+1,F} \\
\vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\
o_{T,1} & \cdots & o_{T,|F_w|} & o_{T,|F_w|+1} & \ddots & o_{T,F}
\end{array} \tag{23}$$

In the example presented in eq. (23), $T_o = T_w$. Therefore, the label starts at the first time step immediately after the object. In addition, F_w and F_h are disjoint. If $T_o < T_w$ objects and associated labels overlap temporally. If F_w and F_h are not disjoint, objects and associated labels have overlapping features.

After processing the raw data set in this manner, $\mathcal{D} = z_{1:n}$ designates the data set consisting of n examples. We note that the data set $\mathcal{D}$ is a ordered sequence of examples.

3.2.2 Semi-off-line setting

Three steps are necessary to apply the idea in the semi off-line setting.

Step 1: Training. After obtaining the data set $\mathcal{D}$, we partition it into a proper training set $\mathcal{D}_{train}$ and a calibration set $\mathcal{D}_{cal}$. $m_0 = |\mathcal{D}_{train}|$ and $m_1 = |\mathcal{D}_{cal}|$ designate the sizes of the proper training set and the test set respectively with $m_0 + m_1 = n$. The partition of the data set is drawn along the time axis and the elements within $\mathcal{D}_{train}$ and $\mathcal{D}_{cal}$ are not shuffled. Next, we derive the underlying model $\hat{\mu}$ on $\mathcal{D}_{train}$.

Algorithm 2 nmtCP in the semi-off-line setting. The underlying model is only trained once but the calibration set grows with every new label that is provided to the method.

Require: $\mathcal{D} \in \mathcal{Z}^{(n)}$ ▷ data set
Require: $m_0 \in \mathbb{N}$ ▷ proper training set size
Require: $\alpha \in [0,1]$ ▷ significance level
Require: $W \in \mathbb{N} \to \mathbb{R}^*$ ▷ weight generating function
$\mathcal{D}_{train} \leftarrow z_{1:m_0}$ from $\mathcal{D}$ ▷ proper training set of the first m_0 examples of $\mathcal{D}$
$\mathcal{D}_{cal} \leftarrow z_{m_0+1:n}$ from $\mathcal{D}$ ▷ calibration set of the remaining $n - m_0$ examples of $\mathcal{D}$
$\hat{\mu} \leftarrow h(\mathcal{D}_{train})$ ▷ fit the underlying model
for $i \leftarrow m_0 + 1, \ldots, n$ **do** ▷ calibrate the nexCPs
  for $t \leftarrow 1, \ldots, T_h$ **do**
    for $f \leftarrow 1, \ldots, F_h$ **do**
      $r_{i,t,f} \leftarrow |y_{i,t,f} - \hat{\mu}(x_i)_{t,f}|$
    end for
  end for
end for
for $i \leftarrow n + 1, \ldots, *$ **do** ▷ loop for every new element x_i
  $w_{m_0:i-1} \leftarrow W(i - m_0)$ ▷ generate weights
  $\tilde{w}_{m_0:i-1} \leftarrow w_{m_0:i-1} / \sum w_{m_0:i-1}$ ▷ normalize weights
  $\hat{y}_i \leftarrow \hat{\mu}(x_i)$
  for $t \leftarrow 1, \ldots, T_h$ **do**
    for $f \leftarrow 1, \ldots, F_h$ **do**
      $\hat{\mathbf{y}}_{i,t,f} \leftarrow \hat{y}_{i,t,f} \mp \mathsf{Q}_{1-\frac{\alpha}{T_h \cdot F_h}}(r_{m_0:i-1,t,f}, \tilde{w}_{m_0:i-1})$
    end for
  end for
  provide $\hat{\mathbf{y}}_i$ to the user ▷ prediction region
  $y_i \leftarrow$ observe_reality
  for $t \leftarrow 1, \ldots, T_h$ **do**
    for $f \leftarrow 1, \ldots, F_h$ **do**
      $r_{i,t,f} \leftarrow |y_{i,t,f} - \hat{\mu}(x_i)_{t,f}|$ ▷ new residuals
    end for
  end for
end for

Step 2: Initial calibration. We use the fitted model $\hat{\mu}$ to generate predictions $\hat{y}_i = \hat{\mu}(x_i)$ for all elements $x_i \in \mathcal{D}_{cal}$. The chosen NCM A as defined in Subsect. 2.1.1 computes a nonconformity score $r_{i,t,f} = A(y_{i,t,f}, \hat{y}_{i,t,f})$ for every dimension of $\mathcal{Y}$. We use the absolute residuals as NCM. These residuals are then used to calibrate $T_h \times F_h$ nexCPs (Barber et al., 2022) denoted by

$$\Gamma^{\alpha'}_{t,f}, \quad \forall t = 1, \ldots, T_h, \ \forall f = 1, \ldots, F_h. \tag{24}$$

For any combination of fixed t and f, the associated nexCP $\Gamma^{\alpha'}_{t,f}$ initially uses the nonconformity scores $r_{m_0:n,t,f}$. This means that the nexCP associated with one dimension of the labels space is calibrated exclusively on the nonconformity scores computed on that particular dimension of the label space. Instead of using α as the targeted miscoverage rate, we apply the Bonferroni Correction resulting in

$$\alpha' = \frac{\alpha}{T_h \cdot F_h} \tag{25}$$

to account for the family-wise error rate.

Step 3: Inference. Presented with a new object $x_i, i > n$, the underlying model generates a prediction for the associated label $\hat{y}_i = \hat{\mu}(x_i)$. Then the $T_h \times F_h$ nexCPs $\Gamma^{\alpha'}_{1:T_h,1:F_h}$ compute an interval for every dimension in the label space. They do this by applying the weighted quantile function to the absolute residuals $r_{m_0:i-1,t,f}$ of the associated dimension (t, f). We combine the predicted intervals to form the prediction region

$$\begin{aligned} \hat{\mathbf{y}}_i = [\, \hat{y}_{i,t,f} \mp Q_{1-\frac{\alpha}{T_h \cdot F_h}}(r_{m_0:i-1,t,f}, \tilde{w}_{m_0:i-1}), \\ \forall t \in \{1, \ldots, T_h\},\ \forall f \in \{1, \ldots, F_h\}\,] \end{aligned} \tag{26}$$

with $\hat{\mathbf{y}}_i \in \mathbb{R}^{T_h \times F_h \times 2}$ and where $\tilde{w}_{m_0:i-1}$ is a vector of weights on the nonconformity scores. In the semi-off-line setting nature then provides the true label y_i, which we use to compute the residuals $r_{i,1:T_h,1:F_h}$. These new residuals are added to the set of known residuals for their respective dimension and will be used when the algorithm generates the prediction region for the next example n_{i+1}.

Algorithm 2 describes these three steps in a way that can directly be implemented.[7] In the presence of a pretrained model $\hat{\mu}$, the first step can be omitted entirely and the entire initial data set $\mathcal{D}$ can be used as calibration set. This holds as long as none of the examples from the data set $\mathcal{D}$ were used during training of the model $\hat{\mu}$.

3.2.3 On-line, semi-on-line and off-line setting

The method also works in the other settings laid out in Subsect. 2.1.2. Only minor changes to the steps shown in Subsect. 3.2.2 are necessary.

In the on-line setting, the partition between proper training set $\mathcal{D}_{train}$ and the calibration set $\mathcal{D}_{cal}$ is redrawn every time a new label becomes available. The underlying model is then fitted to the new training set $\mathcal{D}_{train}$ and the nexCPs $\Gamma^{\alpha'}_{1:T_h,1:F_h}$ are calibrated on $\mathcal{D}_{cal}$. The prediction step omits the computation of the residuals when nature provides a new label as this action is already covered by the calibration step. If the underlying model is computationally expensive to fit, this mode of operation might become prohibitively costly.

The semi-on-line setting is a hybrid of the on-line and semi-off-line setting. As new examples become available, they are periodically split into a proper training set $\mathcal{D}_{train}$ and a calibration set $\mathcal{D}_{cal}$ along the time axis. When this happens, the underlying model is fitted to the new training set $\mathcal{D}_{train}$ and the nexCPs $\Gamma^{\alpha'}_{1:T_h,1:F_h}$ are calibrated on $\mathcal{D}_{cal}$. Then the method operates in the semi-off-line setting until the examples are partitioned again and the underlying model is retrained.

Although the method can operate in the off-line setting, it is not useful, as the information provided by new labels remains unused. Over time, it would not adapt to distribution shifts in the data.

[7] In our implementation we vectorised operations where possible. In addition our implementation allows for the evaluation of multiple values for α in parallel, reducing the number of times a potentially complex underlying models needs to be run.

3.3 Validity

Theorem 1 Let $\hat{\mu}$ be any pre-fitted model. Then the nmtCP method defined in (26) satisfies

$$\mathsf{P}\big(y_{n+1} \notin \hat{\mathbf{y}}_{n+1}\big) \leq \alpha + \sum_{t=1}^{T_h} \sum_{f=1}^{F_h} G_{n_0+1:n,t,f} \tag{27}$$

where $G_{n_0+1:n,t,f} = \sum_{i=n_0+1}^{n} \tilde{w}_i \cdot \mathsf{d}_{\mathsf{TV}}(r_{n_0+1:n,t,f}, r^{[i]}_{n_0+1:n,t,f})$ is the coverage gap and $\tilde{w}_i$ are the weights on the nonconformity scores.

Proof We view each nexCP as having been calibrated on a dedicated calibration set for the associated dimension. For nexCPs Barber et al. (2022) state in Theorem 2a that in the inductive conformal prediction setting

$$\mathsf{P}\big(y_{n+1,t,f} \in \hat{\mathbf{y}}_{n+1,t,f}\big) \geq 1 - \frac{\alpha}{T_h \cdot F_h} - G_{n_0+1:n,t,f} \tag{28}$$

for a new object n_{n+1}, where $r^{[i]}$ swaps elements i and n in r. This is equivalent to

$$\mathsf{P}\big(y_{n+1,t,f} \notin \hat{\mathbf{y}}_{n+1,t,f}\big) \leq \frac{\alpha}{T_h \cdot F_h} + G_{n_0+1:n,t,f} \tag{29}$$

providing an upper bound for the miscoverage rate for every dimension in the label space. Using Boole's inequality,

$$\mathsf{P}\big(y_{n+1} \notin \hat{\mathbf{y}}_{n+1}\big) = \mathsf{P}\Big(\cup_{t=1}^{T_h} \cup_{f=1}^{F_h} \big(y_{n+1,t,f} \notin \hat{\mathbf{y}}_{n+1,t,f}\big)\Big) \tag{30}$$

$$\leq \sum_{t=1}^{T_h} \sum_{f=1}^{F_h} \mathsf{P}\big(y_{n+1,t,f} \notin \hat{\mathbf{y}}_{n+1,t,f}\big) \tag{31}$$

$$\leq \sum_{t=1}^{T_h} \sum_{f=1}^{F_h} \frac{\alpha}{T_h \cdot F_h} + G_{n_0+1:n,t,f} \tag{32}$$

$$= \alpha + \sum_{t=1}^{T_h} \sum_{f=1}^{F_h} G_{n_0+1:n,t,f} \tag{33}$$

bounds the miscoverage rate for the predicted region constructed from the individual nexCPs' intervals. □

Equation (27) shows that the upper bound on the miscoverage rate of the proposed method depends on the targeted miscoverage rate α and the sum of the coverage gaps $G_{n_0+1:n,t,f}$ of the individual nexCPs. The coverage gaps $G_{n_0+1:n,t,f}$ of the individual nexCPs depend on the chosen weights $\tilde{w}_i$. This means that if the weights are chosen carefully, the coverage gaps $G_{n_0+1:n,t,f}$ decrease and the method becomes approximately valid (Oliveira et al., 2022) for slow shifts in distribution. Optimising the weighs $\tilde{w}_i$ directly is not possible, as the computation of the total variation distance d_{TV} requires knowledge of the true label

y_{n+1}. However, assigning higher weights to past examples similar to the one for which the method is predicting the label and lower ones to past examples that are less similar leads to a decrease in coverage gap $G_{n_0+1:n,t,f}$ and give the weights their intuitive interpretation (Barber et al., 2022).

The theoretical upper bound on the method's miscoverage rate presented in Eq. (27) does not require the same weights $\tilde{w}_i$ for every dimension. While this gives the user a lot of control over the method, we did not explore this possibility in the context of this article. Following the advice of Barber et al. (2022), we opted for a few simple weight generating functions for the experiments presented in Sect. 4.

3.4 Comparison

This subsection compares nmtCP to the original CP (Vovk et al., 2005) and highlights some key differences with the methods discussed in Sect. 2.

In contrast to the original ICP, nmtCP does not require the underlying data to be exchangeable, and remains computationally efficient even while being applied in a multi-target regression setting. This comes at the cost of a weakened validity guarantee.

CIbP (Chernozhukov et al., 2018) and CF-RNN (Stankevičiūtė et al., 2021) rely on the presence of exchangeable blocks in the time-series for their respective validity guarantees. nexCP (Barber et al. 2022) and therefore nmtCP makes no such assumption and can therefore be applied to time-series that exhibit long term dependencies or sudden distribution shifts.

EnbPI (Xu and Xie, 2021) is computationally more expensive than nmtCP because it trains an ensemble of underlying models. The benefit to this approach is, that it avoids splitting the known examples into a proper training set and a calibration set. In the semi-off-line setting EnbPI behaves like nexCP and therefore, like nmtCP with a weight function that assigns the same weight to a fixed number of past residuals. The weighted quantile function used in nexCP and nmtCP can be interpreted as a generalization of the sliding window approach that Xu and Xie (2021) chose for the integration of new residuals in EnbPI. Because Xu and Xie (2021) built their validity guarantees on assumptions about the residuals, it is difficult to make a comparison to the one presented in Eq. (27). However, nmtCP offers more flexibility to determine how the method reacts to distribution shifts through the weights w_i in the weighted quantile function, even if the performance of the underlying model varies over time.

ACI (Gibbs & Candès, 2021), being its direct precursor, is quite similar to nexCP. It is capable to adapt to distribution shifts, is computationally efficient, comes with strong validity guarantees and offers the flexibility of the weighted quantile function. Just as nexCP, and in contrast to nmtCP, it lacks the support for multi-target regression.

In comparison to nmtCP, Rolling CI (Feldman et al., 2022) does not need to split the known examples into a proper training set and a calibration set, allowing the underlying model to be fitted to the most recent data points. Rolling CI also includes a training step for the underlying model every time a new label becomes available, making it highly efficient in an on-line setting. It behaves similar to ACI, scaling the size of the prediction region over time. However, for multi-target regression the prediction region is scaled equally in all dimensions. This might not be desirable as not all dimensions necessarily contribute equally to a change in coverage rate. nmtCP does not suffer from this behavior since the intervals for the individual dimensions in the label space are constructed independently and then combined into the prediction region.

Overall nmtCP offers a flexible solution that adapts to distribution shifts in the data. It can be used on top of any multi-target regression model with little to no modifications to the underlying model. The miscoverage rate of nmtCP is bound theoretically. This set of characteristics tailored to multistep-ahead multivariate time-series forecasting is not found in the other methods examined in this article. It comes at the expense of weaker validity guarantees or splitting the data, reducing the effective training set size for the underlying model.

4 Experiments

In this section, we examine the empirical performance of the nmtCP method presented in Sect. 3 on simulated and real-world time-series forecasting tasks and compare it to CF-RNN (Stankevičiūtė et al., 2021). Subsection 4.1 introduces the data sets, and explains the preprocessing steps taken to prepare them for nmtCP. Subsection 4.2 describes how the experiments are conducted and what variants of the method we used. This is followed by the metrics used to evaluate the results in Subsect. 4.3. The results of the experiments are reported in Subsect. 4.4, which we discuss in Subsect. 4.5.

4.1 Data sets

We generate simulated data distributions to analyze our method's behavior. Inspired by Rolnick et al. (2019), we also chose two data sets from the electricity domain, ELEC2 (4.1.2) and Tétouan City (4.1.3), to test our method.

4.1.1 Simulations

All simulation settings consist of six features and 4800 time-steps. They are generated as follows.

- Feature 1: $o_{t,1} = sin(\frac{\pi t}{12}) \quad \forall t \in \{1, \dots, 4800\}$
- Feature 2: $o_{t,2} = sin(\frac{\pi t}{84}) \quad \forall t \in \{1, \dots, 4800\}$
- Feature 3: $o_{t,3} = -sign(sin(\frac{\pi t}{12})) \quad \forall t \in \{1, \dots, 4800\}$
- Feature 4: $o_{t,4} = -sign(92 - \text{mod}(t, 168)) \quad \forall t \in \{1, \dots, 4800\}$
- Feature 5: $o_{t,5} = [o_{t,1}, o_{t,2}, o_{t,3}, o_{t,4}]\beta_{t,5} + \epsilon_{t,5} \quad \forall t \in \{1, \dots, 4800\}$
- Feature 6: $o_{t,6} = [o_{t,1}, o_{t,2}, o_{t,3}, o_{t,4}]\beta_{t,6} + \epsilon_{t,6} \quad \forall t \in \{1, \dots, 4800\}$

The settings differ in their values for $\beta_{t,5}$ and $\beta_{t,6}$ outlined in eqs. (34) to (41) below.

Before feeding the data set to the underlying model, nmtCP, and CFRNN we split it into consecutive windows as shown in Subsect. 3.2.1 with $T_w = T_o = 3$, $T_h = 3$, and a stride of $s = 3$ between consecutive windows, resulting in 1600 examples. The objects are made up of the first 4 features and the corresponding labels are made up of features 5 and 6. Features 5 and 6 each contain a random noise term $\epsilon_{t,5} = \mathcal{N}(0, \frac{1+\text{mod}(t,3)}{10})$ and $\epsilon_{t,6} = \mathcal{N}(0, \frac{1+\text{mod}(t,3)}{10})$ that increases in variance over the span of the prediction horizon for each object. We split the examples along the time axis into a proper training set, a

calibration set, and a test set containing 1200, 1200, and 4800 time-steps, i.e. 400, 400, and 800 examples, respectively. Dashed lines represent these splits in Figs. 1, 2, 3 and 4.

We consider four simulation settings described in detail below.

- **Simulation Setting 1: changepoint on all label features** introduces a changepoint at $t = 3600$, i.e. after 1200 examples, affecting both of the label's features. It uses the following values for β.

$$\beta_{t,5} = \begin{cases} [0.7, 0.3, 0, 0]^T & \text{if } t \in \{1, \dots, 3600\} \\ [0, 0.7, 0.3, 0]^T & \text{if } t \in \{3601, \dots, 4800\} \end{cases} \tag{34}$$

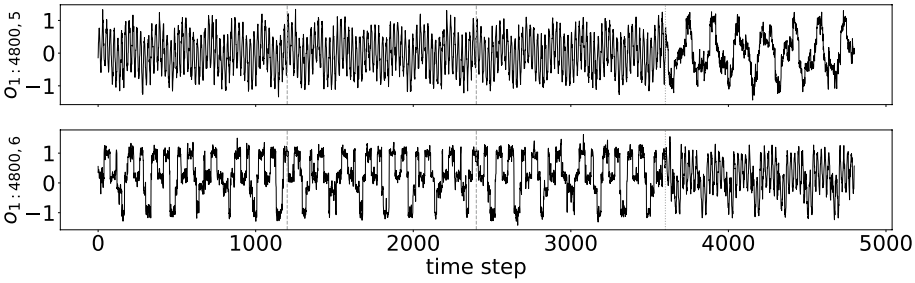

Fig. 1 Features 5 and 6 of simulation setting 1, exhibiting a changepoint at time-step 3601

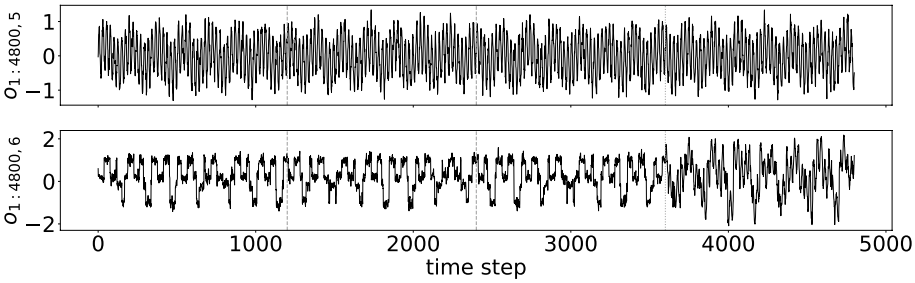

Fig. 2 Features 5 and 6 of simulation setting 2. Feature 6 exhibits a changepoint at time-step 3601

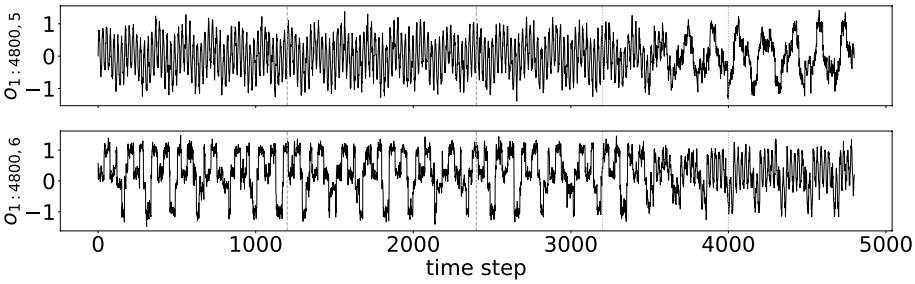

Fig. 3 Features 5 and 6 of simulation setting 3, exhibiting a distribution drift from at time-step 3201 until 4000

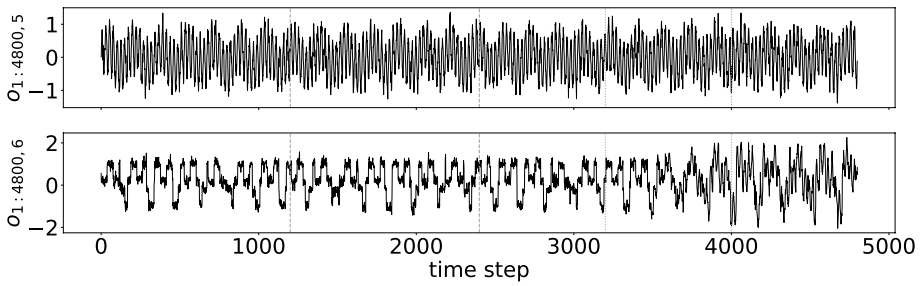

Fig. 4 Features 5 and 6 of simulation setting 4. Feature 6 exhibits a distribution drift from at time-step 3201 until 4000

$$\beta_{t,6} = \begin{cases} [0,0,0.4,0.6]^T & \text{if } t \in \{1,\dots,3600\} \\ [0.6,0,0,0.4]^T & \text{if } t \in \{3601,\dots,4800\} \end{cases} \tag{35}$$

Figure 1 displays features 5 and 6 for setting 1. The changepoint is highlighted by a dotted line. The entire data set for setting 1 is represented in figure 38 in the appendix.

- **Simulation setting 2: changepoint on one label feature** introduces a changepoint at $t = 3600$, i.e. after 1200 examples, affecting only feature 6. It uses the following values for β.

$$\beta_{t,5} = [0.7,0.3,0,0]^T \quad \forall t \in \{1,\dots,4800\} \tag{36}$$

$$\beta_{t,6} = \begin{cases} [0,0,0.4,0.6]^T & \text{if } t \in \{1,\dots,3600\} \\ [0.7,0.3,0.4,0.6]^T & \text{if } t \in \{3601,\dots,4800\} \end{cases} \tag{37}$$

Figure 2 displays features 5 and 6 for setting 2. The changepoint is highlighted by a dotted line.

- **Simulation setting 3: distribution drift on all label features** introduces distribution drift starting at $t = 3200$, i.e. at example 1066, and running until $t = 4000$, i.e. at example 1066, affecting both of the label's features. It uses the following values for β.

$$\beta_{t,5} = \begin{cases} [0.7,0.3,0,0]^T & \text{if } t \in \{1,\dots,3200\} \\ [0.7,0.3,0,0]^T + \frac{t-3200}{800}[-0.7,0.4,0.3,0]^T & \text{if } t \in \{3201,\dots,4000\} \\ [0.7,0.3,0,0]^T & \text{if } t \in \{4001,\dots,4800\} \end{cases} \tag{38}$$

$$\beta_{t,6} = \begin{cases} [0,0,0.4,0.6]^T & \text{if } t \in \{1,\dots,3200\} \\ [0,0,0.4,0.6]^T + \frac{t-3200}{800}[0.6,0,-0.4,-0.2]^T & \text{if } t \in \{3201,\dots,4000\} \\ [0.6,0,0,0.4]^T & \text{if } t \in \{4001,\dots,4800\} \end{cases} \tag{39}$$

Figure 3 displays features 5 and 6 for setting 2. The start-point and end-point of the distribution drift are highlighted by a dotted line.

- **Simulation setting 4: distribution drift on one label feature** introduces distribution drift starting at $t = 3200$, i.e. at example 1066, and running until $t = 4000$, i.e. at example 1066, affecting only feature 6. It uses the following values for β.

$$\beta_{t,5} = [0.7, 0.3, 0, 0]^T \quad \forall t \in \{1, \dots, 4800\} \tag{40}$$

$$\beta_{t,6} = \begin{cases} [0, 0, 0.4, 0.6]^T & \text{if } t \in \{1, \dots, 3200\} \\ [0, 0, 0.4, 0.6]^T + \frac{t-3200}{800}[0.7, 0.3, 0, 0]^T & \text{if } t \in \{3201, \dots, 4000\} \\ [0.7, 0.3, 0.4, 0.6]^T & \text{if } t \in \{4001, \dots, 4800\} \end{cases} \tag{41}$$

Figure 4 displays features 5 and 6 for setting 2. The start-point and end-point of the distribution drift are highlighted by a dotted line.

4.1.2 ELEC2

The ELEC2 data set (Harries, 1999) contains information about the electricity demand and price in the Australian states of New South Wales and Victoria. Measures have been taken every 30 min between May 1996 and December 1998. For the experiments, we chose to use the first 20,000 time steps in the data set and the features representing the electricity demand in New South Wales, the electricity demand in Victoria, and the amount of electricity transferred between the two states. Figure 5 shows this selection and the split between proper training set, calibration set and test set through the vertical dotted lines. The proper training set contains the first 8000 time steps, the calibration set is made up of the following 8000 time steps and the test set comprises the last 4000 time steps.

Before feeding the data set to the underlying model, nmtCP, and CF-RNN we split it into consecutive windows as shown in Subsect. 3.2.1 with $T_w = T_o = 192$, $T_h = 12$, and a stride of $s = 12$ between consecutive windows, resulting in 1650 examples and a prediction horizon of six hours. Both the object and the corresponding label are made up of all three features. We split the examples into a proper training set, a calibration set, and a test set containing 660, 660 and 330 examples, respectively, along the time axis. Note that the features VIC and Transfer have a constant value throughout the proper training and the calibration set and display a strong distribution shift in the test set, visible in Fig. 5.

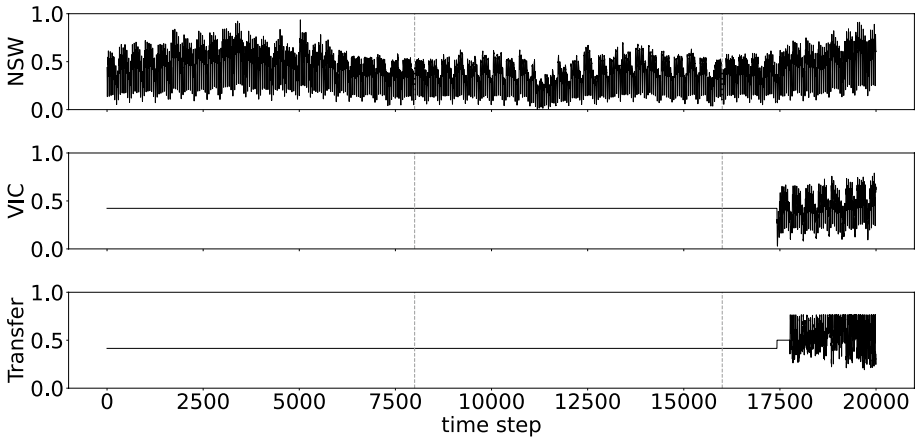

Fig. 5 Features and time steps from the ELEC2 data set (Harries, 1999) used in the experiments

4.1.3 Tétouan City

The Tétouan City data set (Salam & Hibaoui, 2018) contains weather information and the electricity consumption of three different distribution networks of the northern Moroccan city of Tétouan. The measures were taken every 10 min in 2017. For the experiments, we chose to use the first 40,000 time steps in the data set and the features representing the electricity consumption for each of the three distribution networks. Figure 6 shows this selection and the split between proper training set, calibration set, and test set through the vertical dotted lines. The proper training set contains the first 20,000 time steps, the calibration set is made up of the following 5000 time steps and the test set comprises the last 15,000 time steps. The three distribution networks are referred to as Zone 1, Zone 2, and Zone 3 in Fig. 6.

Similar to the ELEC2 data set (Harries, 1999), we split the Tétouan City data set into consecutive windows with $T_w = 144$, $T_h = 6$, and a stride of $s = 6$ between consecutive windows, resulting in 6642 examples and a prediction horizon of one hour. Again, both, the object and the corresponding label are made up of all three features. The proper training set, the calibration set, and the test set contain 3321, 830 and 2491 examples, respectively, split along the time axis. Compared to the ELEC2 data set, the features of the Tétouan City data set exhibit a more gradual distribution shift in the test set, shown in Fig. 6.

4.2 Experimental setup

The experiments follow the process outlined in Subsect. 3.2.2.

Step 1: Training. We chose linear regression for the underlying model, using PyTorch for the implementation and the Adam optimizer during training. While producing similar results in this task, compared to the recurrent neural networks used in Stankevičiūtė et al. (2021) and Schlembach et al. (2022), the linear regression model is much faster to train and allowed for multiple repetitions of every experiment. For both data sets, we train the underlying models on the proper training set for 1000 epochs, using a batch size of 100 and a learning rate of 0.01.

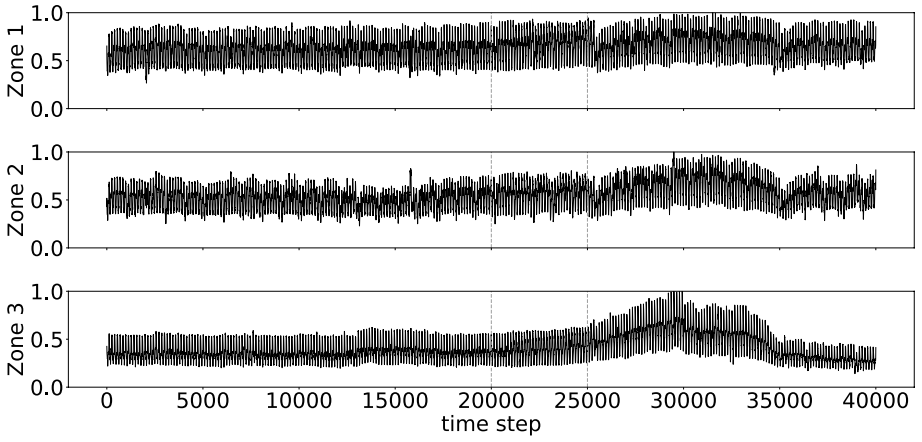

Fig. 6 Features and time steps from the Tétouan City data set (Salam & Hibaoui, 2018) used in the experiments

Step 2: Initial calibration. After the underlying model is fitted to the proper training set, we use the model to generate predictions on the calibration set. Then, the nexCPs (Barber et al., 2022) use these predictions to calibrate for every dimension in the label space. To increase efficiency, we calibrate multiple groups of nexCPs with different weights for the quantile functions simultaneously. Subsection 4.2.1 provides a detailed description of the different functions generating these weights. In addition to increasing efficiency, sharing the same underlying model between multiple instances of nmtCP, with different weights for the quantile function, has the added benefit that, when comparing the validity and efficiency of the produced prediction regions, they are based on the same residuals.

Step 3: Inference. Finally, we present the underlying model with the objects from the test set sequentially in their original order. The prediction is given to each group of calibrated nexCPs for each to generate a prediction region in the form of a hyper-rectangle. Then we compute the corresponding residual for every dimension in the label space using the true label and add the resulting values to the calibration set of each nmtCP.

After computing all examples in the test, set we evaluate them using the metrics described in Subsect. 4.3.[8] All experiments are repeated 20 times and results averaged to account for the random initialization of the underlying model.

4.2.1 Weight functions

In the semi-off-line setting, the number of residuals used by the weighted quantile function in every nexCP (Barber et al., 2022) grows with every new true label y_{i-1} that becomes available. Therefore, instead of assigning static weights $\tilde{w}_i$, we used functions to generate the appropriate number of weights every time a new object x_i is processed. In the experiments, we compared the following weight functions.

Exponential The weight of past residuals decreases exponentially the further they are in the past.

$$W(i-m_0) = [e^{\beta(j-i+m_0)}]_{j=0}^{i-m_0} \tag{42}$$

where β is a parameter controlling how quickly the value of past weights decreases. In the experiments, $\beta = 0.007$.

Soft cutoff This weight function mimics a sliding window approach with a soft transition from the past residuals that are included to the ones that are excluded.

$$W(i-m_0) = \left[\frac{j-i+m_0+\beta_c}{\beta_s + |j-i+m_0+\beta_c|} + 1\right]_{j=0}^{i-m_0} \tag{43}$$

where β_c is the cutoff point, indicating how many past residuals are considered and β_s controls the "softness" of the transition. In the experiments $\beta_c = 200$ and $\beta_s = 50$.

Linear The weight of past residuals decreases linearly the further they are in the past.

$$W(i-m_0) = \left[\frac{j}{i-m_0}\right]_{j=0}^{i-m_0} \tag{44}$$

[8] The software architecture supports on-line processing of new examples and the on-line evaluation of the results. This feature is not used for the experiments presented in this article.

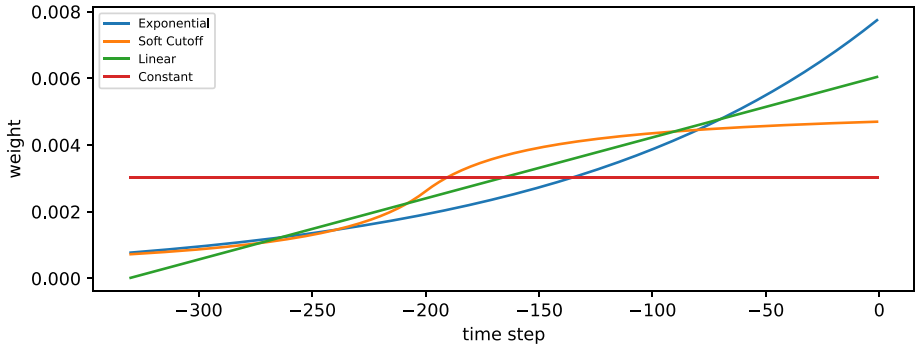

Fig. 7 Normalized weights produced by different weight functions for the past 300 examples

Constant All past residuals are given the same constant weight.

Figure 7 illustrates the weights these functions generate for 300 past residuals.

4.3 Evaluation

To evaluate the method's validity in the experiments, we used two metrics, the *mean coverage rate*[9] and the *rolling coverage rate*. Following the definition of validity given in Subsect. 2.1.1, the coverage rate computes the fraction of prediction regions $\hat{\mathbf{y}}_i$ that contain the true label y_i for a given confidence level $1 - \alpha$. If the coverage rate is equal or greater than $1 - \alpha$, the method is valid. The rolling coverage rate applies the same principle using a sliding window approach hence providing a local coverage rate.

The *mean interval width* provides information about the efficiency of the method for a given confidence level $1 - \alpha$. In the multistep-ahead multivariate time-series forecasting setting, it computes the average difference between the upper and the lower bound of the prediction interval for every dimension in the label space and every example in the test set. A lower value therefor indicates greater information efficiency.[10]

4.4 Results

The experimental results are grouped by data set.

4.4.1 Simulations

Simulation setting 1: changepoint on all label features shows how the method behaves when presented with a sudden change in the data's distribution. In this setting, nmtCP remains globally valid when using the exponential or the soft cutoff weight functions.

[9] Vovk (2015) suggest the use of the calibration plot instead, representing the same information using α and the empirical miscoverage rate instead of the confidence level $1 - \alpha$ and the empirical coverage rate. For the sake of exposition, we opted for the latter.

[10] In addition to applying these metrics globally the user of the nmtCP method can gain further insights into its performance and the behavior of the underlying model by applying them to a subset of the dimensions of the label space.

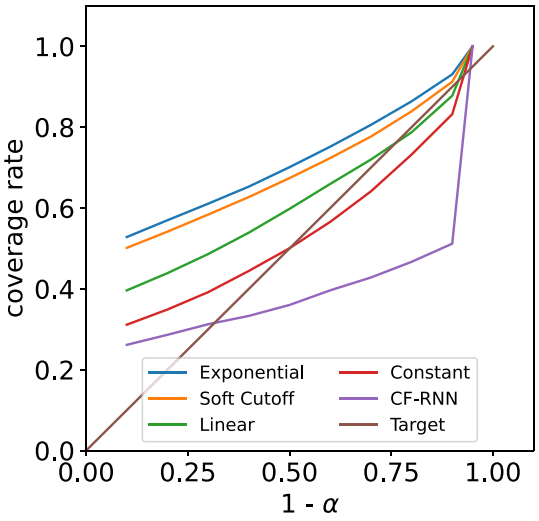

Fig. 8 Mean coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Average taken over 20 trials

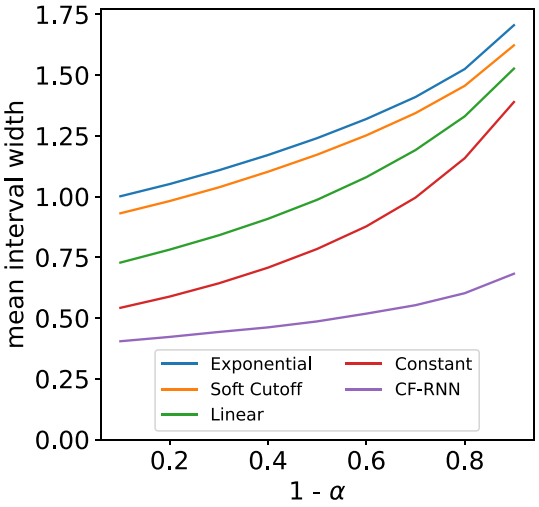

Fig. 9 Mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Average taken over 20 trials

nmtCP remains almost valid in combination with the linear weight function, but loses validity for most confidence levels when all residuals are given the same weight. Figure 8 shows that in all instances, nmtCP outperforms CF-RNN by realizing a greater coverage rate for all values of the confidence level $1 - \alpha$. CF-RNN does not utilize new residuals when they become available. Figure 9 shows that nmtCP achieves this boost in validity by increasing interval widths of the predicted region compared to CF-RNN.

Both, the global coverage rate and the mean interval width are large for small values of the confidence levels $1 - \alpha$ resulting in all methods being valid for $1 - \alpha \leq 0.2$. Figure 9 shows that the mean interval width increases slowly with the confidence levels $1 - \alpha$.

A coverage rate ≥ 0.95 is only achieved through infinite prediction intervals. This behavior cannot be represented in Fig. 9.

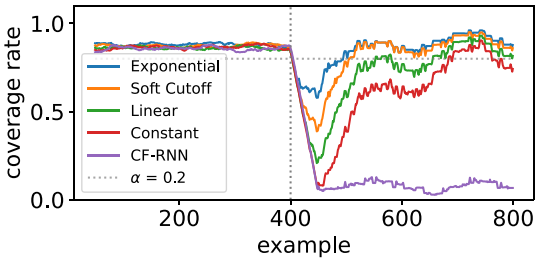

Fig. 10 Rolling coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

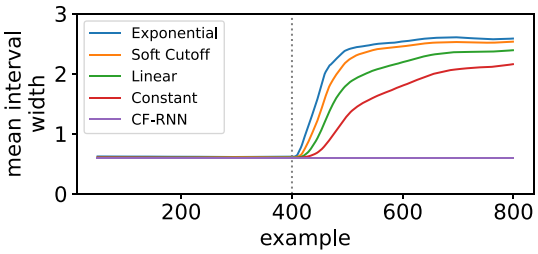

Fig. 11 Rolling mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

While Figs. 8 and 9 provide a global view of the behavior of nmtCP and CF-RNN with different weight functions W, the rolling coverage rate presented in Fig. 10 shows how the coverage rate changes locally. With a window size of 50 it computes the coverage rate over the past 50 examples. Once the distribution shift occurs, nmtCP initially loses validity for all employed weight functions W. When using the exponential weight function and the soft cutoff weight function, the method recovers quickly and remains approximately valid for the remaining examples in the test set. nmtCP achieves this by quickly increasing the width of the prediction regions, as shown in Fig. 11. In combination with the linear weight function and the constant weight function, nmtCP takes much longer to recover and is not able to regain validity consistently after the change point. With these weight functions the mean interval width grows slower. Figure 10 also shows that for CF-RNN only a small number of the real labels y_i are within the prediction region after the distribution shift. It does not adapt the size of the prediction region over time.

Figs. 12 and 13 show the rolling coverage rate and the rolling mean interval width for the two features of the label space. These metrics are aggregated over all time-steps in the prediction horizon T_h. We observe, that the method's behavior for each feature in the label space is similar to its behavior for both features combined. Both, per feature and for both features combined, the weight functions that achieve higher coverage rates are the ones that adapt faster and grow the prediction interval's size more quickly.

Simulation setting 2: changepoint on one label feature only varies the distribution of one of the label's features. Figures 14 and 15 show that the method's general behavior is similar to its behavior in Simulation Setting 1 with respect to the influence of the chosen weight functions.

Compared to simulation setting 1, the behavior differs when looking at the contributions of the individual features of the label space to the rolling coverage rate and the rolling mean interval width, presented in Figs. 18 and 19. We observe that only $y_{1:800,2}$, the feature containing the changepoint, displays changes in both metrics. This influences nmtCP's performance when measured for all dimensions of the label space as shown in Figs. 16 and 17.

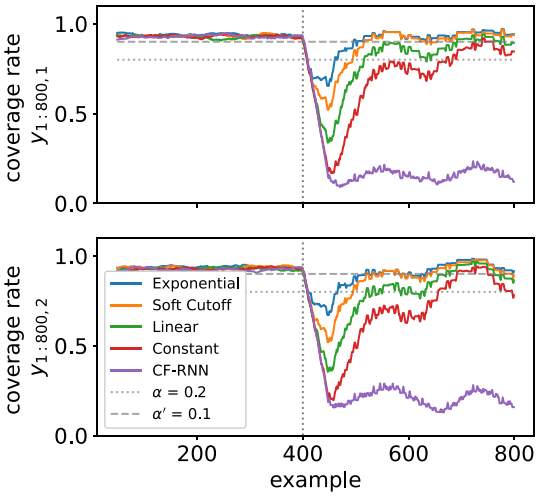

Fig. 12 Rolling coverage rate by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

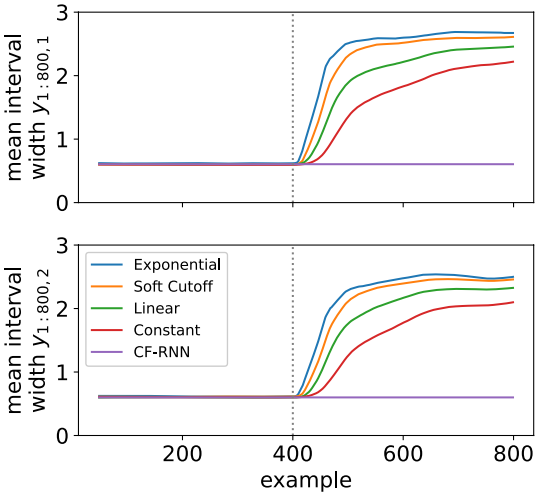

Fig. 13 Rolling mean interval width by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 1. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

The rolling coverage rate and the rolling mean interval width remain stable for $y_{1:800,1}$, the feature who's distribution does not change.

Similar to simulation setting 1, the weight functions that achieve higher coverage rates are the ones that adapt faster and grow the prediction interval's size more quickly.

Simulation setting 3: distribution drift on all label features shows how the method behaves when presented with a progressive change in the data's distribution. Remaining fixed in the first third of the test set, the distribution of both features of the labels space drifts linearly during the second third of the test set and is fixed again during the remainder of the test set.

In contrast to the previous settings, nmtCP only remains globally valid when using the exponential weight function. Figure 20 shows that in combination with other weight functions, nmtCP loses its validity for most confidence levels, as does CF-RNN. Figure 21 shows again that higher coverage rates are achieved by increasing the interval widths of the predicted region.

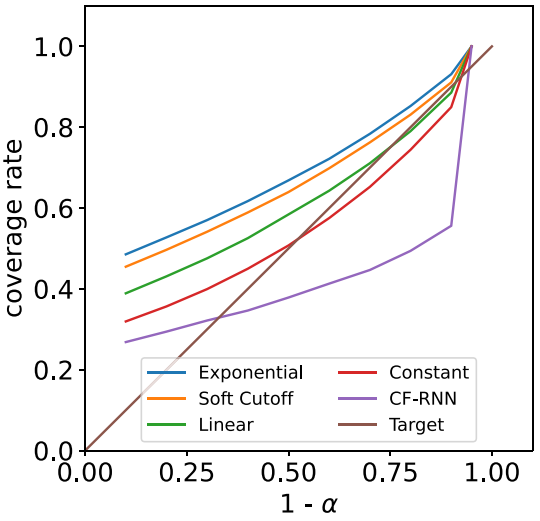

Fig. 14 Mean coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Average taken over 20 trials

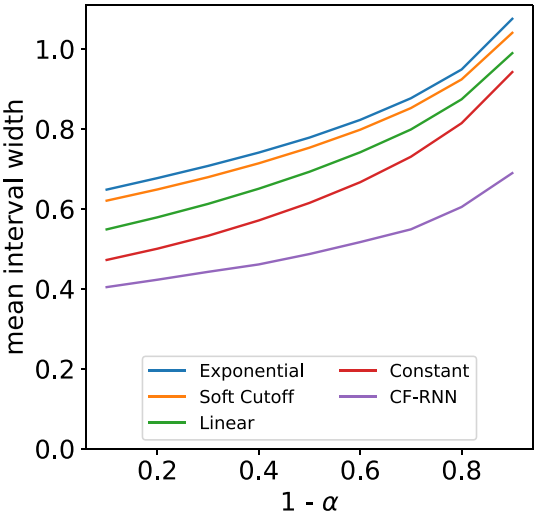

Fig. 15 Mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Average taken over 20 trials

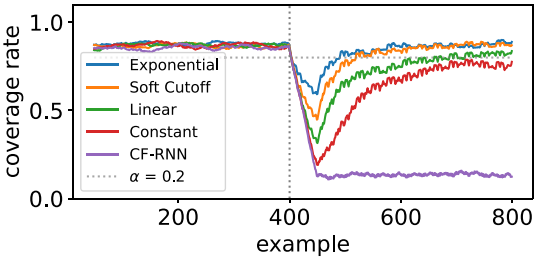

Fig. 16 Rolling coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

The rolling coverage rate in Fig. 22 shows that the method drops below the targeted coverage rate soon after the distribution drift starts for all weight functions. nmtCP only

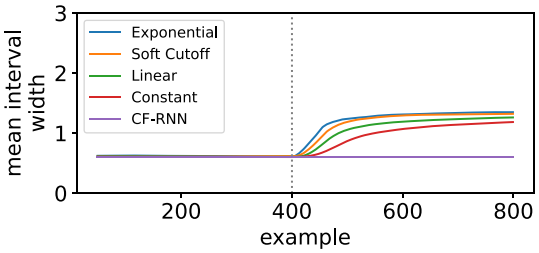

Fig. 17 Rolling mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

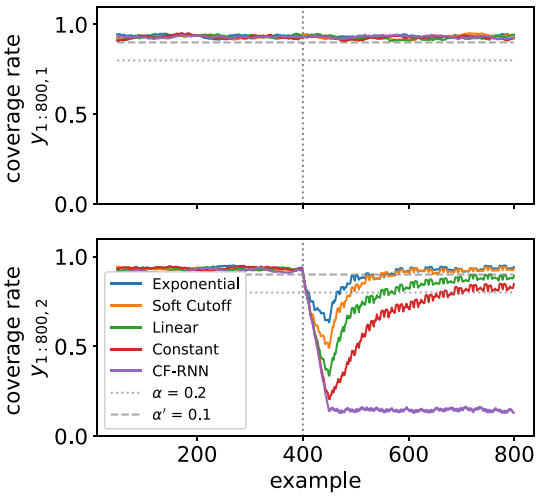

Fig. 18 Rolling coverage rate by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

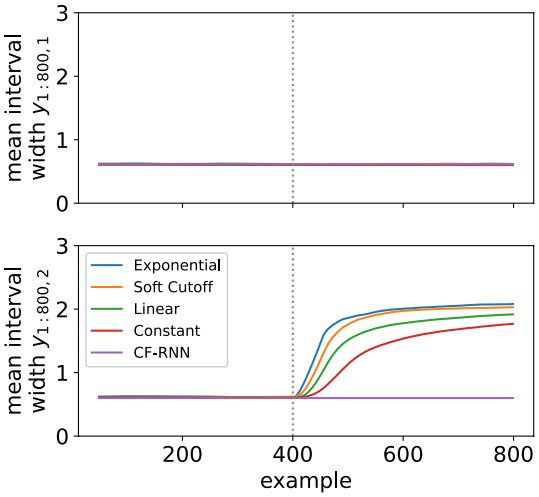

Fig. 19 Rolling mean interval width by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 2. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

manages to recover and stay above the targeted coverage rate for the exponential weight function and the soft cutoff weight function and only after the distribution is no longer

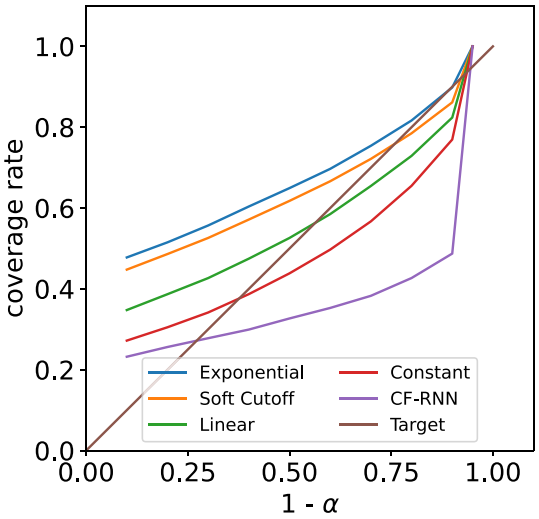

Fig. 20 Mean coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Average taken over 20 trials

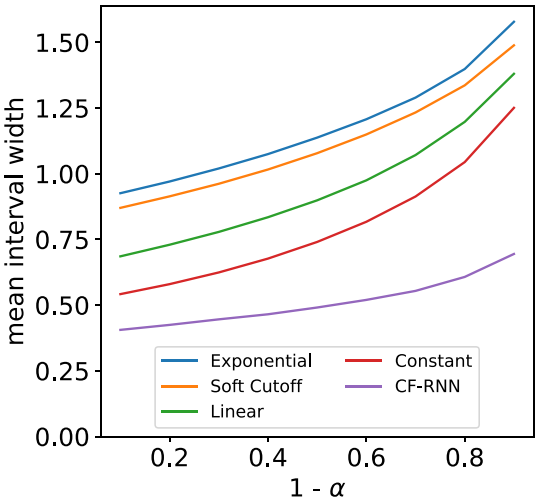

Fig. 21 Mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Average taken over 20 trials

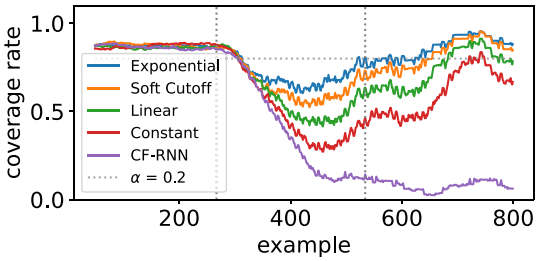

Fig. 22 Rolling coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

shifting. Figure 23 shows that mean interval width also continues growing after the distribution is no longer shifting.

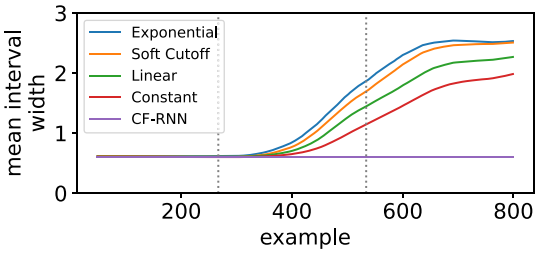

Fig. 23 Rolling mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

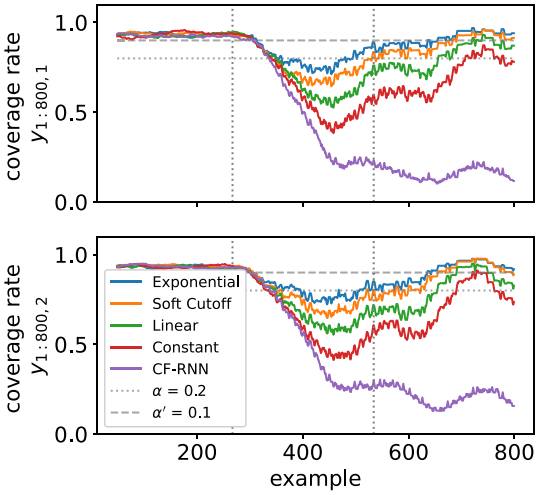

Fig. 24 Rolling coverage rate by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

Similar to simulation setting 1, we observe that the method's behavior for each feature in the label space is similar to its behavior over all. Figures 24 and 25 show the rolling coverage rate and the rolling mean interval with for each feature of the label space. Again, the weight functions that achieve higher coverage rates are the ones that adapt faster and grow the prediction interval's size more quickly.

Simulation setting 4: distribution drift on one label feature progressively changes the distribution of one of the label's features. Remaining fixed in the first third of the test set, the distribution of one of the features in the label space drifts linearly during the second third of the test set and is fixed again during the remainder of the test set. The other feature in the label space is not affected by this drift and retains its original distribution throughout the test set. Figures 26 and 27 show that the method's general behavior is similar to its behavior in Simulation Setting 3 with respect to the influence of the chosen weight functions.

Looking at the metrics' evolution over time in Figs. 28 and 29 it is also similar to the situation encountered in simulation setting 3. For all weight functions the method drops below the targeted coverage rate soon after the distribution drift starts. nmtCP only manages to recover and stay above the targeted coverage rate for the exponential weight function and the soft cutoff weight function and only after the distribution is no longer shifting. The rolling mean interval width also continues growing after the distribution is no longer shifting for all weight functions.

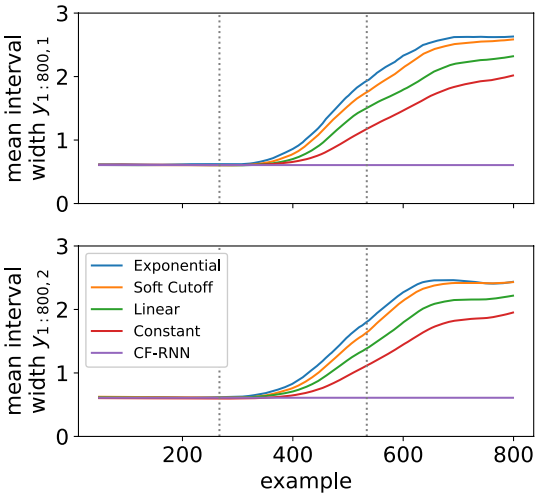

Fig. 25 Rolling mean interval width by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 3. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

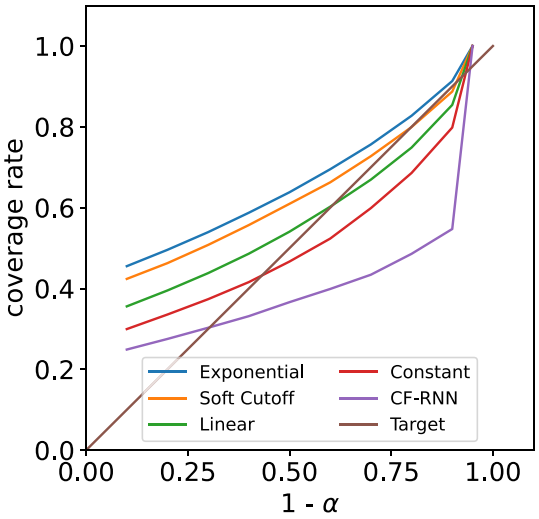

Fig. 26 Mean coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Average taken over 20 trials

Compared to simulation setting 3, only $y_{1:800,2}$, the feature experiencing distribution drift, displays changes in the rolling coverage rate and the rolling mean interval width. It is also the only feature for which the mean interval width grows over time. The rolling coverage rate and the rolling mean interval width remain constant for $y_{1:800,1}$. For this feature, the distribution does not change. Figures 30 and 31 show this behavior.

4.4.2 ELEC2

This subsection presents the results obtained from applying nmtCP to the subset of the ELEC2 data set (Harries, 1999) described in Subsect. 4.1.2. Due to the nature of the data

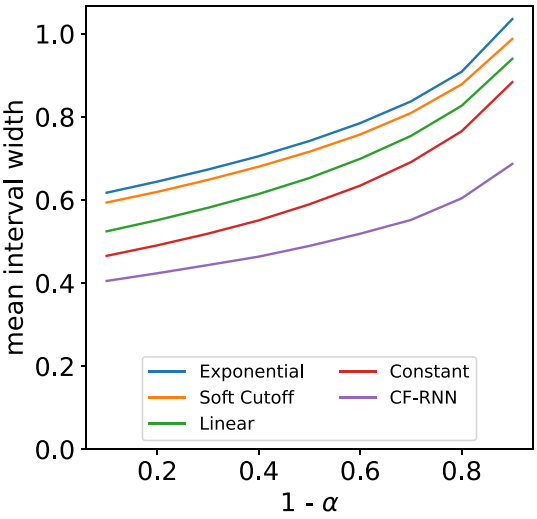

Fig. 27 Mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Average taken over 20 trials

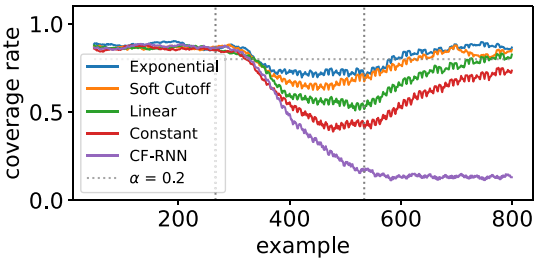

Fig. 28 Rolling coverage rate for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

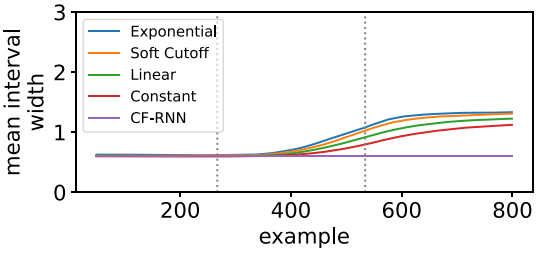

Fig. 29 Rolling mean interval width for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

set, this set of experiments showcases how the method behaves in the presence of a strong distribution shift comparable to a changepoint.

In these adverse conditions, nmtCP remains globally valid when using the exponential or the soft cutoff weight functions. nmtCP remains almost valid in combination with the linear weight function, but loses validity for most confidence levels when all residuals are given the same weight. Table 2 shows that in all instances, nmtCP outperforms CF-RNN (Stankevičiūtė et al., 2021) by realizing a greater coverage rate on average. This behavior can be observed in Fig. 32. It shows the coverage rate of the prediction regions generated by nmtCP on the test set for different confidence levels $1 - \alpha$. In addition to the results of

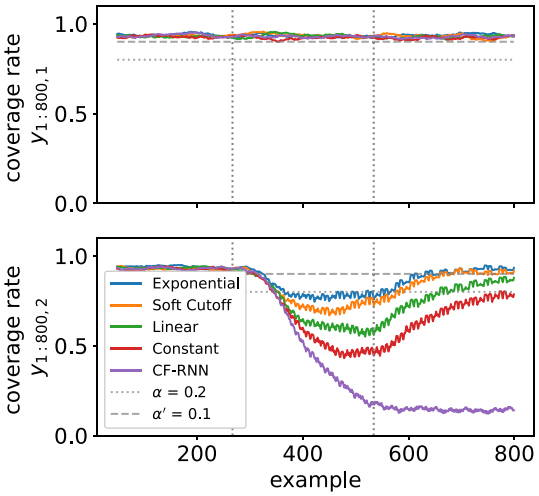

Fig. 30 Rolling coverage rate by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

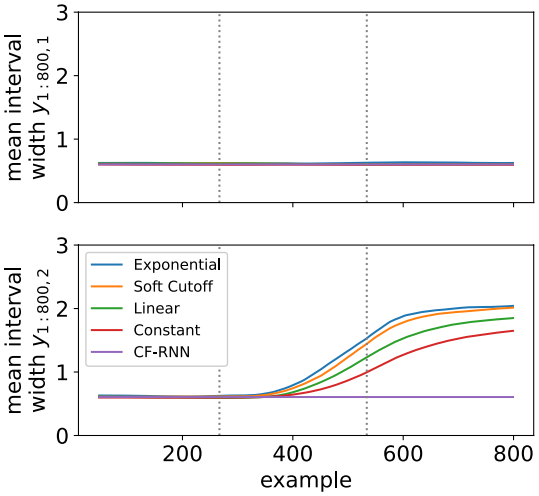

Fig. 31 Rolling mean interval width by feature for nmtCP with different weight functions and CF-RNN on the test set of simulation 4. Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

nmtCP with different weight functions W, it also displays the coverage rate measured for CF-RNN, which does not utilize new residuals when they become available.

Table 3 shows that nmtCP achieves this boost in validity by increasing interval widths of the predicted region compared to CF-RNN Stankevičiūtė et al. (2021). Figure 33 displays the mean prediction interval width for different confidence levels $1 - \alpha$ corresponding to the coverage rates in Fig. 32.

Both, the global coverage rate and the mean interval width are large for small values of the confidence levels $1 - \alpha$ resulting in all methods being valid for $1 - \alpha \leq 0.2$. The mean interval width increases slowly with the confidence levels $1 - \alpha$.

Tables 2 and 3 also show that a coverage rate of 1 is only achieved through infinite prediction intervals. This behavior cannot be represented in Fig. 33.

While Tables 2 and 3 and Figs. 32 and 33 provide a global view of the behavior of nmtCP with different weight functions W, the rolling coverage rate shown in Fig. 34 shows how the coverage rate changes locally. With a window size of 50 it computes the coverage

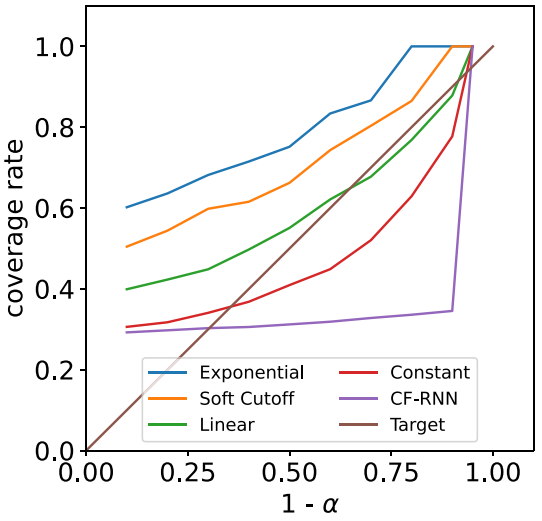

Fig. 32 Mean coverage rate for nmtCP with different weight functions and CF-RNN measured on the ELEC2 data set (Harries, 1999). Average taken over 20 trials

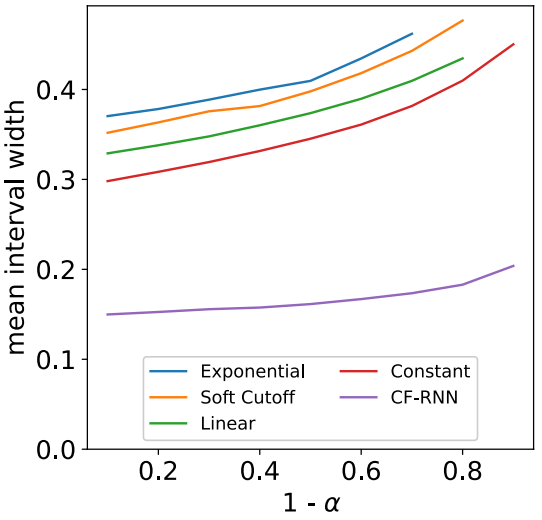

Fig. 33 Mean interval width for nmtCP with different weight functions and CF-RNN measured on the ELEC2 data set (Harries, 1999). Average taken over 20 trials

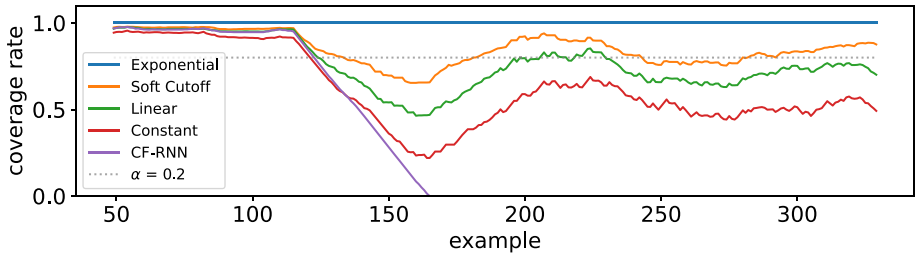

Fig. 34 Rolling coverage rate for nmtCP with different weight functions and CF-RNN on the test set of the ELEC2 data set (Harries, 1999). Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

Table 2 Mean coverage rate for nmtCP with different weight functions and CF-RNN measured on the ELEC2 data set (Harries, 1999). Average taken over 20 trials

1-α	Exponential	Soft cutoff	Linear	Constant	CF-RNN
0.10	0.603	0.502	0.384	0.315	0.285
0.20	0.633	0.545	0.411	0.326	0.289
0.30	0.682	0.599	0.439	0.348	0.294
0.40	0.717	0.617	0.482	0.375	0.297
0.50	0.747	0.665	0.534	0.415	0.303
0.60	0.834	0.743	0.605	0.458	0.311
0.70	0.867	0.806	0.663	0.526	0.319
0.80	1.000	0.868	0.762	0.636	0.329
0.90	1.000	1.000	0.875	0.782	0.341
0.95	1.000	1.000	1.000	1.000	1.000

Table 3 Mean interval width for nmtCP with different weight functions and CF-RNN measured on the ELEC2 data set (Harries, 1999). Average taken over 20 trials

1-α	Exponential	Soft cutoff	Linear	Constant	CF-RNN
0.10	0.370	0.349	0.331	0.293	0.151
0.20	0.378	0.360	0.340	0.304	0.154
0.30	0.389	0.372	0.350	0.315	0.157
0.40	0.401	0.378	0.362	0.327	0.159
0.50	0.412	0.394	0.375	0.340	0.164
0.60	0.439	0.414	0.391	0.356	0.169
0.70	0.468	0.439	0.411	0.377	0.176
0.80	$+\infty$	0.474	0.437	0.406	0.188
0.90	$+\infty$	$+\infty$	$+\infty$	0.447	0.210
0.95	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$

rate over the past 50 examples. Once the distribution shift occurs, only nmtCP using the exponential weight function remains valid for a targeted coverage rate of $1 - \alpha = 0.8$. It does so by continuously producing infinite intervals. In combination with the soft cutoff weight function the method loses validity briefly, remaining approximately valid for the remaining time steps of the test set. In combination with the linear weight function and the constant weight function, the nmtCP is not able to regain validity consistently after the change point. Figure 34 also shows that for CF-RNN (Stankevičiūtė et al. 2021) none of the real labels y_i are within the predicted region after the distribution shift.

4.4.3 Tétouan City

This subsection presents the results obtained from applying nmtCP to the subset of the Tétouan City data set (Salam & Hibaoui (2018) described in Subsect. 4.1.3. In contrast to the results presented in Subsect. 4.4.2, this set of experiments shows how the method behaves in the presence of a slow distribution shift.

Globally all models are valid for all confidence levels $1 - \alpha$. nmtCP with the exponential weight function, the soft cutoff weight function and the linear weight function are the models with the highest coverage rate closely followed by CF-RNN (Stankevičiūtė et al., 2021) and nmtCP with constant weights. Table 4 and Fig. 35 show these results. They also show that the difference in coverage rate between the variants of the method is small and

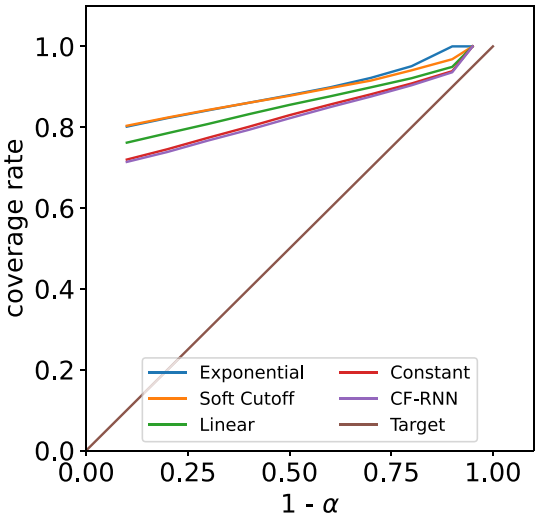

Fig. 35 Mean coverage rate for nmtCP with different weight functions and CF-RNN measured on the Tétouan City data set (Salam & Hibaoui, 2018). Average taken over 20 trials

that their coverage rate is above the targeted coverage rate, especially towards the lower end of the confidence level $1 - \alpha$.

While in Subsect. 4.4.2 models with larger coverage rate also produced larger prediction intervals, this is not the case for this set of experiments. nmtCP paired with the constant weight function has both the lowest coverage rate and produces the smallest prediction regions. However CF-RNN has a similar coverage rate while producing the largest prediction regions. Table 5 and Fig. 36 also indicate that for $1 - \alpha = 0.95$ all of them produce infinitely large prediction regions while nmtCP with the exponential weight function already does so for $1 - \alpha = 0.9$.

The three models with the lowest coverage rate, namely nmtCP with the constant weight function, CF-RNN and nmtCP with the linear weight function all lose their

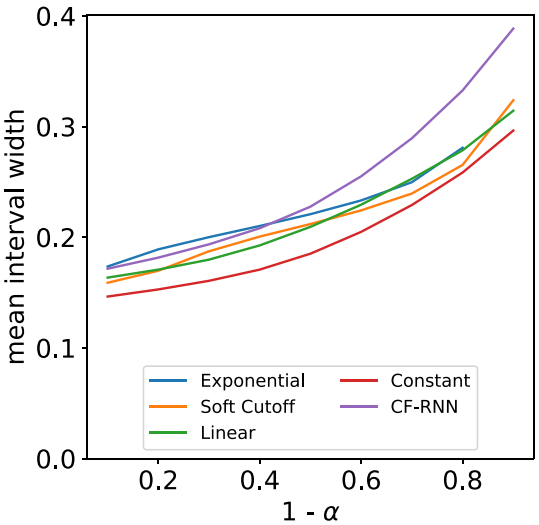

Fig. 36 Mean interval width for nmtCP with different weight functions and CF-RNN measured on the Tétouan City data set (Salam & Hibaoui, 2018). Average taken over 20 trials

coverage rate temporarily as shown in Fig. 37. For most of the time all models reach a coverage rate that is substantially above the targeted coverage rate of $1 - \alpha = 0.8$.

4.5 Discussion

The results presented in Subsects. 4.4.1 and 4.4.2 show that nmtCP is able to quickly adapt to strong distribution shifts such as changepoints, where CF-RNN (Stankevičiūtė et al., 2021) loses its validity. Subsection 4.4.1 also shows that nmtCP takes longer to recover

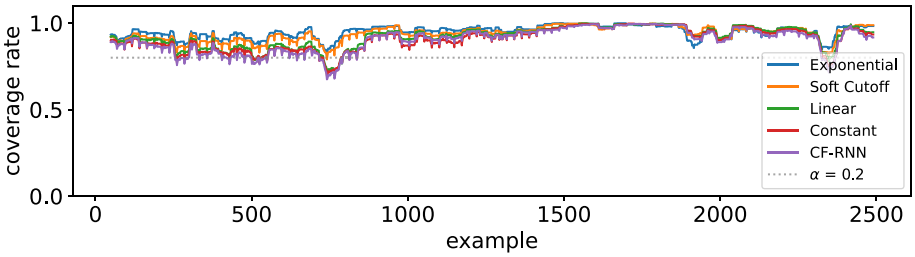

Fig. 37 Rolling coverage rate by feature for nmtCP with different weight functions and CF-RNN on the test set of the Tétouan City data set (Salam & Hibaoui, 2018). Rolling window size = 50, target coverage rate $1 - \alpha = 0.8$, average taken over 20 trials

Table 4 Mean coverage rate for nmtCP with different weight functions and CF-RNN measured on the Tétouan City data set (Salam & Hibaoui, 2018). Average taken over 20 trials

1-α	Exponential	Soft cutoff	Linear	Constant	CF-RNN
0.10	0.800	0.792	0.769	0.700	0.717
0.20	0.822	0.813	0.791	0.726	0.739
0.30	0.841	0.833	0.815	0.754	0.764
0.40	0.861	0.853	0.837	0.783	0.790
0.50	0.881	0.873	0.860	0.815	0.818
0.60	0.900	0.894	0.880	0.844	0.846
0.70	0.923	0.916	0.901	0.872	0.872
0.80	0.950	0.940	0.922	0.901	0.900
0.90	1.000	0.969	0.948	0.934	0.934
0.95	1.000	1.000	1.000	1.000	1.000

Table 5 Mean interval width for nmtCP with different weight functions and CF-RNN measured on the Tétouan City data set (Salam & Hibaoui, 2018). Average taken over 20 trials

1-α	Exponential	Soft cutoff	Linear	Constant	CF-RNN
0.10	0.181	0.153	0.160	0.144	0.197
0.20	0.198	0.163	0.167	0.150	0.206
0.30	0.209	0.180	0.176	0.157	0.216
0.40	0.219	0.192	0.189	0.166	0.228
0.50	0.229	0.203	0.205	0.179	0.243
0.60	0.242	0.215	0.225	0.197	0.264
0.70	0.259	0.230	0.249	0.221	0.292
0.80	0.289	0.254	0.275	0.251	0.326
0.90	$+\infty$	0.313	0.309	0.287	0.367
0.95	$+\infty$	$+\infty$	$+\infty$	$+\infty$	$+\infty$

from a local loss of validity when the distribution shifts progressively. Subsection 4.4.3 showed that nmtCP can produce smaller, more efficient prediction regions when both methods are valid. In all conducted experiments, nmtCP retains its global validity when the right weight functions W are chosen. These experiments confirm that nmtCP is generally suited for the multistep-ahead multivariate time-series forecasting task.

The experiments also show that the global validity, the loss of local validity, the speed of recovery of local validity and the mean interval width all depend on the weight function W that is used. The use of the exponential weight function consistently leads to the highest coverage rates, followed by the soft cutoff weight function. With the chosen parameters for both weight functions, they place larger weights on recent residuals while discounting older ones heavily. This allows nmtCP to quickly adapt to distribution shifts. The drawback of these weight functions is that by heavily discounting a large number of old residuals they reduce the effective sample size of past residuals Barber et al. (2022) leading to larger steps in the weighted quantile function Q. This contributes to a larger number of uninformative intervals of infinite size for high confidence levels $1 - \alpha$. The other two weight functions suffer less from this phenomenon. Discounting older residuals less quickly leads to a slower recovery after a local loss of validity which costs them their global validity in the experiment presented in Subsects. 4.4.1 and 4.4.2. Hence, the choice of the weight function and thereby the speed and severity with which to discount older residuals presents the method's user with a classic bias-variance trade-off.

A notable limitation of nmtCP is its inability to recover from a local loss of validity during a progressive distribution drift, no matter which of the weight functions presented in Subsect. 4.2.1 is chosen. Given the adversarial simulation settings 3 and 4, the quality of the underlying model's predictions worsens during the distribution drift. This means that the nonconformity scores for the dimensions affected by the distribution drift of new example will, on average, be larger than previous than their counterparts of previous examples. Ideally, past nonconformity scores of a particular dimension with a similar magnitude as the new one would be assigned larger weights and those with a dissimilar magnitude would be assigned lower weights to limit the coverage gap in every dimension of the label space. However, none of the selected weight functions with their fixed shape allow for such a dynamic assignment of the weights. This means that, with the weight functions presented in Subsection 4.2.1, the speed at which the prediction region grows in any dimension of the label space cannot surpass the speed at which the nonconformity scores grow in any dimension of the label space. Assuming there were a mechanism to assign the ideal weight to past nonconformity scores, there might simply not be any nonconformity scores that are large enough. If that is the case for even just one dimension in the label space, it would force the imaginary mechanism to assign the largest possible weight to $\delta_{r_{+\infty}}$ in that dimension's quantile function, resulting in a prediction region of infinite size with limited information provided to the method's user. One potential possibility to overcome this limitation is to replace the individual nexCPs in nmtCP with an adaptive method such as ACI, wich also supports a weighted quantile function.

The proof presented in Subsect. 3.3 guarantees the method's validity through an upper bound of the miscoverage rate by relying on the Bonferroni correction. It does, however, not provide information about the method's efficiency. The results of the experiments conducted on the real world data sets show that nmtCP is often too conservative, producing prediction regions exceeding the targeted coverage rate, especially for low confidence levels $1 - \alpha$. While (almost) assumption free validity is the main appeal of the conformal prediction methods, inefficient intervals are undesirable as they underestimate the underlying model's confidence. This conservative behavior

of nmtCP is, in the worst case, another factor contributing to a larger number of uninformative intervals of infinite size for high confidence levels $1 - \alpha$.

To investigate the influence of the correction method for the family-wise error rate on the method's efficiency and its conservative behavior we conducted an additional experiment that uses the portion of the ELEC2 data set (Harries, 1999) described in Subsect. 4.1.2. However, instead of comparing different weight functions W for the weighted quantile function Q, it compares different correction methods for the family-wise error rate. All trials use the soft cutoff weight function presented in Subsect. 4.2.1. The correction methods are

- The Bonferroni correction shown in Equation (25), replicating the results from Subsect. 4.4.2;
- The Šidák Correction, assuming that the errors occur independently in every dimension of the label space, resulting in $\alpha' = 1 - (1 - \alpha)^{T_h \cdot F_h}$, and;
- No correction at all with $\alpha' = \alpha$.

Subsection B.1 in the appendix contains the results for this experiment. Table 6 shows that with no correction at all, nmtCP is not valid, except in the extreme case where the produced interval is of infinite size and therefor uninformative. Using the Šidák Correction, nmtCP remains valid on average for all confidence levels $1 - \alpha$ and compares favorably to the Bonferroni correction by being more efficient. Figure 39a shows that the coverage rate remains closer to the targeted coverage rate, especially for lower confidence levels $1 - \alpha$ and Fig. 39b confirms that the produced intervals are smaller on average. For $1 - \alpha = 0.8$, the two methods are remarkably similar as shown in Fig. 39c. Due to their similarity for high confidence levels $1 - \alpha$, the use of the Šidák Correction does not reduce the generation of uninformative intervals of infinite size.

In addition to being a valid method for generating prediction regions for multistep-ahead multivariate time-series forecasting, nmtCP also provides the user with tools to investigate the contributions of every dimension in the label space to the miscoverage rate and the average interval width. This was already hinted to in Subsect. 4.3. Not only can individual dimensions be scrutinized, but they can also be aggregated. Figures 12, 13, 18, 19, 24, 25, 30, 31, in Subsect. 4.4.1 are examples for this. They allow for the observation that in the Simulation Settings, only the coverage rate and the mean interval width of the features experiencing changes in the distribution are affected by those changes. Fig. 40 in the appendix is another example of this, showing the rolling coverage rate for every feature of the results presented in Subsect. 4.4.2. These detailed insights are easily obtained for nmtCP because the prediction intervals are individually generated and expressed for every dimension in the label space and aggregated to a common prediction region after their generation. This transparency comes at the cost of depriving "the conformal model of profiting from the possible correlations between the different outputs", as Messoudi et al. (2020) note.

5 Conclusion

This article examined the task of conformal multistep-ahead multivariate time-series forecasting. It investigates the difficulties associated with applying conformal prediction to non-exchangeable data and multi-target regression. The proposed method, nmtCP produces

prediction regions for multistep-ahead multivariate time-series forecasting tasks. nmtCP, is computationally efficient and relatively easy to implement, requiring no modifications to the underlying model. For this method, a theoretical analysis showed that it has desirable theoretical properties by proving that the miscoverage rate has an upper bound that depends on the chosen hyperparameters. Because this proof does not rely on the exchangeability of the examples in the data set or a symmetric model fitting algorithm for the underlying model, the method is suitable for time-series applications. In addition, the method's architecture allows for an easy investigation of its behavior. This is possible because the metrics used to evaluate the method can be applied to each dimension in the label space individually or be aggregated over multiple dimensions.

The experimental results on four synthetic and two public data sets from the electricity domain serve to validate nmtCP's theoretical properties and provide insights into the method's behavior. They show that even in adverse conditions the method retains its validity when paired with the right parameters. It recovers quickly after a strong distribution shift and is more efficient than a competing method when both are valid.

The experiments also show that the method is often too conservative, especially for small confidence levels, producing prediction intervals that are larger than they need to be. Given the chosen weights on the nonconformity scores, another limitation of nmtCP revealed by the experiments is the method's inability to grow the prediction region quickly enough for the duration of a distribution drift.

5.1 Future work

We have identified the following key areas for future research: addressing the method's overly conservative behavior, providing practical guidance for tuning the weighted quantile function's weights, expanding support to quantile regression models, and developing a benchmarking platform. This section elaborates on these areas, proposes possible next steps, and highlights related research for each. Addressing the method's overly conservative behavior is a pressing next step. Integrating the work of Messoudi et al. (2021) offer a promising opportunity to do that by leveraging the correlations between the different dimensions in the label space using copulas. Instead of a hyper-rectangle, in their subsequent publication, Messoudi et al. (2022) present a method that produces a multidimensional ellipsoidal uncertainty region. Another possible approach to address the method's overly conservative behavior is building upon Ajroldi et al. (2023) and Diquigiovanni et al. (2024). If the values for multiple dimensions in the label space are sampled from the same continuous phenomena over a spatial or a temporal domain, Ajroldi et al. (2023) and Diquigiovanni et al. (2024) take a functional time series approach and generate prediction bands for functional data. This eliminates the reliance on family-wise error rate correction methods at the cost of predicting functions instead of matrices.

While the weights w of the weighted quantile function Q offer the user a lot of flexibility to tune nmtCP, we provide no concrete advice on how to exploit this opportunity. Based on the suggestion made by Barber et al. (2022), an intuitive interpretation of the task assumes that newer residuals should carry a higher weight. Additional experiments, such as the ones presented by Guan (2023) and Hore and Barber (2023) are required to offer more practical guidance to the user on how to set the weights.

In its current form, nmtCP only supports underlying models that produce point forecasts. It might be of interest to explore if the advances made for quantile regression used in

(Barber et al., 2022; Gibbs and Candès, 2021) can be applied to nmtCP. This would add support for underlying quantile regression models.

Finally, to test and compare nmtCP and future methods, constructing a set of synthetic benchmarks that simulate different scenarios such as change points in the distribution or slow distribution shifts in the data occurring for all or a subset of the features would be useful. These benchmarks should ideally come with a set of metrics that allow for an objective and comparable quantification of the method's properties, such as the time it takes for the method to recover after falling below the targeted coverage rate. They should also go beyond the adversarial approach taken in this article and put greater emphasis on the information efficiency of a tested method. Applying nmtCP to a larger set of real-world data sets, such as the one compiled by Messoudi et al. (2020) might also help to further understand the method's behavior and provide the user with practical guidance.

Appendix A Complete data set

See Fig. 38

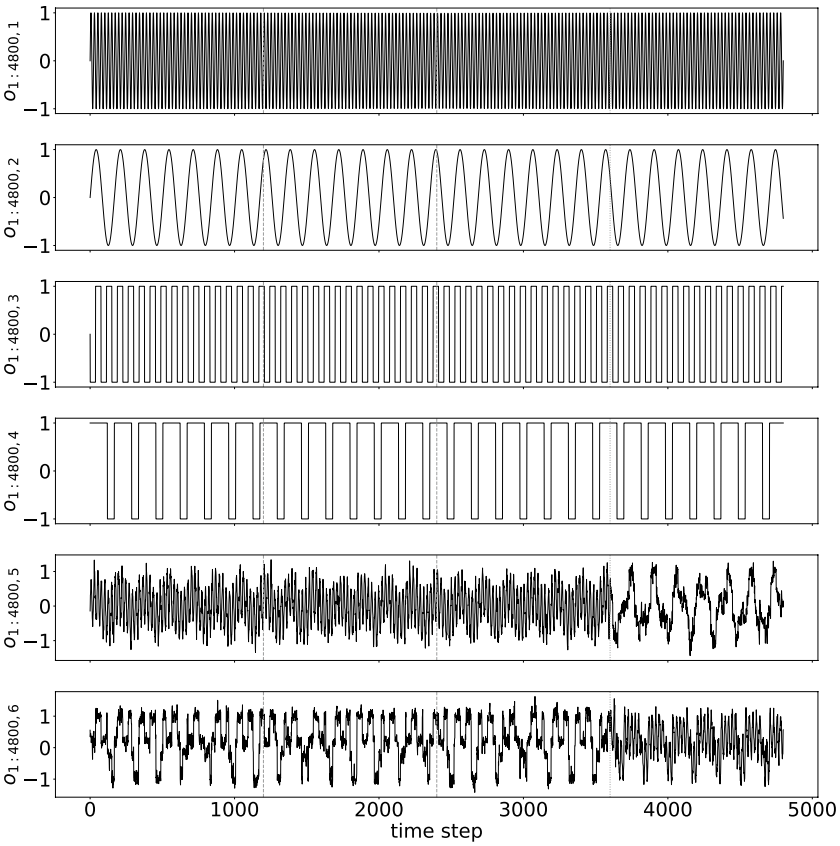

Fig. 38 Complete data set for simulation setting 1 with change-point used in the experiments

Appendix B Additional experimental results

B.1 Correction method comparison on ELEC2

This experiment uses the same setup as the experiment presented in Subsect. 4.4.2, applying nmtCP to the subsection of the ELEC2 data set (Harries, 1999) presented in Subsect. 4.1.2. However, instead of comparing different weight functions, the aim of this experiment is to compare different ways to correct for the family-wise error rate (FWER). Using the soft-cutoff weight function for all runs, the results presented here showcase the behavior of nmtCP using the Bonferroni correction, reproducing the results presented in Subsect. 4.4.2, using a correction based on the assumption that errors occur independently of each other, and using no correction for the family-wise error rate at all. These

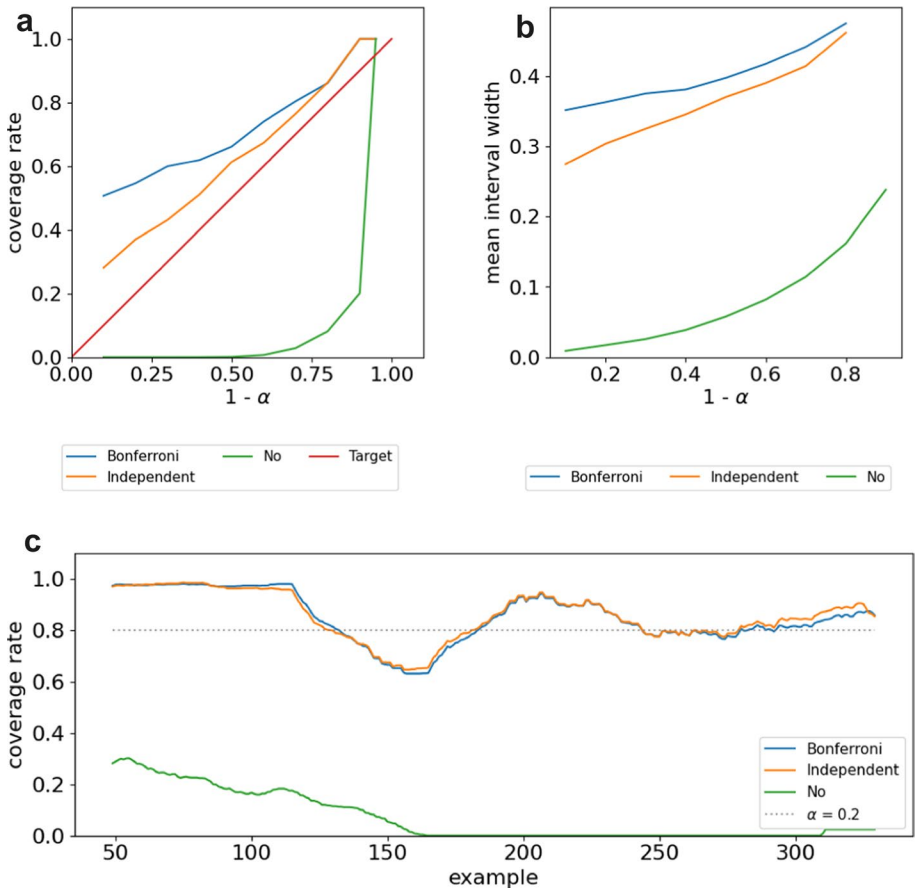

Fig. 39 Rolling coverage rate with a window size of 50 for different weight functions. Target coverage rate $1 - \alpha = 0.8$. Average taken over 20 trials. **a** Mean coverage rate for nmtCP with different FWER corrections. Average taken over 20 trials. **b** Mean interval width for nmtCP with different FWER corrections. Average taken over 20 trials. **c** Rolling coverage rate with a window size of 50 for different FWER corrections. Target coverage rate $1 - \alpha = 0.8$. Average taken over 20 trials

Table 6 Empirical coverage rate and mean interval width measured on the ELEC2 data set (Harries, 1999). Average taken over 20 trials

	Coverage rate			Mean interval width		
1-α	Bonferroni	Independent	No	Bonferroni	Independent	No
0.10	0.506	0.282	0.000	0.351	0.274	0.009
0.20	0.546	0.370	0.000	0.363	0.304	0.017
0.30	0.599	0.431	0.000	0.375	0.325	0.026
0.40	0.619	0.511	0.000	0.381	0.345	0.039
0.50	0.661	0.613	0.001	0.397	0.370	0.058
0.60	0.740	0.673	0.007	0.417	0.390	0.082
0.70	0.803	0.764	0.028	0.441	0.414	0.114
0.80	0.861	0.862	0.081	0.475	0.462	0.162
0.90	1.000	1.000	0.201	$+\infty$	$+\infty$	0.238
0.95	1.000	1.000	1.000	$+\infty$	$+\infty$	$+\infty$

three correction schemes are labeled as *Bonferroni*, *Independent* and *No*, respectively, in Fig. 39a, b and c as well as in Table 6.

B.2 Weight function comparison on ELEC2

See Fig. 40.

Additional results related to the ones presented in Subsection 4.4.2.

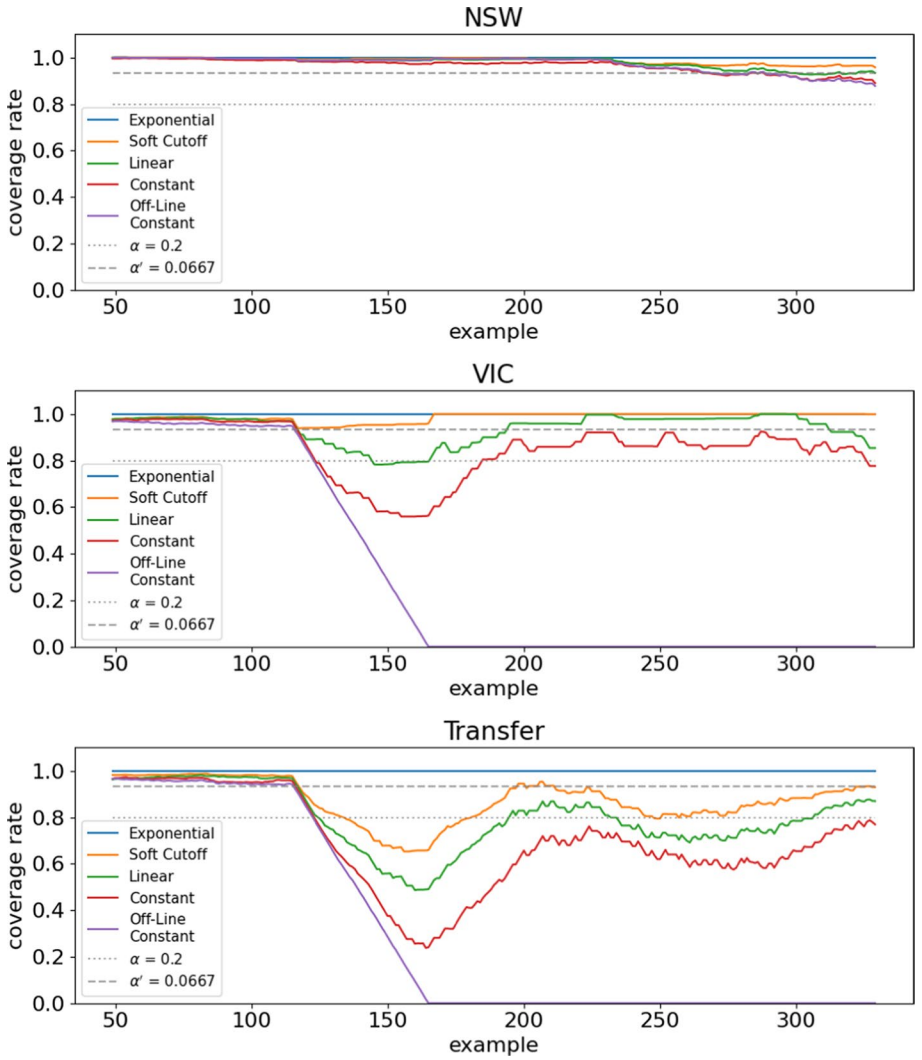

Fig. 40 Rolling coverage rate with a window size of 50 for all features in the data set and different weight functions. Target coverage rate $1 - \alpha = 0.8$. Average taken over 20 trials

Author contributions F.S. made substantial contributions to the conception or design of the work and wrote the main manuscript text. F.S. also wrote the software to conduct the experiments, analyzed the data and interpreted the results. E.S., I.K. and M.W. made substantial contributions to the analysis and the interpretation of the results and revised the manuscript critically.

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Data availability The ELEC2 data set (Harries, 1999) is available at https://www.kaggle.com/datasets/yashsharan/the-elec2-dataset . The Tétouan City data set (Salam & Hibaoui, 2018) is available at https://archive.ics.uci.edu/ml/datasets/Power+consumption+of+Tetouan+city

The code written to produce the results presented in this article is made available at https://doi.org/10.5281/zenodo.14833627.

Declarations

Conflict of interest I declare that the authors have no Conflict of interest as defined by Springer, or other interests that might be perceived to influence the results and/or discussion reported in this article.

Ethical approval Not Applicable

Consent for publication Not Applicable

Consent to participate Not Applicable

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Uncertainty quantification in Bayesian physics-informed deep learning-based traffic state prediction

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Abstract

Accurate and reliable traffic state prediction (TSP) is an essential task for intelligent transportation systems. However, achieving this goal is challenging due to the high-dimensional and coupled nature of traffic feature evolution patterns, which are deeply recessive and make it difficult to effectively characterize and model TSP using purely data-driven methods. Furthermore, a significant limitation of existing TSP methods is their inability to estimate data and model uncertainty, which is crucial for understanding inherent data variations and model limitations. To address these challenges, this study proposes a novel TSP model that combines the diffusion convolutional recurrent neural network (DCRNN) with physical prior knowledge within a Bayesian framework. Specifically, DCRNN captures the spatiotemporal correlation among various sensors. Furthermore, this approach leverages Monte Carlo dropout and heteroskedasticity modeling to quantify epistemic and aleatoric uncertainties. The model's efficacy is evaluated using the Xuancheng China urban dataset and the PeMS04 US highway dataset. Empirical results show that the proposed method outperforms state-of-the-art methods in both prediction accuracy and uncertainty quantification. These findings highlight the advantages of a data-model hybrid-driven approach to achieve accurate and reliable TSP. This study effectively quantifies and mitigates both aleatoric and epistemic uncertainties, holding significant implications for the control and management of real traffic flow.

1 | INTRODUCTION

Traffic state prediction (TSP), which involves forecasting traffic flow, speed, and density, has emerged as a crucial focus within intelligent transportation systems (Li et al., 2020; Liu et al., 2020). Accurate TSP provides robust decision support for urban traffic planners and drivers, facilitating effective traffic management and travel planning (Xue et al., 2024). However, despite extensive research on TSP, limited attention has been given to the quantification of uncertainty within this domain (Mallick et al., 2024). The reliability of TSP is significantly challenged by the inherent uncertainty in transportation systems, compounded by external environmental shocks. Additionally, neural network models, which are commonly employed in TSP, are susceptible to overfitting, thereby limiting their capacity to generalize (Jiang & Adeli, 2005; Samant & Adeli, 2001; Szegedy, 2013).

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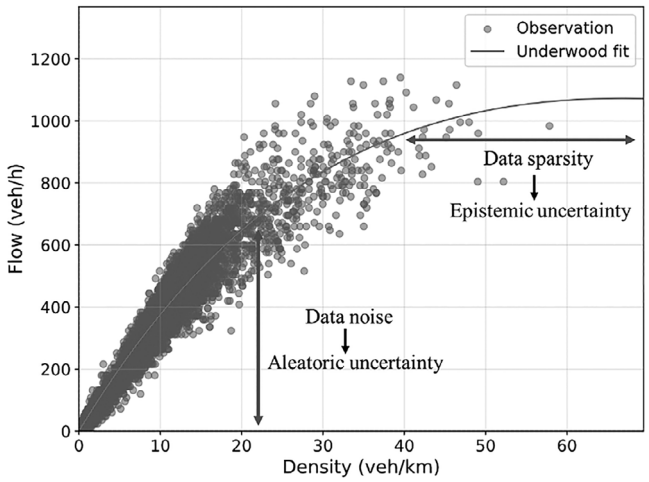

FIGURE 1 The sources of epistemic and aleatoric uncertainties in the traffic flow-density relationship using the fitted Underwood model. The dots represent observed traffic flow under varying densities; the curve illustrates the fitted Underwood model, which effectively captures the nonlinear relationship between traffic flow and density while maintaining mathematical simplicity. The vertical arrow indicates aleatoric uncertainty (AU), representing the inherent data noise or variability at a given density. The horizontal arrow denotes epistemic uncertainty (EU), reflecting the model's limited knowledge in high-density regimes due to data sparsity.

These challenges collectively hinder the accuracy and reliability of TSP models (Munikoti et al., 2023). Ignoring uncertainties in TSP can lead to overconfidence among decision-makers, potentially resulting in overlooked risks, exacerbated traffic congestion, or even traffic safety issues (Cruz & Sarmento, 2020). Therefore, quantifying uncertainty and improving the reliability of TSP are critical for advancing both practical traffic applications and scientific research.

Two primary sources of uncertainty in TSP have been identified: aleatoric uncertainty (AU) and epistemic uncertainty (EU; Abdar et al., 2021). AU, as illustrated in Figure 1, refers to the inherent variability or noise in traffic data (Qian et al., 2023). This type of uncertainty arises from uncontrollable factors such as driver behavior, unexpected traffic incidents, or environmental conditions. It is considered irreducible, meaning that neither increasing the volume of data nor enhancing model complexity can eliminate it (Bao et al., 2020). In contrast, EU stems from a lack of knowledge or insufficient data (Shridhar et al., 2019). For example, it may occur when predicting traffic flow in areas with sparse sensor coverage or when the model encounters scenarios it was not trained on, such as road closures or unusual weather conditions. Unlike AU, EU can be mitigated through the acquisition of additional data or improvements in model design. Accurate quantification of both aleatoric and epistemic uncertainties is crucial for enhancing model interpretability and reliability (Tagasovska & Lopez-Paz, 2018). Understanding the distinct sources of uncertainty and their causal relationships enables the development of targeted strategies to improve model performance (Qian et al., 2024). However, despite its significance, most existing studies only focus on the total uncertainty. The failure to distinguish and separately quantify aleatoric and epistemic uncertainties has limited the ability to fully understand of the reliability of traffic state predictions.

Current research mainly employs Bayesian-based and frequentist-based deep learning methods for quantifying uncertainty in traffic prediction (Wu et al., 2021). Among these methods, the Bayesian approach has a systematic advantage of addressing various sources of uncertainty along with prior beliefs (Munikoti et al., 2023). However, the above approaches are fundamentally data-driven and encounter significant limitations in practical applications (Abdar et al., 2021). First, deep learning models require large datasets and are highly sensitive to missing or noisy data, which undermine their prediction robustness and generalizability (Wang et al., 2024). Second, these models often lack a solid theoretical foundation, relying predominantly on empirical heuristics and trial-and-error processes. This lack of interpretability in model parameters poses challenges in transferring models and their results to other application scenarios (Di et al., 2023; K. Xu et al., 2018). Furthermore, for decision-makers, merely quantifying uncertainties in data and models is inadequate; the crucial task is to effectively reduce these uncertainties to narrow prediction intervals, thereby enhancing the model's utility and decision-support capabilities.

Motivated by the aforementioned issues, this study proposes a novel model that integrates the diffusion convolutional recurrent neural network (DCRNN) with physics-informed knowledge in a Bayesian framework (named B-PI-DCRNN) for network-level TSP and uncertainty quantification. Both aleatoric and epistemic uncertainties are captured in a unified approach. AU is quantified through heteroskedasticity modeling, and Monte Carlo (MC) dropout is used to capture EU in model parameters. Both uncertainties are then combined to estimate the total uncertainty in TSP. The sensor network is represented as a graph, and the relationships between various sensors are captured based on the DCRNN. Furthermore, a traffic flow model is incorporated to develop a physics-informed deep learning model (PIDL), which establishes linkages between traffic flow and speed for each sensor in the network. This provides an essential prior knowledge for predicting traffic state. The contributions of this study are as follows:

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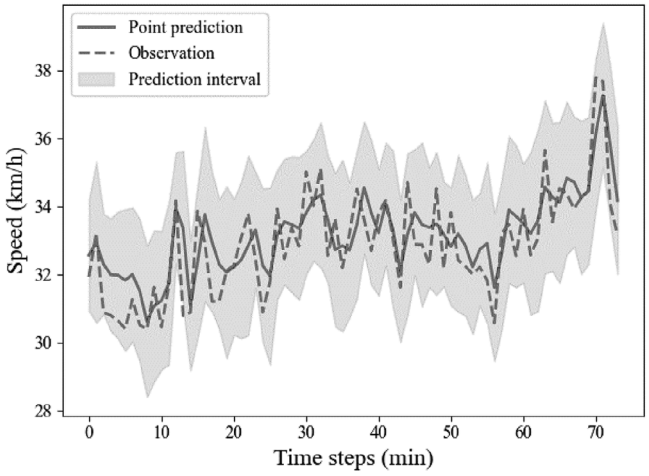

FIGURE 2 Illustration of speed prediction with deterministic model and uncertainty quantification.

1. This study presents a Bayesian framework with MC dropout and physical prior knowledge to quantify aleatoric and epistemic uncertainties in TSP, offering a nuanced understanding of inherent data variability and model limitations in the field of transportation.
2. By integrating a physical model that encapsulates the fundamental laws of traffic flow, the data-model hybrid-driven approach ensures that the predicted traffic state variables—speed and flow in this study—adhere to these laws, achieving an effective fusion of prediction accuracy and interpretability.
3. To the best of our knowledge, the B-PI-DCRNN framework is the first to quantitatively demonstrate that integrating physics-based constraints into deep-learning architectures markedly improves the reliability of TSP, thereby establishing a new paradigm for the comprehensive performance evaluation of traffic prediction models.

This paper is structured as follows: Section 2 reviews the related literature on traffic state prediction, deep learning models, and uncertainty quantification techniques. Section 3 outlines the methodology, including the proposed model for AU and EU quantification and the B-PI-DCRNN model. Section 4 presents the experimental results, while Section 5 concludes the paper and outlines future directions for research in this area.

2 | RELATED WORK

This section reviews existing literature related to B-PI-DCRNN from two perspectives: traffic prediction with a deterministic model and uncertainty quantification. Traditional deterministic models, as illustrated in Figure 2, generate point predictions without assessing predictive confidence. In contrast, uncertainty-aware framework provides both predicted values and their corresponding prediction intervals.

2.1 | Traffic prediction with deterministic model

In recent years, TSP has become a crucial research area within intelligent transportation systems, prompting the development of various methodologies grounded in statistics and machine learning algorithms (Qian et al., 2024; Zou & Chung, 2024). Statistical methods like the autoregressive integrated moving average and Kalman filters are grounded in robust mathematical principles but often struggle with highly nonlinear, complex data and face challenges from the curse of dimensionality (Pan et al., 2013). With advancements in machine learning, models like support vector machines (SVMs) and random forests were introduced, capable of capturing non-linear relationships (Mingheng et al., 2013). For example, SVM can classify and regress complex traffic data by constructing decision boundaries in high-dimensional feature spaces (Wang & Shi, 2013). Nonetheless, the performance of machine learning models is heavily contingent upon the quality of feature engineering, posing challenges in maintaining stable prediction performance under high-dimensional and heterogeneous traffic data.

Deep learning techniques have significantly advanced spatiotemporal data processing for TSP (Ahmed et al., 2024; Hooshdar & Adeli, 2004; Li et al., 2020). For instance, convolutional neural networks (CNNs) have been employed to capture spatial dependencies, while recurrent neural networks (RNNs) are utilized for modeling temporal relationships (Song et al., 2017). Recently, graph neural networks have emerged as powerful tools capable of modeling both spatial and temporal dependencies in TSP (Jiang & Luo, 2022; Jiang et al., 2023). Yu et al. (2024) proposed a spatial–temporal autoformer, which leverages spatial–temporal sequence autocorrelation for TSP. Deng (2025) developed a network congestion prediction method combining a long short-term memory (LSTM) network with association rules. Similarly, Zou and Chung (2024) proposed a CNN-LSTM model to predict traffic volumes for each detector. However, these deep learning models typically require substantial volumes of data and, due to their black-box nature, suffer from limited interpretability (Wang et al., 2024).

To address the limitations of purely data-driven models, recent research has focused on PIDL (Raissi et al., 2019; Z. Shi et al., 2023), a promising approach that leverages the strengths of both model-driven and data-driven

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methodologies to improve traffic prediction accuracy (Deshpande & Park, 2025; Wang et al., 2024; Xu et al., 2024). For example, Xing et al. (2025) developed an origin–destination prediction model that integrated a deep learning framework with traffic engineering domain knowledge. Similarly, Gao et al. (2025) proposed a Graph Neural Network (GNN) model for speed prediction that incorporated some prior knowledge such as vehicle dynamics, highway geometry, and driver expectations. Integrating such domain knowledge ensures alignment with established physical laws and enhances prediction accuracy (Shi et al., 2021; Zhang et al., 2024). Although existing models have demonstrated good performances in TSP, they predominantly generate point predictions, neglecting the inherent randomness in traffic data and the uncertainties associated with the models themselves (Sengupta et al., 2024). Therefore, incorporating uncertainty quantification into TSP is crucial. This would provide more reliable information regarding the confidence levels of predictions, thereby enhancing the overall utility and robustness of these models.

2.2 | Uncertainty quantification

Traffic prediction, similar to other complex prediction tasks, is fundamentally influenced by uncertainty (Allen et al., 2022; Qian et al., 2023). This uncertainty stems from the stochastic nature of traffic state evolution within a road network, which remains non-deterministic for any observer (Li et al., 2024). The limited observability inherent in traffic systems necessitates that predictive models not only provide point predictions but also evaluate the associated uncertainty (Wu et al., 2021). To accurately quantify this uncertainty, some methods are utilized, including probability distribution functions, statistical methods, and ensemble prediction techniques (Wang et al., 2024). Guo et al. (2014) addressed the limitation of traditional Kalman filters, which assume constant process variance, by proposing an adaptive Kalman filter that updates the process variance for traffic flow prediction and generates prediction intervals. Similarly, Wu et al. (2021) applied both frequentist and Bayesian methods to predict traffic speeds and quantify the associated prediction uncertainty. Their experimental results demonstrated that Bayesian methods generally achieve better mean predictions, whereas frequentist approaches provide confidence intervals that better cover data variations. Recently, Wang et al. (2024) introduced a framework based on probabilistic graph neural networks to quantify uncertainty in spatiotemporal travel demand. Overall, these studies focus on quantifying total uncertainty but fail to distinguish and separately quantify aleatoric and epistemic uncertainties (Der Kiureghian & Ditlevsen, 2009). This limitation constrains the ability to comprehensively understand and evaluate the reliability of traffic predictions.

As shown in Table 1, current research mainly employs Bayesian and frequentist methods for quantifying aleatoric and epistemic uncertainties (Wu et al., 2021). Bayesian methods, as probabilistic models for uncertainty quantification, offer a principled framework for incorporating both prior beliefs and various sources of uncertainty (Munikoti et al., 2023). For example, Sengupta et al. (2024) proposed a Bayesian RNN for uncertainty quantification in traffic flow prediction. On the other hand, frequentist methods quantify uncertainty by training multiple neural networks with varied initializations and analyzing the statistical properties of their prediction outputs (Lakshminarayanan et al., 2017). For instance, Mallick et al. (2024) proposed a deep ensemble spatiotemporal graph neural network to enhance the accuracy and reliability of traffic flow predictions by quantifying uncertainties. In summary, these methods do not directly model different sources of uncertainty. Instead, they infer uncertainty indirectly through model aggregation.

Uncertainty quantification in TSP is an emerging area, yet current models suffer from limitations that undermine their reliability and decision-support value, particularly in complex traffic scenarios. First, existing models mainly focus on the total uncertainty quantification, failing to

TABLE 1 Comparison of recent studies for uncertainty quantification in traffic state prediction (TSP).

Author	Traffic variables		Method	Method type
	Speed	Flow		
Wu et al. (2021)	√	/	Frequentist/ Bayesian	DM
Qian et al. (2024)	/	√	**Ensemble**	**DM**
Sengupta et al. (2024)	√	/	Bayesian	DM
Mallick et al. (2024)	√	/	**Ensemble**	**DM**
This study	√	√	B-PI-DCRNN	DMHM

Abbreviations: B-PI-DCRNN, diffusion convolutional recurrent neural network (DCRNN) with physics-informed knowledge in a Bayesian framework; DM, data-driven method; DMHM, data-model hybrid-driven method.

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differentiate between the EU (resulting from model limitations) and AU (inherent data variability) within a unified framework. Second, current studies primarily emphasize uncertainty evaluation, neglecting the development of targeted mechanisms to actively reduce the impact of uncertainty on model performance. Third, the prevalent use of purely data-driven approaches in existing uncertainty quantification models leads to a reduced generalization capability when faced with low data quality, small sample sizes, and non-stationary traffic patterns. Furthermore, most TSP studies treat speed and flow as independent variables, overlooking their intrinsic physical relationship, which compromises the models' ability to maintain prediction accuracy alongside physical consistency. To enhance the accuracy, robustness, and interpretability of TSP models, it is crucial to develop a novel data-model hybrid-driven framework that is capable of quantifying and mitigating multiple sources of uncertainty simultaneously.

3 | METHODOLOGY

3.1 | Problem definition

An urban road network can be modeled as a graph $G = (V, L)$, where V represents the set of nodes corresponding to N sensors where traffic states are measured, and L denotes the set of links connecting these nodes. From this graph, the adjacency matrix $A (A \in \mathbb{R}^{N \times N})$ is constructed to represent the connectivity between sensors. The traffic state of the network over t time steps is denoted as $X = [X_1, \dots, X_t]$, and the goal is to predict future traffic states $X^f = [X_{t+1}, \dots, X_{t+H}]$, with H representing the forecasting horizon. In this study, the traffic state at time step t, $X_t = [p(n,t), v(n,t)]^{\mathrm{T}} \in \mathbb{R}^{2 \times N}$, is defined for each sensor $n (1 \le n \le N)$, where $p(n,t)$ and $v(n,t)$ denote the traffic flow and travel speed, respectively, at sensor $n \in V$ and time step $t \in T$. Additionally, the traffic density $k(n,t)$ can be derived using the relationship $k(n,t) = p(n,t)/v(n,t)$. This study aims to develop a function $f(X, A) \to X^f$ that predicts traffic state while quantifying both aleatoric and EU within a unified framework.

3.2 | Model structure

The overall framework of B-PI-DCRNN is illustrated in Figure 3. Within the physical calibration layer, training data are processed through a fundamental diagram model to systematically calibrate the physical characteristics of each sensor. The calibrated results are subsequently fed into a Bayesian neural network (BNN) that employs MC dropout for model training. Unlike CNN, BNN applies prior distributions to their weight parameters. By employing MC dropout, the posterior distribution of these weights can be approximated, enabling the inference of EU. AU, based on input and probabilistic weights, is quantified using heteroskedasticity modeling. Integrating aleatoric and epistemic uncertainties offers a comprehensive measure of total uncertainty.

In the loss function component, the data loss accounts for the residuals between the model's predicted values and the observed values, while also incorporating a regularization term. The regularization term is crucial to prevent the model from minimizing the loss by excessively increasing prediction variance and to avoid overfitting. Second, the physical loss evaluates the discrepancy between predicted values and underlying physical traffic flow model. This component ensures that the model adheres to the physical laws of traffic flow, thereby mitigating risks of overfitting and deviations. Finally, the total loss function is constructed by combining the data loss and physical loss with appropriate weighting factors. This aggregation ensures that the model achieves a balance between accurate data fitting and adherence to physical laws. By following this approach, the model generates predictions of spatiotemporal traffic flow and travel speed, accompanied by confidence intervals that reflect the associated aleatoric and epistemic uncertainties.

3.3 | Quantifying EU and AU in a unified Bayesian framework

This section details the specific methods for modeling epistemic and aleatoric uncertainties of the TSP within a unified Bayesian framework. EU is addressed by assigning a prior distribution to the model's weights and examining their variation through MC sampling. On the other hand, AU is quantified by modeling the output as a heteroskedastic Gaussian distribution, where the variance captures input-dependent data noise. Additionally, this study also considers the dependency between multiple traffic variables in TSP.

3.3.1 | EU

By applying Bayesian inference in a deep learning architecture, EU can be effectively quantified through the probabilistic modeling of model parameters. The uncertainty arises from the variability in the weight matrix W, represented by the conditional probability distribution $p(W|X, S)$, where S and X are dependent and independent variables, respectively. Hence, EU can be quantified

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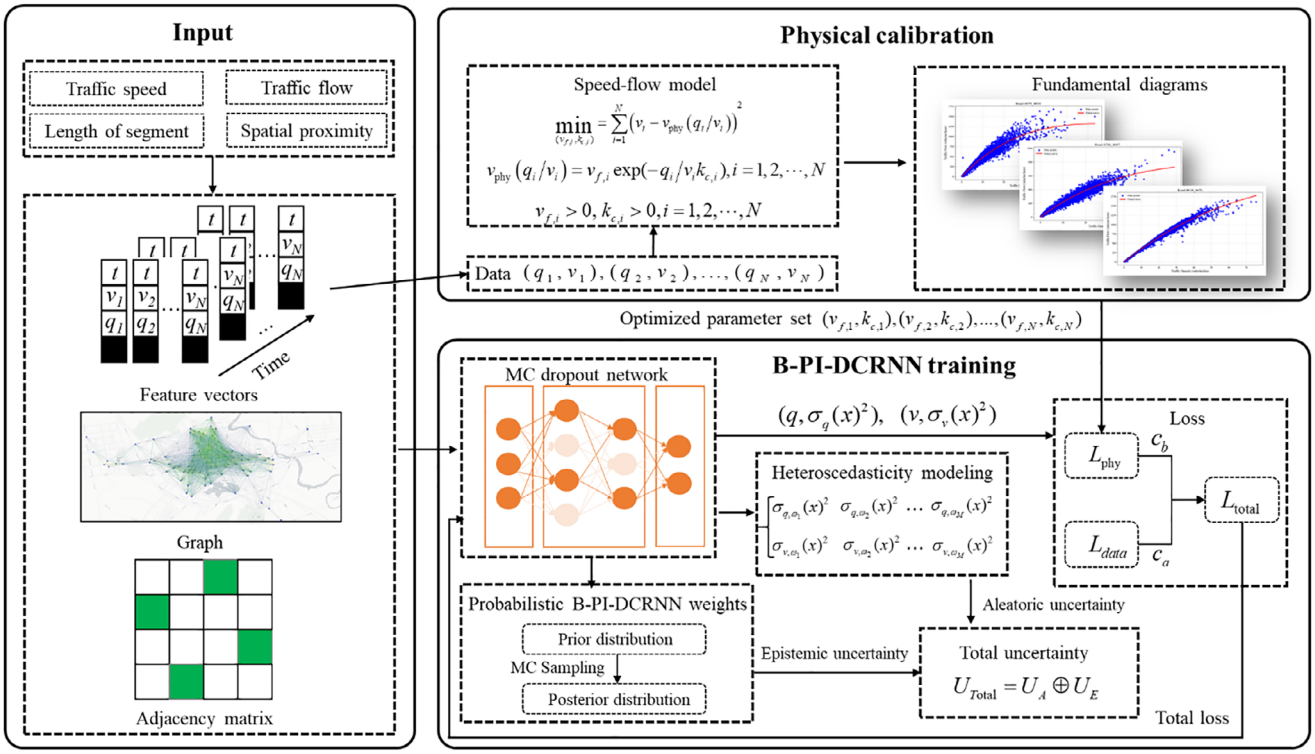

FIGURE 3 The architecture of diffusion convolutional recurrent neural network (DCRNN) with physics-informed knowledge in a Bayesian framework (B-PI-DCRNN).

within a Bayesian framework by introducing a prior $p(W)$ as follows (Li et al., 2020).

$$p(W|X,S) = \frac{p(S|X,W)p(W)}{p(S|X)} \tag{1}$$

When making predictions for a new input x, the model's prediction distribution is given by:

$$p(s|x,X,S) = \int p(s|x,X,S)\, p(W|X,S)\, \mathrm{d}W \tag{2}$$

where s represents the predictive mean, $p(s|x,X,S)$ denotes the predictive likelihood, and $p(W|X,S)$ refers to the posterior of the weights. Precise Bayesian inference in deep learning architecture is computationally expensive. To mitigate this challenge, this study uses the MC dropout, a technique that utilizes dropout as a regularization term to compute predictive uncertainty (Gal & Ghahramani, 2016).

MC dropout is used to approximate the posterior distribution of parameters by employing dropout during both the training and inference phases. During training, neurons are randomly dropped with a probability p, which imposes a variational distribution $q(W)$ on the network weights. This approach aims to approximate the true posterior distribution by minimizing the divergence between the true posterior and the variational distribution. The loss function associated with this process is given by:

$$L_{\text{dropout}} := \frac{1}{2N}\sum_{i=1}^{N}\|q_i - \hat{q}_i\|_2^2 + \frac{1}{2N}\sum_{i=1}^{N}\|v_i - \hat{v}_i\|_2^2 + \lambda\sum_{l=1}^{L}\left(\|W_l\|_2^2 + \|b_l\|_2^2\right) \tag{3}$$

where N denotes the number of data points, q_i and $\hat{q}_i$ represent the observed and predicted values of flow at sensor i, respectively. Similarly, v_i and $\hat{v}_i$ correspond to the observed and predicted values of speed at sensor i, respectively. The parameter λ signifies the weight decay. Furthermore, b_l and W_l denote the bias vector and weight matrix of layer l, respectively.

During the inference phase, MC dropout is consistently applied, and M random forward passes are executed for the same data. In each forward pass, different neurons are randomly dropped, leading to a variety of predictions. By statistically analyzing these predictions, the mean and variance of the predicted values can be calculated. This process allows us to approximate the model's posterior distribution and quantify the uncertainty in TSP.

$$\left[\mu_m, {\sigma_m}^2\right] = f^{W_m}(X) \tag{4}$$

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where the weight matrix W_m is sampled through dropout in the mth stochastic forward pass. This process results in the new predictions for traffic flow and speed, and mean values denoted as $\mu_m = [\hat{q}_m, \hat{v}_m]$. Additionally, the variance of these predicted values is represented as $\sigma_m{}^2 = [\sigma_{q,m}{}^2, \sigma_{v,m}{}^2]$. The function f is parameterized by the model weights. By conducting multiple forward passes and examining the outcomes, we can derive both the mean and variance of TSP.

AU stems from the inherent stochasticity within the data itself. In traffic systems, this uncertainty is often introduced by external and uncontrollable variables, including accidents, inclement weather, and variations in road infrastructure. For instance, traffic conditions during peak hours exhibit greater volatility and noise, compared to off-peak periods. AU can be further classified into homoskedastic and heteroskedastic types (Qian et al., 2024). The distribution of a sensor's measurement of a traffic variable (e.g., flow and speed) at a specific time can be modeled as a Gaussian distribution with a predicted mean $\mu(x)$ and variance σ (Kendall & Gal, 2017; Sengupta et al., 2024). Homoskedastic models assume a constant variance σ across all inputs. Conversely, heteroskedastic models allow the variance $\sigma(x)$ to vary as a function of the input, thereby capturing differing levels of noise associated with different traffic states. This dynamic variance modeling enables heteroskedastic models to more precisely represent the AU inherent in TSP. Directly optimizing the model parameters W based on the Gaussian probability density function is intractable. To address this, we adopt the equivalent formulation of maximizing the log-likelihood of the Gaussian distribution:

$$\log p\left(s_i \mid \mu\left(x_i\right), \sigma\left(x_i\right)\right) = -\frac{1}{2}\log\left(2\pi\sigma(x_i)^2\right) - \frac{\left(s_i - \mu\left(x_i\right)\right)^2}{2\sigma_v(x_i)^2} \tag{5}$$

Accordingly, the negative log-likelihood is used as the loss function following Equation (5), and the learning objective for AU estimation is to minimize this likelihood-based loss.

$$L(\theta) = \frac{\zeta_q}{N}\sum_{i=1}^{N}\frac{1}{2\sigma_q(x_i)^2}\left\|q_i - \hat{q}_i\right\|^2 + \frac{\zeta_v}{N}\sum_{i=1}^{N}\frac{1}{2\sigma_v(x_i)^2}\left\|v_i - \hat{v}_i\right\|^2 + \frac{\zeta_q}{2}\log\sigma_q(x_i)^2 + \frac{\zeta_v}{2}\log\sigma_v(x_i)^2 \tag{6}$$

where ζ_q and ζ_v are weight coefficients. To improve numerical stability, we train the network to predict the logarithmic variances $s_{q,i} := \log\sigma_q(x_i)^2$ and $s_{v,i} := \log\sigma_v(x_i)^2$ rather than directly regressing the variances $\sigma_q(x_i)^2$ and $\sigma_v(x_i)^2$(Kendall & Gal, 2017). This approach leverages the properties of the exponential function, which ensures that the predicted variance values remain positive by regressing the unconstrained scalars $\exp(-\sigma_q(x_i)^2)$ and $\exp(-\sigma_v(x_i)^2)$.

$$L(\theta) = \frac{\zeta_q}{N}\sum_{i=1}^{N}\frac{1}{2e^{s_{q,i}}}\left\|q_i - \hat{q}_i\right\|^2 + \frac{\zeta_v}{N}\sum_{i=1}^{N}\frac{1}{2e^{s_{v,i}}}\left\|v_i - \hat{v}_i\right\|^2 + \frac{\zeta_q}{2}s_{q,i} + \frac{\zeta_v}{2}s_{v,i} \tag{7}$$

Finally, considering both the epistemic and aleatoric uncertainties, as indicated by Equations (3) and (7), the combined loss function can be described as follows:

$$L_{\text{data}} = \frac{\zeta_q}{N}\sum_{i=1}^{N}\frac{1}{2e^{s_{q,i}}}\left\|q_i - \hat{q}_i\right\|^2 + \frac{\zeta_v}{N}\sum_{i=1}^{N}\frac{1}{2e^{s_{v,i}}}\left\|v_i - \hat{v}_i\right\|^2 + \frac{\zeta_q}{2}s_{q,i} + \frac{\zeta_v}{2}s_{v,i} + \lambda\sum_{l=1}^{L}\left(\|W_l\|_2^2 + \|b_l\|_2^2\right) \tag{8}$$

The quantification of predictive uncertainty in our model is derived from the fundamental principles of Bayesian inference and the law of total variance. Within the Bayesian framework, the total predictive uncertainty for a given input x_i can be decomposed into AU (U_{A}), which captures inherent data noise, and EU (U_{E}), which arises from model parameter uncertainty. This decomposition is expressed mathematically as

$$\text{Var}\left(s_i\right) = E\left[\text{Var}\left(s_i \mid x_i\right)\right] + \text{Var}\left[E\left(s_i \mid x_i\right)\right] \tag{9}$$

Here, the combination of aleatoric variance $E[\text{Var}(s_i|x_i)]$ and epistemic variance $\text{Var}[E(s_i|x_i)]$ forms the overall predictive variance, aligning with Equation (2). In the context of our specific architecture, which predicts two physical quantities (denoted q and v), the AU is estimated as the average of the predicted variances across all MC samples:

$$U_{\text{A}} = E\left[\text{Var}\left(s_i \mid x_i\right)\right] \approx \frac{1}{M}\sum_{m=1}^{M}\left(\zeta_q\sigma_{q,m}(x_i)^2 + \zeta_v\sigma_{v,m}(x_i)^2\right) \tag{10}$$

$$U_{\text{E}} = E\left[E(s_i \mid x_i)^2\right] - E[E\left(s_i \mid x_i\right)]^2 \approx \frac{1}{M}\sum_{m=1}^{M}\left(\zeta_q\hat{q}_m^2 + \zeta_v\hat{v}_m^2\right) - \zeta_q\left(\frac{1}{M}\sum_{m=1}^{M}\hat{q}_m\right)^2 - \zeta_v\left(\frac{1}{M}\sum_{m=1}^{M}\hat{v}_m\right)^2 \tag{11}$$

$$U_{\text{T}} = U_{\text{A}} + U_{\text{E}} \tag{12}$$

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where M denotes the number of stochastic forward passes performed during the model's inference phase. By integrating the MC dropout method with Bayesian inference, we can effectively quantify both aleatoric and epistemic uncertainties in TSP. This integration allows for the approximation of Bayesian reasoning within the deep neural networks, thereby providing a robust tool for modeling uncertainty in traffic predictions. Notably, U_{E} would diminish to zero when there is no uncertain parameter, meaning that all draws W_m take the same constant value.

3.4 | Physics-informed module

Predicting traffic flow and travel speed separately often leads to outcomes that violate fundamental physical laws, especially when the data are noisy or insufficient (Wang et al., 2024). To address this issue and enhance the physical consistency and generalization of predictive models, this study proposes an integration of fundamental diagrams of the traffic flow into the BNN framework. Fundamental diagrams play a crucial role in both constructing feature vectors for each sensor and establishing physical consistency constraints. These constraints are subsequently incorporated into the neural network's loss function.

To begin, it is essential to calibrate the parameters for each sensor within the traffic network. The primary objective of this calibration process is to determine a corresponding set of physical parameters for each sensor using a mathematical model. Specifically, by solving model (13)–(15), we can derive the optimal set of physical parameters for each sensor, where $v_{f,i}$ represents the free flow speed of the ith sensor and $k_{c,i}$ denotes the critical density. Given the superior fitting performance of the Underwood model on the urban traffic dataset (as shown in Table A1), this study adopts the Underwood model to characterize the fundamental relationship between traffic speed and flow (Underwood, 1960). Consequently, the problem of parameter calibration for the ith sensor is defined as follows:

$$\min_{(v_{f,i},k_{c,i})} = \sum_{i=1}^{N} \left(v_i - {}_{\mathrm{phy}}\left(q_i/v_i\right)\right)^2 \tag{13}$$

Subject to:

$$v_{\mathrm{phy}}\left(q_i/v_i\right) = v_{f,i}\exp\left(-q_i/v_i k_{c,i}\right), i = 1, 2, \ldots, N \tag{14}$$

$$v_{f,i} > 0, k_{c,i} > 0, i = 1, 2, \ldots, N \tag{15}$$

By solving the aforementioned model, the optimal parameters $(v_{f,i}, k_{c,i})$ for each sensor i are derived, thereby establishing the physical parameter set. This parameter set encapsulates the traffic flow characteristics specific to each sensor. The relationship between traffic flow and speed is incorporated into the model as part of the loss function, acting as a soft physical constraint. This integration guides the model towards generating predictions that are consistent with physical principles. The physical loss is defined based on Equation (16) and expressed as follows:

$$L_{\mathrm{phy}} = \frac{1}{N}\sum_{i=1}^{N}\left|v_i - v_{f,i}\exp\left(-q_i/v_i k_{c,i}\right)\right|^2 \tag{16}$$

3.5 | Combined loss function

The optimization of the B-PI-DCRNN model is achieved by integrating the data-driven methods with physical prior knowledge. This approach ensures that the model not only fits the observed data but also adheres to the established physical laws of traffic flow. The update process of the parameters is guided by the minimization of a composite loss function. This loss function is designed to incorporate both data error terms and physical consistency constraints. The formulation of the total loss function is as follows:

$$L_{\mathrm{total}}(\theta) = c_a L_{\mathrm{data}} + c_b L_{\mathrm{phy}} \tag{17}$$

where θ represents the set of neural network parameters, c_a and c_b serve as weighting factors that balance the data-driven loss term L_{data} and the loss term L_{phy} associated with physical constraints. The weight of physical loss needs to be smaller than the data-driven loss, as overly strong physical constraints can sometimes hinder the model's ability to fit observed data (Wang et al., 2024). In this study, we set the coefficients c_a and c_b to 0.7 and 0.3, respectively.

4 | CASE STUDIES

4.1 | Data description

This case study employs a real-world dataset, initially released by Wang et al. (2023), collected from automated vehicle identification detectors deployed throughout the urban area of Xuancheng city, China. This dataset, which has been adopted in TSP studies (He et al., 2025; Zhang & Li, 2025), spans a 30-day period from September 1 to September 30, 2020. It comprises a 5-min aggregated traffic flow and speed measurements collected from a sensor network of 100 nodes as illustrated in Figure 4. These detectors cover over 97% of the road network, ensuring a comprehensive acquisition of traffic flow and speed data. The dataset also includes the information of

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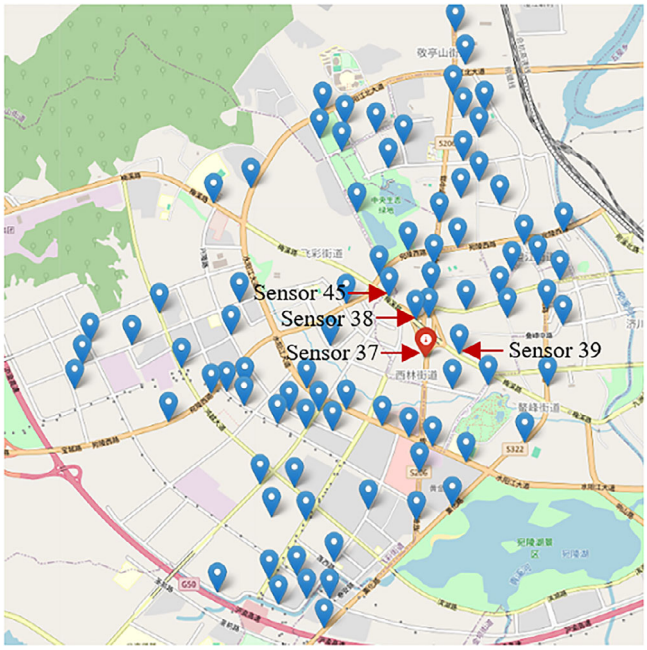

FIGURE 4 Spatial distribution of study area with sensor locations.

the geographic coordinates of each sensor and the length of the corresponding road segment. Hence, we can model the spatio-temporal relationship of segment traffic state within the urban network.

4.2 | Experimental design and data processing

To enhance prediction accuracy, this study leverages the DCRNN model to capture both temporal dynamics and spatial dependencies inherent within the sensor network. Specifically, spatial relationships are modeled through a spatial correlation matrix, constructed using a Gaussian kernel based on pairwise Euclidean distances between sensors (Wu et al., 2021), thereby facilitating a more realistic representation of spatial interactions. The DCRNN is configured with the following hyperparameters. The Adam optimizer, with an initial learning rate of 0.003, is employed for training. The model utilizes a batch size of 64 and incorporates two diffusion steps. The architecture comprises two RNN layers, each containing 64 units, with the dual random walk serving as the filter type. The model is trained over 75 epochs, incorporating a weight decay coefficient of 1×10^{-6} and a dropout rate of 0.2. The model uses the past 12 time steps of time series data (representing 60-mine interval at 5-min resolution) to forecast the traffic state for subsequent 12 time steps. In our experiment, we used the first 80% of the data for training and validation and the remaining 20% for testing.

We observed frequent instances of zero-flow conditions on low-demand segments during off-peak periods, such as early mornings and late nights. Under these conditions, sensor-reported speed was often recorded as zero values in the dataset. To ensure physical realism in these scenarios, we replaced zero-speed values occurring under zero-flow conditions with the segment-specific calibrated free-flow speed. In addition, missing data and outliers were filled using the historical average imputation and linear interpolation (He et al., 2025; Tang et al., 2017).

Within the unified framework, we compare two strategies for uncertainty quantification in TSP that either incorporate or exclude physical prior knowledge, referred to as B-PI-DCRNN and B-DCRNN. Specifically, B-DCRNN is implemented by setting the physical loss weight to 0 and the data loss weight to 1, while maintaining the same overall architecture as B-PI-DCRNN. All models are executed on an Intel(R) Core i5-12600KF CPU and an NVIDIA RTX 4070 Ti GPU, utilizing the PyTorch framework. The performance of the models is assessed by examining accuracy and uncertainty. To comprehensively evaluate the obtained results, multiple statistical measures are employed. The overall error of the method is quantified using the root mean square error (RMSE; Wang et al., 2024). To detect any systematic under- or over-prediction, the mean absolute percentage error (MAPE) is utilized (Antoniou et al., 2013). Furthermore, the reliability of the prediction intervals is gauged through the prediction interval coverage probability (PICP) and the mean prediction interval width (MPIW). The indexes of PICP and MPIW both reflect the uncertainty in TSP (Sengupta et al., 2024).

4.3 | Results and discussion

4.3.1 | Performance evaluation

Table 2 summarizes the key performance metrics for traffic flow and speed predictions obtained from five independent runs of both B-DCRNN and B-PI-DCRNN under identical experimental conditions. The mean values and standard deviations are reported to ensure the robustness of the results. Compared to B-DCRNN, B-PI-DCRNN consistently demonstrates superior performance in both flow and speed predictions. Specifically, for flow prediction, B-PI-DCRNN achieves a 21.2% reduction in RMSE and a 9.0% reduction in MAPE, alongside a 34.0% decrease in EU, indicating enhanced accuracy and reliability. Similarly, for speed prediction, RMSE and MAPE are reduced by 13.8% and 17.4%, respectively, while EU and AU decrease by 17.8% and 13.6%. This enhanced performance can be attributed to the incorporation of physical constraints, which prevent the model from generating physically implausible

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TABLE 2 Prediction performance comparison of TSP.

Metric	Flow (veh/5 min)		Speed (km/h)	
	B-DCRNN	B-PI-DCRNN	B-DCRNN	B-PI-DCRNN
Root mean square error (RMSE)	5.2 ± 0.2	4.1 ± 0.1	5.8 ± 0.2	5.0 ± 0.2
Mean absolute percentage error (MAPE; %)	23.3 ± 0.7	21.2 ± 0.4	13.2 ± 0.9	10.9 ± 1.0
Prediction interval coverage probability (PICP; %)	98.5 ± 0.2	99.2 ± 0.1	95.6 ± 0.2	95.1 ± 0.2
Mean prediction interval width (MPIW)	33.0 ± 0.3	31.9 ± 0.4	38.9 ± 0.6	33.6 ± 0.7
Epistemic uncertainty (EU)	5.3 ± 0.3	3.5 ± 0.3	4.5 ± 0.5	3.7 ± 0.5
Aleatoric uncertainty (AU)	22.9 ± 0.3	22.5 ± 0.3	27.2 ± 0.3	23.5 ± 0.4

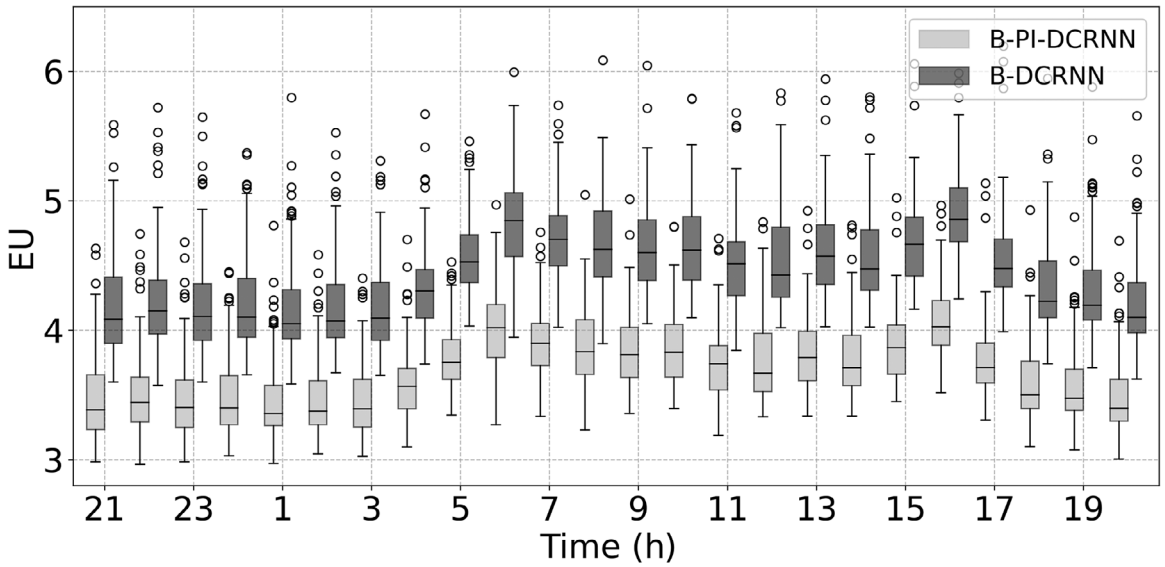

FIGURE 5 EU across models over time.

predictions, thereby mitigating systematic errors and enhancing predictive accuracy. Furthermore, the inclusion of physical priors derived from the fundamental diagram serves as an additional source of information, effectively mitigating overfitting, particularly in data-sparse or noisy regions. Consequently, B-PI-DCRNN demonstrates narrower MPIW coupled with higher PICP in both prediction tasks, signifying more reliable predictive intervals. By more effectively capturing the underlying traffic dynamics, the integration of such constraints reduces model uncertainty and ultimately improves overall prediction performance. In addition, Table A2 presents a comparative evaluation of traffic flow prediction performance against state-of-the-art uncertainty quantification methods for TSP on the public PeMS04 high-way dataset, which further demonstrates the superiority and effectiveness of the proposed approach on a larger-scale traffic network.

Using Equations (10) and (11), we calculate AU and EU, respectively. Figures 5 and 6 provide a comparative analysis of these uncertainties for B-DCRNN and B-PI-DCRNN across various hours of the day. In these figures, the median of the distribution is represented by the horizontal line in the middle of each box, while the ends of the boxes indicate the 25% (bottom) and 75% (top) quantiles of the distribution. Figure 5 reveals that B-PI-DCRNN consistently exhibits lower EU, compared to B-DCRNN throughout the day. This observation further highlights the effectiveness of incorporating physical constraints and multi-variable predictions in reducing EU. Notably, both models exhibit increased EU during peak traffic hours, with B-DCRNN experiencing a more pronounced rise. This pattern suggests that the complexity and challenges of accurately capturing traffic dynamics are heightened during peak times. The reduced fluctuations in EU for B-PI-DCRNN

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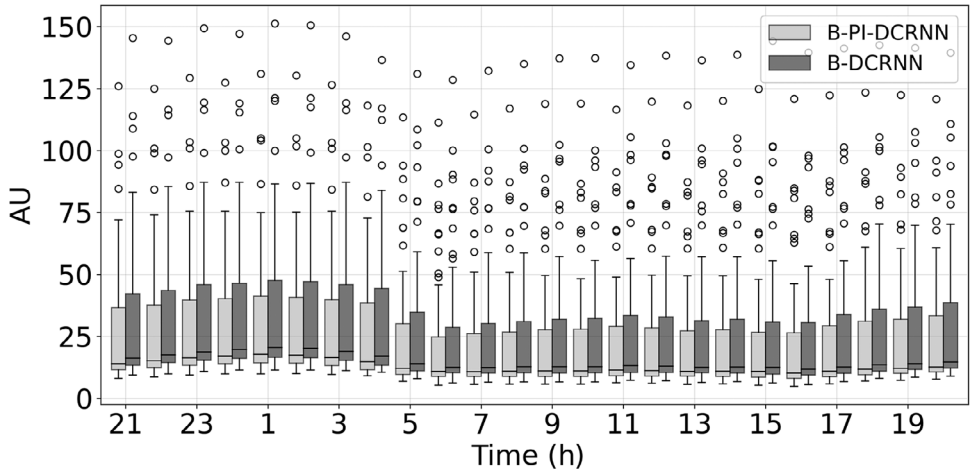

FIGURE 6 AU across models over time.

TABLE 3 Prediction performance comparison of weekdays and weekends.

	Flow (veh/5 min)		Speed (km/h)	
Metric	**Weekdays**	**Weekends**	**Weekdays**	**Weekends**
RMSE	5.86	5.17	4.58	4.14
MAPE (%)	23.38	23.5	21.23	21.53
PICP (%)	97.71	98.62	99.02	99.32
MPIW	33.18	32.98	32.08	31.89
EU	5.39	5.34	3.53	3.49
AU	23.05	22.92	22.63	22.5

imply its superior capability in managing these variations by leveraging the physical relationship between flow and speed. Figure 6 illustrates the AU, representing the inherent randomness present in traffic data. This randomness is attributable to various factors, including variations in driver behavior and the influence of external disturbances on traffic flow. The analysis indicates that AU exhibits a consistently high level across different hours of the day, indicating persistent noise in traffic conditions. Although B-PI-DCRNN also encounters increased AU, it consistently exhibits narrower uncertainty bounds, compared to B-DCRNN. This reduction in AU suggests that B-PI-DCRNN is more adept at distinguishing between signal and noise within the traffic data.

Table 3 compares the B-PI-DCRNN framework's predictive performance on weekdays and weekends. To improve the model's capacity to accommodate variations in traffic patterns between these distinct periods, temporal features differentiating days of the week are incorporated into the feature vector. Empirical results indicate that B-PI-DCRNN achieves comparable predictive accuracy and uncertainty quantification in both weekday and weekend scenarios. These findings support the temporal generalization capability of the B-PI-DCRNN framework, demonstrating its effectiveness in providing reliable predictions and maintaining appropriate uncertainty estimates across the fluctuating traffic conditions that typically characterize weekdays and weekends in traffic networks. Additionally, the computational cost performance metrics for both B-PI-DCRNN and B-DCRNN across each dataset are presented in Table A3.

4.3.2 | Prediction and uncertainty analysis at Sensor 37

Figures 7 and 8 illustrate the comparison between the B-DCRNN and the B-PI-DCRNN, showcasing flow and speed prediction results along with quantified uncertainties at Sensor 37 over a 24-h prediction horizon. Sensor 37 is selected for performance evaluation due to its location near major traffic intersections results in complex spatiotemporal dynamics as illustrated in Figure 4. This area is known for its highly volatile traffic patterns, which are driven by frequent and unpredictable variations in vehicle movement (Chand, 2021). Consequently, it serves as an ideal location for assessing the performance of TSP models. Such dynamics introduce significant aleatoric and epistemic uncertainties into traffic predictions. The B-DCRNN exhibits larger prediction errors and wider uncertainty during periods of high traffic, a consequence of its purely data-driven nature. Although the B-DCRNN effectively captures spatiotemporal correlations under regular traffic conditions, it struggles to generalize during peak or congested periods. These periods are characterized by abrupt fluctuations in flow and speed that introduce strong nonlinearities. Moreover, historical data from these periods tends to be sparse or noisy, further hindering the model's performance. Consequently, the B-DCRNN's ability to

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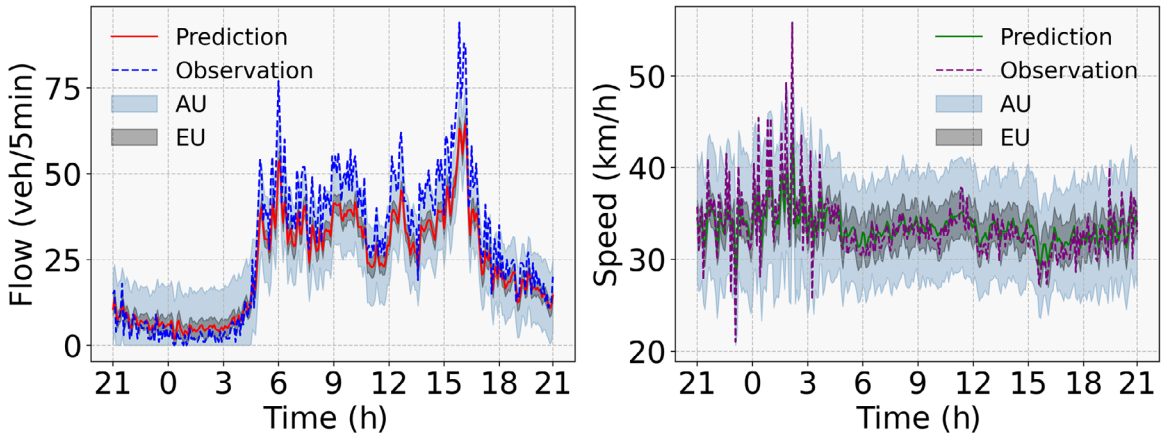

FIGURE 7 Flow and speed prediction by B-DCRNN versus observation at Sensor 37.

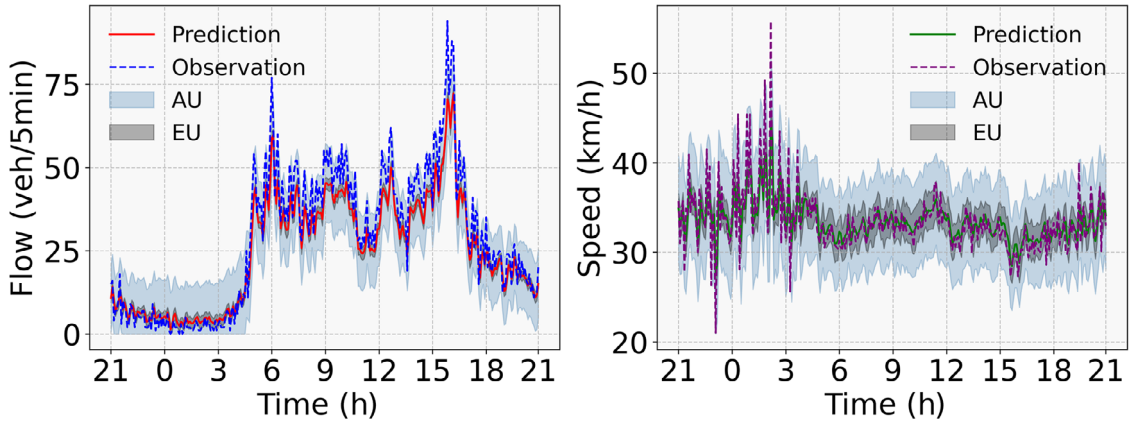

FIGURE 8 Flow and speed prediction by B-PI-DCRNN versus observation at Sensor 37.

TABLE 4 Prediction performance at Sensor 37.

	Flow (veh/5 min)		Speed (km/h)	
Metric	**B- DCRNN**	**B-PI-DCRNN**	**B-DCRNN**	**B-PI-DCRNN**
RMSE	9.4	6.9	2.4	2.3
PICP (%)	**91.8**	**96.8**	**98.7**	**99.7**
MPIW	32.2	31.2	19.4	16.7
AU	**22.4**	**22.0**	**13.1**	**11.4**
EU	5.3	3.5	4.4	3.6

accurately predict and quantify uncertainty diminishes in these complex traffic scenarios. Conversely, the B-PI-DCRNN integrates physical constraints derived from the fundamental diagram of traffic flow. By embedding the nonlinear flow–density relationship as prior knowledge, the model is biased toward physically plausible predictions, even when data are limited or highly variable.

Table 4 provides a comparative analysis of key performance indicators for flow and speed prediction at Sensor 37, contrasting B-DCRNN and B-PI-DCRNN. The results demonstrate that B-PI-DCRNN significantly outperforms B-DCRNN in both overall prediction accuracy and uncertainty quantification. Considering flow prediction, B-PI-DCRNN achieves a 26.6% reduction in RMSE and a 34.0% reduction in EU, compared to B-DCRNN. Although the decrease in AU is marginal, the PICP improves by 5.2%, accompanied by a narrower MPIW. This indicates the generation of sharper and more reliable prediction intervals. Regarding speed prediction, B-PI-DCRNN reduces RMSE by 4.2%, AU by 13.0%, and EU by 18.2%. Notably, the PICP increases by 1.0% while the MPIW decreases by 13.9%, thereby emphasizing enhanced reliability of the predictive intervals, even under conditions of high variability. In summary, these findings suggest that the incorporation

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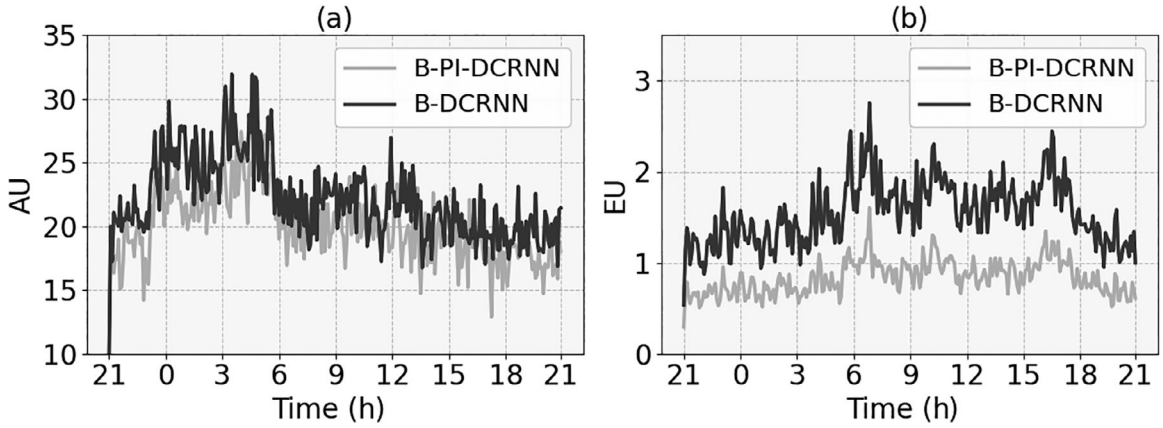

FIGURE 9 Aleatoric and epistemic uncertainties at Sensor 37.

of physical constraints enables B-PI-DCRNN to maintain robustness and precision, even at highly dynamic locations, effectively diminishing both error and uncertainty while simultaneously improving the reliability of the prediction intervals.

Figure 9 compares the AU and EU of B-DCRNN and B-PI-DCRNN at Sensor 37. In Figure 9a, both models demonstrate relatively high AU, indicating the inherent variability and noise present in the traffic data at this sensor. However, it is noteworthy that the B-PI-DCRNN model consistently exhibits lower AU than the B-DCRNN model. This observation suggests that B-PI-DCRNN has a superior ability to capture the stochastic nature of traffic flow, thereby enabling it to model the inherent randomness in the data more effectively. Figure 9b illustrates the variation of EU over time. It is particularly significant that during traffic peak periods, the EU is markedly higher. During these periods, the EU of the B-DCRNN is elevated and exhibits considerable fluctuation. In contrast, the B-PI-DCRNN effectively reduces EU throughout the prediction horizon by learning the specific characteristics of the traffic flow at this sensor.

Overall, the findings demonstrate that the hybrid approach, which integrates data-driven learning with physical modeling, significantly improves both predictive accuracy and uncertainty quantification. The B-PI-DCRNN model, by effectively distinguishing between inherent noise and knowledge gaps, provides more reliable and interpretable traffic forecasts. This capability makes the B-PI-DCRNN particularly well-suited for application in complex traffic environments.

4.3.3 | Robustness evaluation

Figures 10–12 illustrate the impact of Gaussian noise, periodic noise, and burst noise, on the prediction performance of both B-DCRNN and B-PI-DCRNN for traffic flow and speed. To evaluate the robustness of these models, three common noise types encountered in real-world traffic data were introduced into the input features. These noise conditions are represented as $F\prime = F + w * N$, where $w \in [0, 1]$ represents the noise level, and N denotes the noise type. The results indicate that increasing noise levels lead to a rise in RMSE, EU, and AU for both models. Notably, under all three noise conditions, B-DCRNN exhibits higher RMSE, greater total uncertainty, and a faster growth rate, compared to B-PI-DCRNN, demonstrating B-PI-DCRNN's superior noise robustness. Specifically, in terms of AU, both models exhibit an increase as noise levels rise. However, B-PI-DCRNN shows a slower rate of increase, compared to B-DCRNN, indicating its superior ability to handle inherent noise. Regarding EU, B-DCRNN demonstrates a more significant rise, particularly at elevated noise levels, suggesting a greater challenge in generalizing under such conditions. Conversely, the incorporation of physical prior knowledge in B-PI-DCRNN helps stabilize the growth of EU, reflecting an improved comprehension of the system's dynamics despite the presence of noise.

Nevertheless, the sensitivity of B-PI-DCRNN to noise escalates with increasing noise levels, underscoring the necessity for enhanced robustness. Existing research suggests that data preprocessing techniques can mitigate the impact of noise in traffic data. For instance, Chen et al. (2020) demonstrated that the Butterworth filter, wavelet filter, and moving average model effectively cleaned anomalies in traffic data and improved the prediction performance of artificial neural networks under noisy conditions. Consequently, incorporating lightweight denoising strategies into future uncertainty-aware frameworks holds significant potential for enhancing the robustness of B-PI-DCRNN.

Table 5 illustrates the sensitivity of model performance to different weight configurations between the physical

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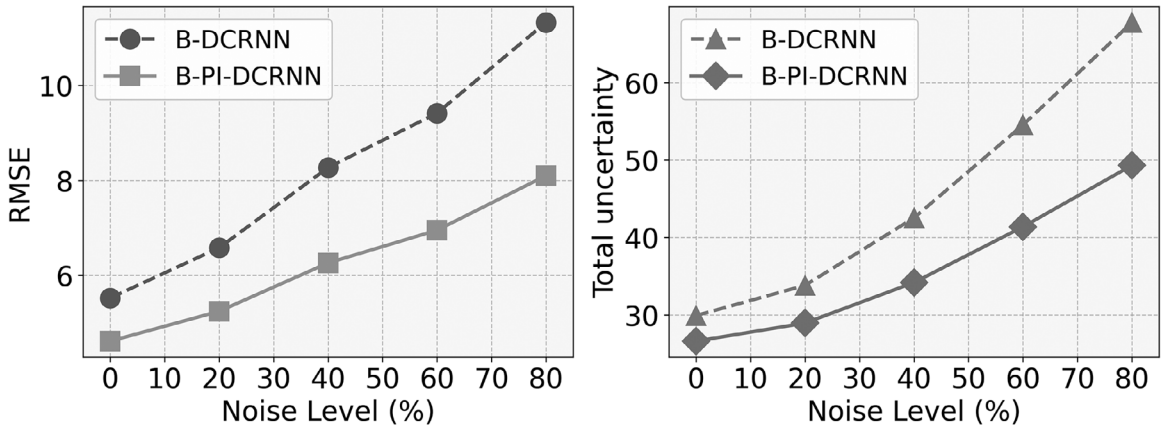

FIGURE 10 Comparison of prediction performance as increasing Gaussian noise levels. RMSE, root mean square error.

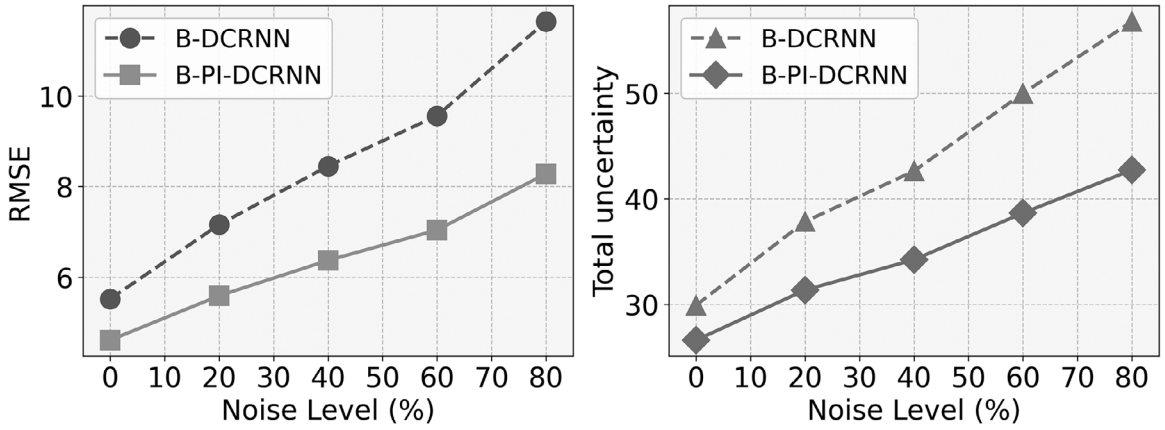

FIGURE 11 Comparison of prediction performance as increasing periodic noise levels. RMSE, root mean square error.

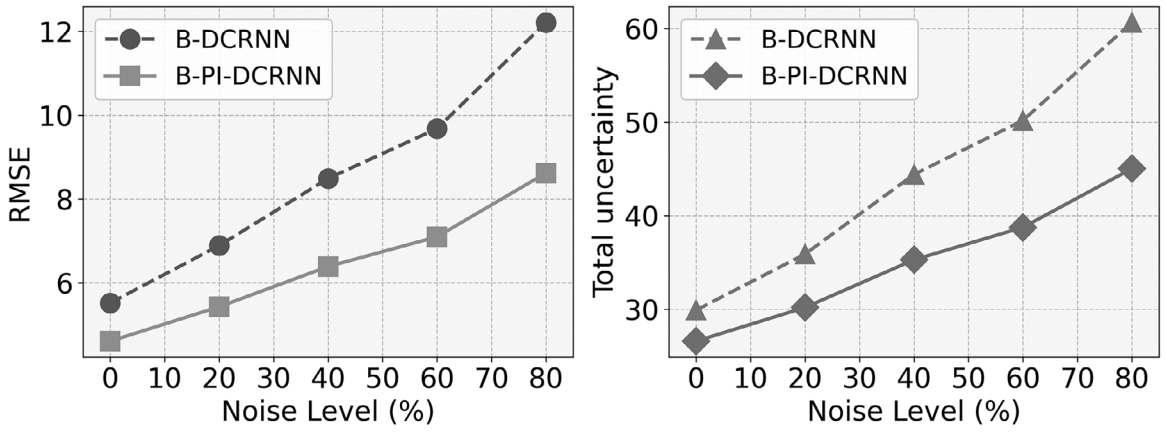

FIGURE 12 Comparison of prediction performance as increasing burst noise levels. RMSE, root mean square error.

loss weight and the data loss weight. The results reveal that the optimal predictive accuracy is achieved at the configuration ($c_a = 0.7$, $c_b = 0.3$), where RMSE reaches its minimum value of 4.6. This finding suggests that moderate incorporation of physical priors creates a synergistic effect with data-driven learning, enhancing the model's capacity to capture underlying traffic dynamics without overly constraining the learning process. Moreover, both AU and EU demonstrate monotonic decreases as the physical loss weight increases. This uncertainty reduction

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TABLE 5 Impact of physical-data loss weight ratios on model performance.

(c_a, c_b)	RMSE	AU	EU
(0, 1)	5.5	25.1	4.9
(0.1, 0.9)	5.1	24.2	4.1
(0.2, 0.8)	4.7	23.7	3.8
(0.3, 0.7)	4.6	23.2	3.7
(0.4, 0.6)	5.0	23.0	3.6
(0.5, 0.5)	5.4	22.7	3.5

TABLE 6 Evaluation of prediction performance as the number of samples increases.

	Flow (veh/5 min)			Speed (km/h)		
Metric	RMSE	AU	EU	RMSE	AU	EU
10	5.4	29.2	4.6	6.5	30.6	4.8
20	4.6	24.6	3.8	5.6	26.1	4.1
50	4.2	22.6	3.5	5.0	23.3	3.7
100	4.1	22.2	3.5	5.0	23.2	3.7
500	4.1	22.2	3.5	5.0	23.1	3.7
1000	4.1	22.2	3.5	5.0	23.1	3.6

underscores the positive effect of physical knowledge, which reduces both data-inherent variability and model uncertainty. However, the analysis reveals a fundamental trade-off: While stronger physical constraints consistently reduce uncertainty, excessive emphasis on physical loss leads to degraded predictive accuracy. This phenomenon indicates that overly rigid adherence to physical priors may limit the model's flexibility to capture complex patterns that deviate from idealized physical assumptions.

In the context of complex datasets, the intricacy of underlying features often necessitates a thorough examination of the appropriate sample size to enhance prediction accuracy and model credibility. Table 6 illustrates the effects of increasing MC sample sizes on RMSE, AU, and EU in flow and speed predictions using the B-PI-DCRNN model. As the number of MC samples increases, the model exhibits reductions in both error and uncertainty, indicating improved prediction reliability. Notably, the reduction in RMSE, EU, and AU is particularly significant when the sample size increases from 10 to 50. This suggests that smaller sample sizes were insufficient to fully capture the complexities of the data. Beyond 50 samples, the decrease in these metrics becomes less pronounced, implying that the model's predictive performance has reached a satisfactory level. This finding indicates that a larger sample size enables the model to develop a more comprehensive understanding of the data and traffic characteristics, thereby reducing both epistemic and aleatoric uncertainties. The experimental results thus support the selection of 50 MC samples as a reasonable and effective parameter in this case, achieving a balance between computational efficiency and predictive accuracy.

4.3.4 | Uncertainty in traffic dynamics

In this section, we explore the relationships between uncertainty and traffic dynamics. To achieve a comprehensive understanding of traffic behavior at a specific location, it is essential to assess traffic dynamics through a weighted combination of traffic flow and speed. We introduce a composite traffic dynamics indicator D, which is defined as a weighted linear combination of flow and speed:

$$D = w_q \frac{\sigma_q}{\mu_q} + w_v \frac{\sigma_v}{\mu_v} \tag{18}$$

where w_q and w_v are the weights assigned to the flow and speed components, respectively, reflecting their relative importance in the analysis. These weights can be adjusted based on specific application requirements, such as prioritizing speed dynamics in urban settings or flow dynamics on highways. The terms σ_q/μ_q and σ_v/μ_v represent the coefficients of variation (CV) for flow and speed, respectively, where σ denotes the standard deviation and μ denotes the mean of the corresponding time series data.

Following the calculation of D, we group the indicator values into four bins to classify sensor locations based on their overall traffic dynamics: $0 < D < 0.3$ (low variability), $0.3 < D < 0.5$ (moderate variability), $0.5 < D < 0.7$ (high variability), and $0.7 < D$ (very high variability). The binning approach facilitates a spatially detailed analysis by enabling the identification of locations characterized by varying levels of composite traffic dynamics. Specifically, lower CV values is indicative of stable and predictable traffic conditions, typically observed in free-flow situations. Conversely, higher CV values signify highly volatile conditions, frequently associated with transitions between different traffic states or the formation and dissipation of congestion.

Figures 13–15 illustrate the distribution of total uncertainty across varying levels of traffic dynamics, examined under three distinct weighting configurations: flow-dominant ($w_q = 1, w_v = 0$), equal weights ($w_q = 0.5, w_v = 0.5$), and speed-dominant ($w_q = 0,\ w_v = 1$). The results consistently demonstrate that total uncertainty increases in tandem with heightened traffic dynamics, irrespective of the weighting strategy employed. This observation aligns with established traffic flow theory, which posits that volatile traffic conditions are inherently less pre-

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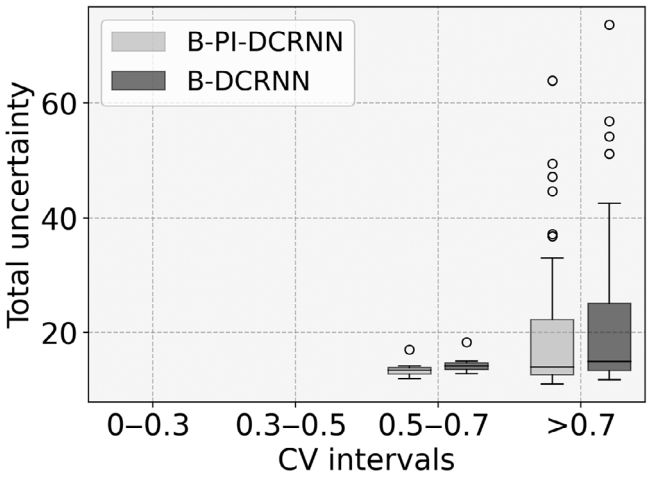

FIGURE 13 Total uncertainty distribution at different levels of traffic dynamics ($w_q = 1$, $w_v = 0$). CV, coefficients of variation.

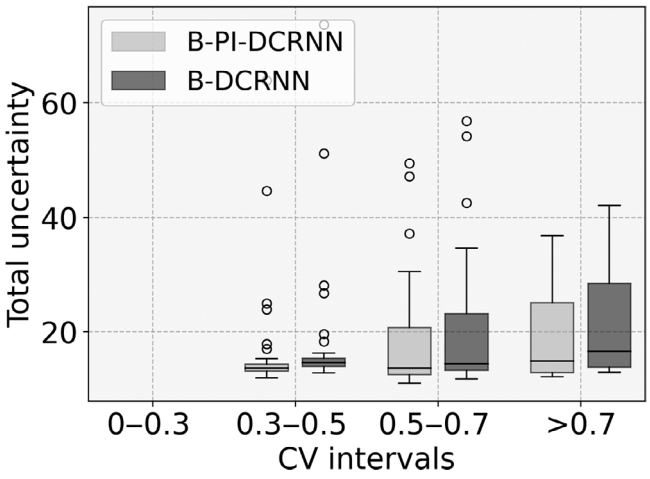

FIGURE 14 Total uncertainty distribution at different levels of traffic dynamics ($w_q = 0.5$, $w_v = 0.5$).

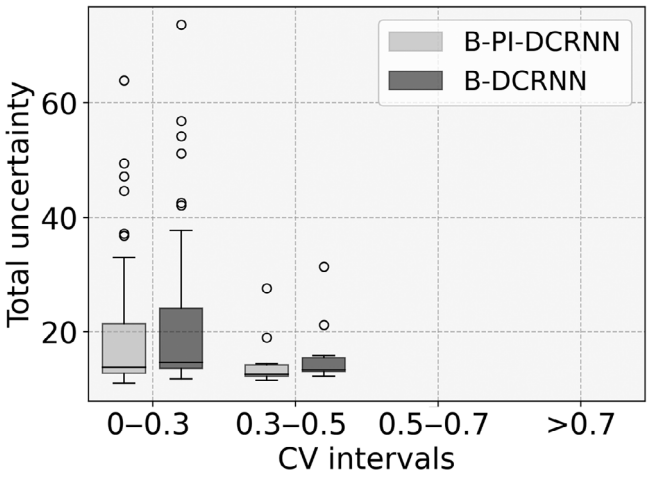

FIGURE 15 Total uncertainty distribution at different levels of traffic dynamics ($w_q = 0$, $w_v = 1$).

dictable than stable states. Moreover, results reveal that flow variation plays a more significant role in characterizing traffic dynamics within the urban traffic dataset. Notably, across all traffic dynamics levels, B-PI-DCRNN consistently exhibits lower 25%, median, and 75% quantiles of total uncertainty, compared to B-DCRNN. These findings suggest that B-PI-DCRNN possesses greater robustness and reliability in managing fluctuating traffic conditions.

4.3.5 | Uncertainty propagation

Our framework employs the DCRNN to model spatial dependencies in the traffic network. This mechanism inherently propagates both traffic state features (flow, speed) and predictive uncertainties, potentially leading to cascading prediction errors or uncertainty amplification. To evaluate the framework's capacity to regulate this uncertainty propagation, we conducted a targeted analysis focusing on three adjacent sensors (45, 38, and 39) situated along a major arterial road as depicted in Figure 4.

Table 7 presents the prediction performance for traffic flow and speed at three adjacent sensors, along with their corresponding aleatoric and EU estimates as generated by the B-PI-DCRNN framework. The results indicate that elevated prediction errors and uncertainty at specific sensor locations do not lead to cascading prediction errors or uncertainty amplification in our approach. For instance, despite sensor 45 exhibiting relatively high prediction errors and uncertainty in flow prediction, no discernible degradation in prediction accuracy or uncertainty amplification is observed at its neighboring sensors, 38 and 39. A similar trend is seen in speed prediction: despite the higher error at sensor 38, the prediction errors and uncertainty levels remain stable at adjacent sensors. This robustness can be attributed in part to the physics-informed module embedded in B-PI-DCRNN, which enforces consistency with the fundamental flow-speed relationship. By serving as a soft constraint, this component regularizes the learning process and stabilizes the propagation of uncertainty across the graph. Overall, these findings indicate that the

TABLE 7 Prediction performance for adjacent Sensors 45, 38, and 39.

	Flow (veh/5 min)			Speed (km/h)		
Metric	**45**	**38**	**39**	**45**	**38**	**39**
RMSE	4.6	4.3	7.2	2.5	5.0	1.6
PICP (%)	**96.2**	**97.6**	**92.4**	**99.3**	**88.4**	**99.6**
MPIW	19.5	19.1	22.8	14.5	10.9	12.2
EU	**3.5**	**3.3**	**3.6**	**3.5**	**3.5**	**3.7**
AU	19.2	18.9	22.6	14.0	10.2	11.5

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B-PI-DCRNN framework effectively supports robust and stable uncertainty propagation across the spatial graph structure, mitigating the risk of cascading predictive errors.

4.4 | Applications of uncertainty quantification in TSP

Quantifying uncertainty in TSP offers substantial practical value for real-world traffic management and control, providing quantifiable support for confidence evaluation, resource allocation, and strategy selection within intelligent transportation systems.

First, uncertainty quantification serves as a key indicator of prediction confidence, enabling traffic authorities to identify spatiotemporal regions where model outputs may be unreliable. This identification is crucial for avoiding over-reliance on point predictions when making critical decisions. For example, in areas exhibiting elevated uncertainty, targeted interventions such as manual oversight, alert activation, or the application of conservative control strategies can be employed to enhance system stability. Conversely, in regions characterized by low uncertainty, more assertive strategies—such as dynamic speed regulation, signal timing adjustments, or lane assignment—can be implemented to improve operational efficiency.

Second, uncertainty quantification enhances the quality and utility of traffic information dissemination. Supplementing predicted travel times with confidence intervals empowers travelers to make more informed route choices, especially during adverse weather conditions or special events. This additional dimension transforms traditional point predictions into comprehensive decision support, better reflecting the dynamic and stochastic nature of traffic systems.

Third, the ability to distinguish between epistemic and AU facilitates the development of targeted traffic management strategies. Regions with high EU may necessitate sensor deployment optimization or model refinement to reduce reducible uncertainty. Conversely, areas exhibiting high AU may require the implementation of more robust stochastic optimization methods to account for inherent randomness. This differentiation enables traffic authorities to implement context-specific interventions tailored to the specific nature of the uncertainty.

5 | CONCLUSION

This paper introduces a novel Bayesian framework that integrates DCRNN and physical prior knowledge to address TSP, as well as aleatoric and EU in road networks. AU, stemming from inherent noise and imprecise data, is quantified through heteroskedasticity modeling. Meanwhile, EU, associated with the probabilistic parameters of the B-PI-DCRNN model, is quantified using MC dropout. By incorporating traffic flow laws from classical models as prior knowledge, B-PI-DCRNN model enhances the accuracy and reliability, thereby mitigating uncertainty in TSP. Experimental results indicate that the proposed model surpasses the baseline model in both prediction accuracy and uncertainty quantification across different noise levels and varying sample sizes. Moreover, the framework is highly generalizable and easily adaptable to diverse domains including energy systems, crowd dynamics, logistics, and environmental monitoring by substituting traffic flow theory with domain-specific knowledge. This integrated approach that combines both data-driven patterns and domain-specific insights is essential for achieving robust predictions.

While the B-PI-DCRNN framework effectively captures and mitigates both aleatoric and epistemic uncertainties in TSP, its current reliance on a deterministic fundamental diagram as prior physical knowledge constitutes a limitation. Specifically, this assumption, although analytically tractable, may not accurately reflect the inherent stochasticity observed in real-world traffic flow dynamics. To address this, future research will explore the incorporation of stochastic fundamental diagrams (Qu et al., 2017). By representing the flow–density relationship as a probabilistic distribution rather than a fixed functional form, the framework can better accommodate uncertainties stemming from diverse traffic conditions. Moreover, future research could integrate a stacked autoencoder into the framework to learn latent representations of event-specific features, potentially improving predictive performance under non-recurrent traffic conditions such as accidents or localized disruptions (Ren et al., 2022; Yu et al., 2017).

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APPENDIX

Table A1 presents the fitting performance of three classical traffic flow models—Greenshields, Greenberg, and Underwood—using the Xuancheng dataset. The results

TABLE A1 Fitting performance of different fundamental diagrams on the urban traffic dataset.

Metric	Underwood	Greenberg	Greenshields
R^2	0.9355	0.9343	0.9332
RMSE	**19.2811**	**19.7748**	**19.4368**
MAE	13.3659	13.8801	13.4794

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TABLE A2 Traffic flow prediction performance comparison of different uncertainty quantification methods on PeMS04.

Metric	Point	Quantile	MIS	Ensemble	B-DCRNN	B-PI-DCRNN
RMSE	38.1	29.6	29.7	30.7	25.8	22.3
MAPE (%)	17.1	15.5	13.6	15.5	15.5	13.6
PICP (%)	/	85.1	86.5	89.1	94.7	95.4
MPIW	/	78.3	78.6	88.7	89.4	82.2

TABLE A3 Computation time and memory cost on Xuancheng and PeMS04 datasets.

	Xuancheng		PeMS04	
Metric	**B-DCRNN**	**B-PI-DCRNN**	**B-DCRNN**	**B-PI-DCRNN**
Training time (s/epoch)	82.1	88.3	148.3	163.0
Inference time (s)	**10.9**	**11.0**	**20.2**	**20.3**
Memory cost (MB)	277.0	294.0	443.0	476.0

indicate that the Underwood model exhibits the best fit among those tested, as evidenced by its highest R^2 value and lowest RMSE and mean abslute error (MAE). Therefore, the Underwood fundamental diagram is adopted as the physical prior knowledge, reflecting its superior ability to represent the observed flow–density relationship within this dataset.

Table A2 presents a comparative evaluation of traffic flow prediction performance using state-of-the-art uncertainty quantification methods for TSP on the public PeMS04 highway dataset. This dataset, collected from the Caltrans Performance Measurement System, comprises data from 307 sensors aggregated at 5-min intervals. To benchmark our proposed B-PI-DCRNN framework, we select four representative uncertainty-aware methods from Table 1: Quantile (Wu et al., 2021), the mean interval score (MIS; Wu et al., 2021), an ensemble method (Mallick et al., 2024), and a Bayesian framework. Given that most existing studies focus on single-variable prediction tasks (e.g., traffic speed or flow), our experiments focus specifically on traffic flow prediction using the PeMS04 dataset. The results demonstrate that the B-PI-DCRNN framework consistently outperforms these advanced baselines in both point prediction accuracy and uncertainty quantification quality. Specifically, B-PI-DCRNN achieves a higher PICP while simultaneously maintaining a lower MPIW. These findings suggest that, in comparison to existing uncertainty-aware methods, B-PI-DCRNN provides improved accuracy, enhanced reliability, and greater interpretability for TSP.

Table A3 details the computational cost of the proposed B-PI-DCRNN and B-DCRNN models, evaluated on the Xuancheng and PeMS04 datasets. The metrics examined include training time, inference time, and memory cost. As shown, integrating physical priors into B-PI-DCRNN leads to a slight increase in training time and memory cost relative to B-DCRNN. Crucially, inference time remains nearly constant across a dataset. Both models achieve a per-step inference time of under 6.44 ms on both datasets, demonstrating efficient inference.

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